



AKTU

B.Tech Sem-6

CS IT & CS Allied

CN: Computer Networks

ONE SHOT Revision

Unit-2 Data Link Layer

ME EE Sem-6



By Dr. Nidhi Parashar Ma'am

- M.Tech Gold Medalist
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AKTU

B.Tech 6th Semester

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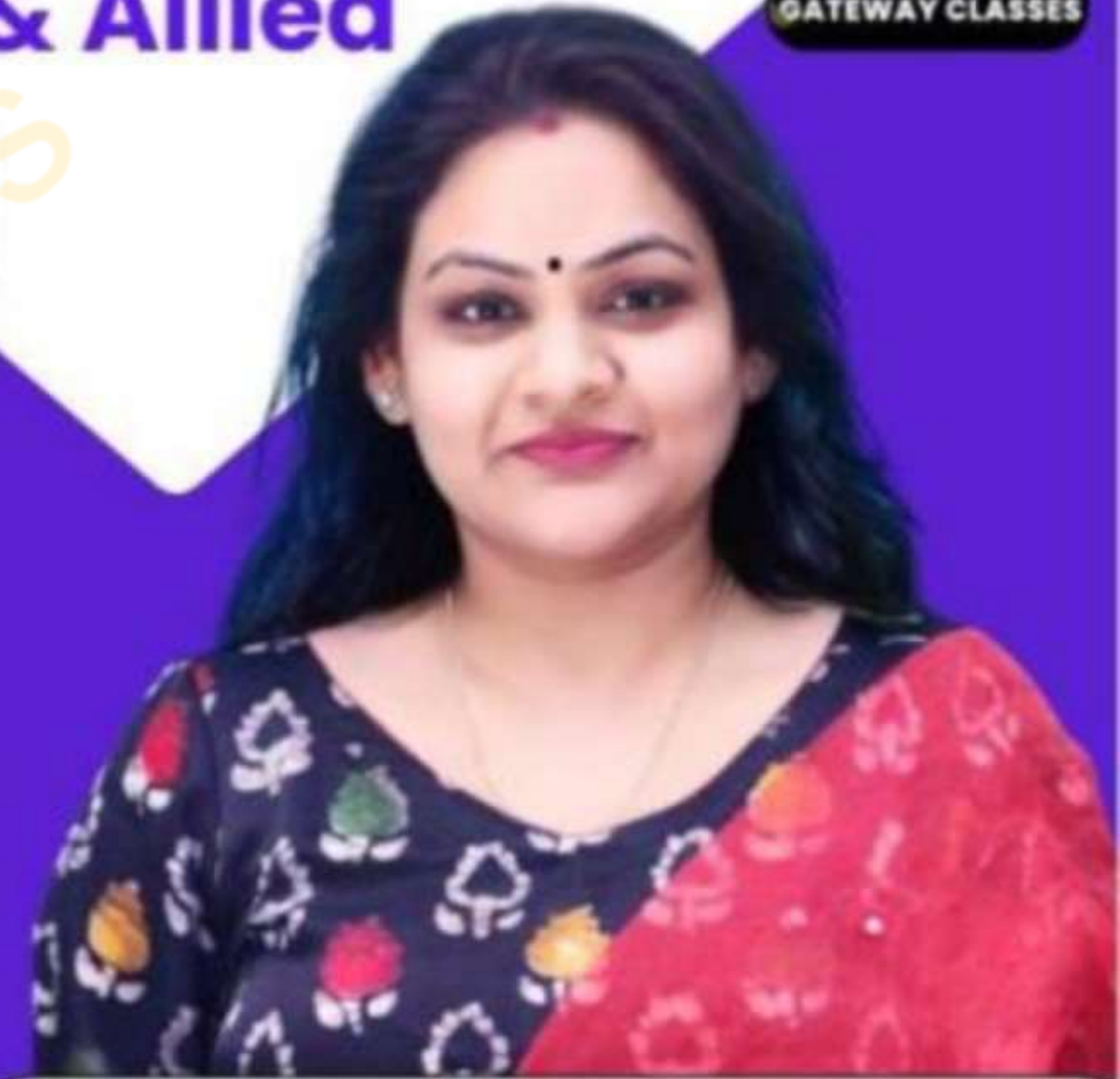


COMPUTER NETWORKS

Unit-2 Lecture-1

Today's Target

- Data Link layer Design Issues
- Framing
- Bit Stuffing
- Byte Stuffing
- AKTU PYQs

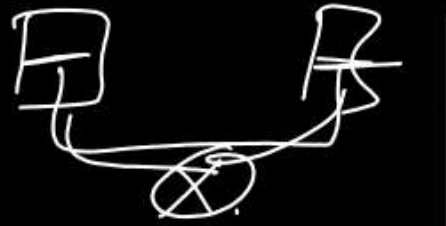
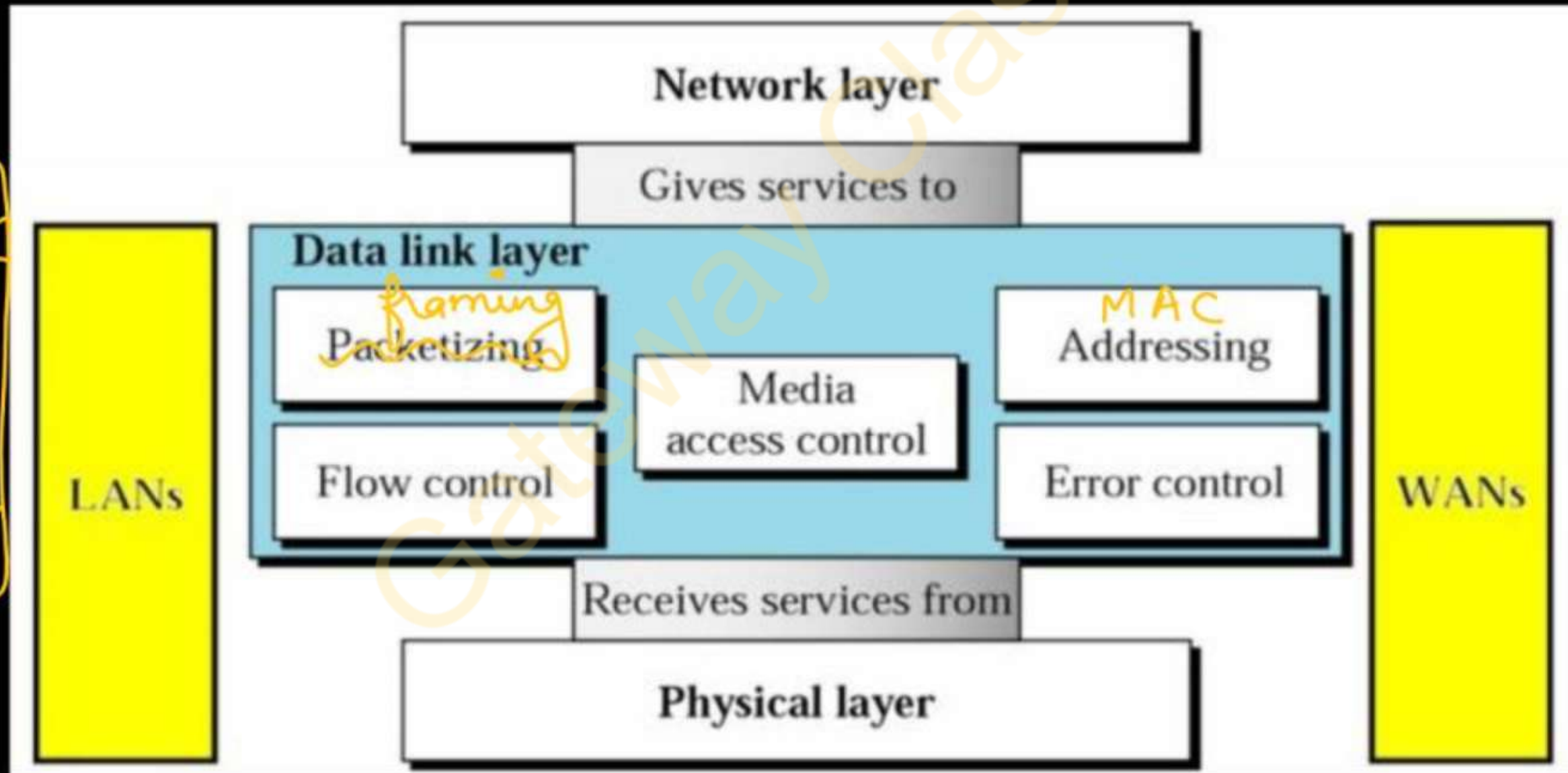


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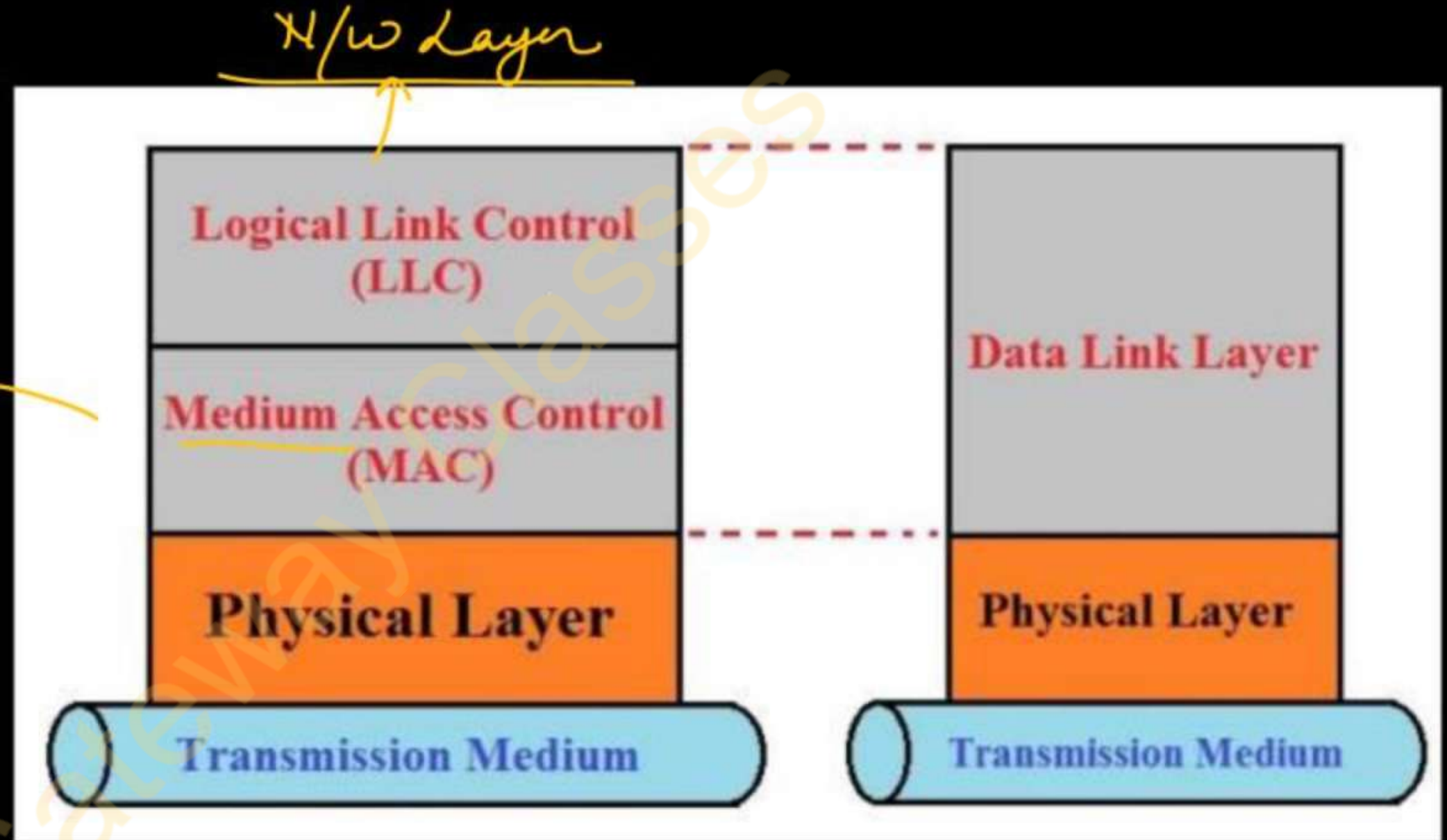
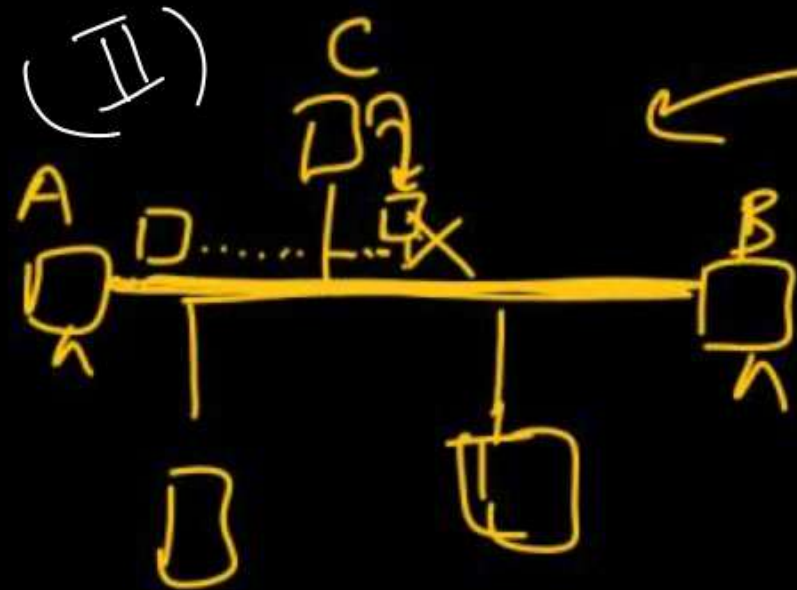
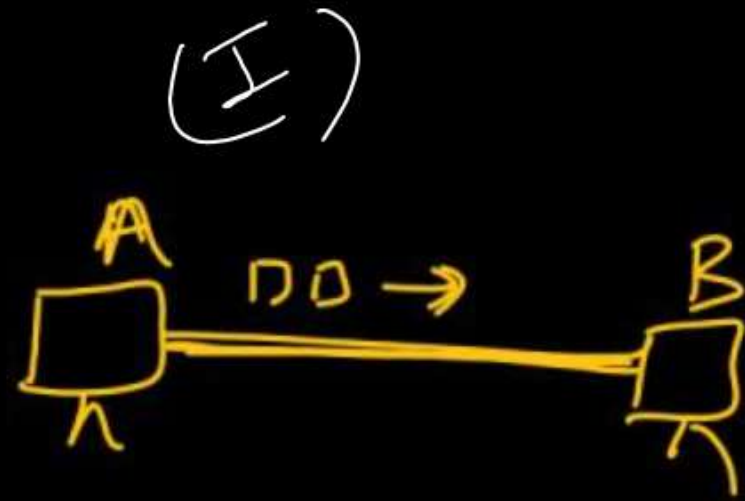
Data Link Layer

- The data link layer is a layer in the networking model that transforms the raw transmission facility (Hub, Repeater, Cables, Modem, and Connectors) provided by the physical layer into a reliable transmission link that is responsible for node-to-node or hop-to-hop communication.



Application
Transport
N/w
DLU
Physical

Data Link Layer Sublayers



Logical Link Control (LLC)

- Manages flow control, error checking, and synchronization.
- Ensures data packets are framed and transmitted correctly.
- Performs multiplexing and demultiplexing.
- Acknowledges received frames.
- Requests retransmission of lost data.
- Facilitates interaction between the data link layer and the network layer.

Media Access Control (MAC)

- Manages access to the physical transmission medium.
- Prevents data collisions.
- Encapsulates frames for transmission.
- Controls the transmission of data packets via remotely shared channels.
- Ensures data integrity by verifying the sender's frame check sequences.
- Strips off the sender's preamble and padding before passing the data upto the higher layers.
- Initiates retransmission if a jam signal is detected.

Design Issues of Data Link Layer(DLL) (AKTU)

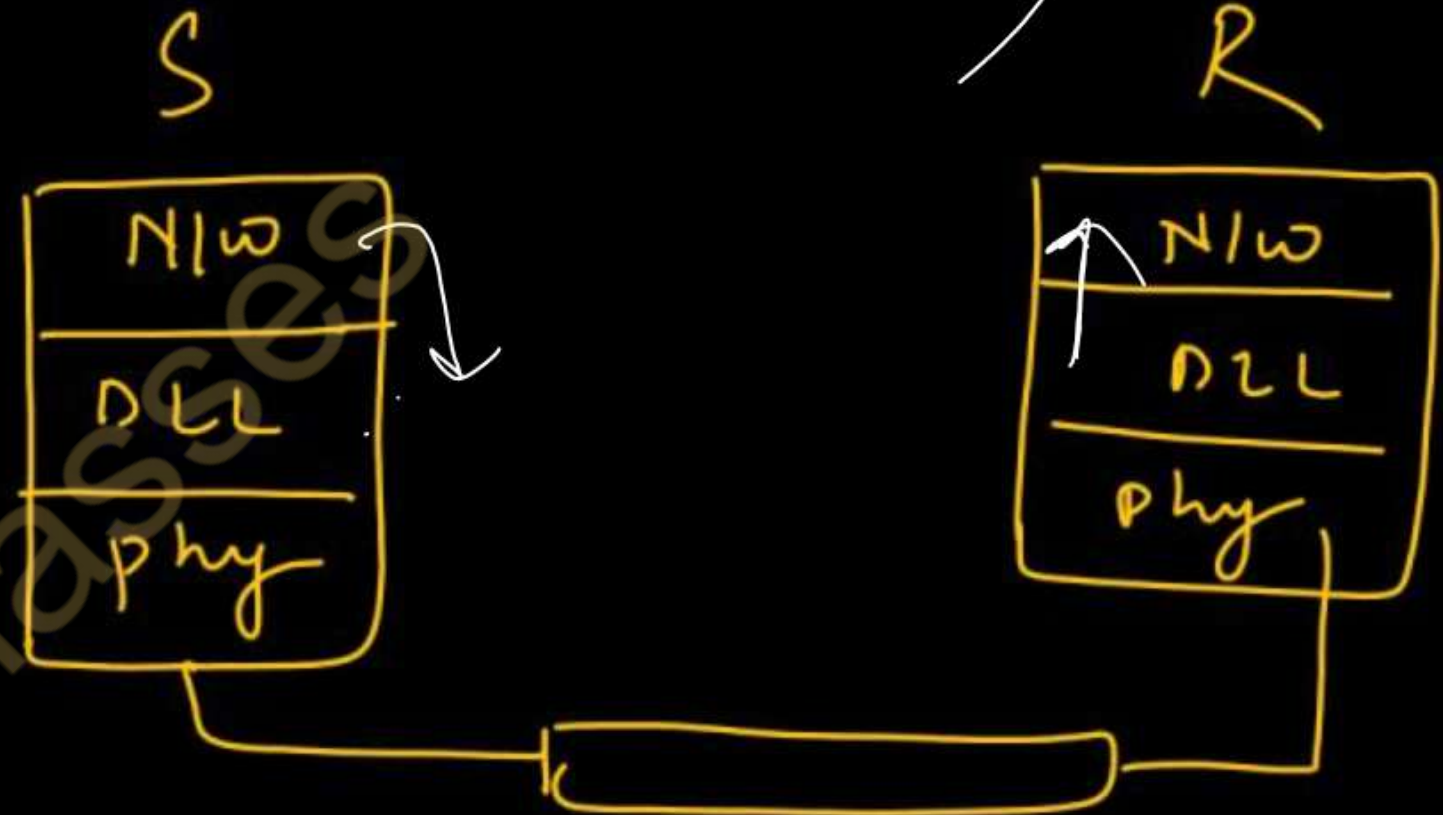
1. Service Provided to Network Layer

2. Framing

3. Error Control

4. Flow Control

5. Medium Access Control



1. Services Provided to Network Layer

- Transfer data from the source machine's network layer to the destination machine's network layer.
- The network layer at the source machine transfers packets to the data link layer. Now the data link layer transmits them to the data link layer at the destination machine. The destination machine's data link layer delivers them to the network layer at the destination machine.

Types of services Provided to Network Layer

1. Unacknowledged connectionless service (Example: **Ethernet**):

- No logical connection between Sender and receiver
- Source m/c delivers data frames to destination m/c without expecting acknowledgement from them.
- In case any frame is lost, the sender does not make any effort to detect or recover the lost frame.
- This kind of connection is appropriate where the error rate of the channel is minimal.

2. Acknowledged Connectionless Service (Example: **WiFi**):

- No logical connection between Sender and receiver.
- Receiver provides acknowledgement for received frames.
- If a frame is lost and the sender does not receive acknowledgement for the same within a specified time interval – the sender sends the frame again.

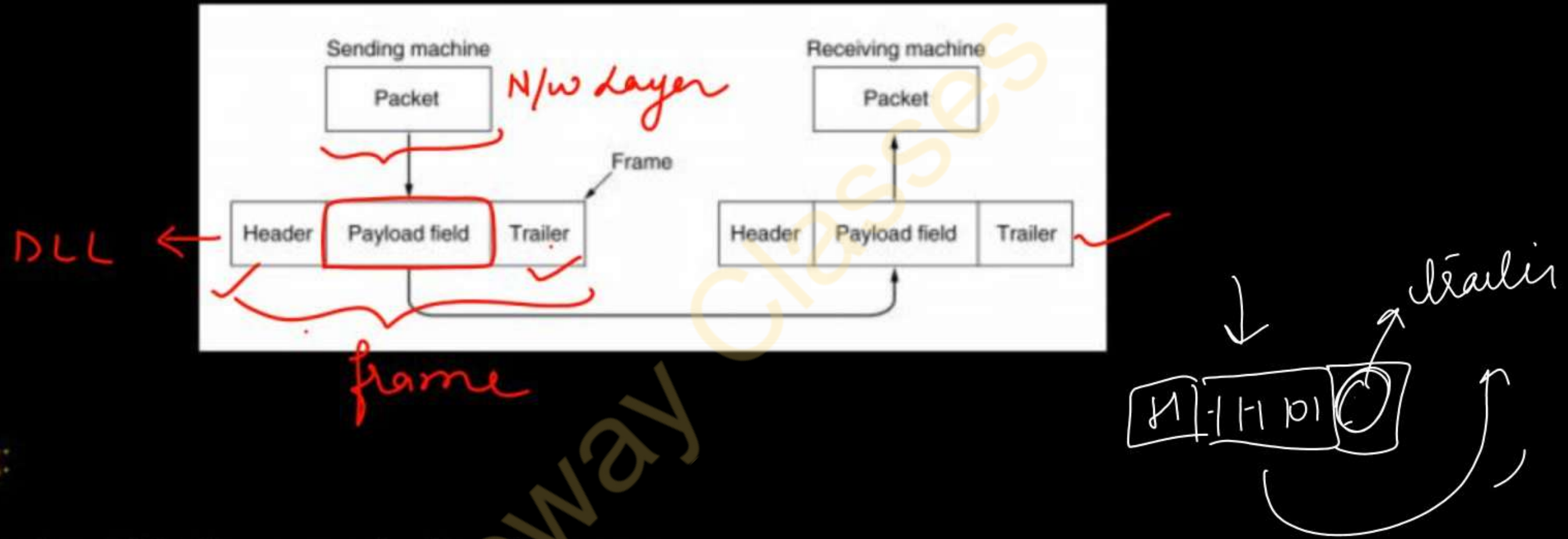
3. Acknowledged Connection-Oriented Service (Example: long distance telephone circuits):

- Sender and receiver establish a logical connection between them.
- Each outgoing frame is sequentially numbered, guaranteeing that each frame will be delivered only once and in the right order.
- Receiver send the acknowledgement after receiving the frame.
- Once all the frames get delivered, the sender and receiver release the connection between them.



2. Framing

DLL encapsulates each data packet from the network layer into frames that are then transmitted.



Why Framing:

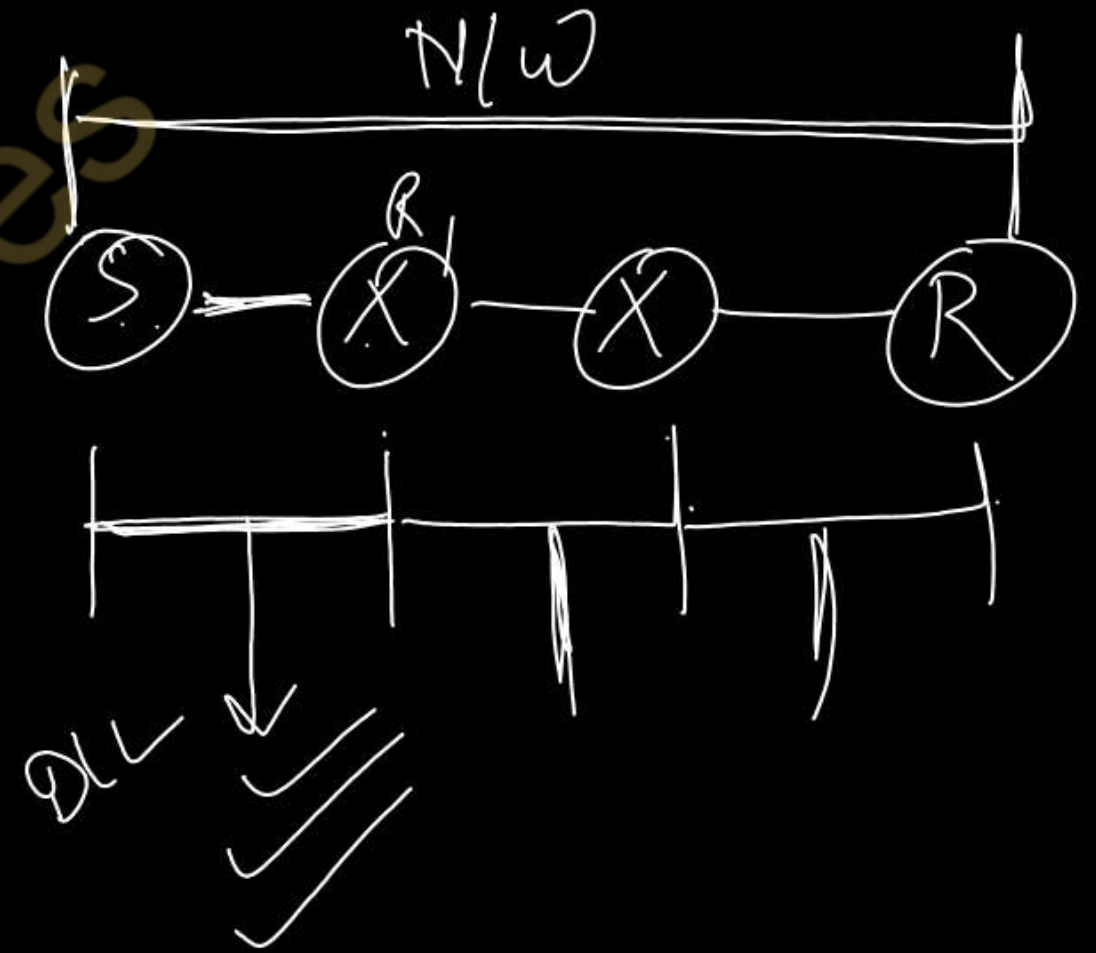
- To detect and rectify the errors in the bit stream.
- For each frame, DLL calculates the checksum and includes this checksum into the frame while transmitting it to the destination.
- When the frame arrives at the destination, the receiver recalculates this checksum, and if it comes out to be different, it indicates errors in the arrived frame.

3. Error Control

The data link layer ensures error free link for data transmission.

The issues it caters to with respect to error control are:

- Dealing with transmission errors ✓
- Sending acknowledgement frames in reliable connections ✓
- Retransmitting lost frames ✓
- Identifying duplicate frames and deleting them. ✓



4. Flow Control ✓

- DLL regulates flow control so that a fast sender does not drown a slow receiver.
- When the sender sends frames at very high speeds, a slow receiver may not be able to handle it.
- There will be frame losses even if the transmission is error-free.

4. Multiple Access Control

- Multiple Access Protocols are methods used in computer networks to control how data is transmitted when multiple devices are trying to communicate using a common link.
- These protocols ensure that data packets are sent and received efficiently, without collisions or interference.
- They help manage the network traffic so that all devices can share the communication channel smoothly and effectively.

Framing:

- DLL pack bits into frames, framing separates a message from one source to a destination, or from other messages to other destinations, by adding a sender address and a destination address.
- Packing whole message in one frame can make it very large, making flow and error control very inefficient.
- When a message is carried in one very large frame, even a single-bit error would require the retransmission of the whole message.
- When a message is divided into smaller frames, a single-bit error affects only that small frame.

Types of Framing



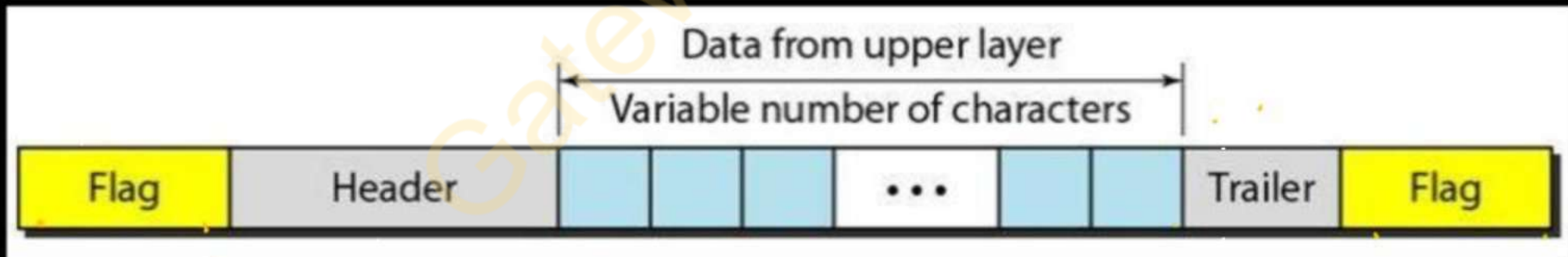
Variable-Size Framing:

- Size of each frame to be transmitted may be different.
- A pattern of bits is used as a delimiter to mark the end of one frame and the beginning of the next frame.
- If the pattern occurs in the message, then mechanisms need to be incorporated so that this situation is avoided.
- The two common approaches are –
 - **Byte – Stuffing** – A byte is stuffed in the message to differentiate from the delimiter. This is also called (character-oriented framing.)
 - **Bit – Stuffing** – A pattern of bits of arbitrary length is stuffed in the message to differentiate from the delimiter. This is also called (bit – oriented framing.)



Character-Oriented Protocols:

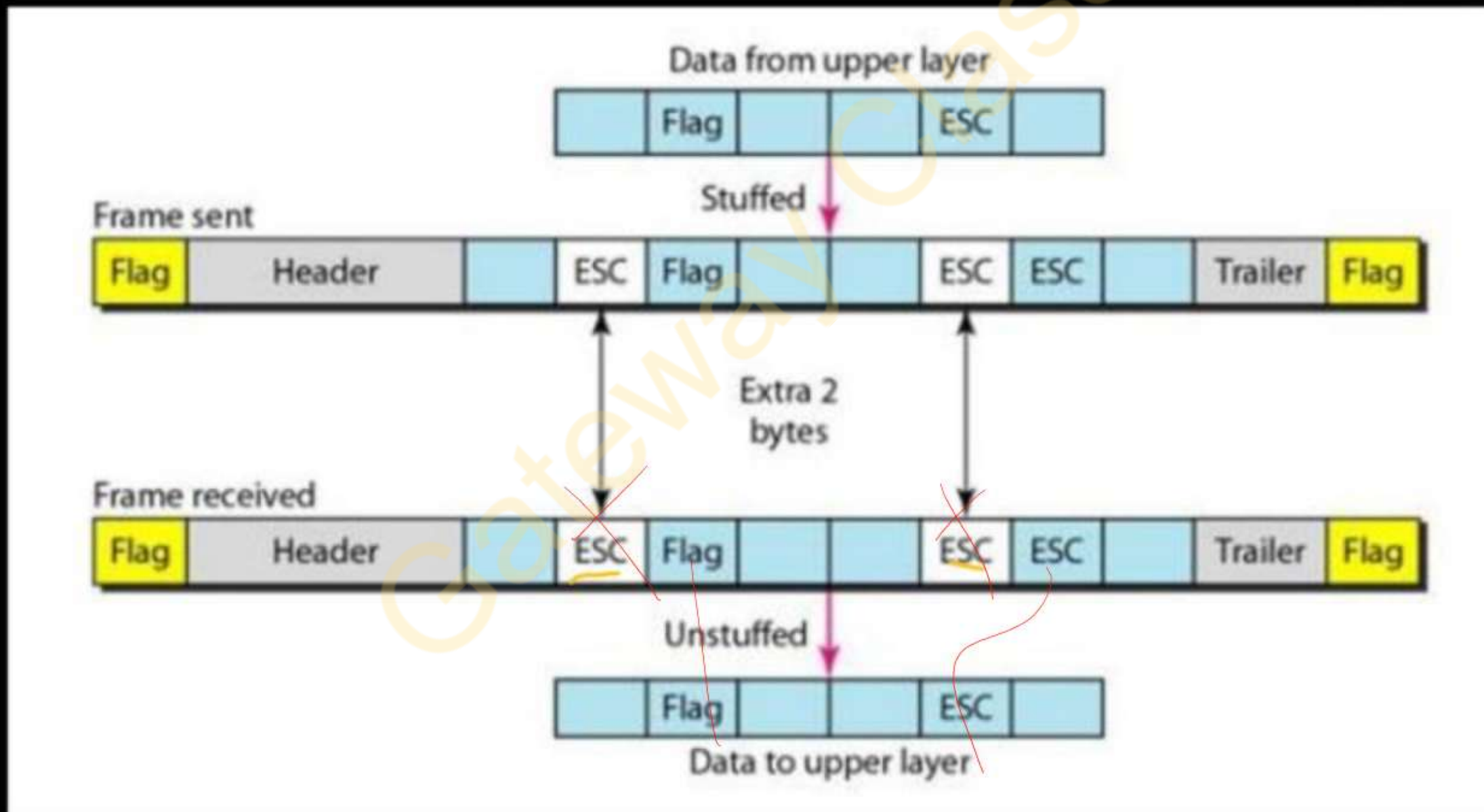
- Data to be carried are 8-bit characters from a coding system such as ASCII.
- The header carrying the source and destination addresses and other control information, and the trailer, which carries error detection or error correction redundant bits, are also multiples of 8 bits.
- To separate one frame from the next, an 8-bit (1-byte) flag is added at the beginning and the end of a
- Frame to signal the start or end of a frame.



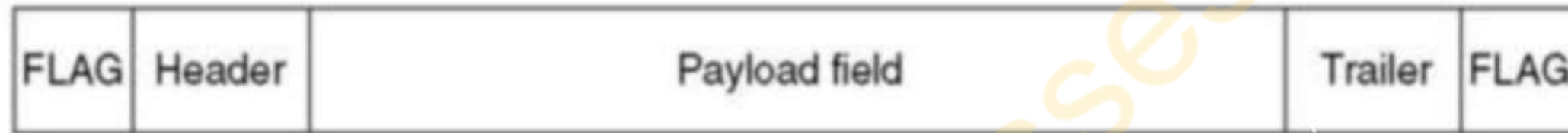
Byte stuffing :

- Process of adding 1 extra byte(usually called escape character) in the data section whenever there is a character with the same pattern as the flag in the text.
- This ESC has a predefined bit pattern. Whenever the receiver encounters the ESC character, it removes it from the data section and treats the next character as data, not a delimiting flag.

flag ABCD flag ESC H flag



Four Scenarios

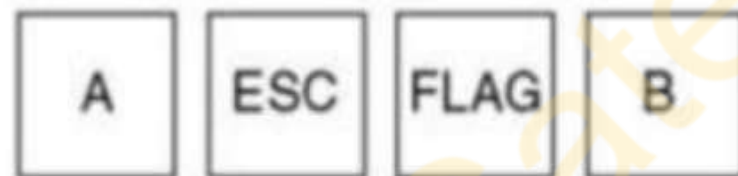


(a)

Original characters

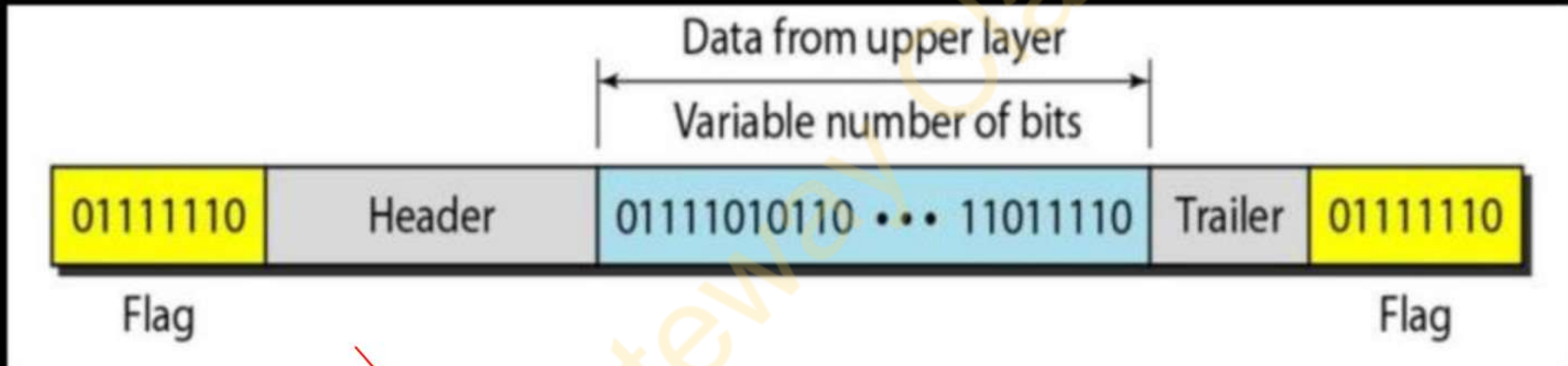


After stuffing



Bit-Oriented Protocols

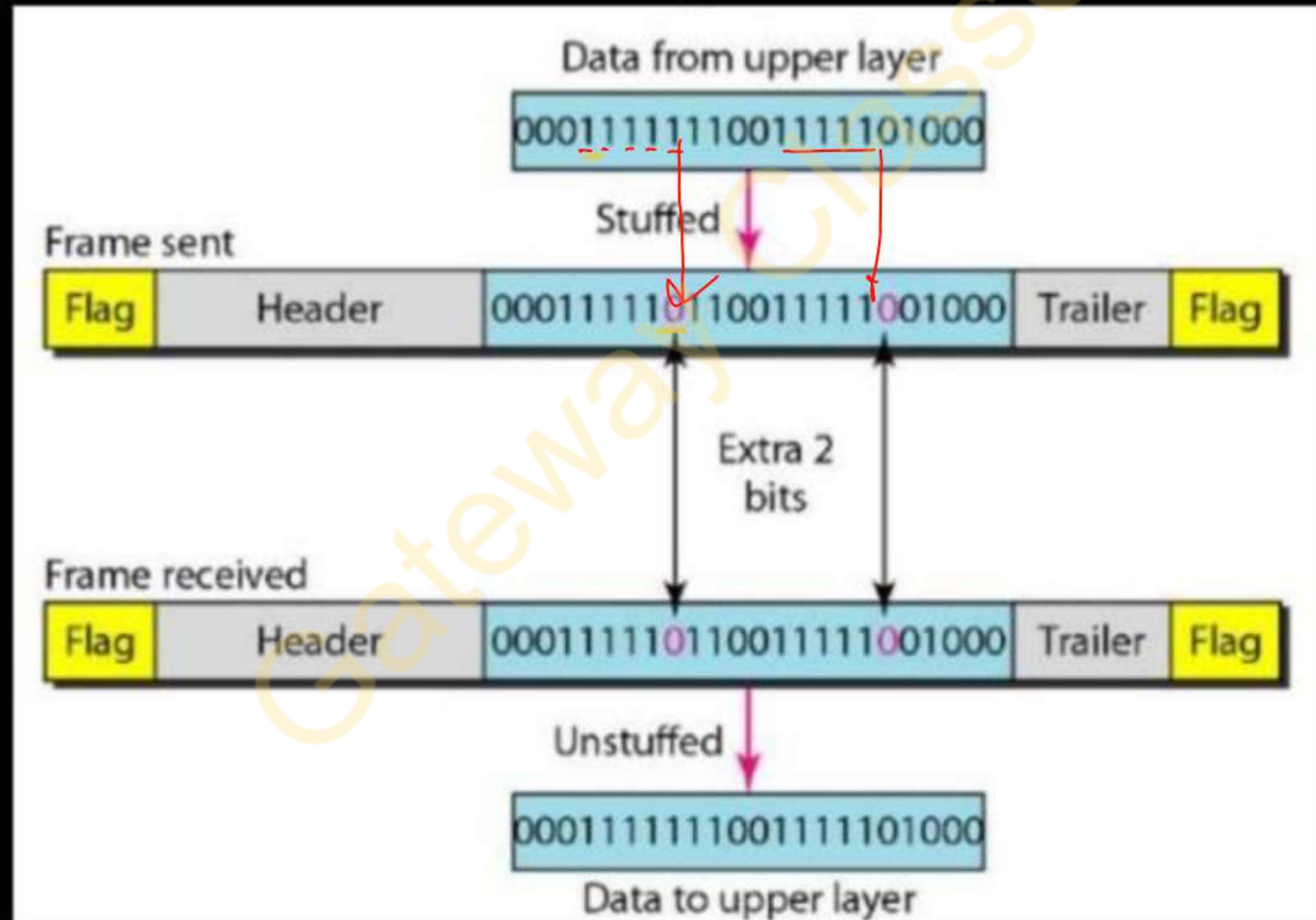
- In a bit-oriented protocol, the data section of a frame is a sequence of bits to be interpreted by the upper layer as text, graphic, audio, video, and so on.
- Most protocols use a special 8-bit pattern flag 01111110 as the delimiter to define the beginning and the end of the frame.



Bit stuffing:

- Adding one extra 0 whenever five consecutive 1's follow a 0 in the data, so that the receiver does not mistake the pattern 0111110 for a flag.
- This extra stuffed bit is removed from the data by the receiver.

0111110
↓
011111010



AKTU PYQs

1. ✓ Define fixed and variable size framing. (AKTU 2023-24)
2. ~~Define fixed and variable size framing~~ (AKTU 2017-18, 2016-17)
3. Discuss Design issues of data-link layer based on the layering principle. (AKTU 2017-18, 2016-17)



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Computer Networks

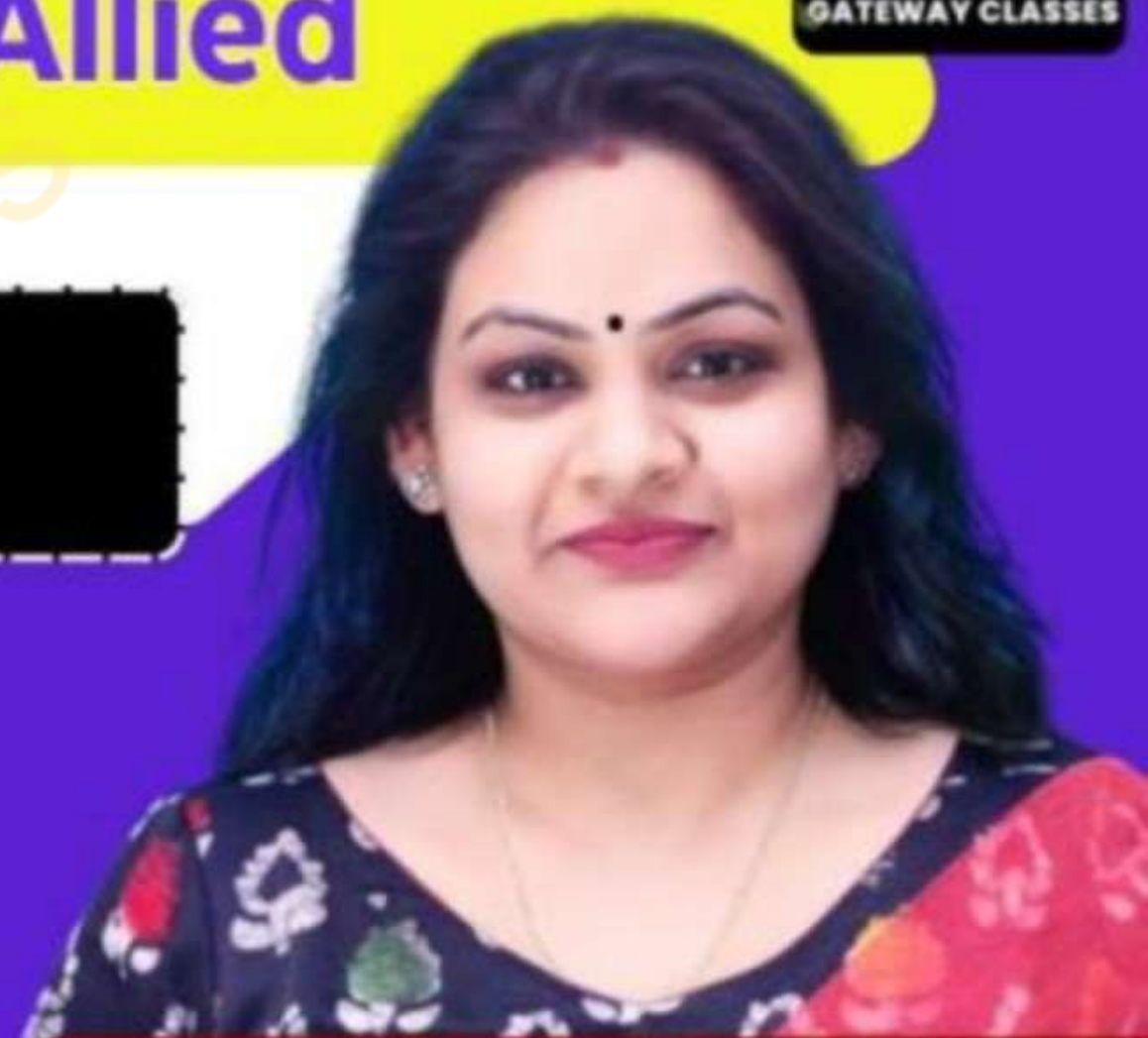
Unit-2 Lecture-2

Today's Target

- Multiple access control
- Random Access Protocols:
 - ALOHA
 - Slotted ALOHA
- AKTU PYQs

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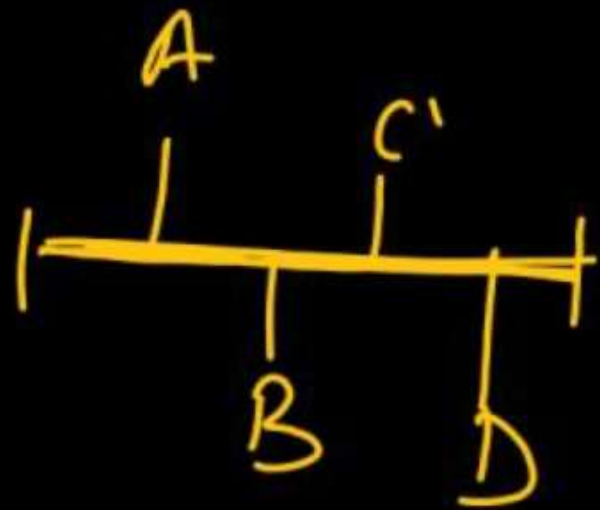
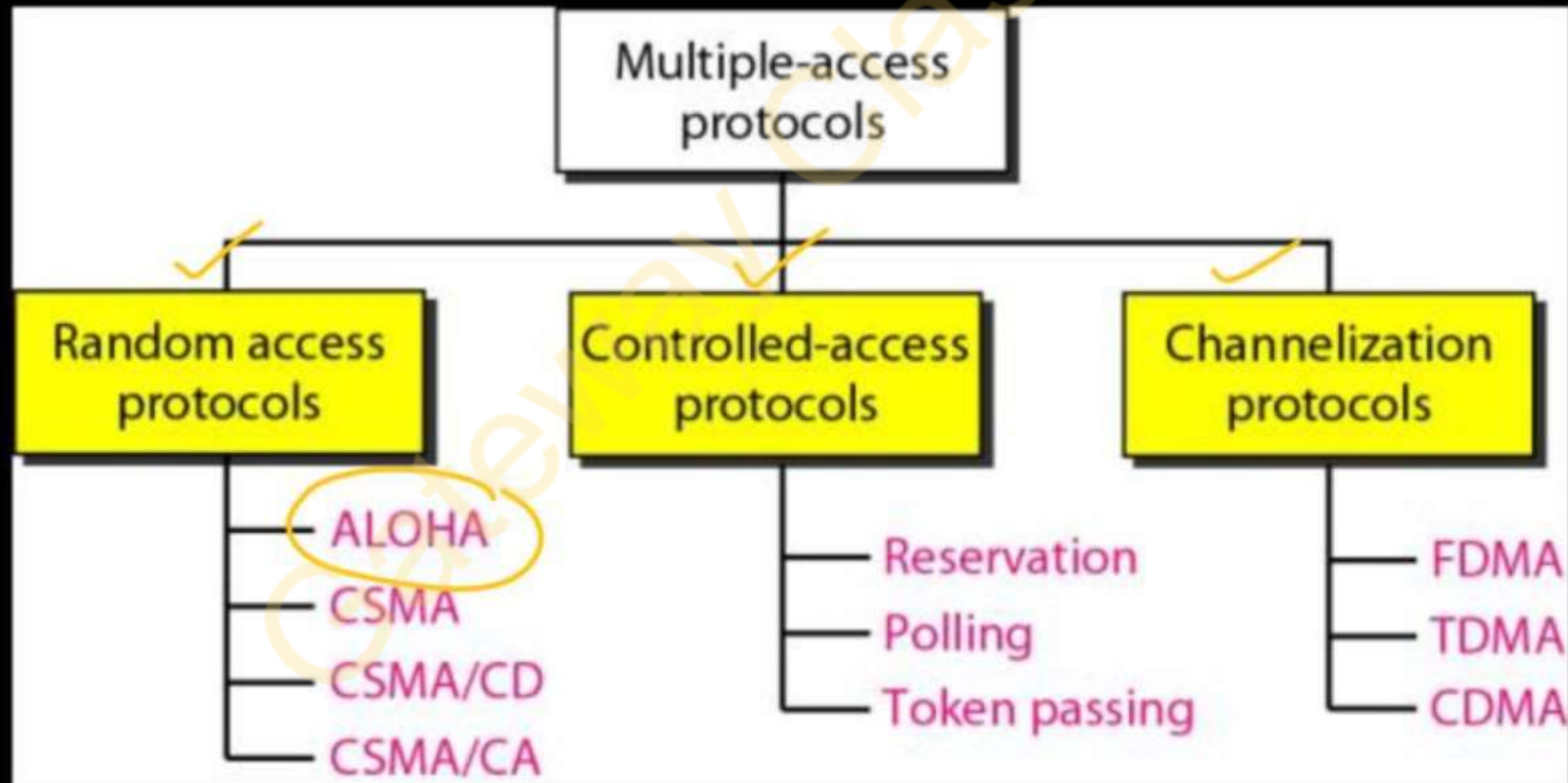
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Multiple Access Protocols

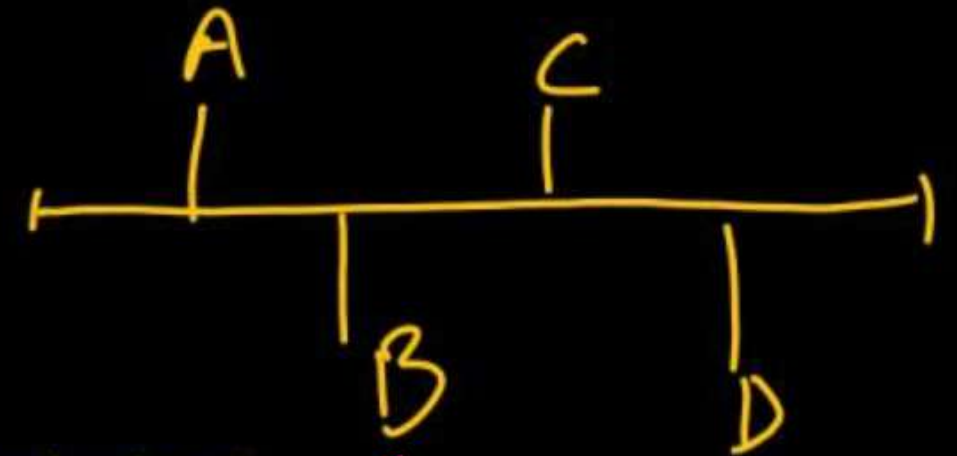
(V. Imp)

When nodes or stations are connected and use a common link, called a multipoint or broadcast link, we need a multiple-access protocol to coordinate access to the link. Many formal protocols have been devised to handle access to a shared link.



Random Access (contention methods)

- No station is superior to another station and none is assigned the control over another.
- Two features:
 - No scheduled time for a station to transmit.
 - No rules specify which station should send next. Stations compete with one another to access the medium.
- If more than one station tries to send, there is an access conflict-collision-and the frames will be either destroyed or modified. To avoid it or to resolve it when it happens, each station follows a procedure that answers the following questions:
 1. When can the station access the medium? ✓
 2. What can the station do if the medium is busy? ✓
 3. How can the station determine the success or failure of the transmission? ✓
 4. What can the station do if there is an access conflict? ✓



Types of Random-Access protocols:

1. ALOHA

2. Carrier Sense Multiple Access

(CSMA)

3. Carrier Sense Multiple Access with Collision Detection

4. Carrier Sense Multiple Access with Collision Avoidance

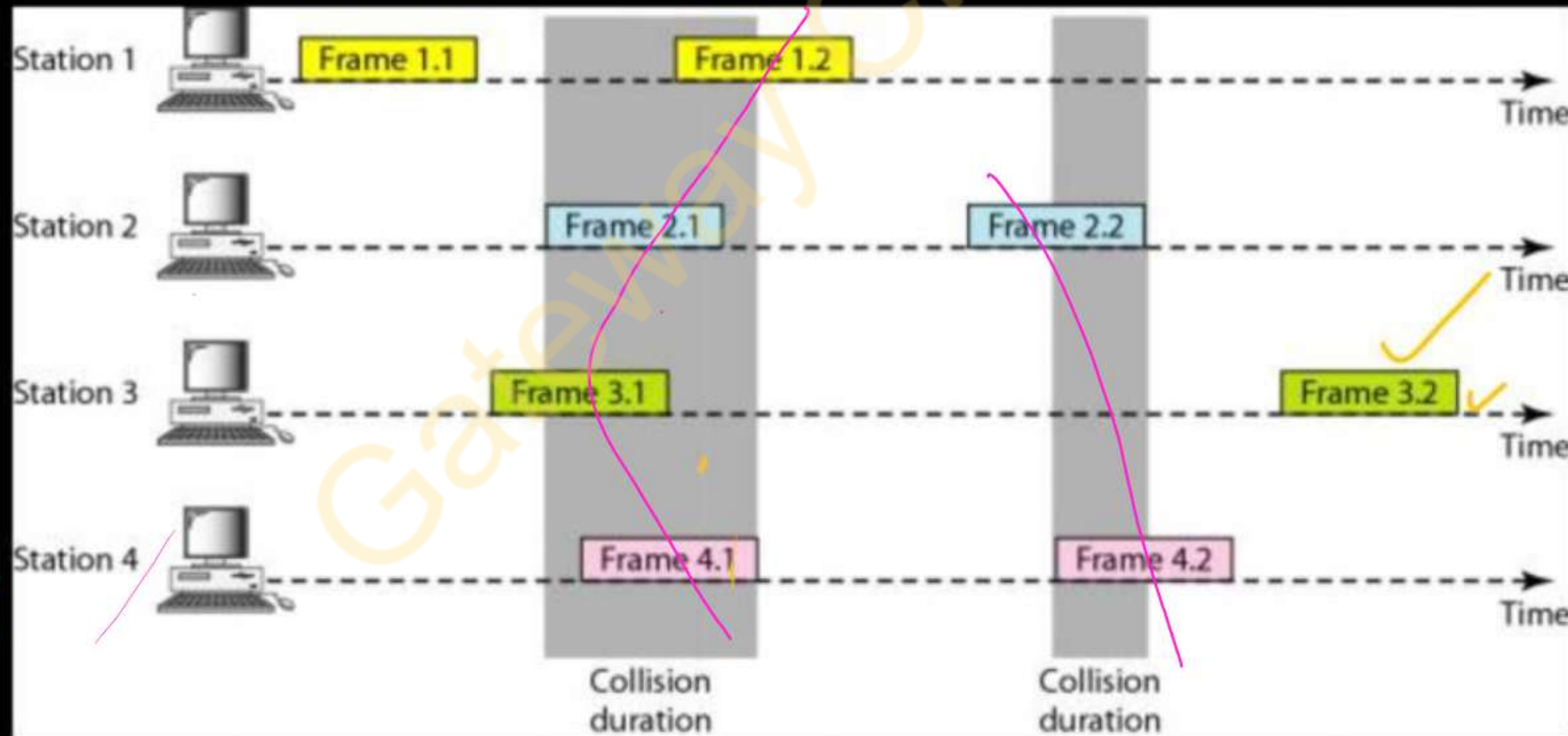
ALOHA

(If station has data to send → Send Immediately.)

- Earliest random-access method to be used on any shared medium.
- When a station sends data, another station may attempt to send at the same time.
- The data from the two stations collide and become garbled.

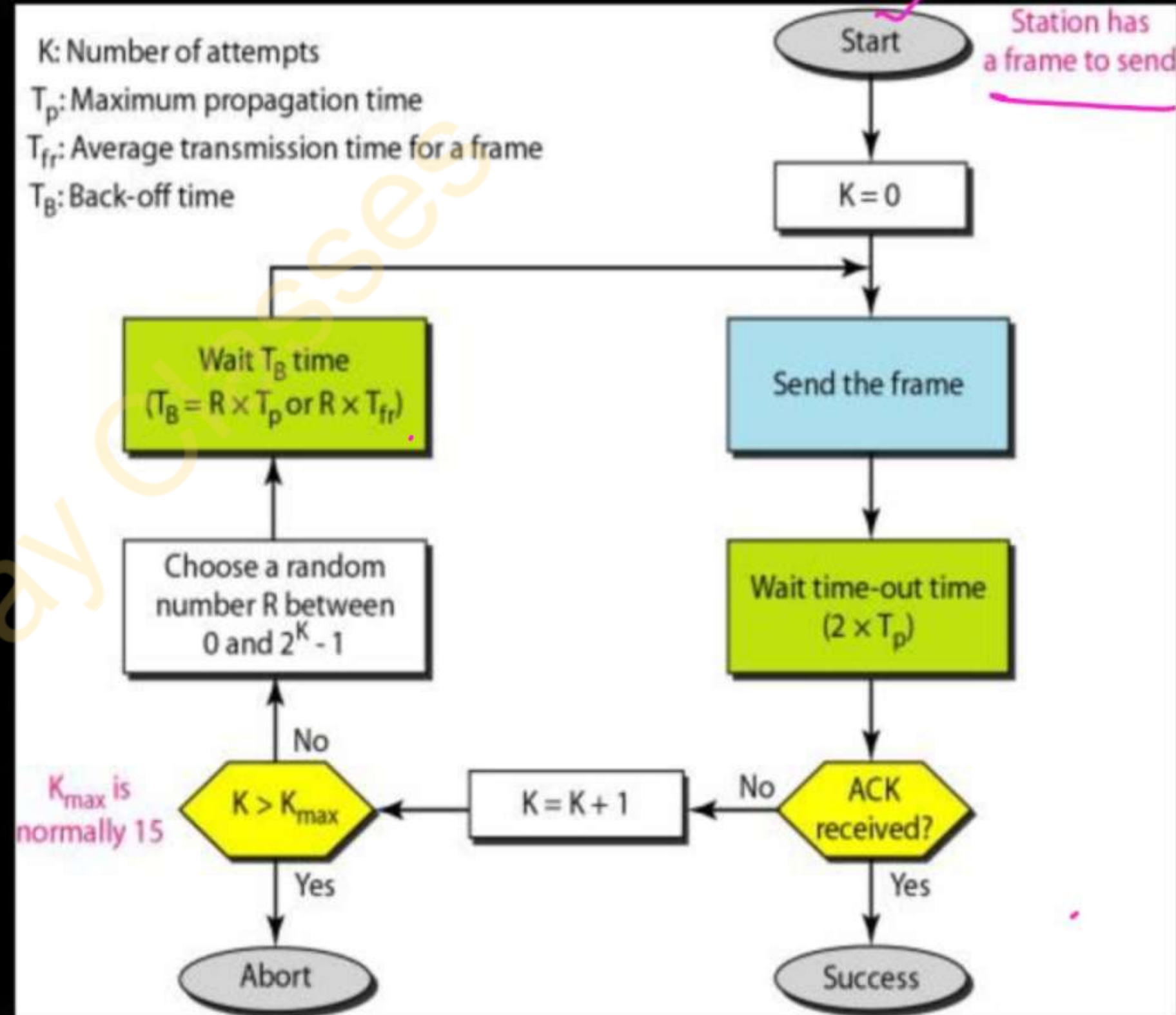
Pure ALOHA

- Each station sends a frame whenever it has a frame to send.
- Relies on acknowledgments from the receiver.
- When a station sends a frame, it expects the receiver to send an acknowledgment. If ack does not arrive after a time-out period; the station assumes that the frame (or the ack) has been destroyed and resends the frame.



Procedure for pure ALOHA protocol

- If all these stations resend their frames after the time-out, the frames will collide again.
- To avoid more collisions, when the time-out period passes, each station waits a random amount of time (back-off time) before resending its frame.
- To prevent congesting the channel with retransmitted frames, after a max number of retransmission attempts ($K_{\max}=15$) a station must give up and try later.
- T_B is calculated using **binary exponential back-off**.



Q:1 The stations on a wireless ALOHA network are a maximum of 600 km apart. If we assume that signals propagate at 3×10^8 m/s, Find T_B for different values of K .

Solⁿ distance = 600 km
propagation speed = 3×10^8 m/s

$$T_p = \frac{600 \times 1000}{3 \times 10^8} \text{ secs.}$$

$$= 2 \times 10^{-3} \text{ sec}$$

$$= \underline{2 \text{ ms}}$$

$$R = 0 - 2^{K-1}$$

$$T_B = \frac{R \times T_p}{R \times T_{fl}}$$

$$\begin{array}{r} K=1 \\ R \Rightarrow \begin{array}{r} 0 - 2^K - 1 \\ 0 - 2^1 - 1 \\ 0 - 1 \end{array} \\ R = 0/1 \checkmark \end{array}$$

$$T_B = 2 \times 0 = 0 \checkmark$$

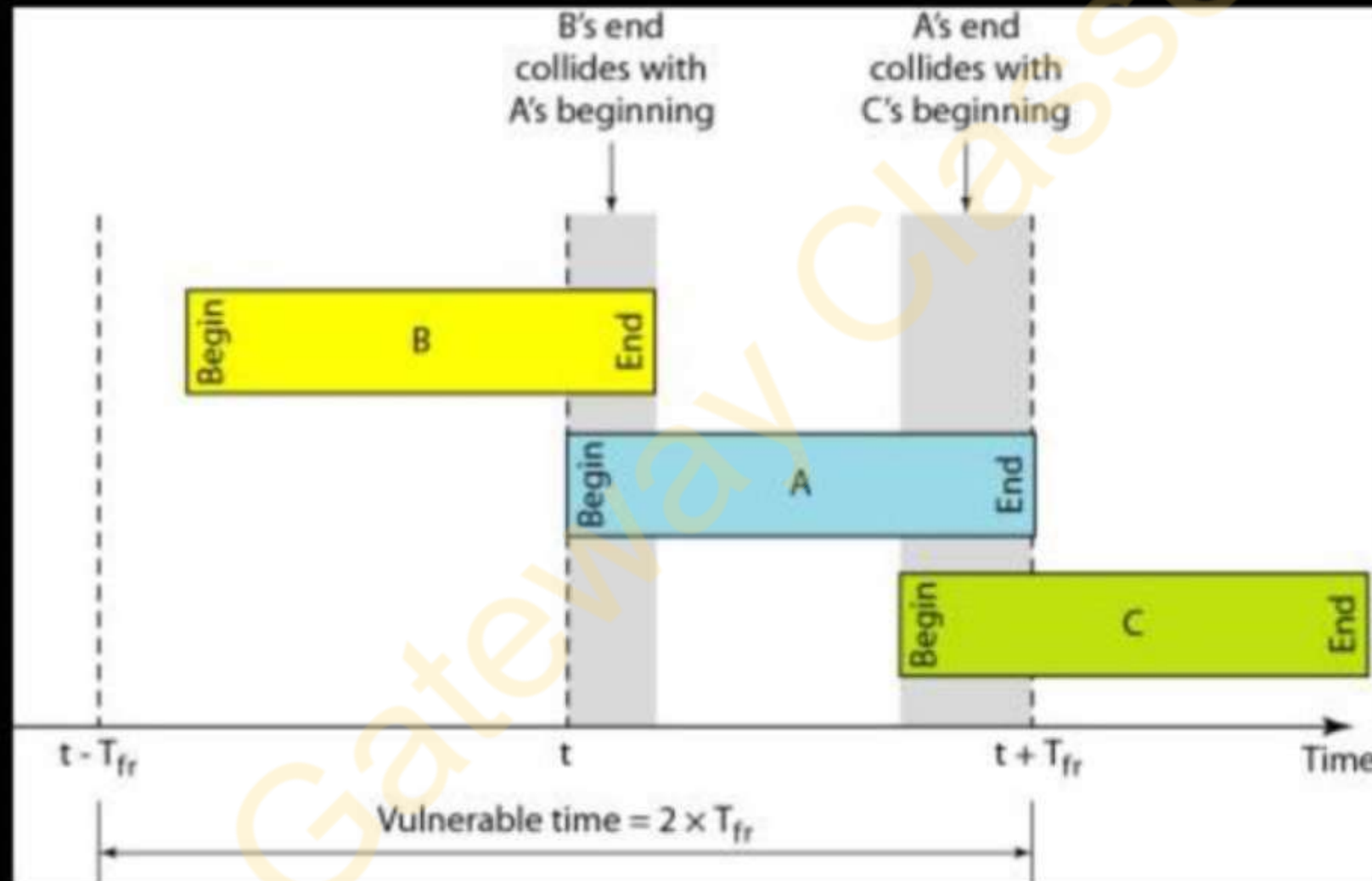
$$T_B = 2 \times 1 = 2 \text{ ms} \checkmark$$

Vulnerable time:

The time in which there is a possibility of collision.

$$\text{Pure ALOHA vulnerable time} = 2 \times T_{fr}$$

Numerical



Vulnerable time for pure ALOHA protocol

Q 2: A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the requirement to make this frame collision-free?

Solⁿ.

$$\text{Vulnerable time} = 2 \times T_{fr}$$

$$T_{fr} = \frac{\text{frame size}}{\text{bandwidth}} = \frac{200}{200 \times 10^3} = \underline{1 \text{ ms}}$$

$$\checkmark \text{ Vulnerable time} = 2 \times 1 \text{ ms} = \boxed{2 \text{ ms}}$$

Throughput:

The throughput for pure ALOHA is

Throughput

$$S = G \times e^{-2G}$$

channel load

The maximum throughput $S_{\max} = 0.184$

when $G = 1/2$

G : average number of frames generated by the system during one frame transmission time.

- It means that if one-half a frame is generated during one frame transmission time (in other words, one frame during two frame transmission times), then 18.4 percent of these frames reach their destination successfully.

Q3. A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces (AKTU)

a) 1000 frames per second (Ans. 0.135)

b. 500 frames per second (Ans. 0.184)

c. 250 frames per second (Ans. 0.152) $\Rightarrow G = \frac{1}{4}$

$$\text{Throughput (S)} = \underline{\underline{G \times e^{-2G}}}$$

Solⁿ

$$T_{fr} = \frac{200}{200 \times 10^{-3}} = \underline{1 \text{ ms}}$$

a) In 1 sec no. of frames generated = 1000

$$\text{In } \underline{1 \text{ ms}} \quad \text{||} \quad \text{||} \quad \text{||} \quad \text{||} = 1000 \times 10^{-3} = \underline{1 \text{ frame}}$$

$$\boxed{G = 1}$$

$$S = G \times e^{-2G} = 1 \times e^{-2} = \underline{0.135} \approx \boxed{13.5\%}$$

It means system is 13.5% efficient.

$$\frac{1000 \times 13.5}{100} = 135 \text{ frames out of 1000 frames are successfully transmitted.}$$

b) frames generated in 1 sec = 500

$$1 \text{ ms} \rightarrow 500 \times 10^{-3} = \frac{500}{1000} = \frac{1}{2} \Rightarrow \boxed{G = \frac{1}{2}}$$

$$\checkmark S = 0.184 \Rightarrow 18.4\% \text{ efficient}$$

$$\checkmark \frac{500 \times 18.4}{1000} = \frac{184}{2} = 92 \text{ frames are successfully transmitted.}$$

Q4: We have a pure ALOHA network with 100 stations. If $T_{fr} = 1 \mu s$, what is the number of frames/s each station can send to achieve the maximum efficiency. (Ans. 5000 Frames)

$$G = N \times FPS \times T_{fr}$$

no. of frames generated in one frame transmission time

No of stations

frames generated/sec

no. of frames generated in one frame transmission time

No of stations

frames generated/sec

Solⁿ

$$N = 100$$

$$T_{fr} = 1 \mu s$$

$$FPS = ?$$

For Max efficiency $G = \frac{1}{2}$

$$FPS = \frac{G}{N \times T_{fr}} = \frac{1}{2 \times 100 \times 10^{-6}} = \frac{1000000}{2 \times 100} = \underline{\underline{5000}}$$

$$= 5000 \text{ frames/sec. Ans}$$

Q5: One hundred stations on a pure ALOHA network share a 1-Mbps channel. If frames are 1000 bits long, find the throughput if each station is sending 10 frames per second. (Ans. 0.135)

Solⁿ

$$T_{fr} = \frac{\text{frame size}}{\text{bandwidth}} = \frac{1000}{1 \times 10^6} = 10^{-3} \text{ sec}$$

$$\text{FPS} = 10$$

$$G = N \times \text{FPS} \times T_{fr}$$

$$= 100 \times 10 \times 10^{-3} = 1$$

$$\text{Throughput (S)} = G \times e^{-2G}$$

$$= 1 \times e^{-2}$$

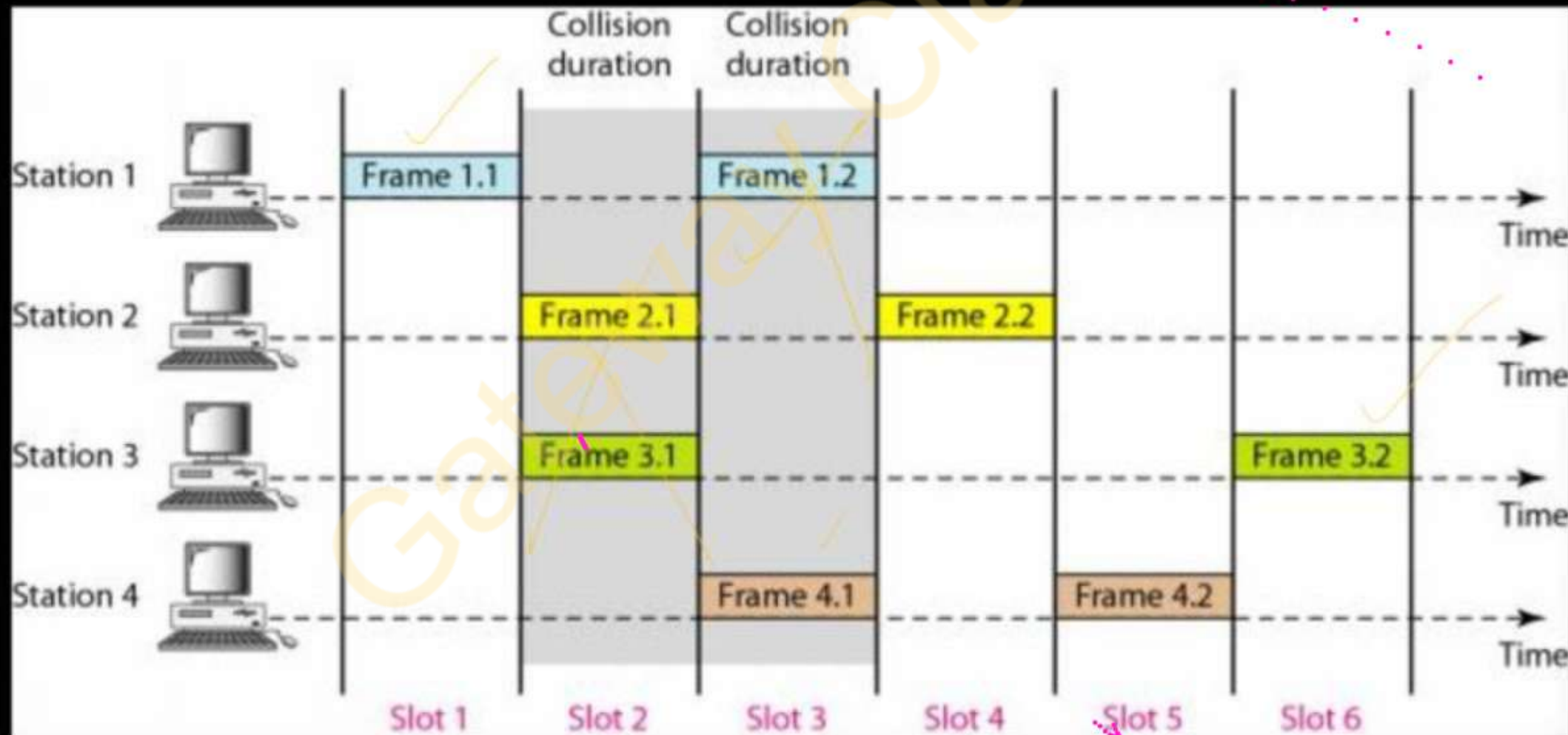
$$= \underline{\underline{0.135}} \text{ or } 13.5\% \text{ efficient}$$

Ans

$$G = N \times \text{FPS} \times T_{fr}$$

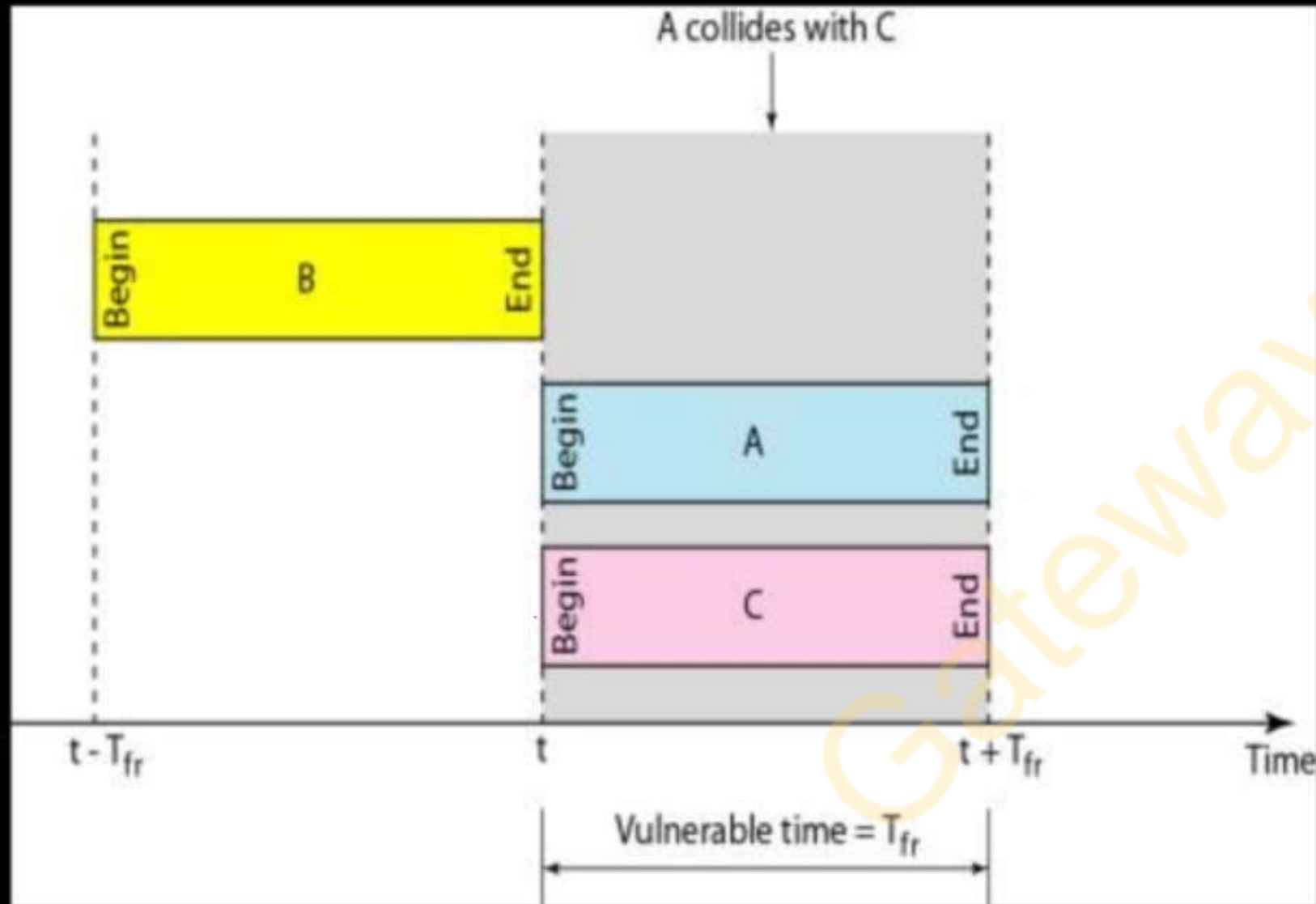
Slotted ALOHA

- Slotted ALOHA was invented to improve the efficiency of pure ALOHA.
- We divide the time into slots of T_{fr} and force the station to send only at the beginning of the time slot.
- If a station misses this moment, it must wait until the beginning of the next time slot.
- There is still the possibility of collision if two stations try to send at the beginning of the same time slot.



Vulnerable time is now reduced to one-half,
equal to T_{fr}

Slotted ALOHA vulnerable time = T_{fr}



The throughput for slotted ALOHA is:

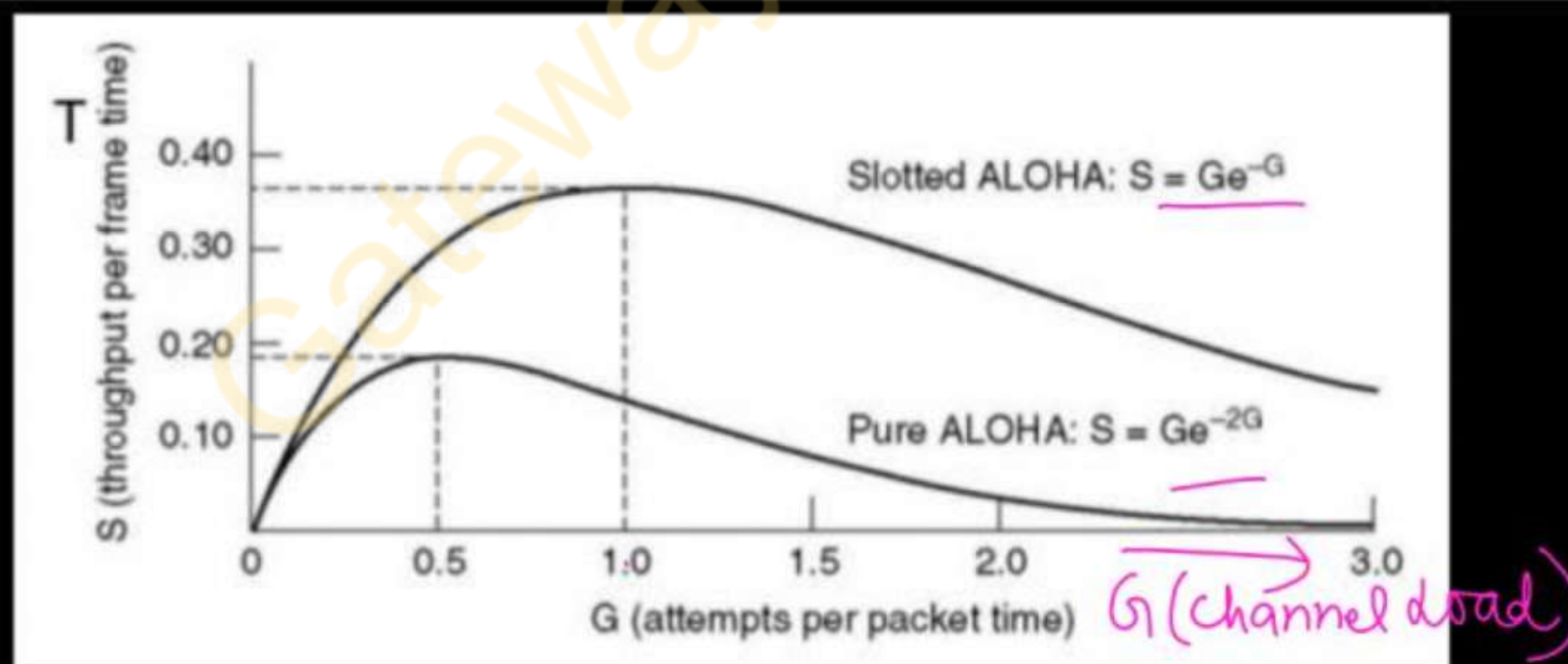
$$S = G \times e^{-G}$$

max throughput $S_{\max} = 0.368$ when $G = 1$

➤ If a frame is generated during one frame transmission time, then 36.8 percent of these frames reach their destination successfully.

Pure ALOHA Vs Slotted ALOHA

Pure ALOHA	Slotted ALOHA
User can transmit data frame when it has data to send.	User has to wait till next time slot start to transmit data frame.
Time is continuous.	Time is discrete.
Probability of successful transmission of data frame is: $S = G * e^{-2G}$	Probability of successful transmission of data frame is: $S = G * e^{-G}$
Time is not globally synchronized.	Time is globally synchronized.
The Max throughput occurs at $G = 1/2$ which is 18%.	The Max throughput occurs at $G = 1$ which is 37%.



Q6: A slotted ALOHA network transmits 400-bit frames using a shared channel with a 400-kbps bandwidth. Find the throughput if the system (all stations together) produces

a. 1000 frames per second (Ans. 0.368) ✓

b. 500 frames per second (Ans. 0.303)

c. 250 frames per second (Ans. 0.195)

0.22.7

AKTU (2017-18)

Solⁿ. $T_{fr} = \frac{400}{400 \times 10^3} = \underline{1 \text{ ms}}$

a) In 1 sec no. of frames generated = 1000

$$\frac{1000}{1 \text{ ms}} = \frac{1000 \times 10^{-3}}{1} = 1 \Rightarrow \boxed{G = 1}$$

$$S = G \times e^{-G}$$

$$S = \underline{0.368} = \underline{36.8\%} \Rightarrow \frac{1000 \times 368}{1000} = \underline{368} \text{ frames will be successfully transmitted.}$$

b) In 1 sec. no. of frames transmitted = 500

1 ms —————

$$500 \times 10^{-3} = \frac{1}{2} \Rightarrow \boxed{G = \frac{1}{2}}$$

$$S = G \times e^{-G} = (0.5) \cdot e^{-0.5}$$

$$= 0.303 \approx 30.3\% \approx \frac{500 \times 30.3}{1000} \approx 30.3 \text{ frames}$$

c) 1 sec — 250 frames

$$1 \text{ ms} \rightarrow 250 \times 10^{-3} = \frac{1}{4} \Rightarrow \boxed{G = \frac{1}{4}}$$

$$S = G \times e^{-G} = (0.25) \cdot e^{-0.25} = 0.195 \approx 19.5\% \approx \frac{250 \times 19.5}{1000} \approx 4.875$$

will be transmitted successfully

Q7: Measurement of Slotted ALOHA channel with an infinite number of users show that 10% of slots are idle. Find

- 1) What is channel load? (Ans. 2.3)
- 2) What is throughput? (Ans. 0.23)
- 3) Is channel under load or overload?

(AKTU 2017-18)

$$P_{\text{succ}} = e^{-G}$$

YES
10% slots are Idle means transmission during these slots will be successful.

$$P_{\text{succ}} = 0.10$$

1)

$$0.10 = e^{-G}$$

$$G = -\left(\log_e 0.10\right)$$

$$G = -(-2.3)$$

$$G = 2.3$$

$$\begin{aligned} 2) S &= G \times e^{-G} \\ &= 2.3 \times 0.10 \\ &= 0.23 \end{aligned}$$

3) In slotted ALOHA, system has highest efficiency when $G=1$. so $G > 1$ will produce overloaded system.

AKTU PYQs

1. Compare ALOHA with slotted ALOHA.. (AKTU 2017-18)
2. Measurement of Slotted ALOHA channel with an infinite number of users show that 10% of slots are idle. Find
 - 1) What is channel load?
 - 2) What is throughput?
 - 3) Is channel under load or overload?(AKTU 2017-18)
3. A slotted ALOHA network transmits 400-bit frames using a shared channel with a 400-kbps bandwidth. Find the throughput if the system (all stations together) produces
 - a. 1000 frames per second
 - b. 500 frames per second
 - c. 250 frames per second(AKTU 2017-18)
4. Explain all Random-Access protocols. (AKTU 2023-24)



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Computer Networks

Unit-2: Lecture 3

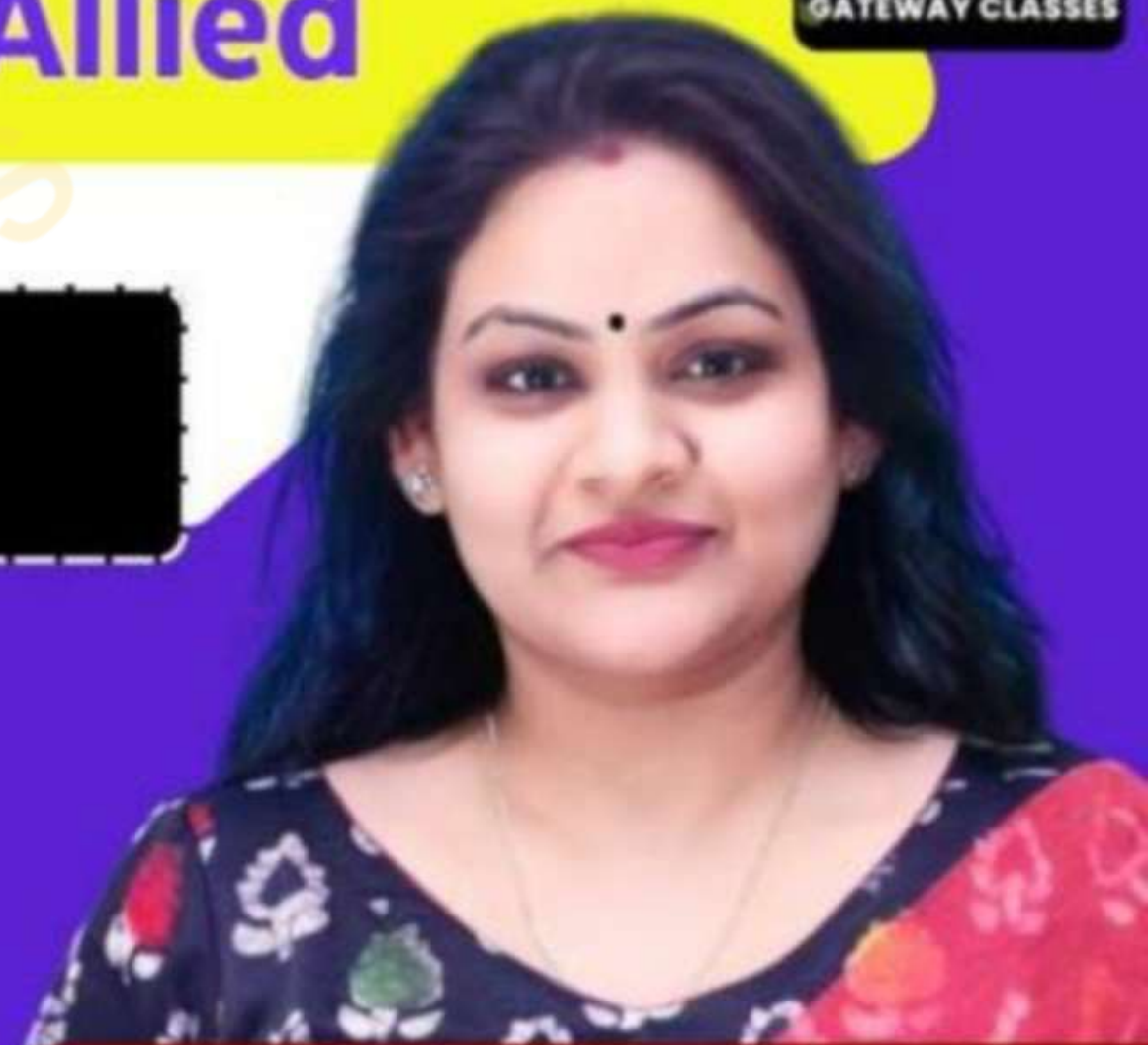
Today's Target

- ✓ CSMA
- CSMA/CD
- CSMA/CA

AKTU PYQs

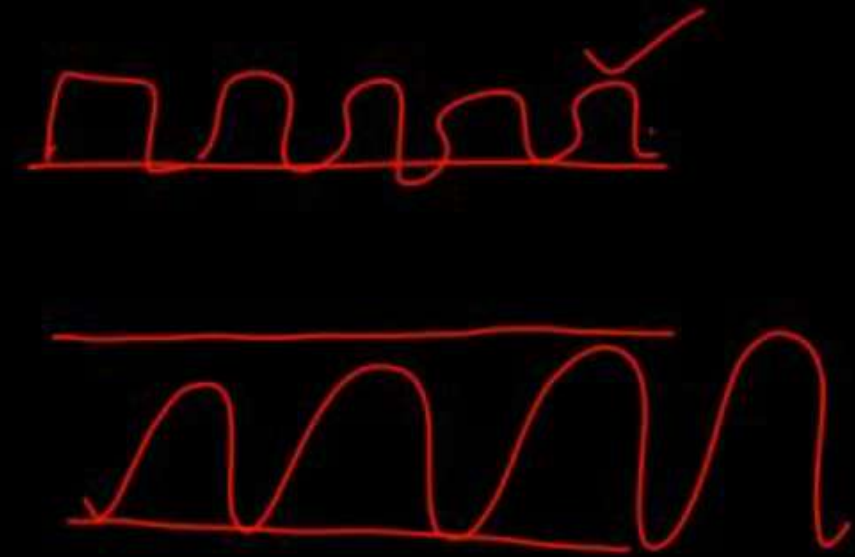
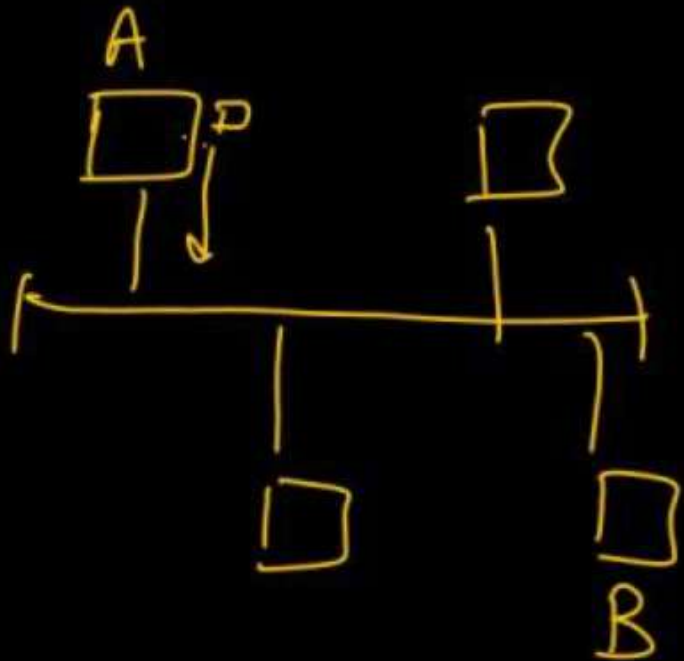
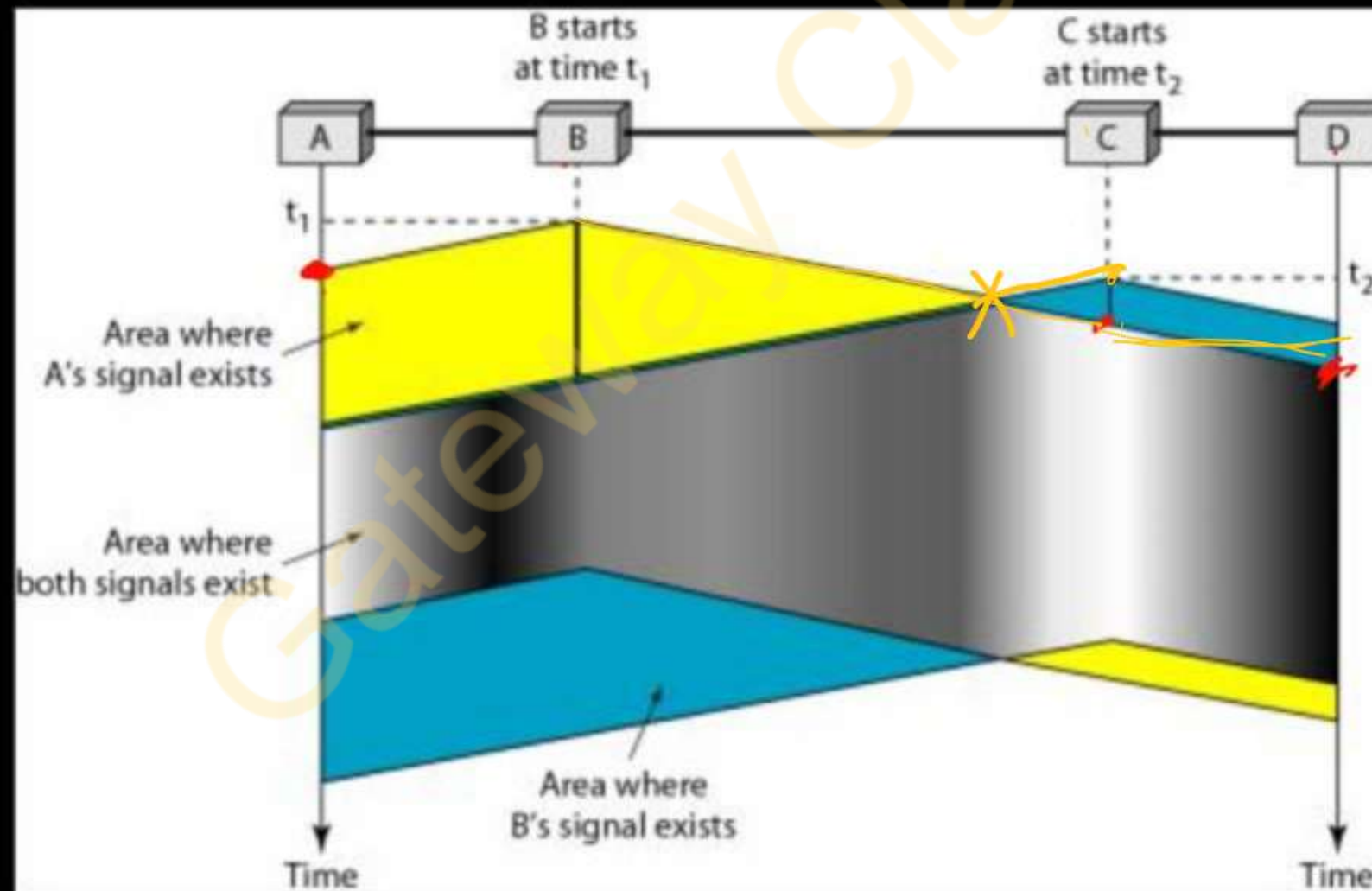
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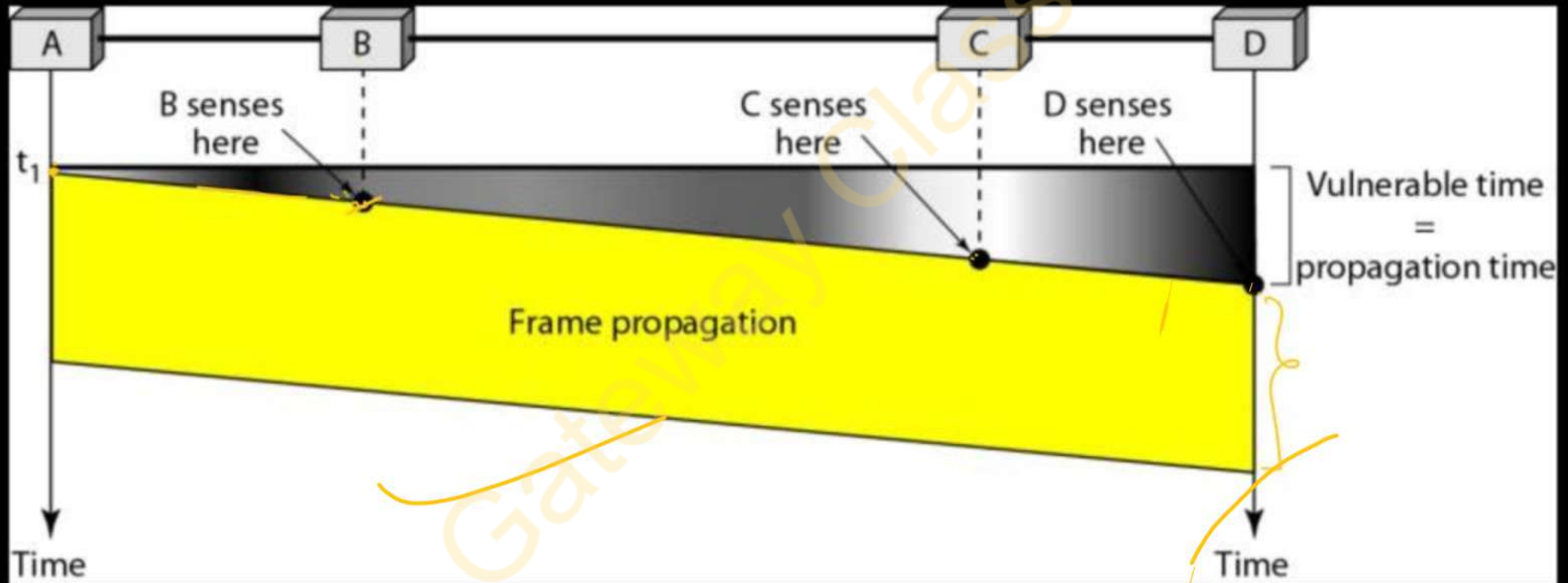
Carrier Sense Multiple Access (CSMA)

- CSMA is based on the principle "sense before transmit" or "listen before talk."
- Each station first listen to the medium (or check the state of the medium) before sending.
- The possibility of collision still exists because of propagation delay; when a station sends a frame, it still takes time for the first bit to reach every station and for every station to sense it.



Vulnerable Time

The vulnerable time for CSMA is the propagation time T_p . If the first bit of the frame reaches the end of the medium, every station will already have heard the bit and will refrain from sending.



Persistence Methods

- What should a station do if the channel is busy?
- What should a station do if the channel is idle?
- Three methods have been devised to answer these questions:

1-Persistent: (100% probability)

- After the station finds the line idle, it sends its frame immediately.
- Highest chance of collision as two or more stations may find the line idle and send frames immediately.

Nonpersistent:

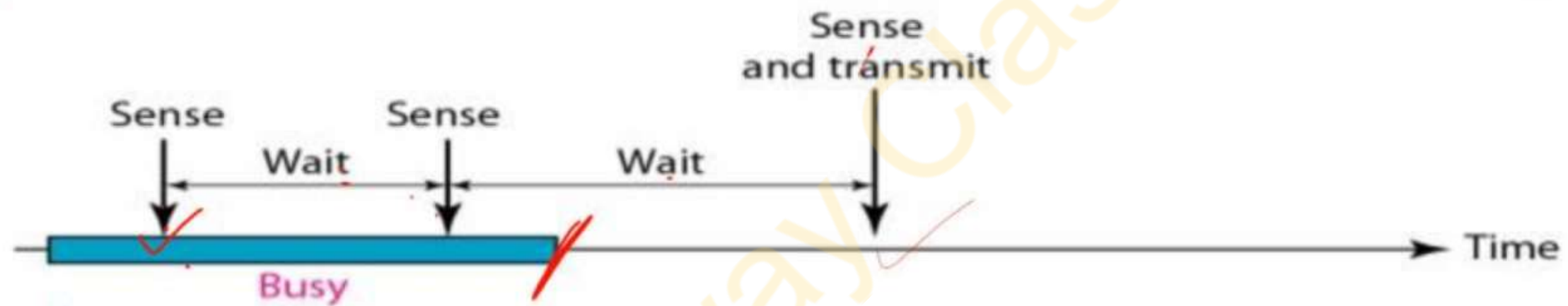
- A station that has a frame to send senses the line. If the line is idle, it sends immediately.
- If not idle, it waits a random amount of time and then senses again.
- Reduces the chance of collision as it is unlikely that two or more stations will wait the same amount of time and retry to send simultaneously.
- Reduces the efficiency as medium remains idle when there may be stations with frames to send.

p-Persistent :

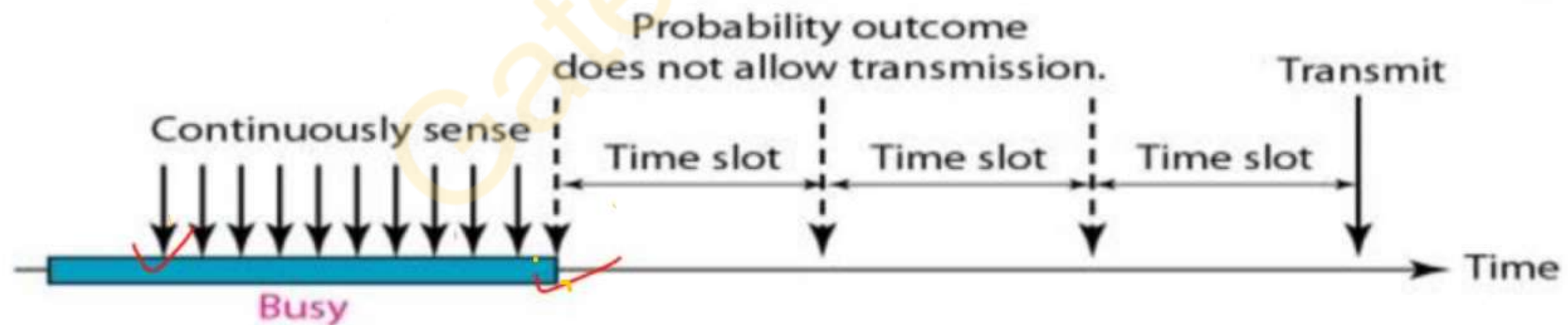
- Used if the channel has time slots with a slot duration equal to or greater than the maximum T_p
- After the station finds the line idle it follows these steps:
 1. With probability p , the station sends its frame.
 2. With probability $q = 1 - p$, the station waits for beginning of next time slot and checks the line again.
 - a. If the line is idle, it goes to step 1.
 - b. If the line is busy, it acts as though a collision has occurred and uses the backoff procedure.



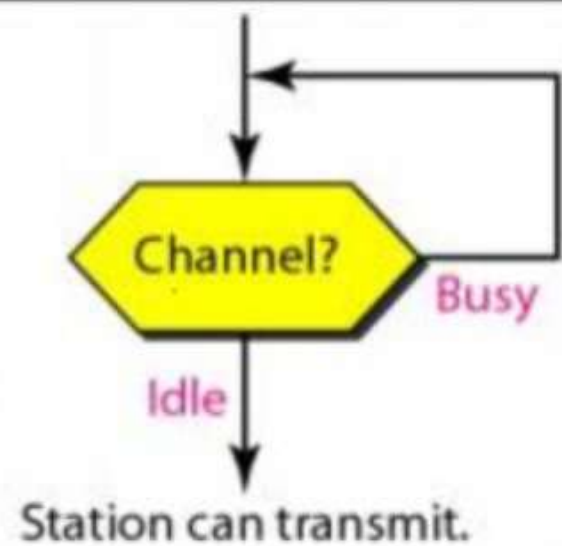
a. 1-persistent



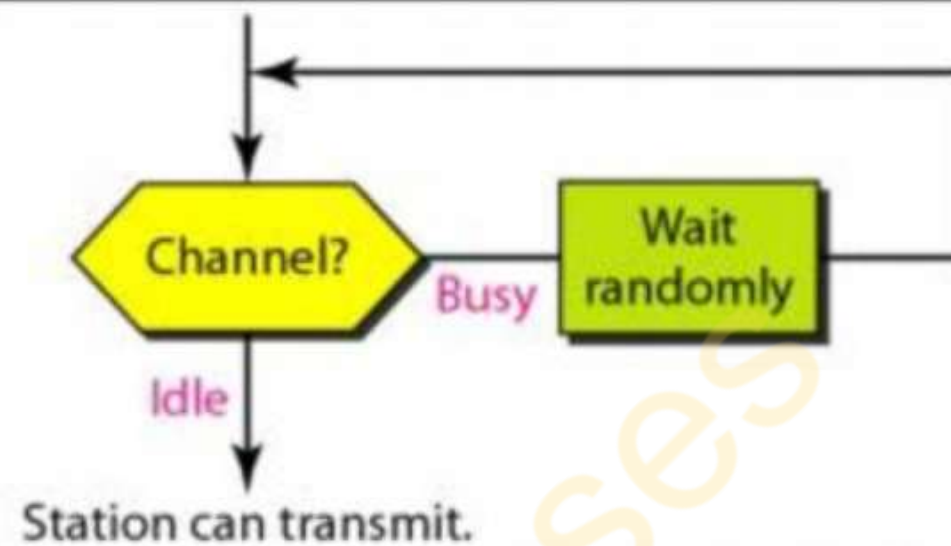
b. Nonpersistent



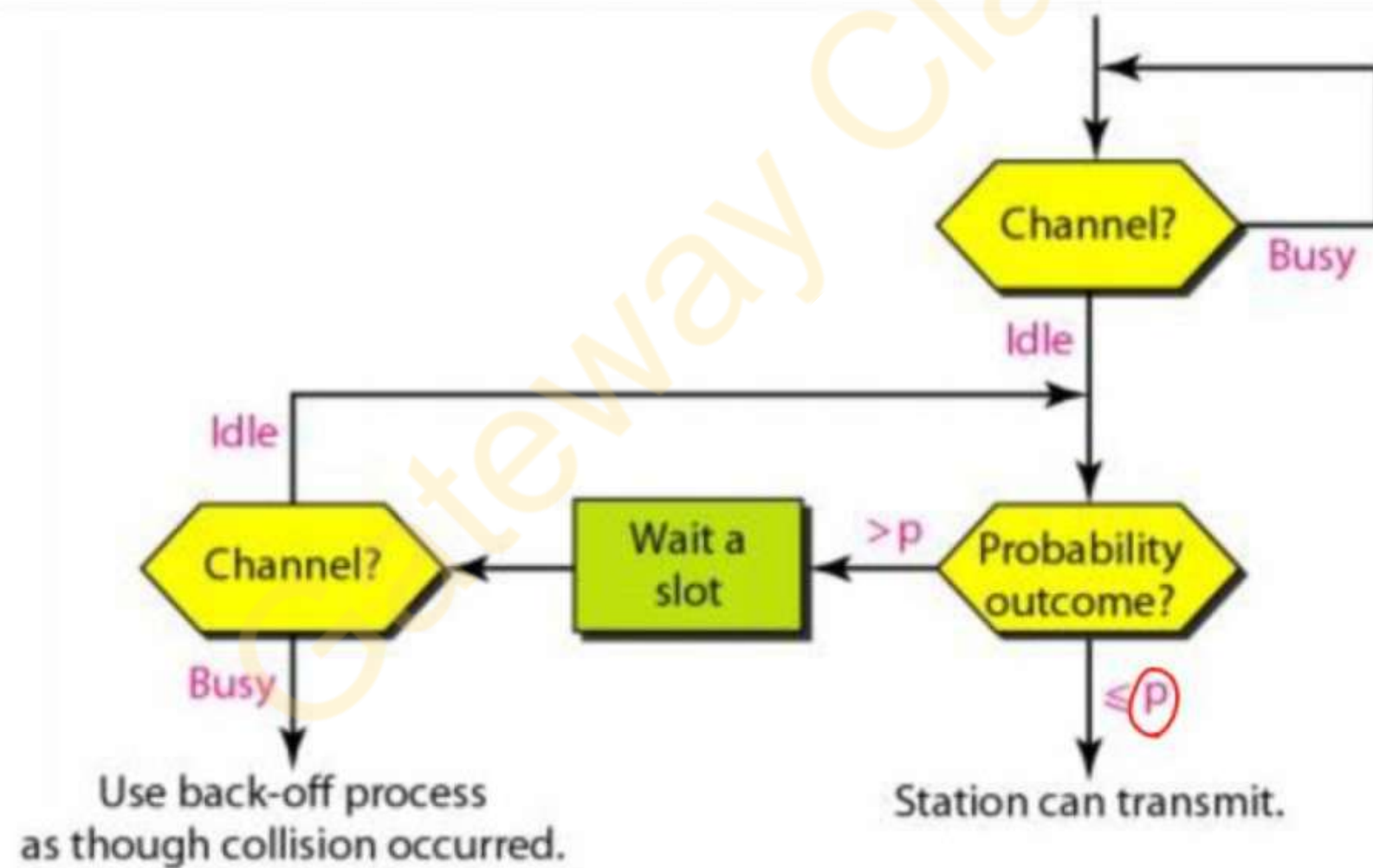
c. p-persistent



a. 1-persistent



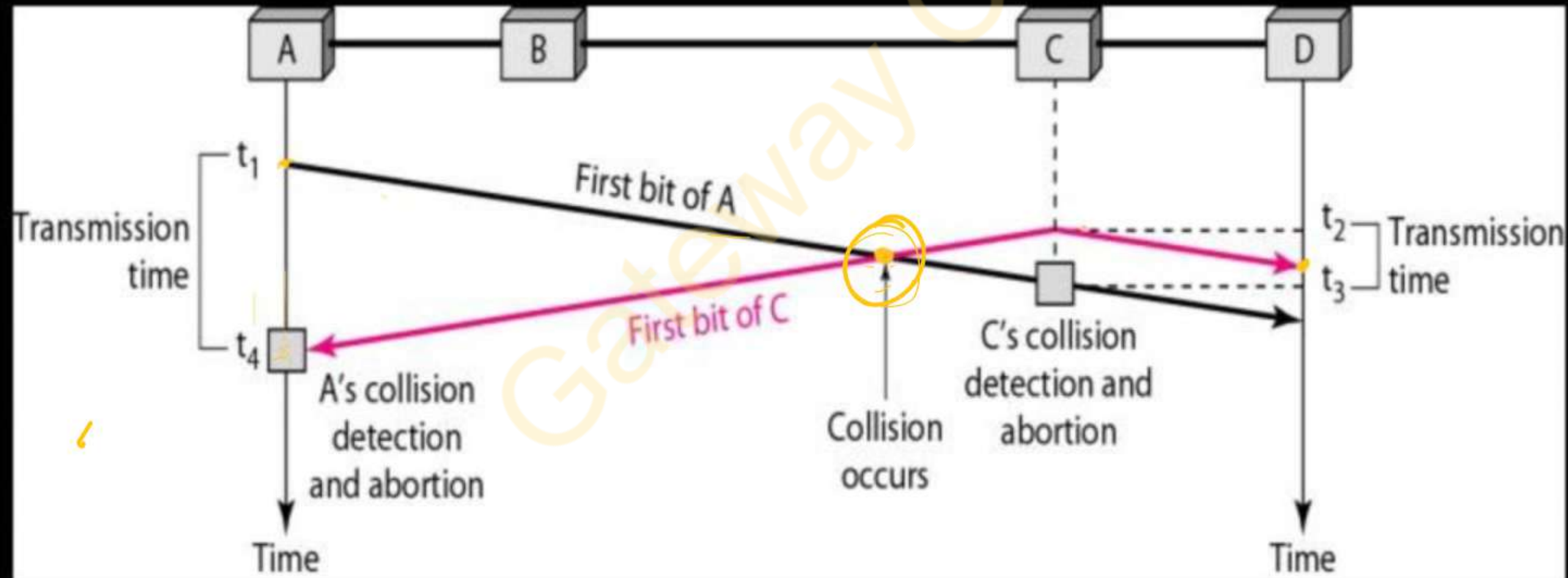
b. Nonpersistent

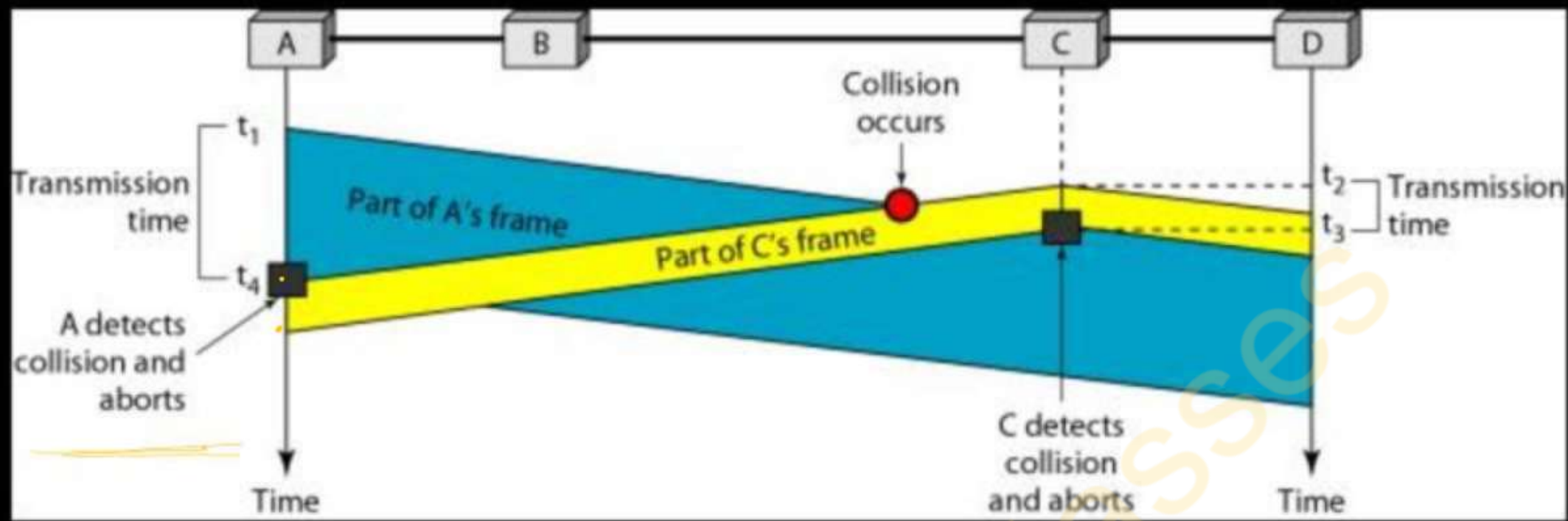


c. p-persistent

Carrier Sense Multiple Access with Collision Detection (CSMA/CD)

- Augments CSMA algorithm to handle the collision.
- A station monitors the medium after sending frame to see if the transmission was successful.
- If so, the station is finished. If there is a collision, the frame is sent again.
- Keeps copy of the frame until entire frame is sent.



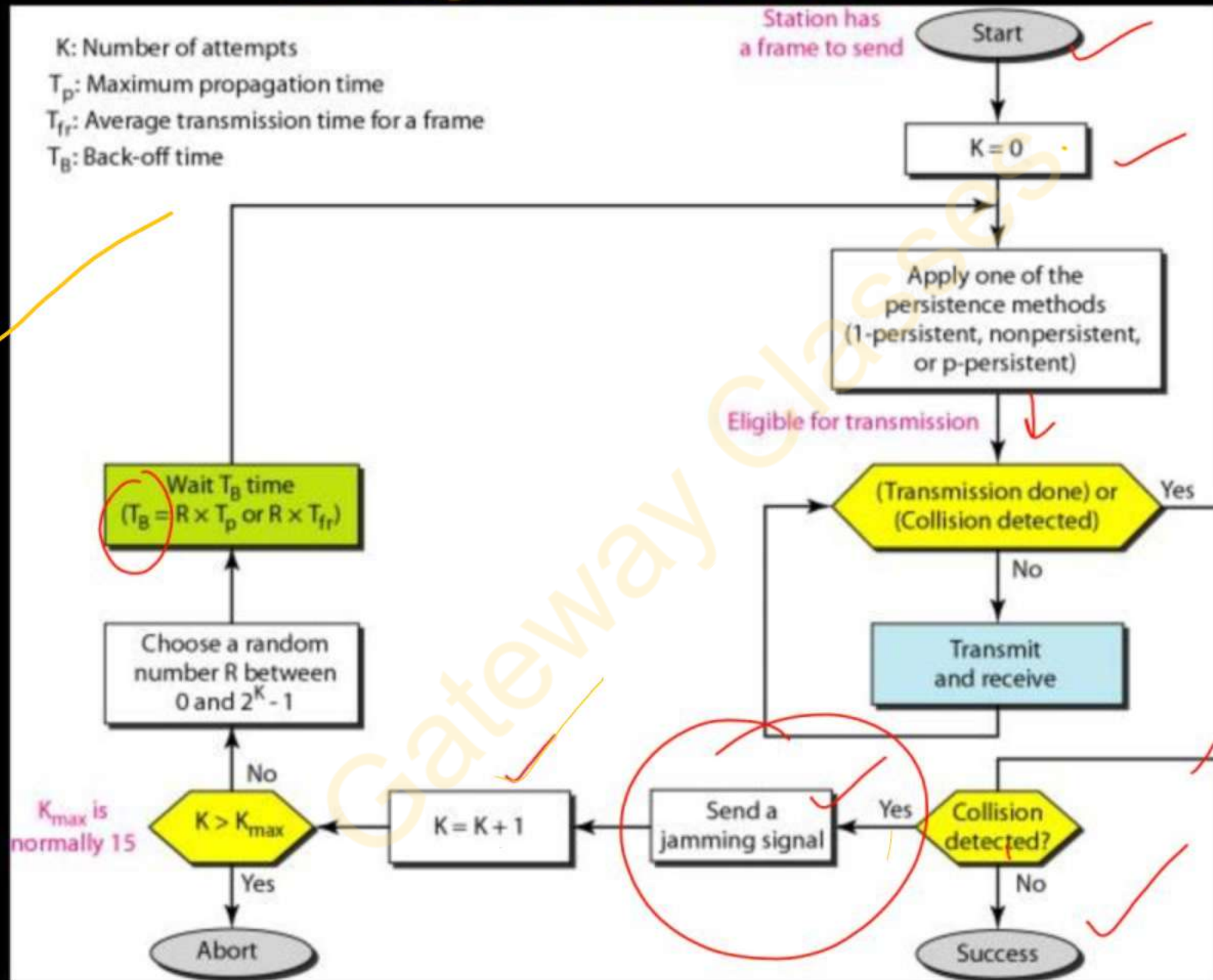


Minimum Frame Size

- For *CSMA/CD* to work, before sending the last bit of the frame, the sending station must detect a collision, if any, and abort the transmission.
- This is so because the station, once the entire frame is sent, does not keep a copy of the frame and does not monitor the line for collision detection.

$$T_{fr} \geq 2 * T_p$$

Flow diagram for the CSMA/CD



Q1 : A network using *CSMA/CD* has a bandwidth of 10 Mbps. If the maximum propagation time (including the delays in the devices and ignoring the time needed to send a jamming signal, as we see later) is 25.6 micro seconds, what is the minimum size of the frame?

Solⁿ $T_p = 25.6 \mu\text{sec.}$

✓ $T_{fr} = 2 \times T_p$

$$\frac{\text{Min. frame size}}{\text{Bandwidth}} = 2 \times T_p$$

$$\text{Min frame size} = 2 \times T_p \times \text{Bandwidth}$$

$$= 2 \times 25.6 \times 10^{-6} \times 10 \times 10^6$$

$$= 51.2 \times 10 = \underline{512 \text{ bits}}$$

$$\text{Min. frame size} = 512 \text{ bits} \quad \underline{\text{Ans}}$$

Q2: In a CDMA/CD network with a data rate of 10 Mbps, the minimum frame size is found to be 512 bits for the correct operation of the collision detection process. What should be the minimum frame size if we increase the data rate to 100 Mbps? To 1 Gbps? To 10 Gbps?

Solⁿ. Bandwidth = 10 Mbps = 10×10^6 bps. min frame size = 512 bits

$$T_{fr} = 2 \times T_p$$

$$T_{fr} = \frac{512}{10 \times 10^6} = 51.2 \mu s$$

$$T_p = \frac{51.2 \mu s}{2} = 25.6 \mu s$$

$$\text{min frame size} = T_{fr} \times \text{Bandwidth}$$
$$= 2 \times T_p \times \text{Bandwidth}$$

propagation time
of the n/w will remain
constant if distance
and speed are same.

1) If data rate is 100 Mbps.

$$\begin{aligned}\text{Min frame size} &= 2 \times T_p \times \text{data rate} \\ &= 2 \times 25.6 \times 10^{-6} \times 100 \times 10^6 \\ &= 51.2 \times 100 = 5120 \text{ bits } \underline{\text{Ans.}}\end{aligned}$$

2) If data rate is 1 Gbps

$$\begin{aligned}\text{Min. frame size} &= 2 \times 25.6 \times 10^{-6} \times 10^9 \\ &= 51.2 \times 10^3 \\ &= 51200 \text{ bits } \underline{\text{Ans.}}\end{aligned}$$

3) If data rate is 10 Gbps

$$\begin{aligned}\text{Min. frame size} &= 2 \times 25.6 \times 10^{-6} \times 10 \times 10^9 \\ &= 51.2 \times 10^4 \\ &= \underline{512000 \text{ bits } \underline{\text{Ans.}}}\end{aligned}$$

Q3: In a CDMA/CD network with a data rate of 10 Mbps, the maximum distance between any station pair is found to be 2500 m for the correct operation of the collision detection process. What should be the maximum distance if we increase the data rate to 100 Mbps? To 1 Gbps? To 10 Gbps?

Solⁿ.

$$T_{fr} = 2 \times T_p \quad \checkmark$$

$$\frac{\text{min frame size}}{\text{data rate}} = \frac{2 \times \text{distance}}{\text{propagation speed}}$$

$$100 \times 10^6$$

$$10^9$$

$$\text{distance} = \frac{\text{min. frame size} \times \text{propagation speed}}{2 \times \text{data rate}} \quad \checkmark$$

1) If data rate is increased to 100 Mbps from 10 Mbps.

$$\text{distance} \propto \frac{1}{\text{data rate}}$$

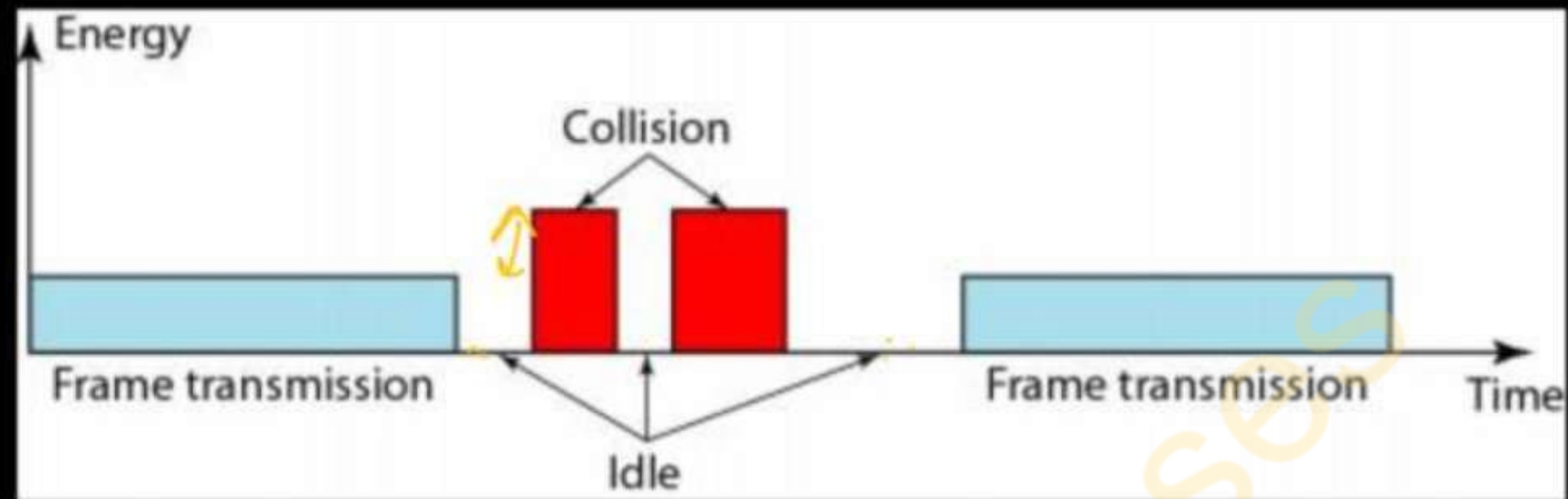
$$\text{distance} = 250 \text{ m.}$$

2) For data rate = 1 Gbps.

$$\text{distance} = 25 \text{ m}$$

3) For data rate = 10 Gbps

$$\text{distance} = 2.5 \text{ m.}$$



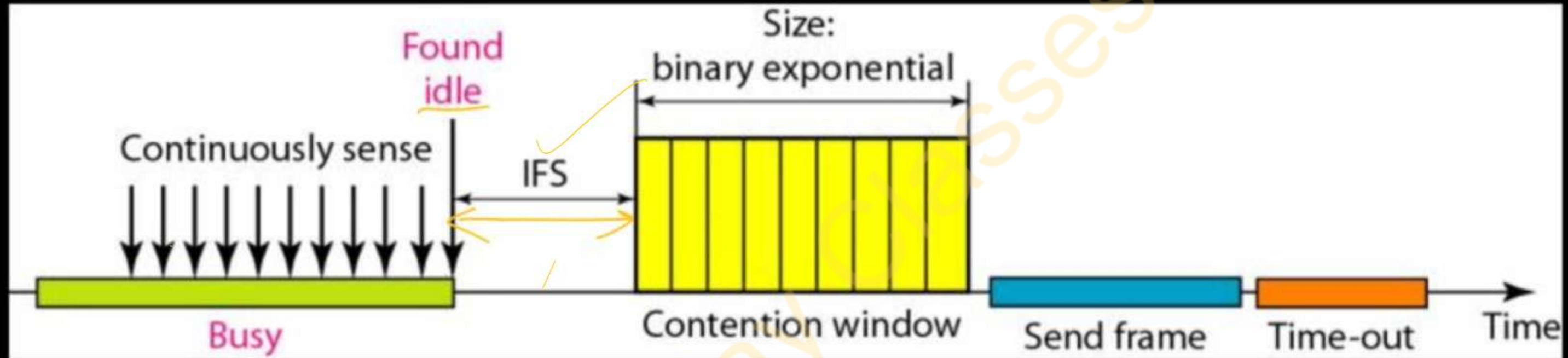
Throughput

- Throughput of CSMA/CD is greater than that of pure or slotted ALOHA.
- For 1-persistent method the maximum throughput is around 50% when $G=1$.
- For nonpersistent method, the maximum throughput can go up to 90% when G is between 3 and 8.

$$G = (3 - 8)$$

CSMA/CA

Collisions are avoided through the use of CSMA/CA's three strategies:
① the interframe space, ② the contention window, and ③ acknowledgments



Interframe Space (IFS)

- When an idle channel is found, the station does not send immediately. It waits for a period of time called the interframe space or IFS.
- If after the IFS time the channel is still idle, the station can send, but it still needs to wait a time equal to the contention time.

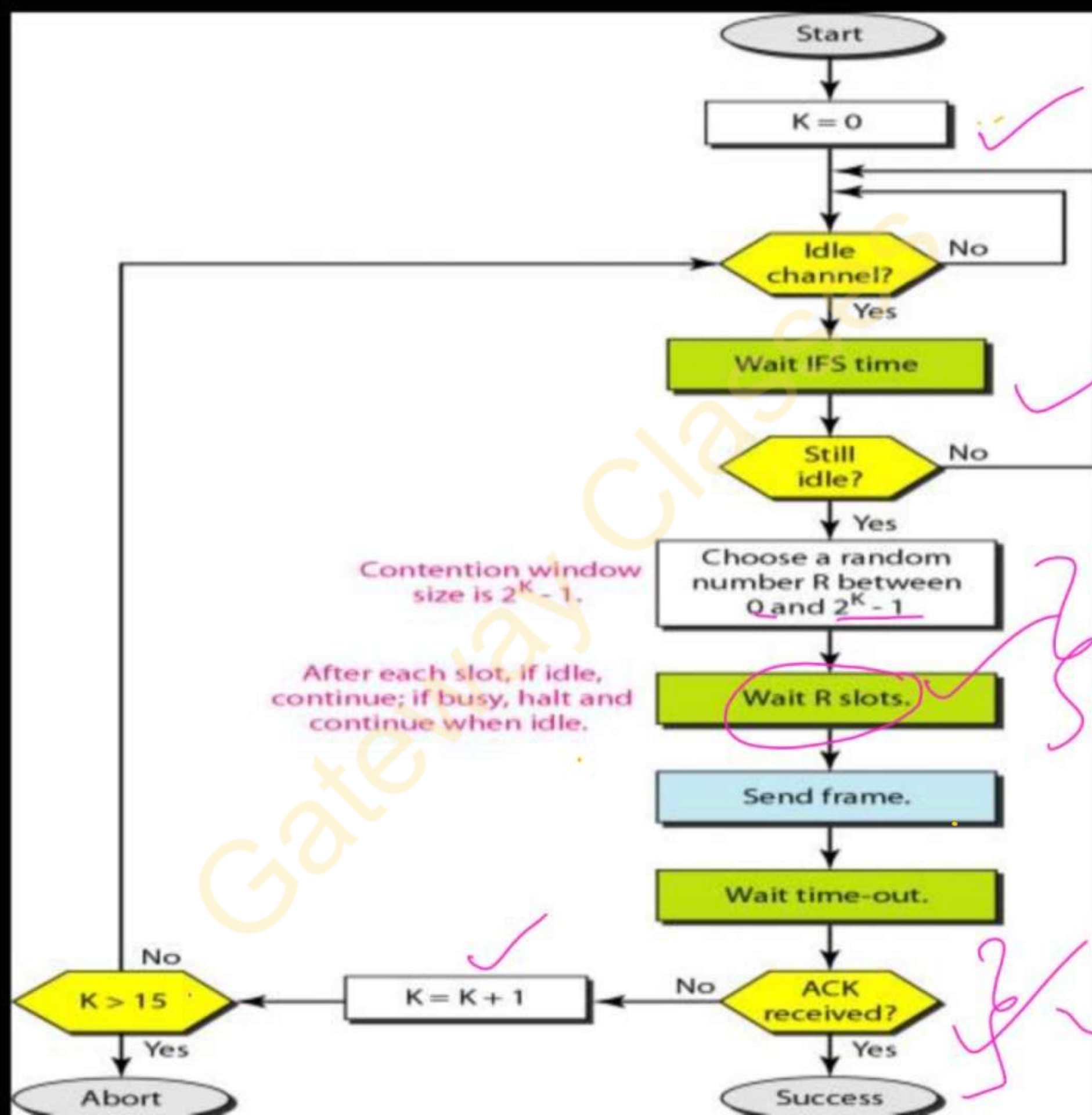
Contention Window

- The contention window is an amount of time divided into slots.
- Station that is ready to send wait for a random no. of slots(chosen using binary exponential back-off).
- Station needs to sense the channel after each time slot.
- If the station finds the channel busy; it just stops the timer and ^{resumes} restarts it when the channel is sensed as idle. This gives priority to the station with the longest waiting time.

Acknowledgment

The positive acknowledgment and the time-out timer can help guarantee that the receiver has received the frame.

In CSMA/CA, the IFS can also be used to define the priority of a station or a frame.



AKTU PYQs

1. Explain CSMA/CD ^{and} CSMA/CA with diagram. (AKTU 2022-23)
2. Explain CSMA/CD in detail. (AKTU 2017-18)
3. Discuss different carrier sense protocols. How are they different from collision protocols? (AKTU 2017-18)
4. Explain all Random-Access protocols. (AKTU 2023-24)

Pure ALOHA
slotted ALOHA
CSMA
CSMA/CD
CSMA/CA



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Computer Networks

Unit-2: Lecture 4

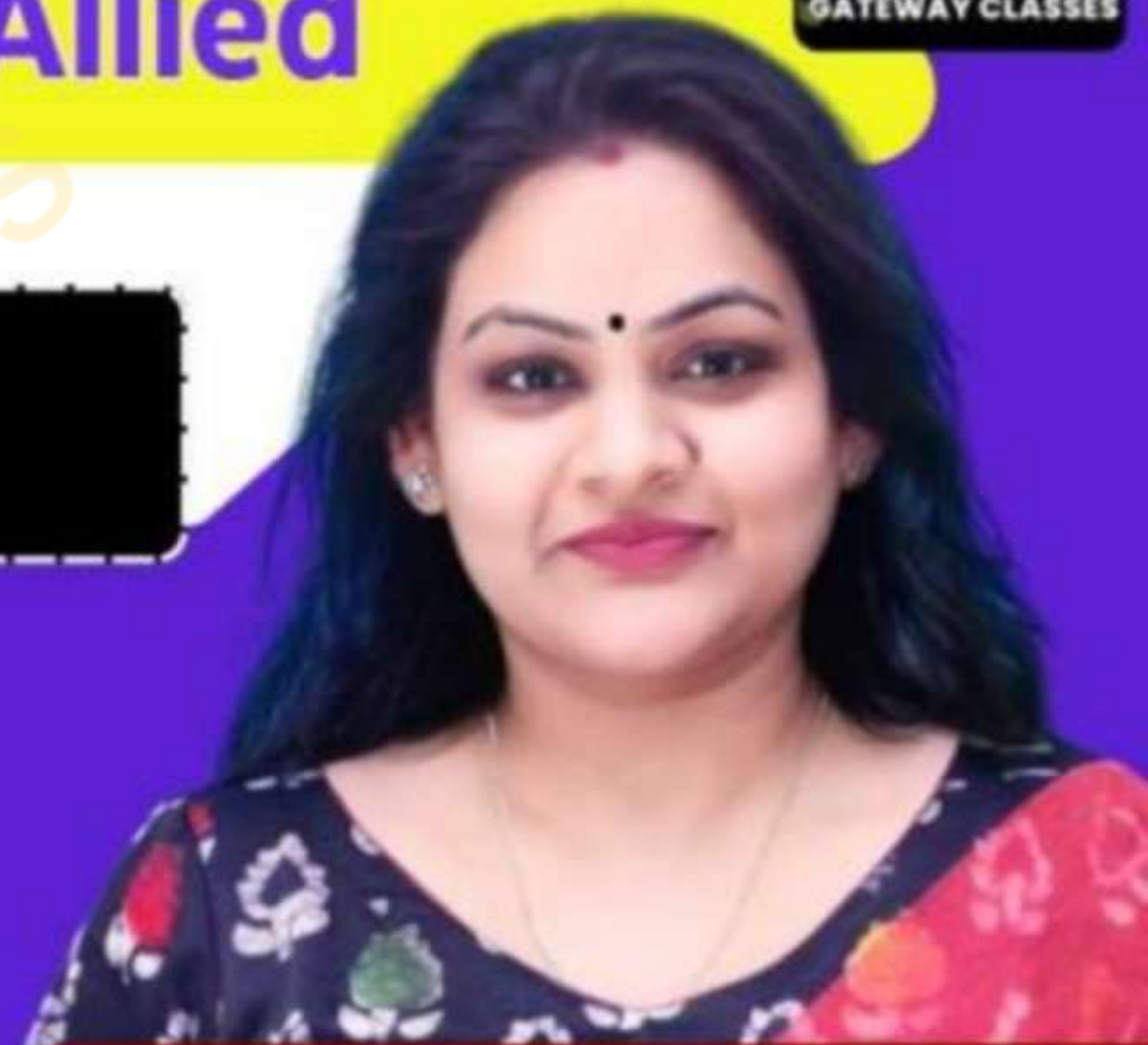
Today's Target

- Controlled Access Protocols
- Channelization Protocols

AKTU PYQs

By Dr. Nidhi Parashar Ma'am

- Ph.D (CSE)
- M.Tech Gold Medalist
- Net Qualified



Controlled Access Protocols

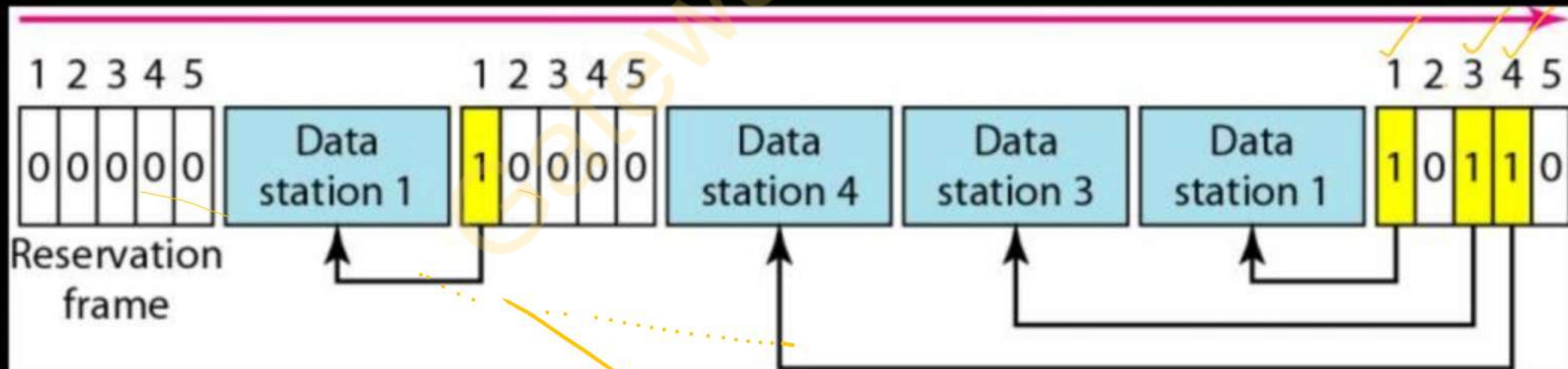
- The stations consult one another to find which station has the right to send.
- A station cannot send unless it has been authorized by other stations.
- The popular controlled-access methods are:

1. Reservation
2. Polling
3. Token Passing

Gateway Classes

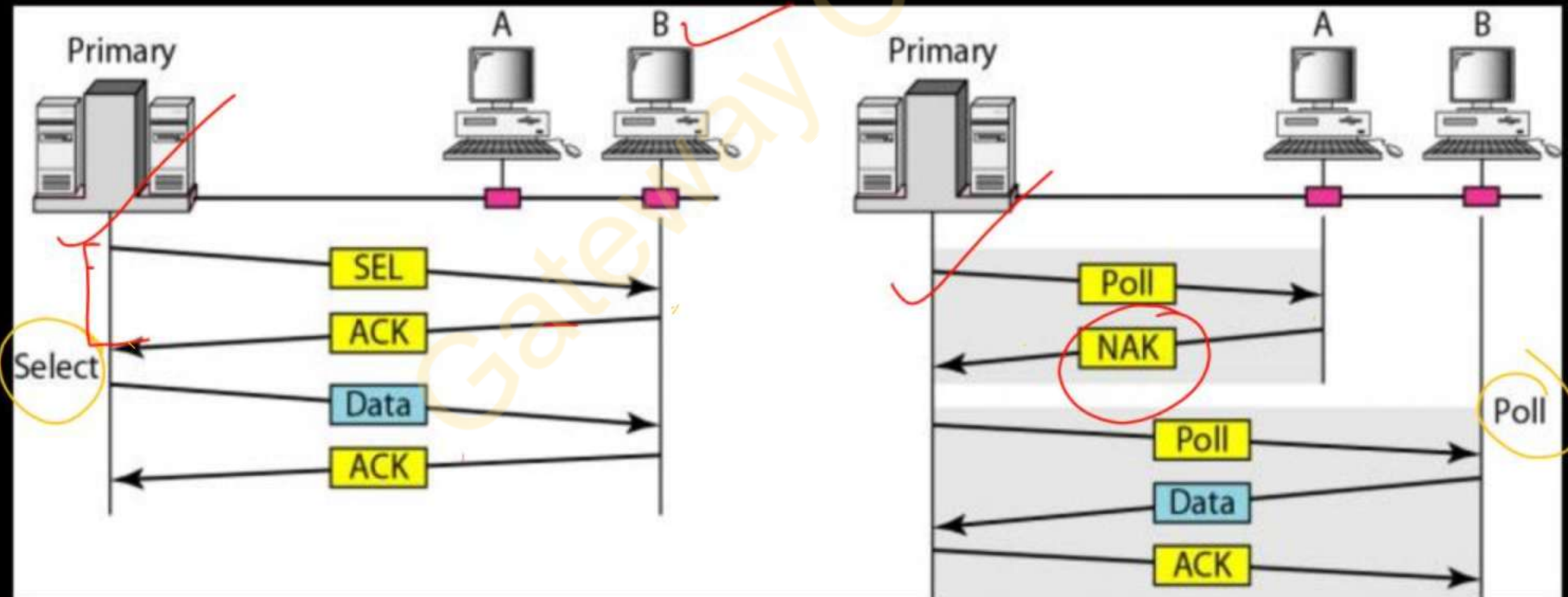
Reservation

- A station needs to make a reservation before sending data.
- Time is divided into intervals.
- In each interval, a reservation frame precedes the data frames sent in that interval.
- If there are N stations in the system, there are exactly N reservation mini slots in the reservation frame. Each mini slot belongs to a station. When a station needs to send a data frame, it makes a reservation in its own mini slot.
- The stations that have made reservations can send their data frames after the reservation frame.



Polling

- Works with topologies where one device is designated primary and other devices are secondary.
- All data exchanges must be made through the primary device.
- The primary device controls the link; the secondary devices follow its instructions.
- Primary device determine which device is allowed to use the channel at a given time.
- The primary device is always the initiator of a session.



Select Function

- Used whenever the primary device has something to send.
- Before sending data, the primary transmits a select (SEL) frame, which includes the address of the intended secondary to alert the secondary to the upcoming transmission and wait for an ack of the secondary's ready status.

Poll Function

- Used by the primary device to solicit transmissions from the secondary devices.
- When primary is ready to receive data, it must ask (poll) each device in turn if it has anything to send.
- Secondary responds either with a NAK frame if it has nothing to send or with data frame if it does.
- If the response is a NAK frame, then primary polls next secondary until it finds one with data to send.
- When response is positive (a data frame), the primary reads the frame and returns an ACK frame.

Token Passing

- Stations in a network are organized in a logical ring.
- The right to channel access is passed from the predecessor to the current station. The right will be passed to the successor when the current station has no more data to send.
- A special packet called a token circulates through the ring. The possession of the token gives the station the right to access the channel and send its data.
- When a station has some data to send, it waits until it receives the token from its predecessor. It then holds the token and sends its data.
- When the station has no more data to send, it releases the token, passing it to the next logical station in the ring.
- In this process, when a station receives the token and has no data to send, it just passes the token to the next station.

Token management is needed for this access method due to the following reasons:

- Stations must be in possession of the token for limited time.
- The token must be monitored to ensure it has not been lost or destroyed.
- To assign priorities to the stations and to the types of data being transmitted.
- To make low-priority stations release the token to high priority stations.

Logical Ring

- In a token-passing network, stations do not have to be physically connected in a ring; the ring can be a logical one.

Physical ring

- When a station sends the token to its successor, the token cannot be seen by other stations.
- If one of the links between two adjacent stations fails, the whole system fails.

Dual ring :

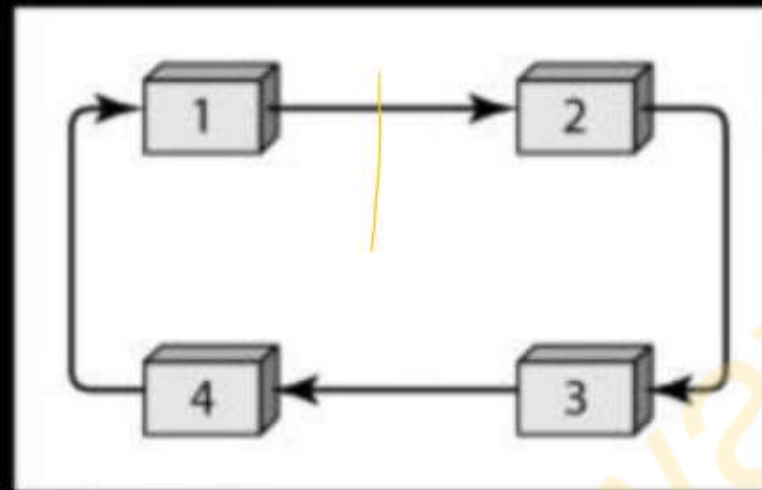
- A second ring (used for emergencies only) operates in reverse direction compared with the main ring.
- If one of the links in main ring fails, system automatically combines two rings to form temporary ring.
- After the failed link is restored, the auxiliary ring becomes idle again.

Bus ring(token bus):

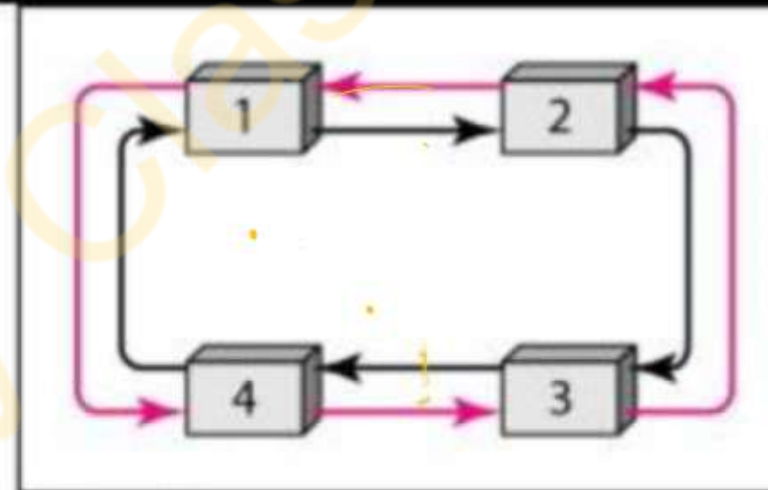
- Stations are connected to a single cable called a bus.
- They make a logical ring as each station knows the address of its successor (and also predecessor for token management purposes).
- When a station has finished sending its data, it releases the token and inserts the address of its successor in the token. Only the station with the address matching the destination address of the token gets the token to access the shared media.

Star ring:

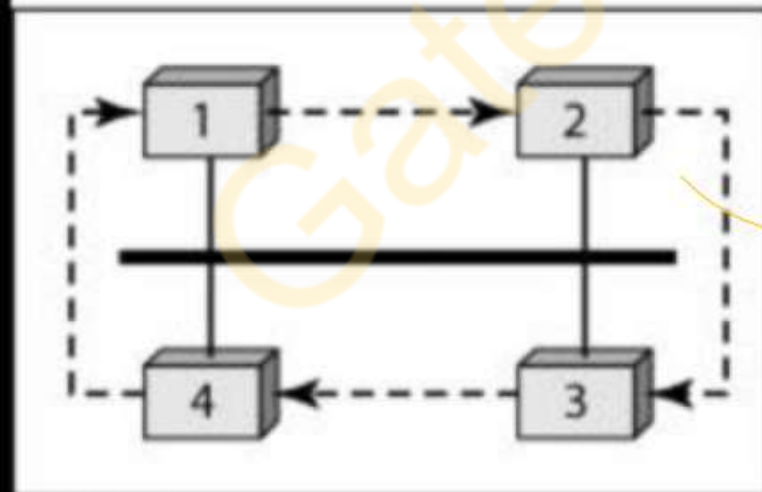
- Wiring inside hub makes the ring; stations are connected to this ring through two wire connections.
- Network is less prone to failure because if a link goes down, it will be bypassed by the hub and rest of the stations can operate.
- Adding and removing stations from the ring is easier.



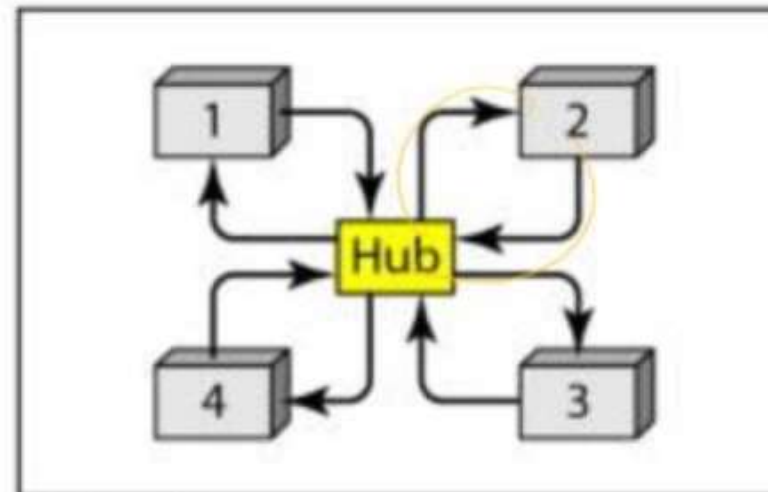
a. Physical ring



b. Dual ring



c. Bus ring



d. Star ring

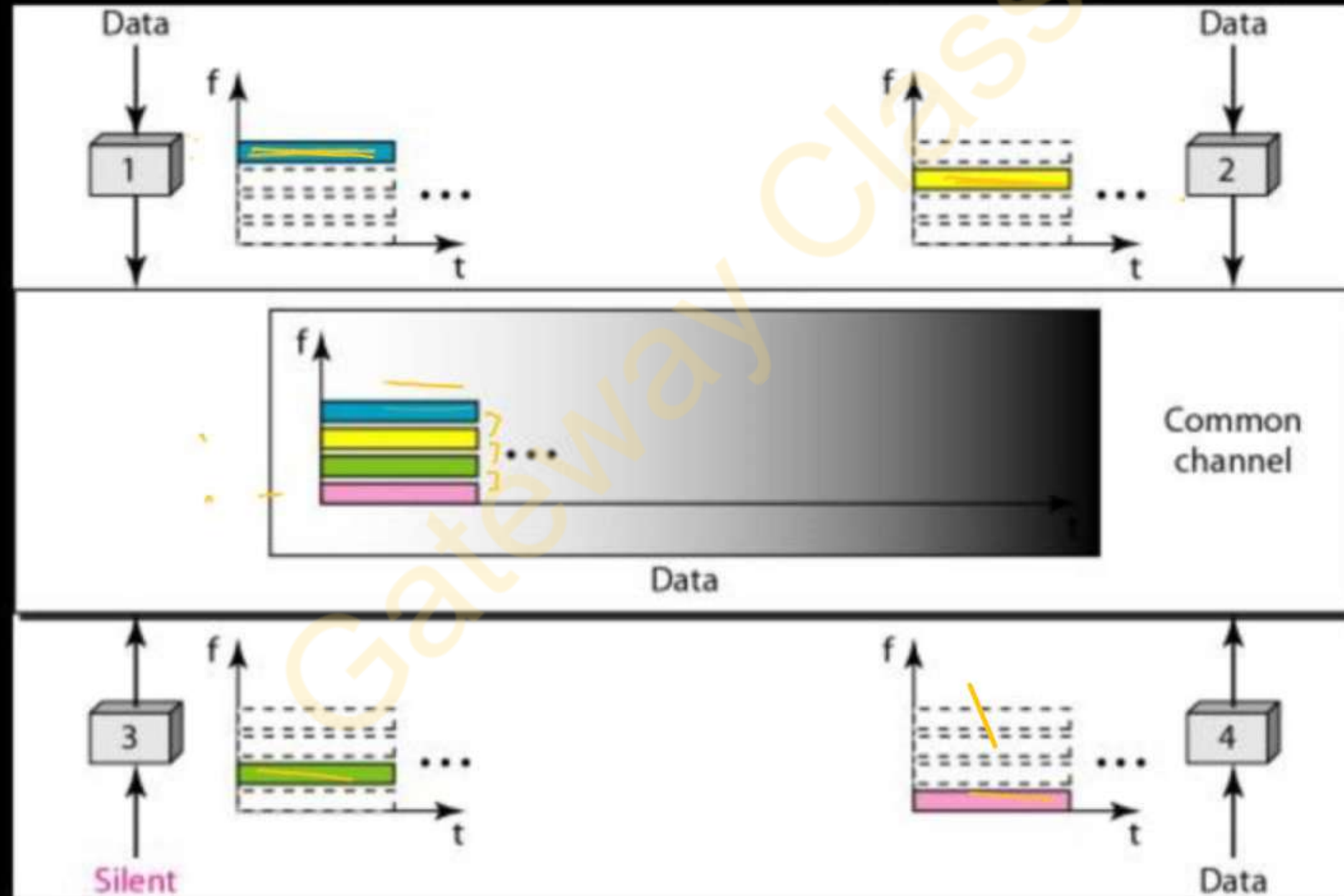
Channelization Protocols

Here available bandwidth of a link is shared in time, frequency, or through code, between different stations. Three channelization protocols:

1. Frequency-Division Multiple Access (FDMA) ✓
2. Time-Division Multiple Access (TDMA) ✓
3. Code-Division Multiple Access (CDMA) ✓

Frequency-Division Multiple Access (FDMA):

- Available bandwidth is divided into frequency bands.
- Each station is allocated a band to send its data for all the time.
- To prevent station interferences, allocated bands are separated by small *guard bands*.

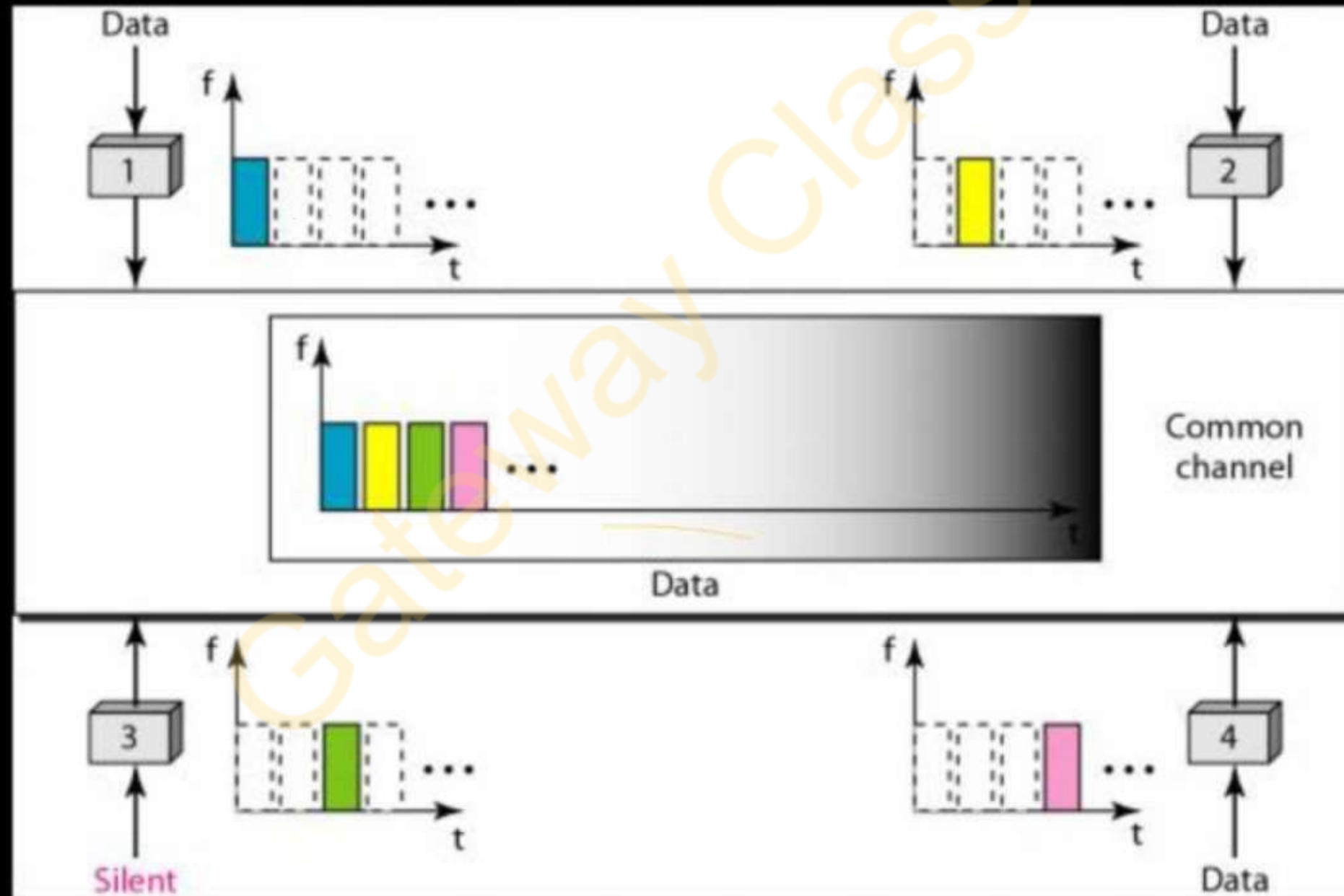


Time-Division Multiple Access (TDMA):

Stations share the bandwidth of the channel in time.

Each station is allocated a time slot during which it can send data.

Each station transmits its data in its assigned time slot.



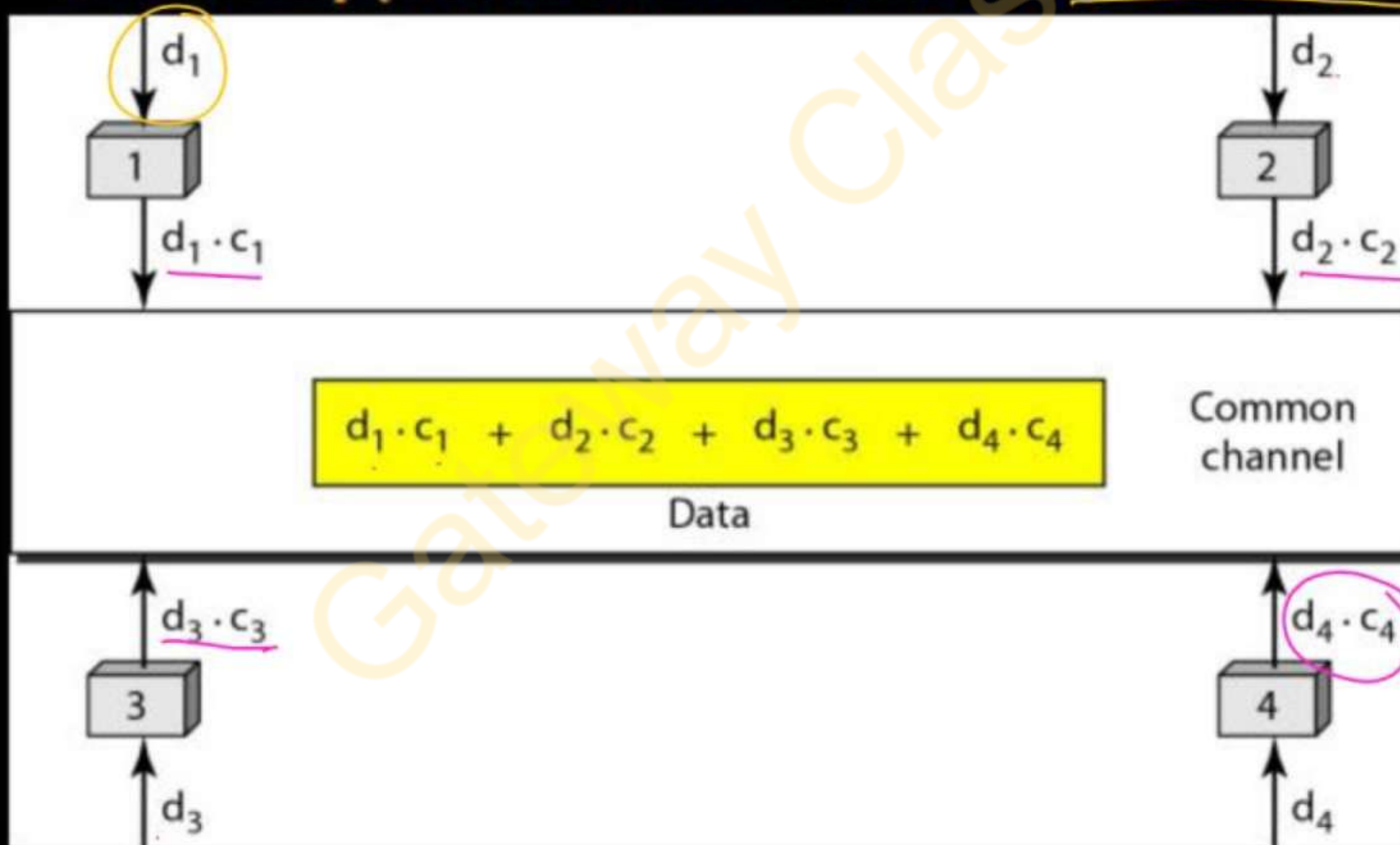
Code-Division Multiple Access (CDMA)

Differs from FDMA because only one channel occupies the entire bandwidth of the link.

Differs from TDMA because all stations can send data simultaneously; there is no timesharing.

In CDMA, one channel carries all transmissions simultaneously.

CDMA simply means communication with different codes.



- CDMA is based on coding theory. Each station is assigned a code, which is a sequence of numbers called chips. They are called orthogonal sequences.



The assigned codes are assumed to have two properties.

1. If we multiply each code by another, we get 0.
2. If we multiply each code by itself, we get 4 (the number of stations).

- Any station that wants to receive data from one of the other three stations, multiplies the data on the channel by the code of the sender.

Code Properties:

1. Each sequence is made of N elements, where N is the number of stations

2. Scaler multiplication For example, 2. $[+1 +1 -1 -1] = [+2 +2 -2 -2]$

3. If we multiply two equal sequences, element by element, and add the results, we get N . This is called the inner product of two equal sequences. $[+1 +1 -1 -1] \cdot [+1 +1 -1 -1] = 1 + 1 + 1 + 1 = \underline{4}$

4. Multiplying two different sequences, element by element (inner product), and add the results, we get 0.

$$[+1 +1 -1 -1] \cdot [+1 +1 +1 +1] = 1 + 1 - 1 - 1 = \underline{0}$$

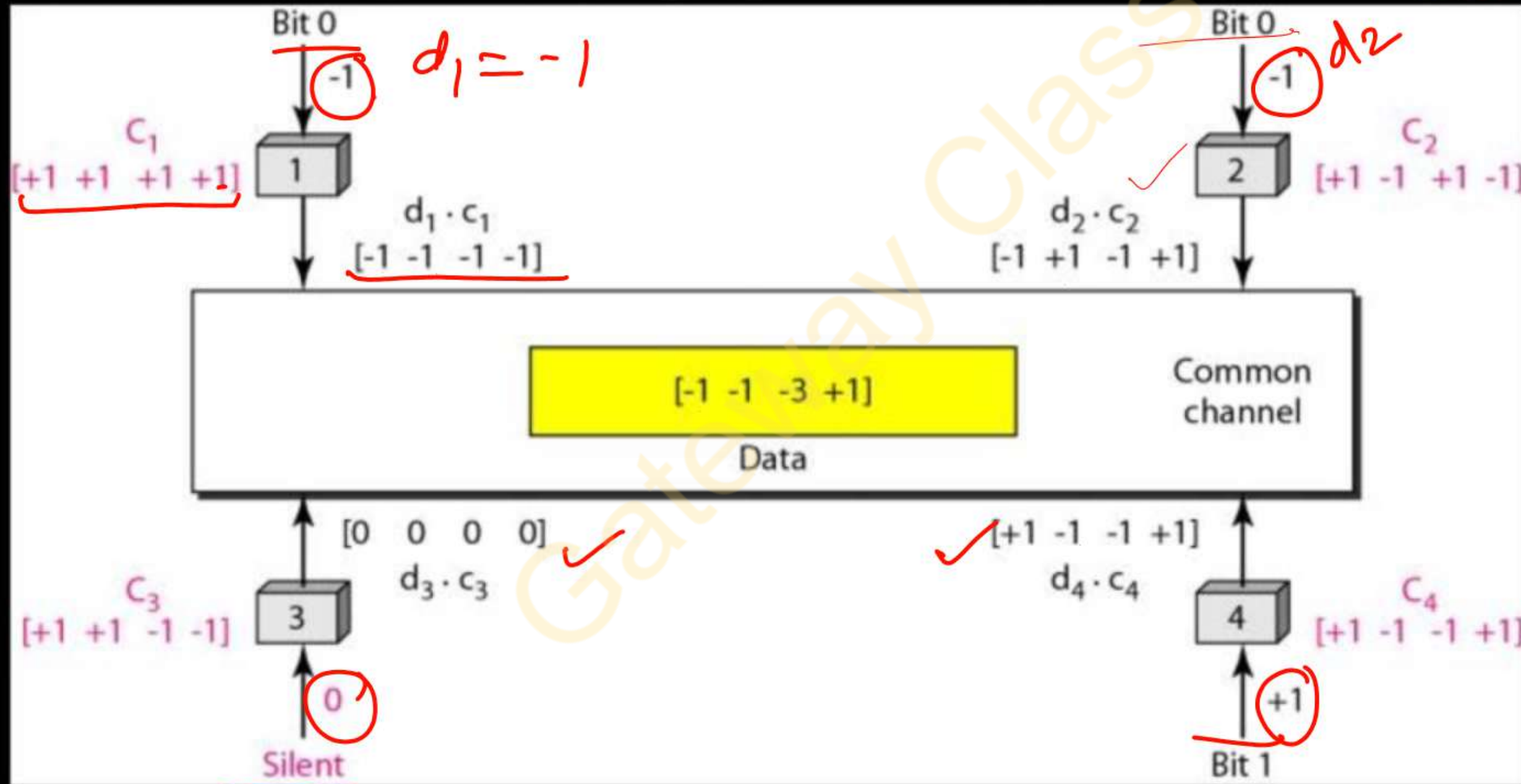
5. Adding two sequences means adding the corresponding elements.

$$[+1 +1 -1 -1] + [+1 +1 +1 +1] = [+2 +2 0 0]$$

Data Representation

- If a station needs to send a 0 bit, it encodes it as -1;
- if it needs to send a 1 bit, it encodes it as +1.
- When a station is idle, it sends no signal, which is interpreted as a 0.

$0 \rightarrow -1$
 $1 \rightarrow +1$ ✓
 $\text{silent} \rightarrow 0$



2) To receive data from station 2

$$[-1 -1 -3 +1] \cdot [+1 -1 +1 -1] \Rightarrow -1 +1 -3 -1 = -4/4 = -1$$

↓
0 is sent

3) To receive data from station 3

$$[-1 -1 -3 +1] \cdot [+1 +1 -1 -1] = -1 -1 +3 -1 = 0/4 = 0$$

↓
silent

4) To receive data from station 4

$$[-1 -1 -3 +1] \cdot [+1 -1 -1 +1] = -1 +1 +3 +1 = 4/4 = +1$$

↓
1 is sent

AKTU PYQs

1. Explain all Random-Access protocols. (AKTU 2023-24)

Gateway Classes



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Computer Networks

Unit-2: Lecture 5

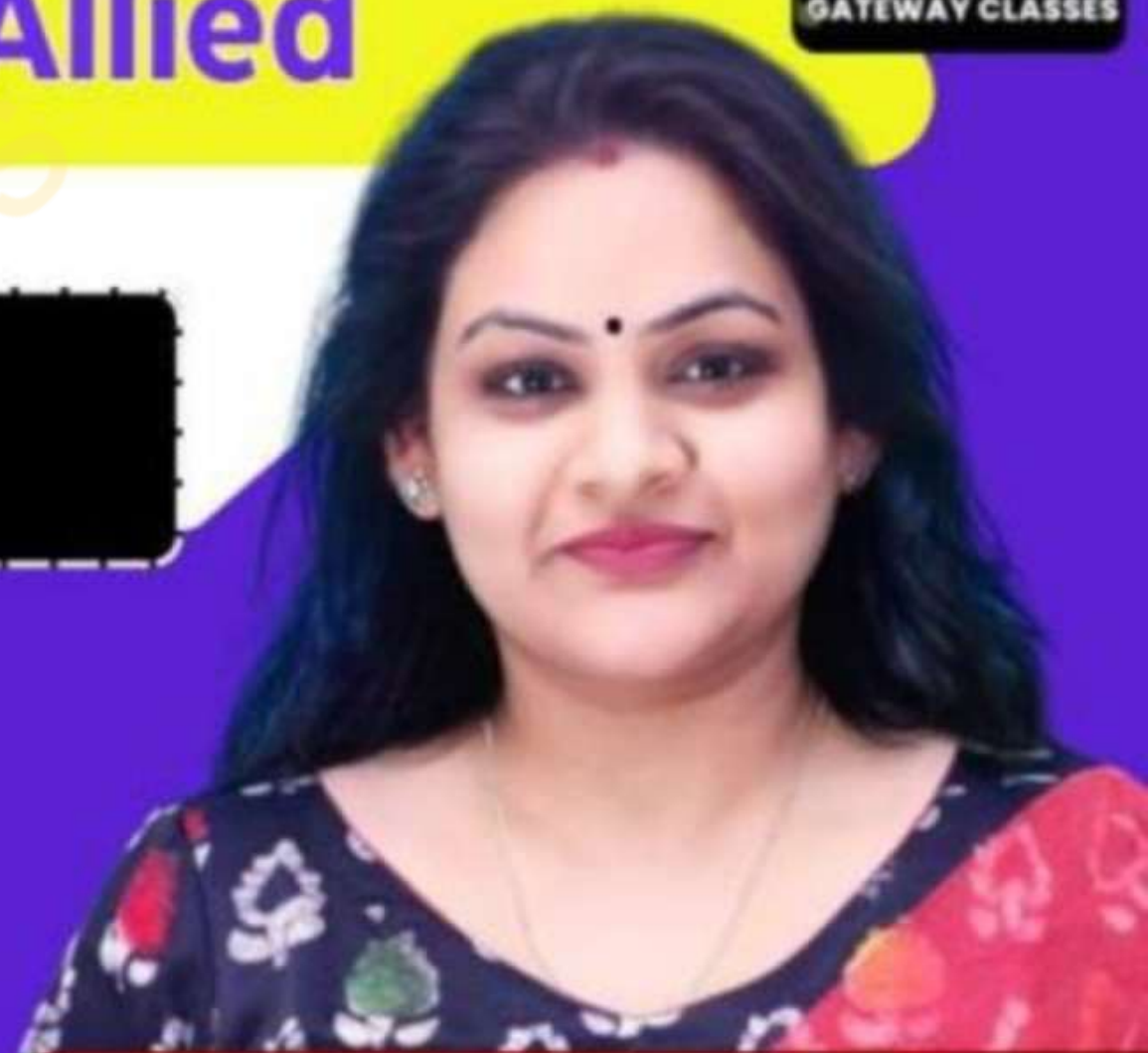
Today's Target

- Flow Control ✓
- Noiseless Protocols ✓
 - Simplest ✓
 - Stop and Wait ✓
- Noisy Protocols ✓
 - Stop and Wait ARQ ✓

AKTU PYQs

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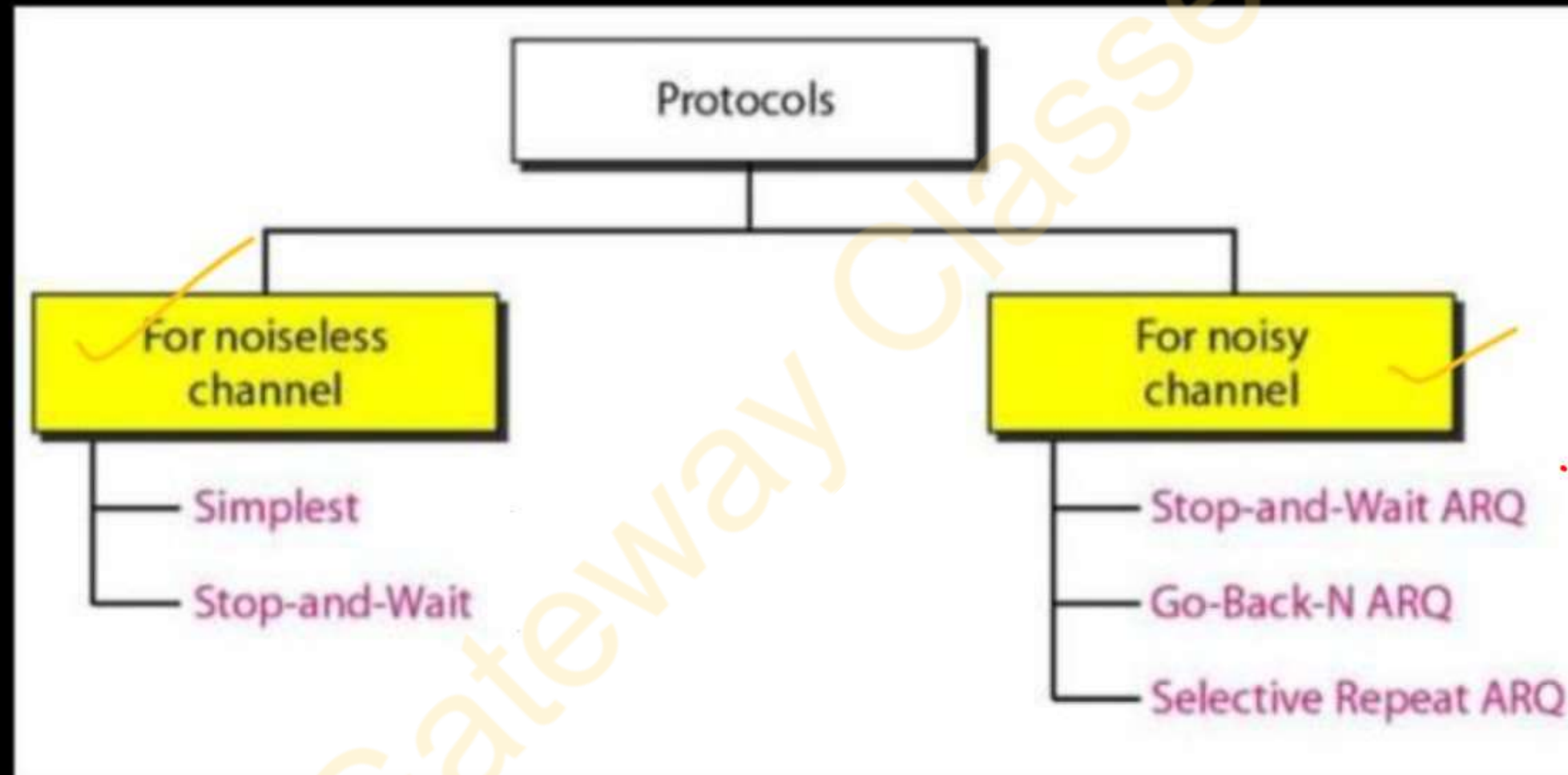


Flow Control

- Flow control refers to a set of procedures used to restrict the amount of data that the sender can send before waiting for acknowledgment.
- Error control in the data link layer is based on automatic repeat request (ARQ), which is retransmission of data.
- What if sender sends the bit pattern too fast?
- Receiver may not be able to process the data as fast as the sender is sending it.
- Receiver can buffer, but buffer has finite size. After that is filled, data is lost.
- Flow control – technique to control data flow between sender and receiver so that sender is blocked if receiver cannot accept any more data
- Sender must get an acknowledgement from the receiver before it can send more data

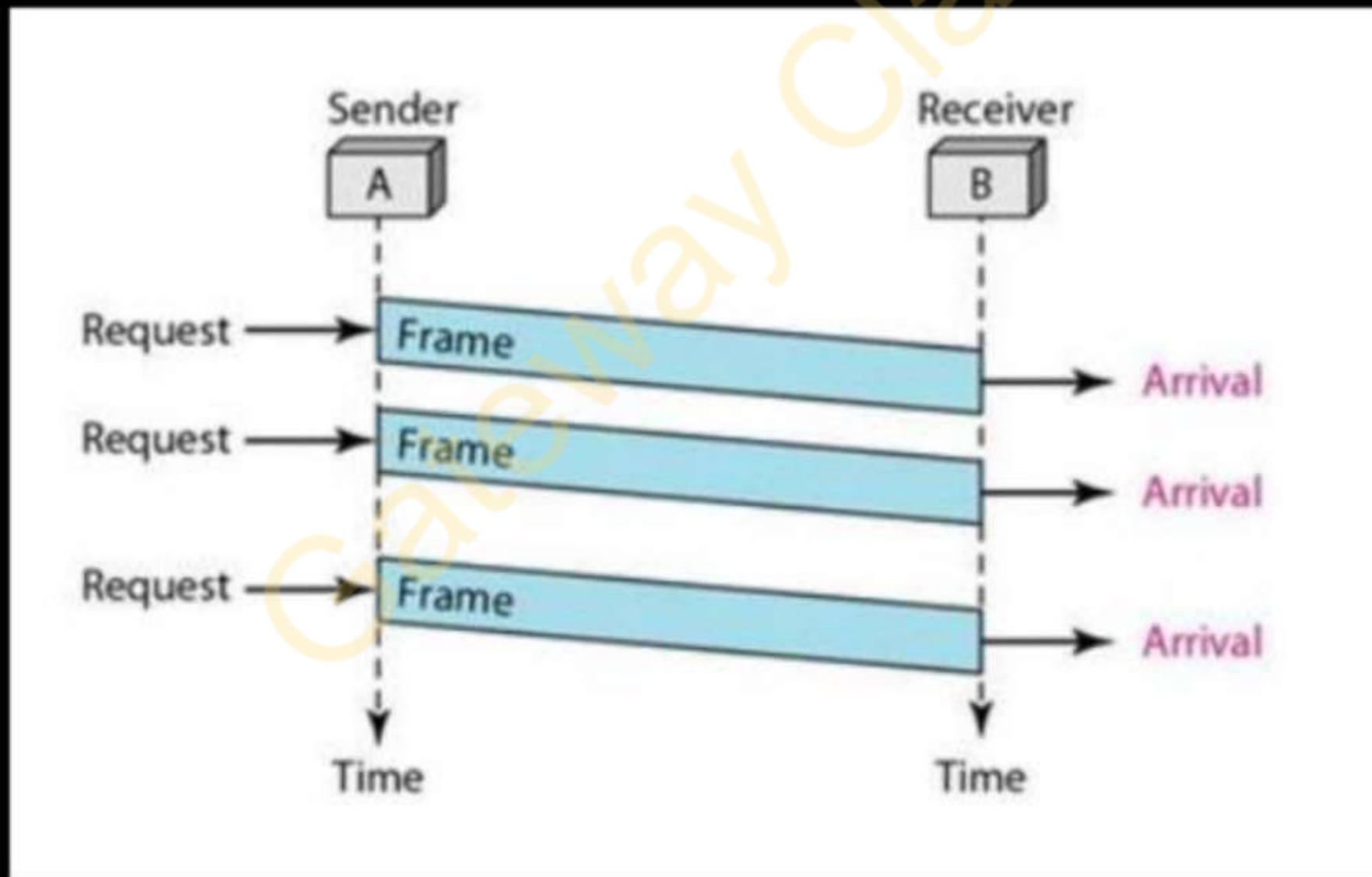
DLC

Error Control
Flow control



Simplest Protocol

The receiver can immediately handle any frame it receives with a processing time that is small enough to be negligible. Receiver can never be overwhelmed with incoming frames.



Stop-and-Wait Protocol

It works under the following assumptions-

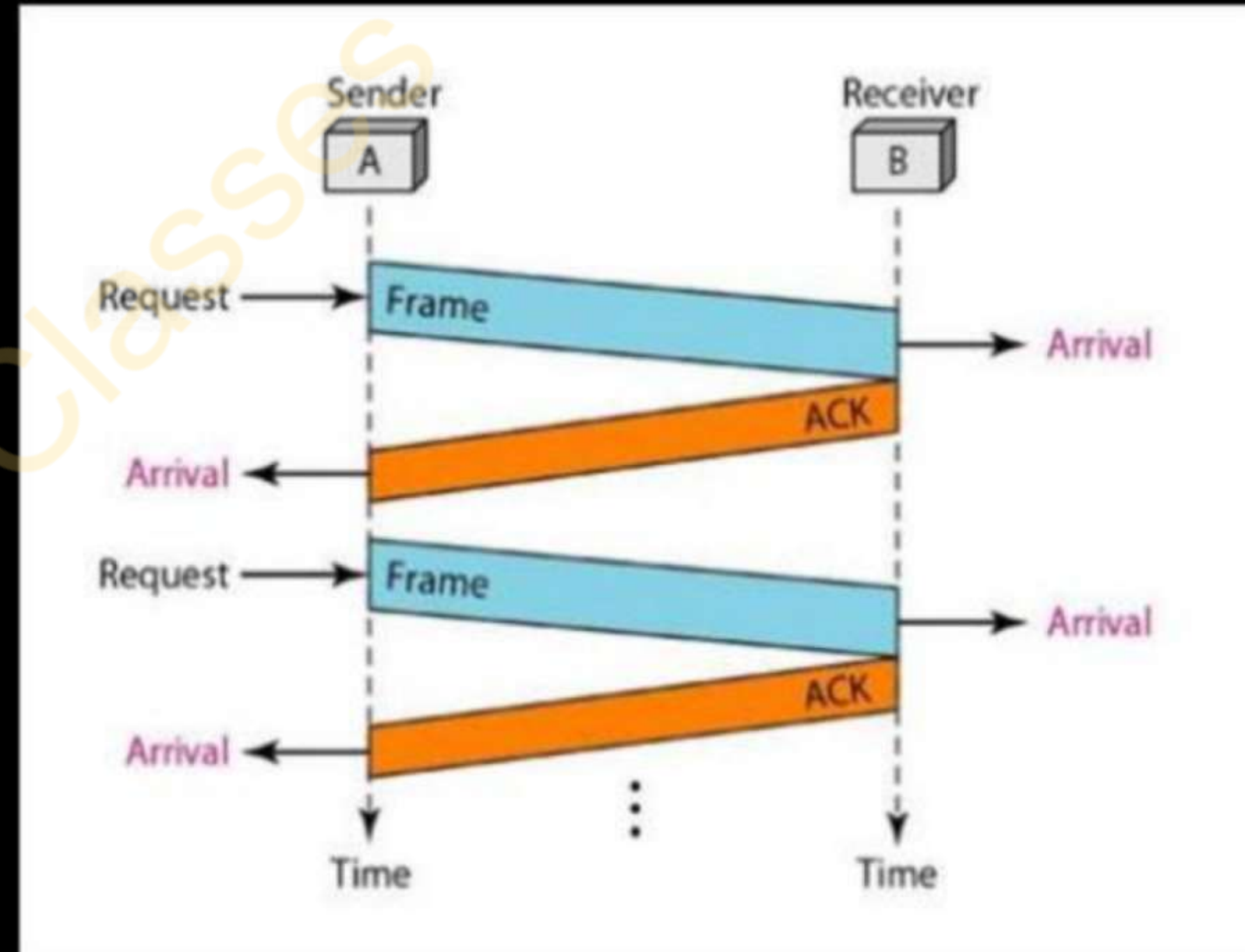
1. Communication channel is perfect.
2. No error occurs during transmission.

At Sender

- Send one data packet at a time.
- Send the next packet only after receiving acknowledgment for the previous.

At Receiver

- Send acknowledgement after receiving and consuming a data packet.
- After consuming packet acknowledgement need to be sent (**Flow Control**)



Problems Associated with Stop and Wait

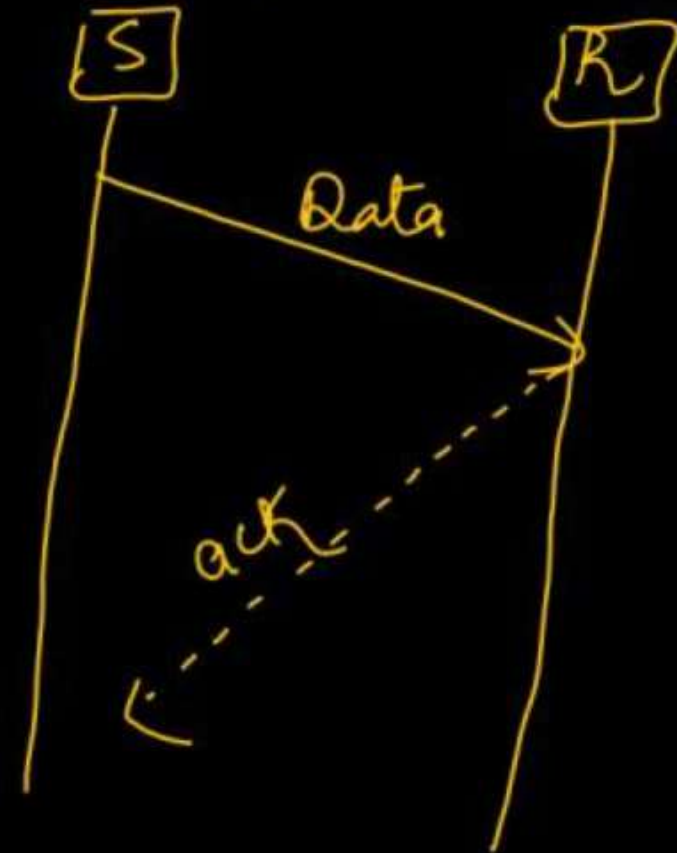
① lost data



② lost Acknowledgement



③ Delayed Ack

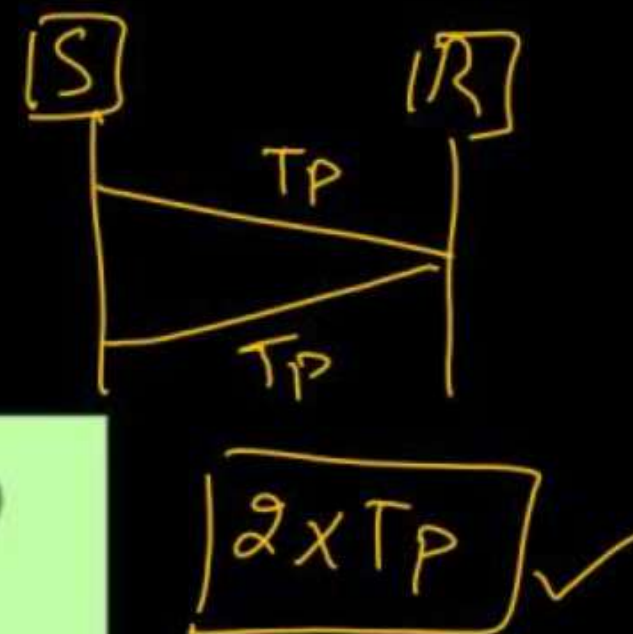
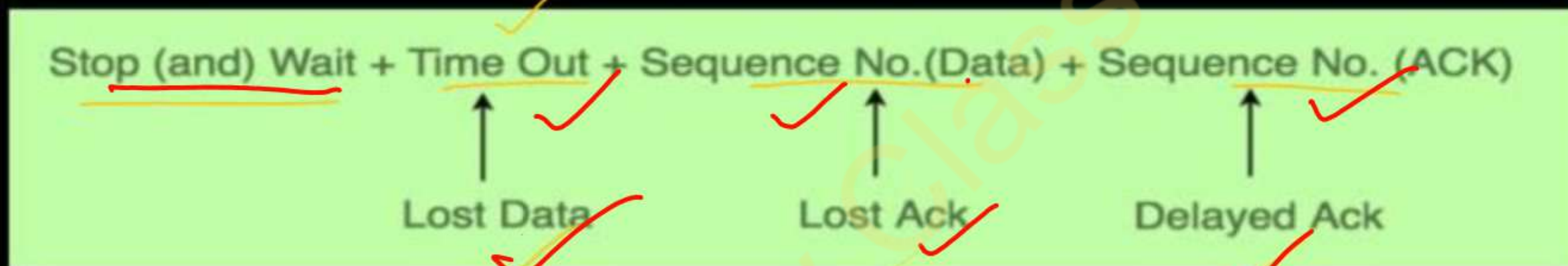


NOISY CHANNELS

- Three protocols that use error control.

1. Stop-and-Wait Automatic Repeat Request

- Resolves all three problems of simple stop and wait protocol.



1. Time Out

- Timeout refers to duration for which sender waits for an ACK from receiver after transmitting a data.
- If the sender does not receive an ACK within this timeout period, it assumes that the frame was lost or corrupted and retransmits the frame.

2. Sequence Number (Data)

- Sender assigns sequence numbers to each data frame it sends.
- Receiver identify and acknowledge each frame individually.
- After sending a frame, the sender waits for an acknowledgment before sending the next frame.



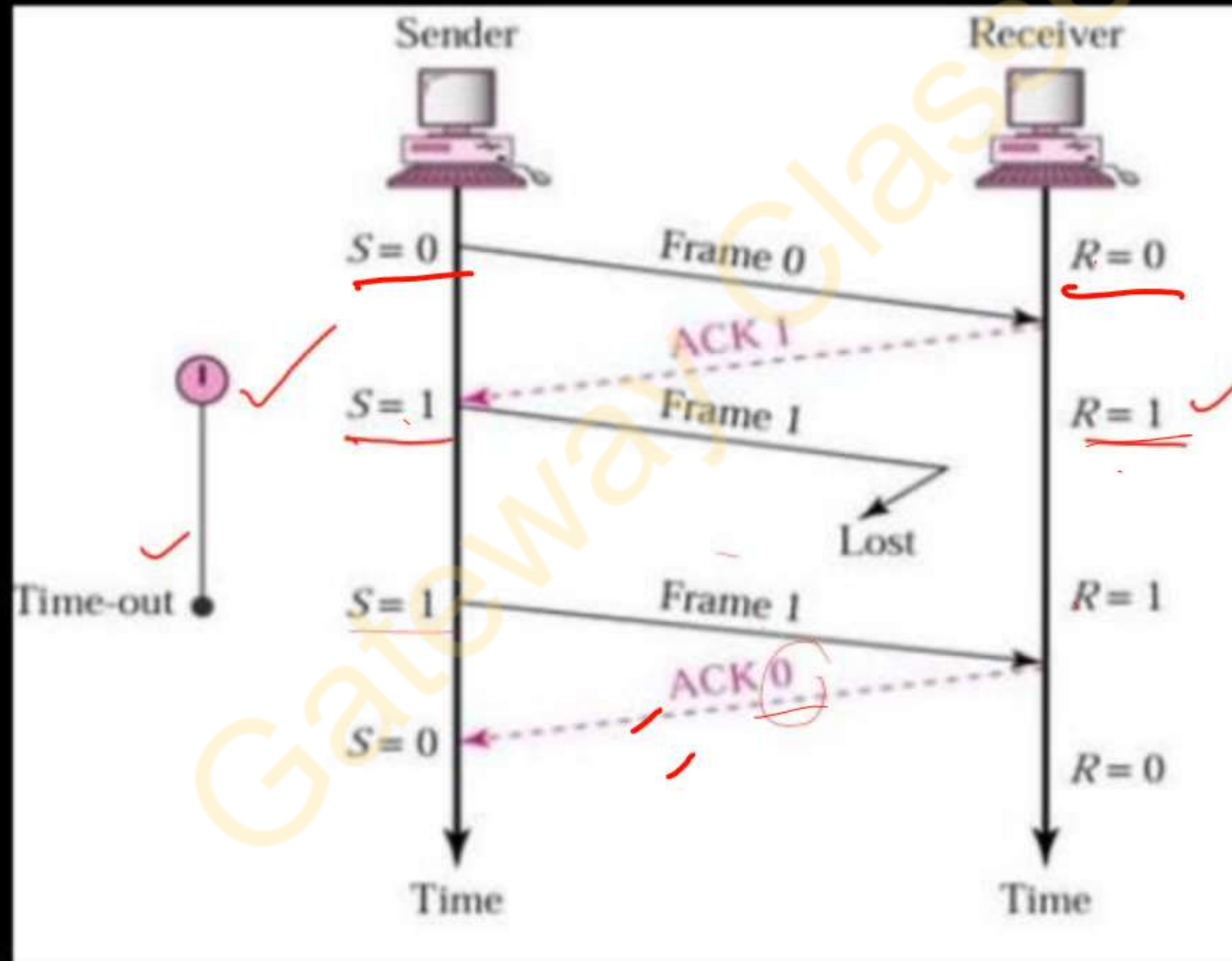
3. Sequence Number(Acknowledgement)

- Sequence numbers are also used in ACKs sent by the receiver.
- When the receiver successfully receives a data frame, it sends an ACK to sender, indicating the sequence number of the next expected frame.
- The sender uses this ACK to determine whether the transmission was successful and whether it can proceed to send the next frame.

In stop and wait ARQ, the sequence numbers are based on modulo-2 arithmetic.

1. Problem of Lost Data Packet

- When the sender sends the data packet, it starts a timer. If the timer goes off before receiving the ack from receiver, the sender retransmits the same data packet.

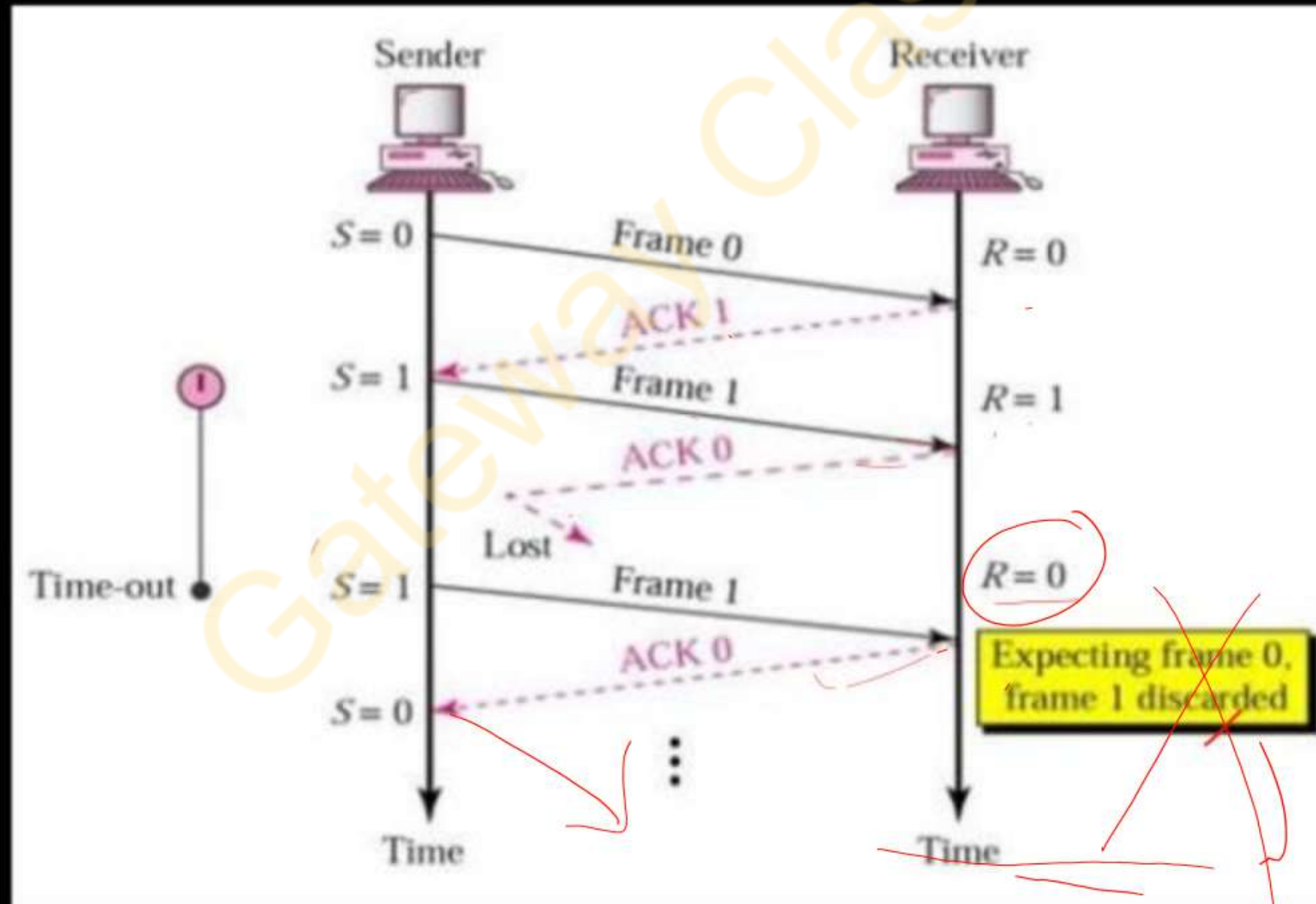


0, 1 ✓

0, 1 ✓

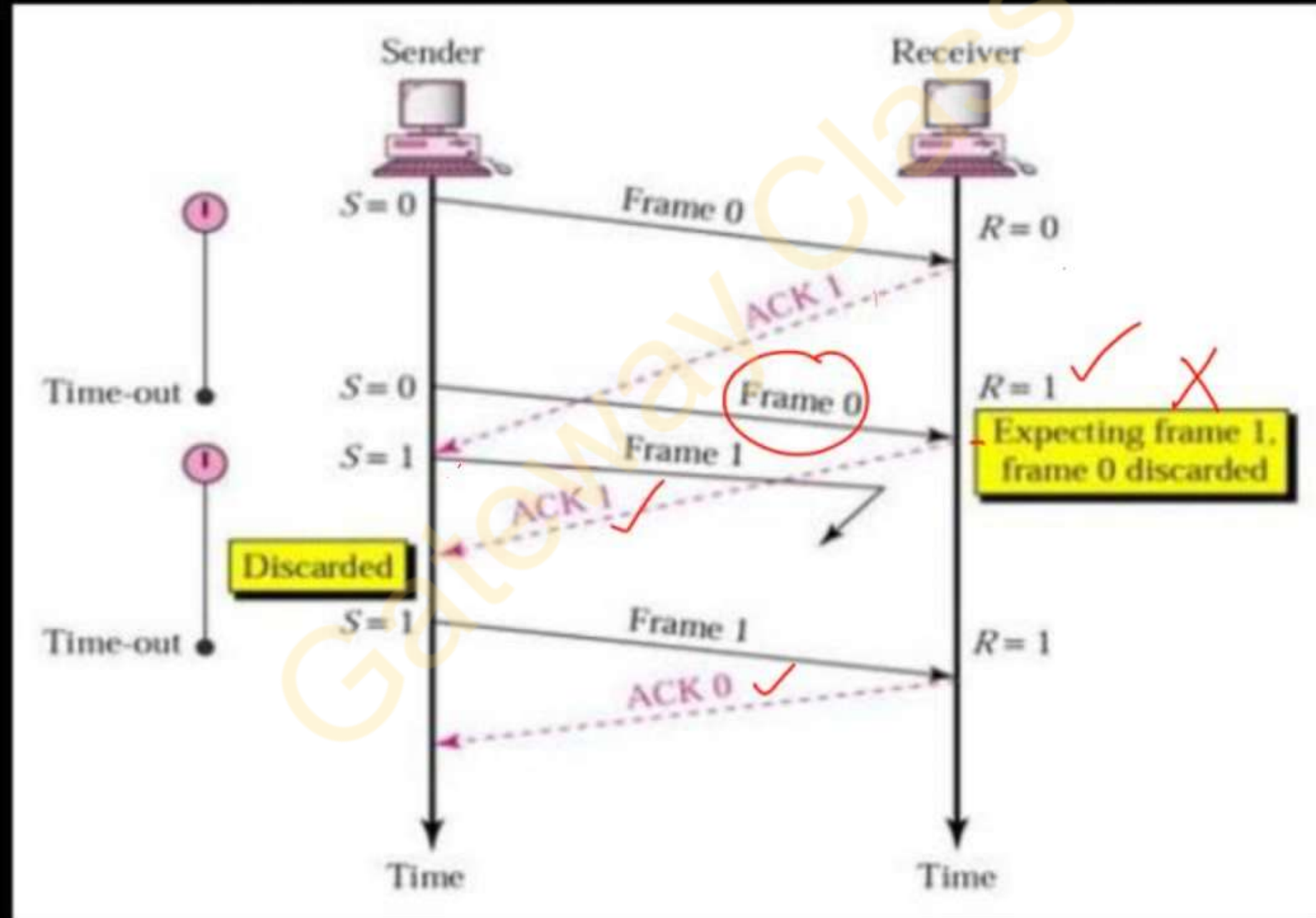
2. Problem of Lost Acknowledgement

- Sender attaches a sequence number with data packet to help the receiver identify the data packet.
- If ack not received within the time the sender retransmits the same data packet.
- Receiver already has the data packet, so it discards it and sends back an ack.
- This tells the sender that the certain data packet is now received correctly.

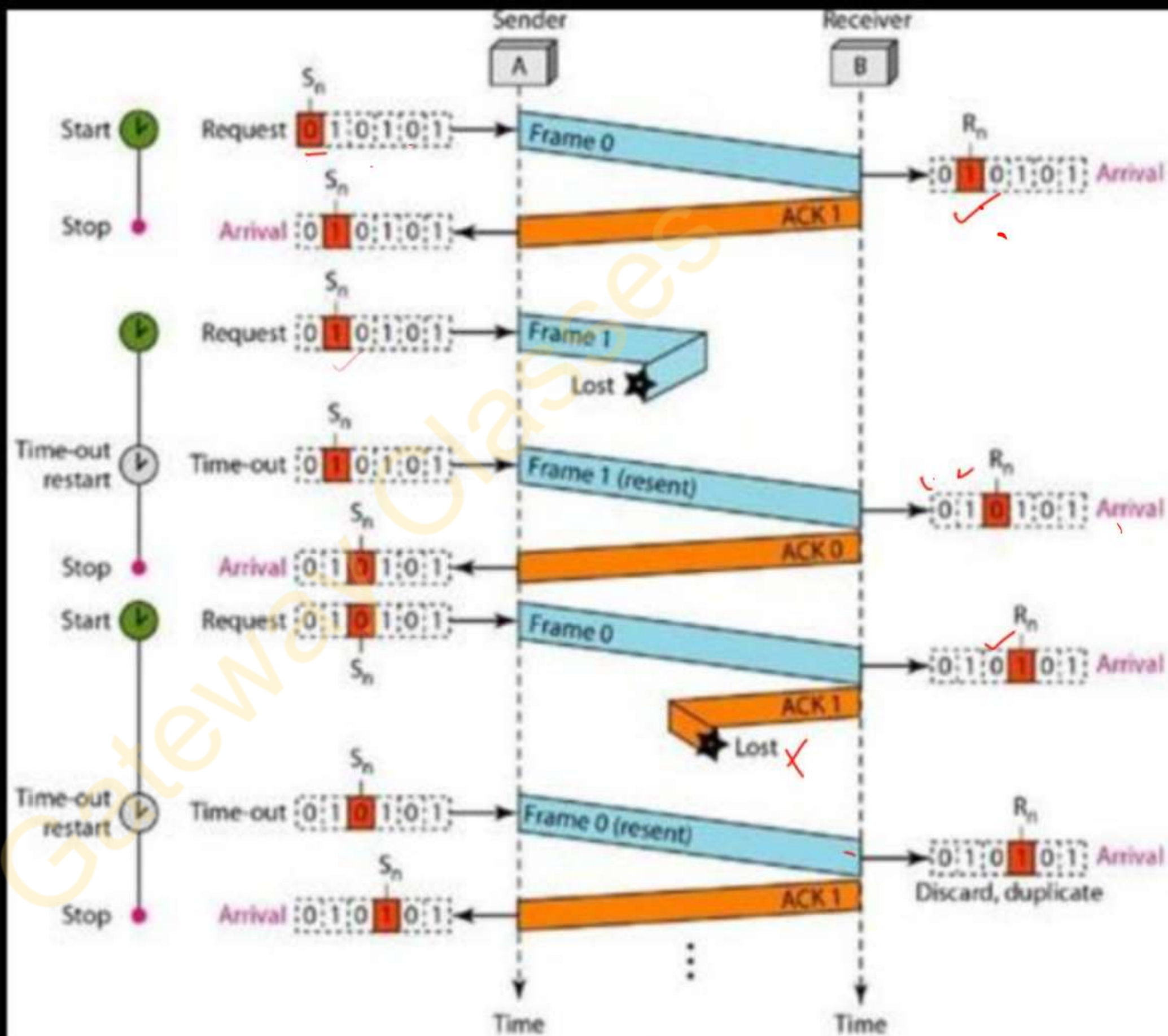


3.Problem of Delayed Acknowledgement

- If timer goes off before receiving the ack from receiver, the sender retransmits the same data packet.
- But the receiver already has the data packet, so it discards the data and sends it back the ack again. if the sender has received the previously sent ack then newer ack is discarded.

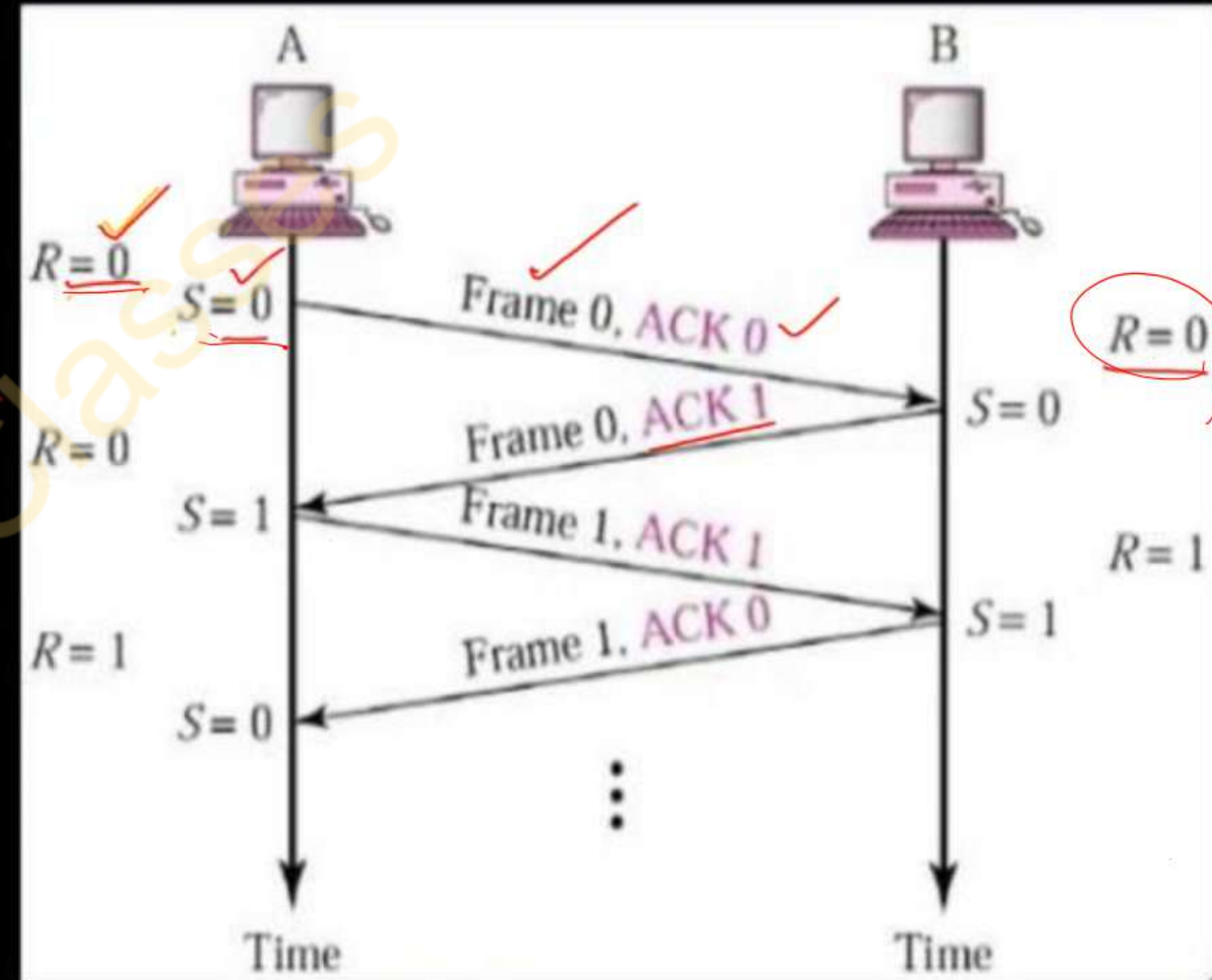


Normal operation of
stop-and-wait ARQ



2mb Piggybacking

- Piggybacking is the protocol to better utilize network bandwidth.
- In this acknowledgment transmission is delayed until we don't have the next data frame to send, ack is sent with the transmission of data frame.
- There is an ack field in the data. This ack bit field size is a few bits, while several bytes are required for the transmission of an acknowledgment frame.
- So, it reduces the bandwidth requirement.



Efficiency – It is defined as the ratio of total useful time to the total cycle time of a packet.

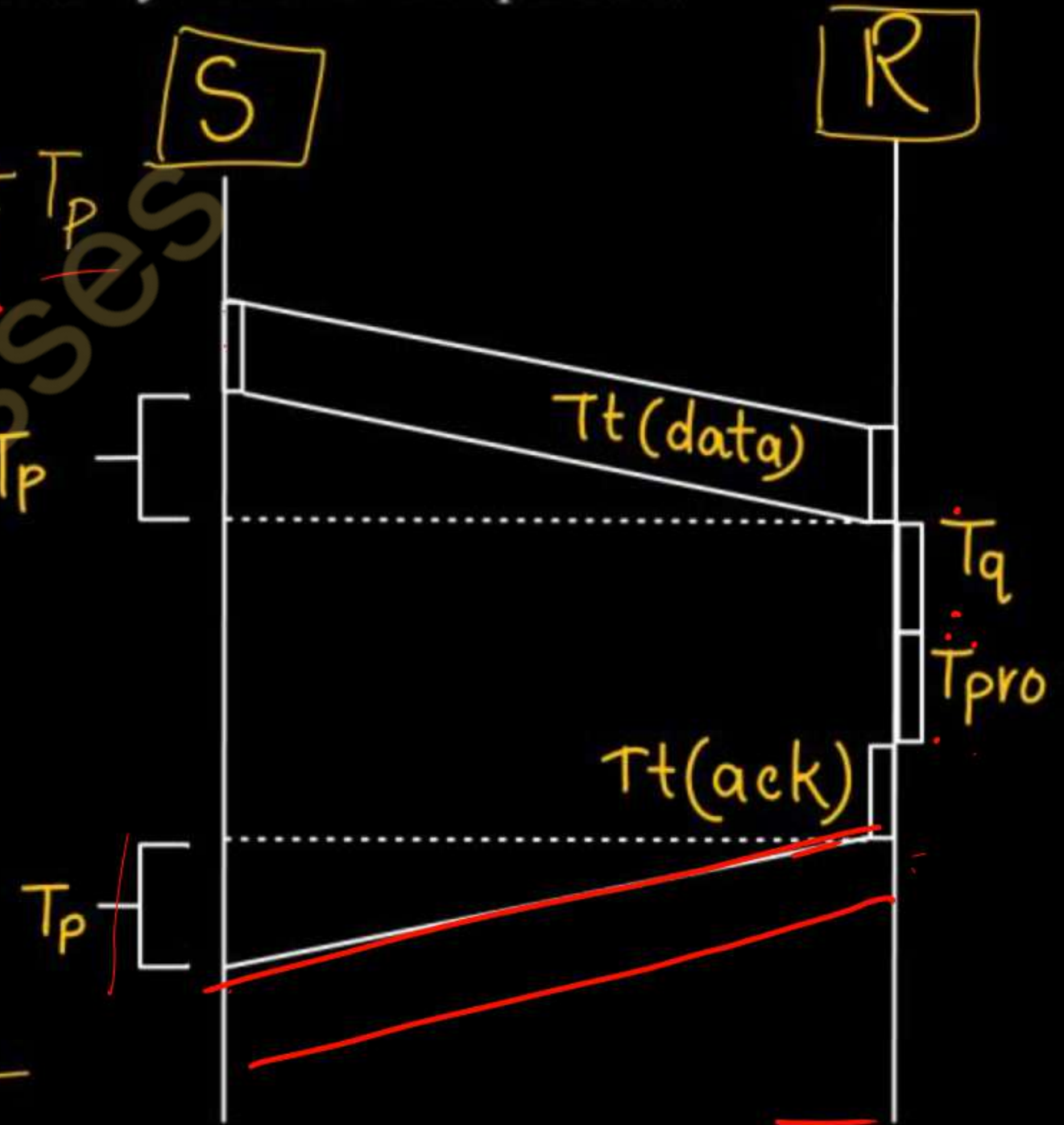
$$\text{Total time} = T_t(\text{data}) + T_p + T_{\text{que}} + T_{\text{proc}} + T_t(\text{ack}) + T_p$$

$$T_t(\text{data}) \gg T_t(\text{ack}) + T_p$$

$$\text{Total time} = T_t(\text{data}) + 2 \times T_p$$

$$\text{useful time} = T_t(\text{data})$$

$$\text{efficiency/link utilization} = \frac{\text{Useful Time}}{\text{Total Time}}$$



$$\text{efficiency } (\eta) = \frac{T_t}{T_t + 2 \times T_p}$$

dividing each term with T_t

$$\eta = \frac{1}{1 + 2 \times \frac{T_p}{T_t}} \rightarrow a$$

$$\eta = \frac{1}{1 + 2 \times a}$$

$$\text{where } a = \frac{T_p}{T_t}$$

Throughput / effective Bandwidth / Bandwidth utilization

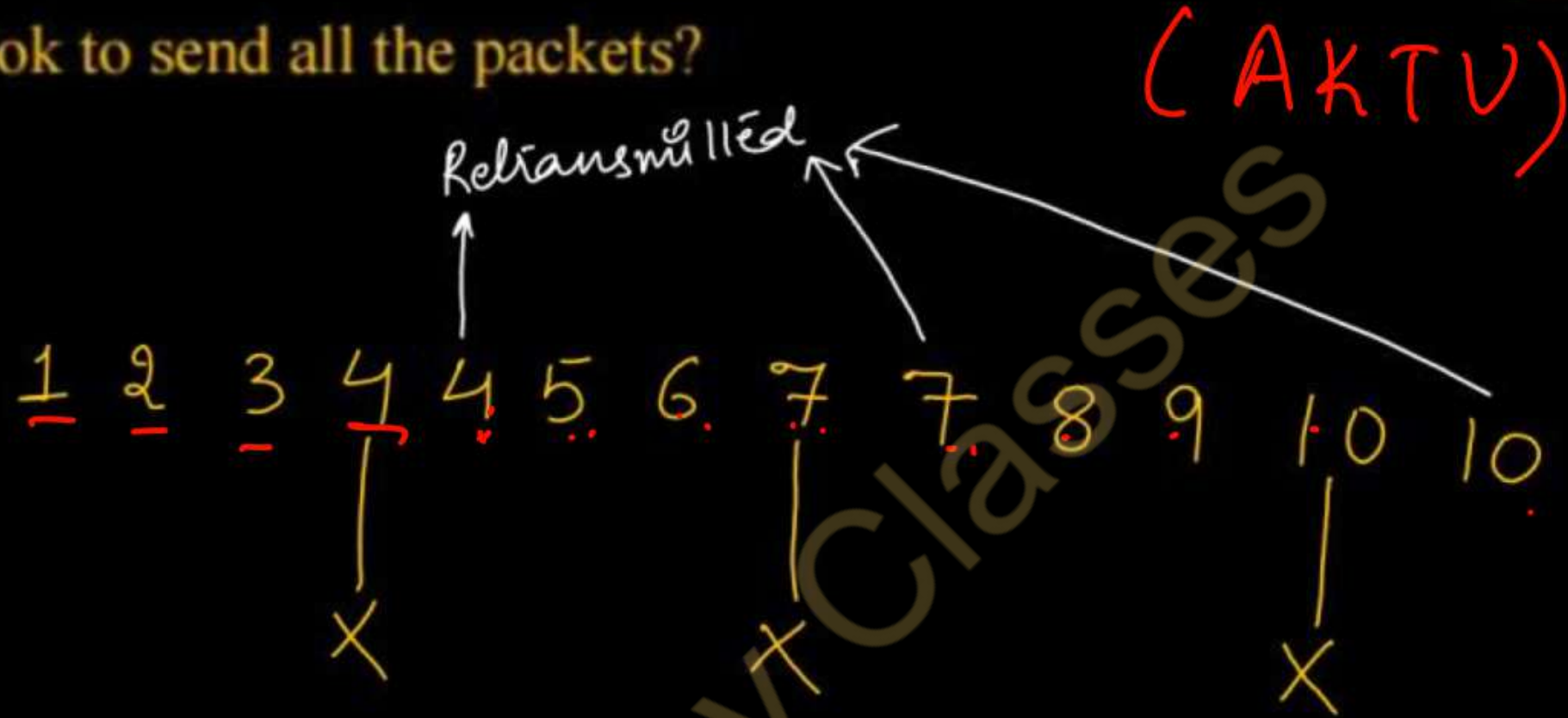
$$\text{Throughput} = \eta * \text{Bandwidth}$$

Summary of Stop and Wait ARQ

- Sender window size (W_s) = 1 ✓
- Receiver window size (W_r) = 1 ✓
- Sequence Number $\geq 1 + 1$
- Uses independent acknowledgement ✓
- Discards out of order packets ✓
- Packet Loss ? Retransmit packet after time out ✓
- Acknowledgement loss ? Resends packet after time out ✓
- **Efficiency = $1/(1+2a)$ where $a = T_p / T_t$**

Q1: In Stop and wait protocol every 4th packet is lost and we need to send total 10 packets so how many transmission it took to send all the packets?

Solⁿ.



Total no. of transmissions = 13 ✓

Q2: If the bandwidth of the line is 1.5 Mbps, RTT is 45 msec and packet size is 1 KB, then find the link utilization in stop and wait.

Solⁿ

$$\text{Bandwidth} = 1.5 \text{ Mbps}$$

$$\text{RTT (2xTP)} = 45 \text{ msec.}$$

$$\text{packet size} = 1 \text{ KB}$$

$$\text{link utilization} = \frac{1}{1+2a}$$

$$a = \frac{T_p}{T_t}$$

$$\therefore T_p = \frac{\text{RTT}}{2} = \frac{45 \text{ msec}}{2} = 22.5 \text{ msec.}$$

$$T_t = \frac{\text{packet size}}{\text{Bandwidth}} = \frac{1024 \times 8}{1.5 \times 10^6} \quad \underline{1 \text{ KB} = 2^{10} \text{ Bytes}}$$
$$= 5.46 \text{ msec.}$$

$$\therefore a = \frac{T_p}{T_t} = \frac{22.5}{5.46} = 4.12$$

$$\text{link utilization} = \frac{1}{1+2 \times 4.12} = \frac{1}{1+8.24}$$
$$= \frac{1}{9.24} = 0.108 \quad \checkmark$$

$$\text{or}$$
$$\underline{\underline{10.8\%}} \quad \checkmark$$

Ans.

Q3: Define the relationship between transmission delay and propagation delay, if the efficiency is at least 50% in STOP N WAIT protocol. (AKTU 2021-22)

Given

$$\eta \geq \frac{1}{2}$$

$$\frac{1}{1+2a} \geq \frac{1}{2}$$

$$(1+2a) \leq 2$$

$$2a \leq 1$$

$$2 \times \frac{T_p}{T_t} \leq 1$$

✓

$$T_t \geq 2 \times T_p$$

Q4: Consider the use of 10 K-bit size frames on a 10 Mbps satellite channel with 270 ms delay. What is the link utilization for stop-and-wait ARQ technique assuming $P=10^{-3}$? (AKTU)

Solⁿ.

$$T_t = \frac{10 \times 1000}{10 \times 10^6} = \underline{1 \text{ ms}}$$

$$T_p = \underline{270 \text{ ms.}}$$

$$a = \frac{T_p}{T_t} = \frac{270}{1} = \underline{270}$$

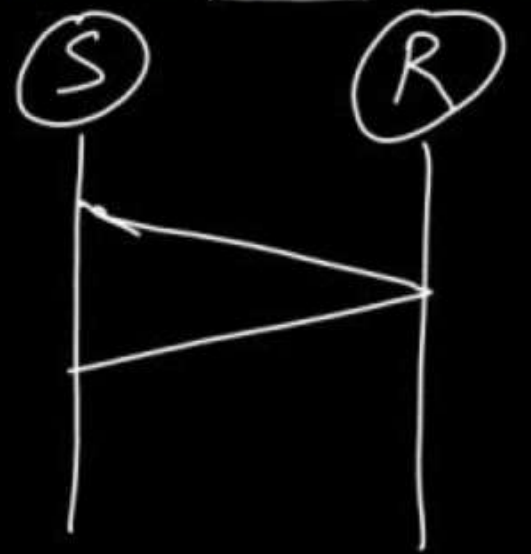
$$\eta = \frac{1}{1 + 2a} = \frac{1}{1 + 2 \times 270} = \frac{1}{541} = \underline{0.18\%}$$

Q5: Assume that, in a Stop-and-Wait ARQ system, the bandwidth of the line is 1 Mbps, and 1 bit takes 20 ms to make a round trip. What is the bandwidth-delay product? If the system data frames are 1000 bits in length, what is the utilization percentage of the link?

$$\text{Bandwidth} = 1 \text{ Mbps}$$

$$\text{Bandwidth} \times \text{delay} = \frac{1 \times 10^6 \times 20 \times 10^{-3}}{1} = 20,000 \text{ bits}$$

$$\text{link utilization} = \frac{\text{useful (data sent)}}{\text{Max. possible data}} = \frac{1000}{20,000} = 5\%$$



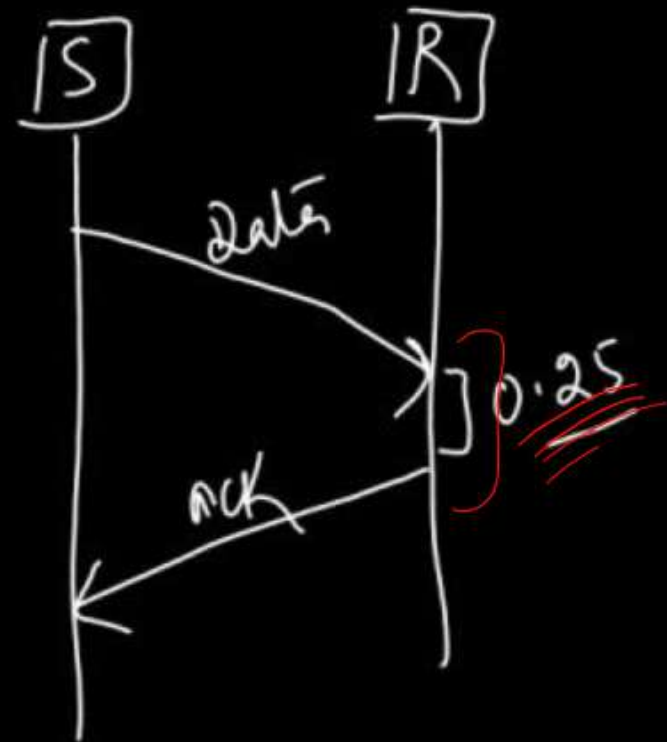
Q.6 A channel has a bit rate of 20 kbps. The stop and wait protocol with frame size 4500 bits is used. The delay for error detection and sending ACK by the receiver is 0.25 seconds because of a fault. Find the maximum efficiency of the channel if the destination is 30000 km away and the speed of the propagation of the signal is 2.8×10^8 m/s. Find the decrease in efficiency due to the fault. [AKTU 2018-19]

Solⁿ.
Bandwidth = 20 Kbps
frame size = 4500 bits
 $T_{\text{processing}} = 0.25$ secs.

distance = 30,000 km

propⁿ speed = 2.8×10^8 m/s

$$\eta = \frac{\text{useful time } (T_t)}{\text{Total time}}$$



$$T_t = \frac{4500}{20 \times 10^3} = 0.225 \text{ secs}$$

$$T_p = \frac{30,000 \times 1000}{2.8 \times 10^8} = 0.10 \text{ sec.}$$

$$\begin{aligned} \text{Total time (Ideal)} &= T_t + 2 \times T_p \\ &= 0.225 + 2 \times 0.10 \\ &= 0.225 + 0.20 \\ &= \boxed{0.425 \text{ secs.}} \end{aligned}$$

$$\begin{aligned} \text{Total time (faulty)} &= \underbrace{T_t + 2 \times T_p} + 0.25 \\ &= 0.425 + 0.25 \\ &= \boxed{0.675 \text{ sec.}} \end{aligned}$$

$$\begin{aligned} \eta_{(\text{ideal})} &= \frac{T_t}{\text{total time (ideal)}} \\ &= \frac{0.225}{0.425} = 0.529 \\ &\approx \underline{52.9\%} \end{aligned}$$

$$\boxed{\eta_{(\text{ideal})} = 52.9\%}$$

$$\begin{aligned} \eta_{(\text{faulty})} &= \frac{0.225}{0.675} = 0.333 \\ &\approx \underline{33.3\%} \end{aligned}$$

$$\text{Decrease in efficiency due to fault} = \eta_{(\text{ideal})} - \eta_{(\text{faulty})}$$

$$52.9 - 33.3$$

$$= 19.6\%$$

$$\underline{\underline{\quad\quad\quad}}$$

Q7: A sender uses the Stop-and-Wait ARQ protocol for reliable transmission of frames. Frames are of size 1000 bytes and the transmission rate at the sender is 80Kbps (1Kbps=1000 bits/second). Size of an acknowledgment is 100 bytes and the transmission rate at the receiver is 8Kbps. The one-way propagation delay is 100 milliseconds. Assuming no frame is lost, find the sender throughput in bytes/ second.

Solⁿ. Frame size = 1000 bytes ✓

Bandwidth (sender) = 80 Kbps. ✓

Bandwidth (Receiver) = 8 Kbps. ✓

Ack size = 100 bytes ✓

(AKTU)

$$\text{Total time} = T_t(\text{data}) + T_p + T_t(\text{ack}) + T_p$$

$$= \frac{1000 \times 8}{80 \times 10^3} + 100 \times 10^{-3} + \frac{100 \times 8}{8 \times 10^3} + 100 \times 10^{-3}$$

$$= 0.1 + 0.1 + 0.1 + 0.1 = 0.4 \text{ secs.}$$

$$\eta = \frac{\text{useful time}}{\text{Total time}} = \frac{T_t}{\text{total time}} = \frac{0.1}{0.4} = \underline{0.25}$$

$$\begin{aligned} \text{Throughput} &= \eta \times \text{Bandwidth} \\ &= \frac{0.25 \times 80 \times 10^3}{800} = 2500 \text{ bytes/sec.} \end{aligned}$$

Advantages of Stop and Wait Protocol

- It is very simple to implement.

Disadvantages of Stop and Wait Protocol

- We can send only one packet at a time.
- If the distance between the sender and the receiver is large then the propagation delay would be more than the transmission delay. Hence, efficiency would become very low.
- After every transmission, the sender has to wait for the acknowledgment and this time will increase the total transmission time.

AKTU PYQs

1. Consider the use of 10 K-bit size frames on a 10 Mbps satellite channel with 270 ms delay. What is the link utilization for stop-and-wait ARQ technique assuming $P=10^{-3}$? (AKTU 2016-17) ✓
2. What is piggybacking? (AKTU 2016-17)
3. List out and discuss the disadvantages in STOP N WAIT protocol (AKTU 2021-22)
4. Define the relationship between transmission delay and propagation delay, if the efficiency is at least 50% in STOP N WAIT protocol. (AKTU 2021-22) ✓



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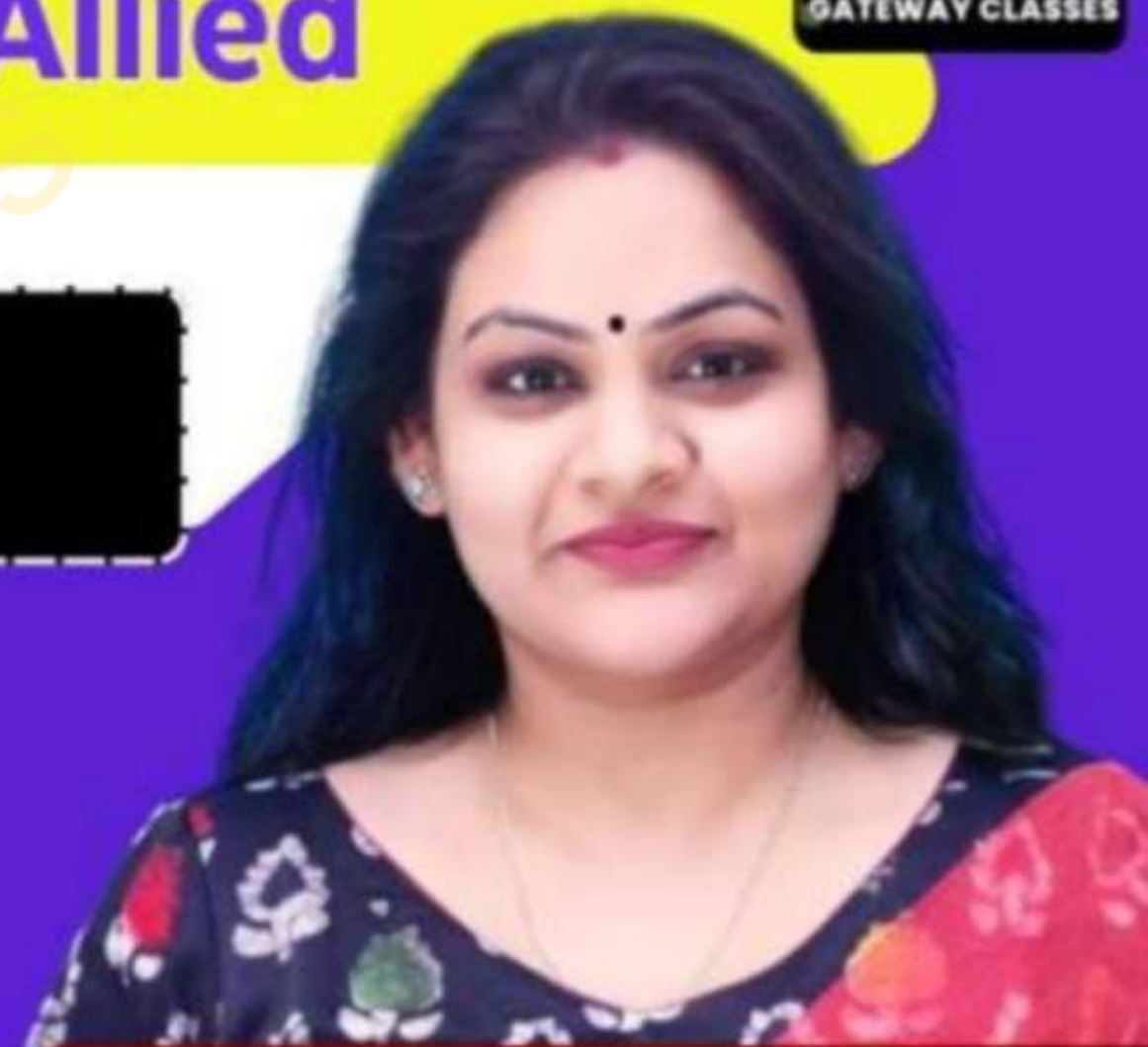


Computer Networks

Unit-2: Lecture 6

Today's Target

- Noisy Protocols
 - Go-Back-N ARQ ✓
 - Selective Repeat ARQ ✓
- AKTU PYQs



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Concept of Pipelining

- In Stop and Wait protocol, sender transmit only 1 packet and waits for ack from receiver.
- Efficiency is very less as we are not filling the channel with more packets after sending 1st packet in total cycle time of $T_t + 2 \cdot T_p$ units

Find out maximum number of packets that sender can transmit on the link before getting an ack:

In T_t units

----> 1 packet is Transmitted.

In 1 units

----> $1/T_t$ packet can be Transmitted.

In $T_t + 2 \cdot T_p$ units

-----> $(T_t + 2 \cdot T_p)/T_t$

-----> $1 + 2a$ [Using $a = T_p/T_t$]

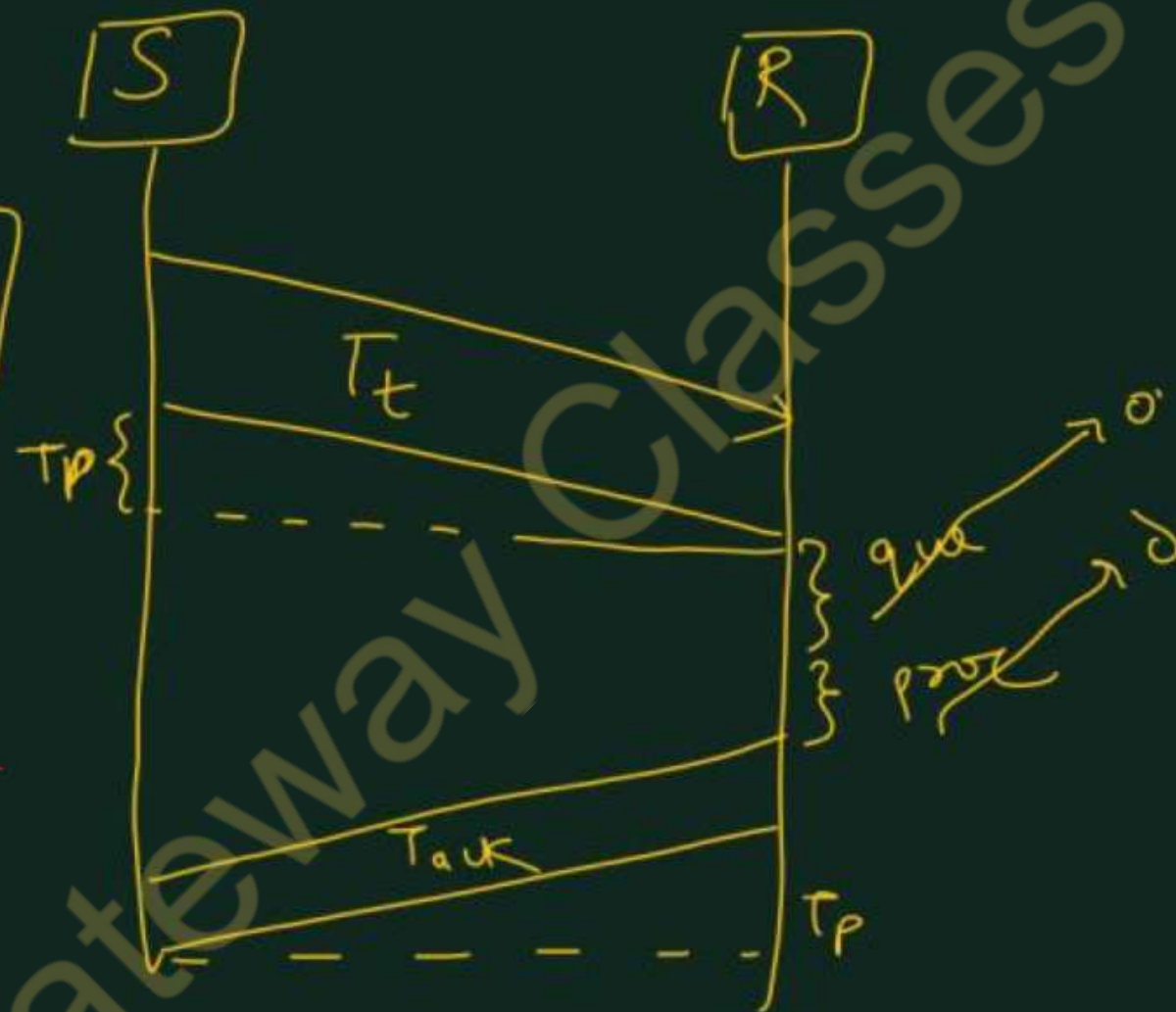
Maximum packets That can be Transmitted in total cycle time = $1 + 2 \cdot a$

$$\frac{T_t + 2 \cdot T_p}{T_t}$$

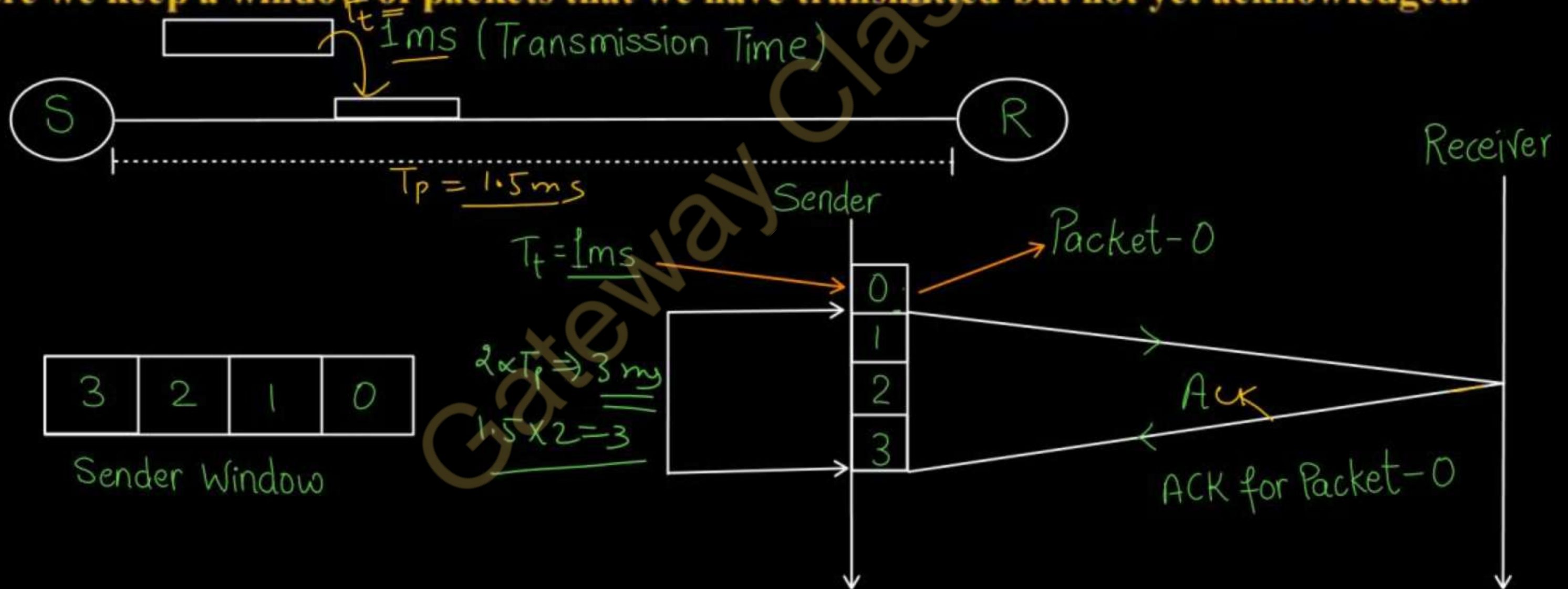
$$1 + 2 \times \left(\frac{T_p}{T_t} \right)$$

$$= \underline{1 + 2a}$$

Total time =

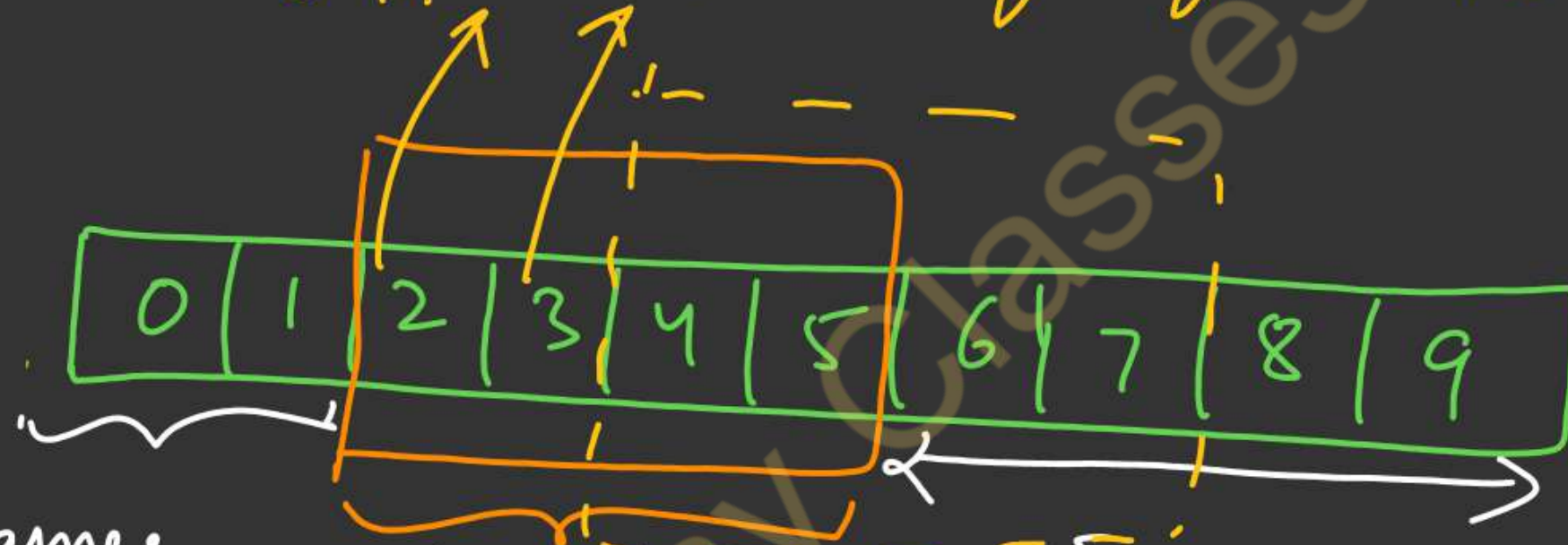


- Consider $T_t = 1\text{ms}$, $T_p = 1.5\text{ms}$.
- After sender has transmitted packet 0, it will immediately transmit packets 1, 2, 3.
- Ack for packet 0 will arrive after $2 \times 1.5 = 3\text{ms}$.
- In Stop and Wait, in time $1 + 2 \times 1.5 = 4\text{ms}$, we were transferring one packet only.
- **Here we keep a window of packets that we have transmitted but not yet acknowledged.**



Sliding window

suppose ack received for frame 2, 3.

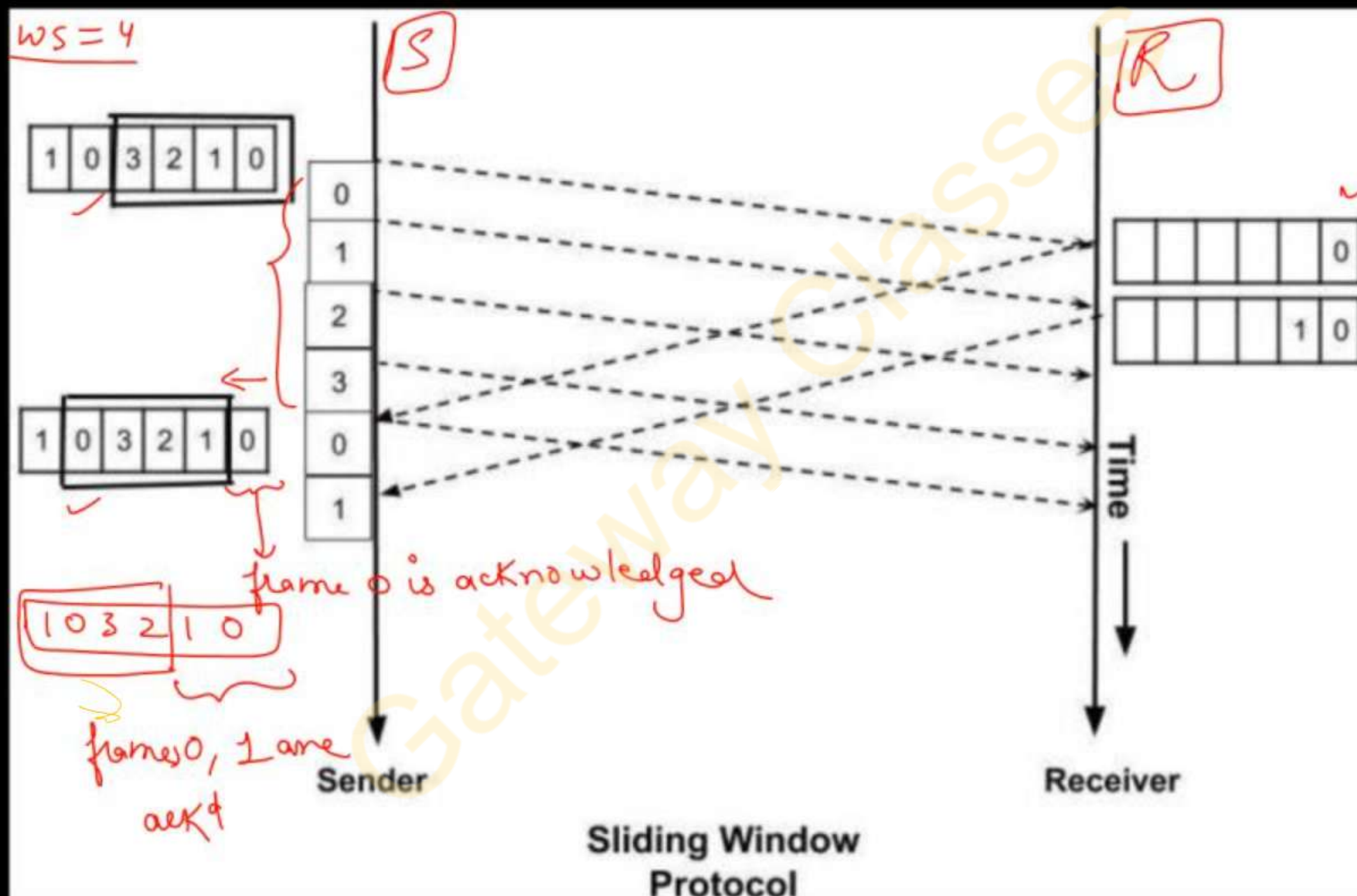


frames
already acknowledged

frames that
can be sent
before receiving
ack.

Frames yet to be sent

- After receiving the Ack for packet 0, window slides and the next packet can be assigned seq no. 0.
- We reuse the sequence numbers which we have acknowledged so that header size can be kept minimum.



Handwritten notes:

0 0 } → 6

0 1 } → 1

1 0 } → 2

1 1 } → 3

Advantages of Sliding Window Protocol

- **Efficiency**: allows multiple packets to be transmitted at the same time so increases throughput of the network.
- **Reliable**: Requiring the receiver to acknowledge receipt of each packet before the next packet can be transmitted to avoid data loss or corruption during transmission.
- **Flexibility**: Can be used with different types of network protocols and topologies.
- **Congestion Control**: By adjusting the size of the window based on the network conditions.

Minimum Number of Bits For Sender Window

Maximum window size = $1 + 2*a$ where $a = T_p/T_t$

Minimum sequence numbers required = $1 + 2*a$ ✓

But sometimes number of bits in the protocol headers is pre-defined.

seq = 4
bits = 2
2 ✓ $\left\lceil \log_2(1+2a) \right\rceil$

If N is the size of sequence number field in the header in bits, then we can have 2^N sequence numbers.

Max Window Size $W_s = \min(1+2*a, 2^N)$ ✓

Minimum Number of bits required to represent the sender window = $\text{ceil}(\log_2(1+2*a))$

4.5 $\rightarrow$ 5

The **window size** determines how many packets the sender can transmit without needing an acknowledgment.

Q: Find out window size and minimum sequence number in sliding window protocol, if

Transmission delay (T_t) = 1 ms, Propagation delay (T_p) = 24.5 ms. (AKTU 2021-22)

Solⁿ

$$ws = 1 + 2a$$

$$a = \frac{T_p}{T_t} = \frac{24.5}{1} = 24.5$$

$$ws = 1 + 2 \times 24.5$$

$$= 1 + 49$$

$$= \underline{50}$$

Minimum seq. no required = 50 Ans

Q: How many bits are required to represent the seq. no. field?

$$\text{bits required} = \lceil \log_2 (1 + 2a) \rceil$$

$$= \lceil \log_2 (50) \rceil$$

$$= 5. \text{ — } \Rightarrow \textcircled{6} \checkmark$$

Q- In the previous question, if no. of bits reserved in the seq. no field is 5 according to the protocol, then what will be the efficiency?

Solⁿ

No. of sequence numbers acc. to protocol = $2^5 = \underline{\underline{32}}$ ✓

$$\text{Efficiency} = \frac{\text{Max window size according to protocol}}{\text{Max. window size possible.}}$$

$$= \frac{32}{50} = \underline{\underline{64\%}} \quad \checkmark$$

Types of Sliding Window Protocol

There are two types of Sliding Window Protocol

1. Go-Back-N ARQ
2. Selective Repeat ARQ

Gateway Classes

Go Back N Protocol

- Sender can send multiple data packets (Equal to window size) without waiting for an ack for each one.
- If one packet is lost or not acknowledged, the sender must go back and resend that packet *and* all the packets that followed it, even if they were received correctly.

The three main characteristic features of GBN are:

1. **Sender Window Size (W_R):** It is N itself.

- If we say the protocol is GB10, then $W_s = 10$. For $N=1$, it reduces to the Stop and Wait protocol.

$$\text{Efficiency} = \frac{N}{1+2a} \text{ where } a = T_p/T_t$$

- Efficiency if processing delay, queuing delay and transmission delay of ack is not zero.

$$\text{Efficiency} = \frac{N * (\text{Useful time})}{(\text{Total Time})}$$

where, useful time = T_t , and $\text{Total time} = T_t + 2 * T_p + P_r + P_q + T_t(\text{ack})$

$$\text{Effective Bandwidth or Throughput} = \text{Efficiency} * \text{Bandwidth}$$

$$\eta = \frac{N \times \left(\frac{1}{1+2a} \right)}$$

2. Receiver Window Size (W_R): W_R is always 1 in GBN.

3. Acknowledgements: There are 2 kinds of acknowledgements:

➤ **Cumulative Ack:**

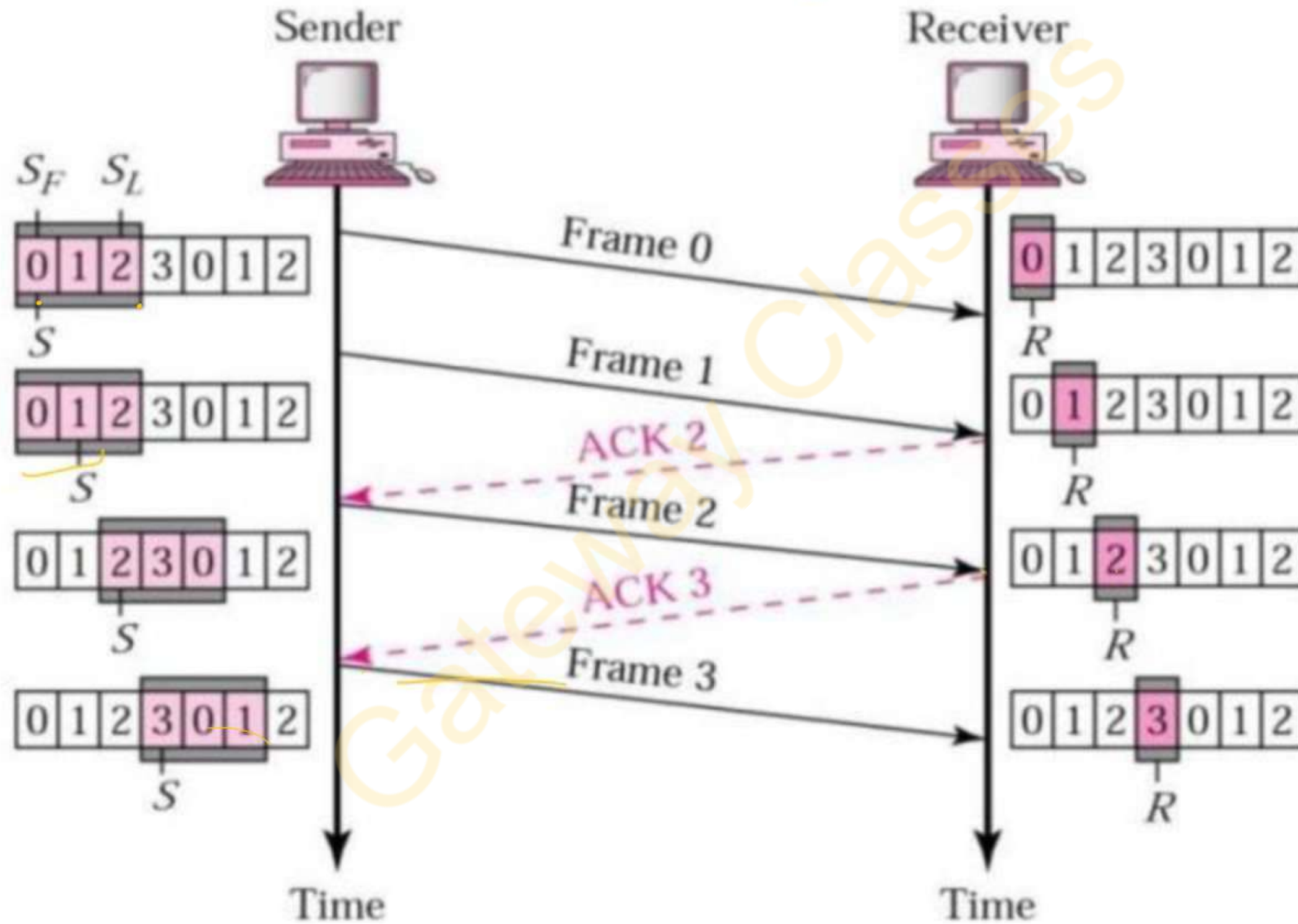
- One ack for many packets.
- traffic is less.
- less reliability as if one ack is the lost that would mean that all the packets sent are lost.

➤ **Independent Ack:**

- If every packet is acknowledged independently.
- High reliability
- High traffic

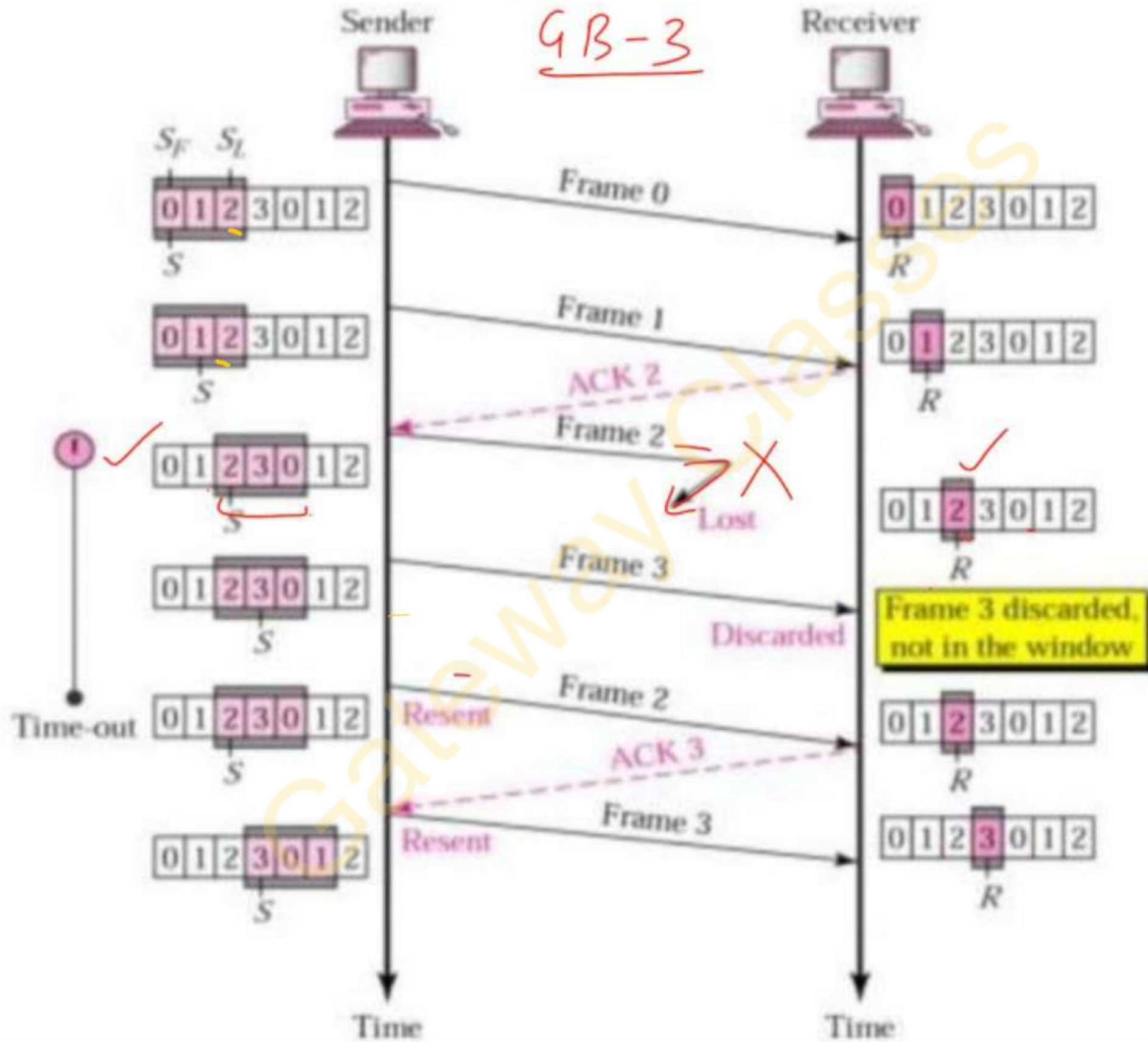
Normal Working of GB-N Protocol

Go back - (3) N



Working of GB-N Protocol in Lost Frame

GB-3



- Assume sender window size of 4.
- Sender has sent the packets 0, 1, 2 and 3.
- After acknowledging packets 0 and 1, receiver is expecting packet 2 and sender window has also slid to further transmit the packets 4 and 5.
- Suppose packet 2 is lost, Receiver will discard all the packets which sender has transmitted after packet 2 as it is expecting seq no. of 2.
- On sender side for every packet sent there is a time out timer which will expire for packet number 2.
- Now from the last transmitted packet 5, sender will go back to the packet number 2 in the current window and transmit all the packets till packet number 5.
- That's why it is called Go Back N. Sender has to go back N places from the last transmitted packet in the unacknowledged window and not from the point where the packet is lost.

GB-4

0 1 2 3 4 5

timeout

0 1 2 3 4 5

S

R

0

1

2

ack 2

4

5

2

3

4

5

0 1 2 3 4 5

X X

discarded

out of order

Retransmitted

Relationship Between Window Size and Sequence Number

The **window size** should be smaller than or equal to the range of available **sequence numbers**.

➤ Relation between window size and sequence number is given by the formula:

$$W_s + W_R \leq \text{ASN}$$

where W_s is sender window size and W_R is receiver window size, and ASN is available sequence number.

$$N + 1$$

$$W_s + 1 \leq \text{ASN}$$

because $W_R = 1$ in GB-N protocol

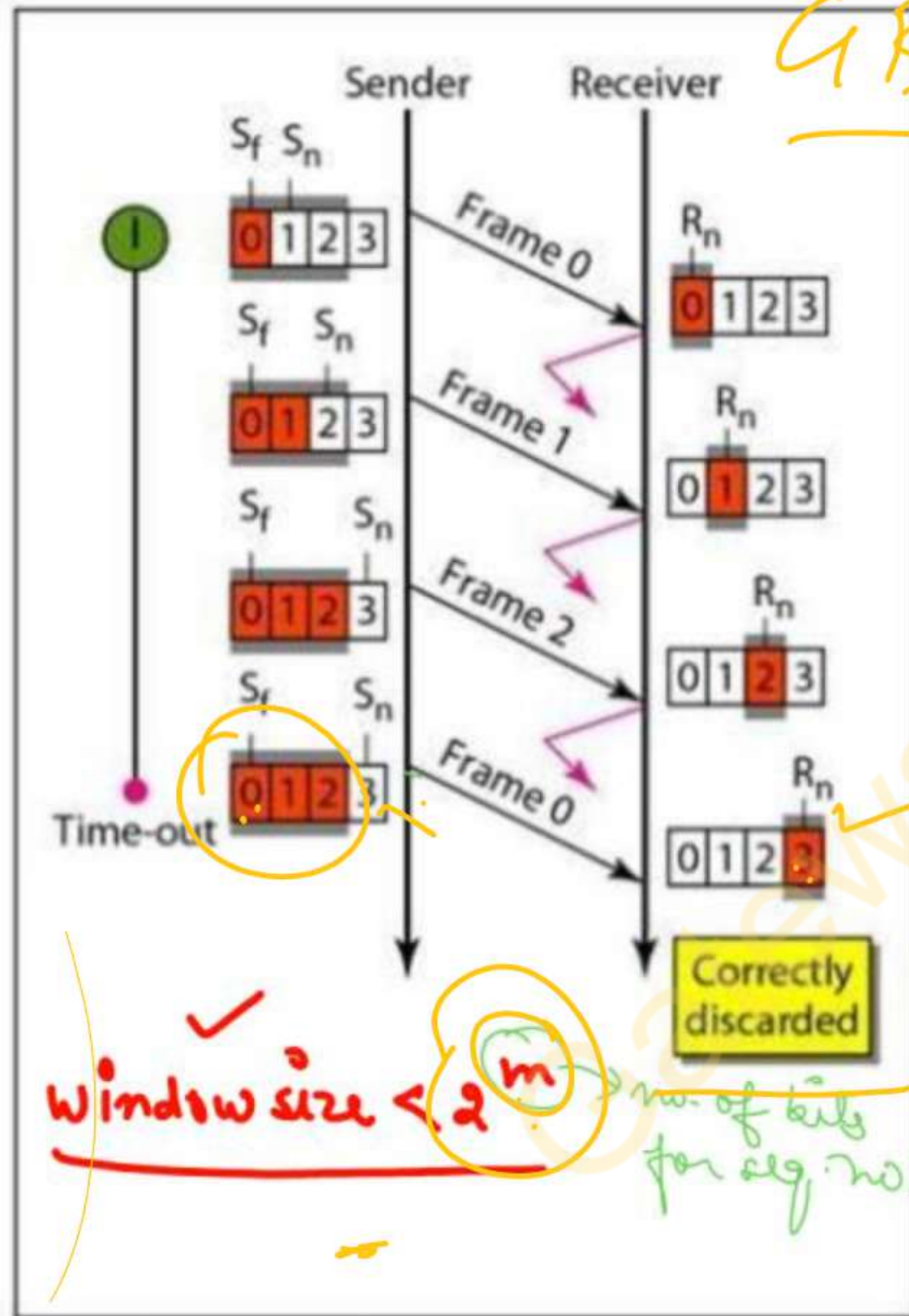
So minimum sequence numbers required in GBN = $N + 1$

ARB(N)

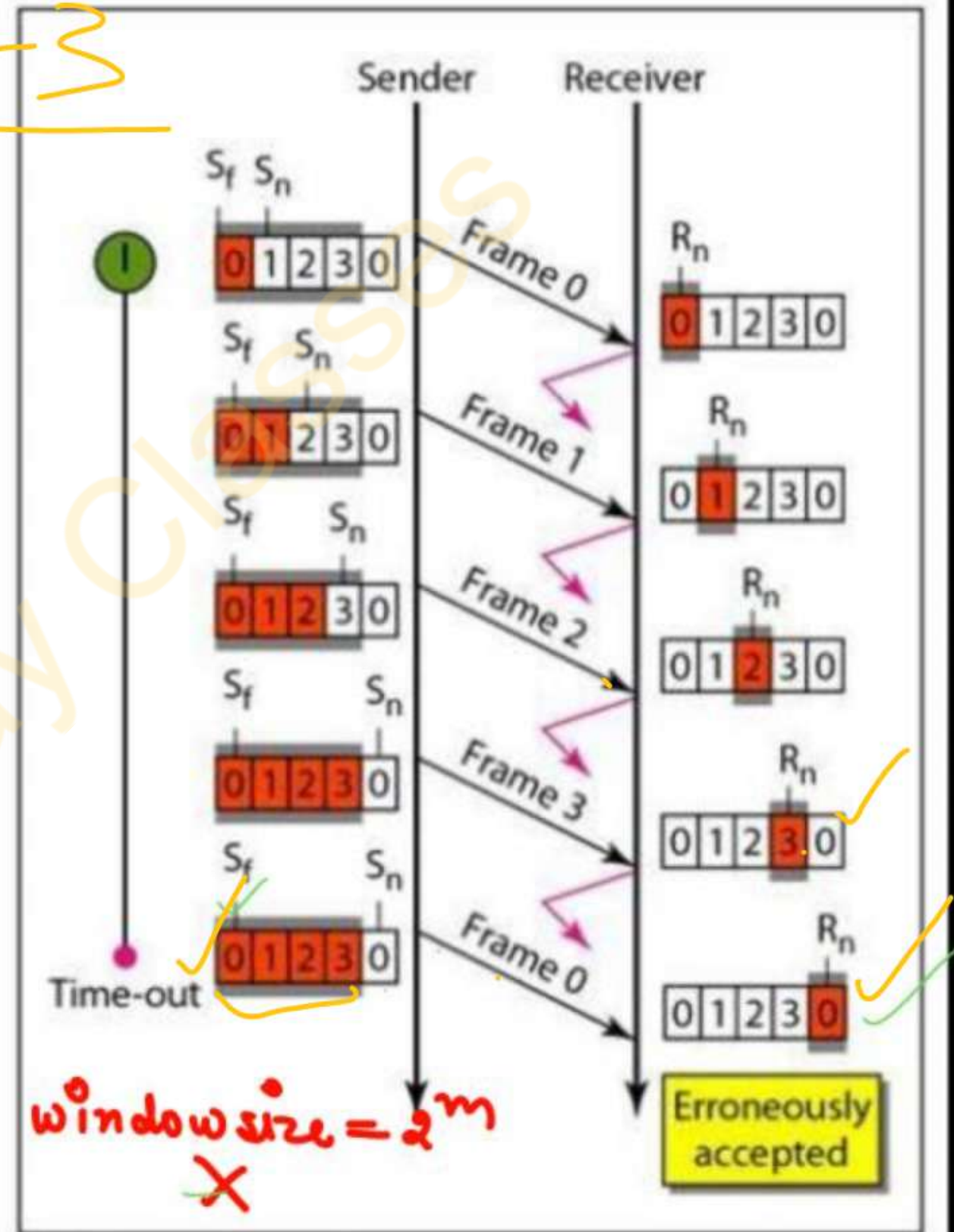
Bits Required in GBN = $\text{ceil}(\log_2 (N + 1))$

The 1 extra sequence number is required in order to avoid the problem of duplicate packets

The 1 extra sequence number is required in order to avoid the problem of duplicate packets



a. Window size $< 2^m$



b. Window size $= 2^m$

Summary of Go-Back-N ARQ

- Sender window size $W_s = N$
- Receiver window size $W_r = 1$
- Sequence number $\geq N + 1$
- Can use both cumulative or independent acknowledgement depends on acknowledge timer
- Discards out of order packets
- Packet Loss ? Track back N size from the last packet within the window limit to the lost packet and retransmit them
- Acknowledgement loss ? If not received before timeout the entire window N size is resend
- **Efficiency = $N/(1+2a)$ where $a = T_p / T_t$**

Advantages of GBN Protocol

- Simple to implement and effective for reliable communication.
- Better performance than stop-and-wait protocols for error-free or low-error networks.

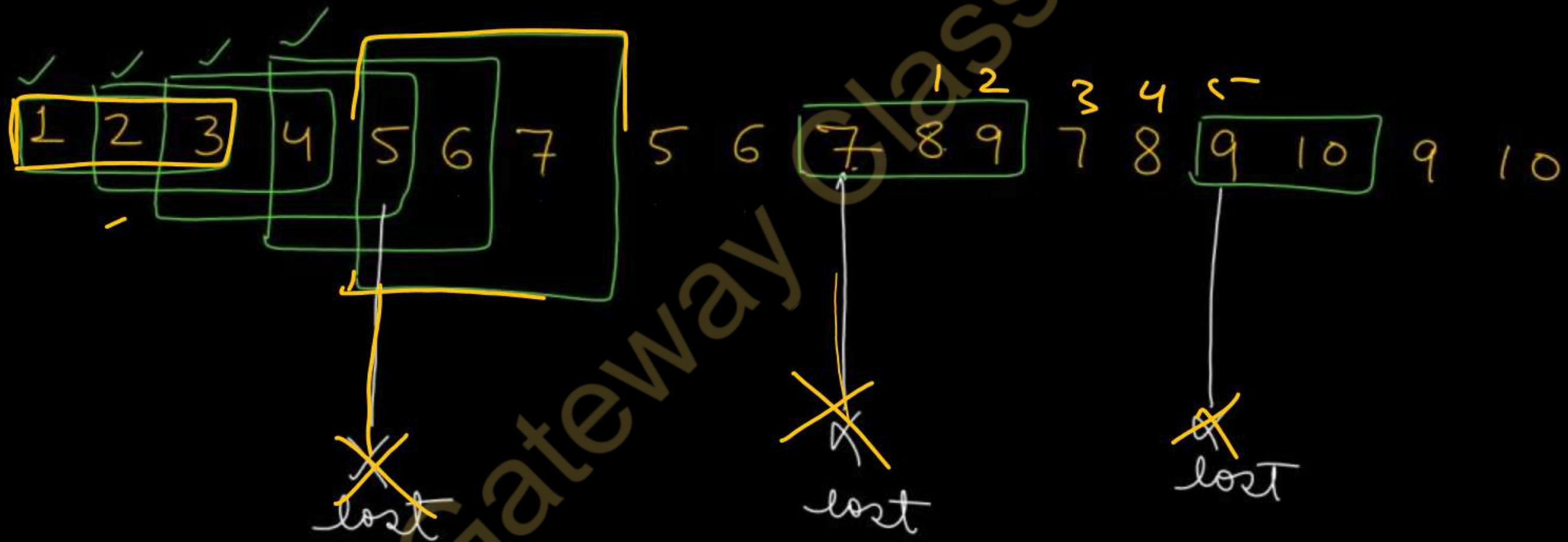
Disadvantages of GBN Protocol

- Inefficient if errors are frequent, as multiple frames might need to be retransmitted unnecessarily.
- Bandwidth can be wasted due to redundant retransmissions.

Q: Calculate the total number of transmissions that are required to send 10 data packets through GBN-3 and every 5th packet is lost. (AKTU 2021-22)

Total transmissions = 18 GB-3

Sender window size = 3



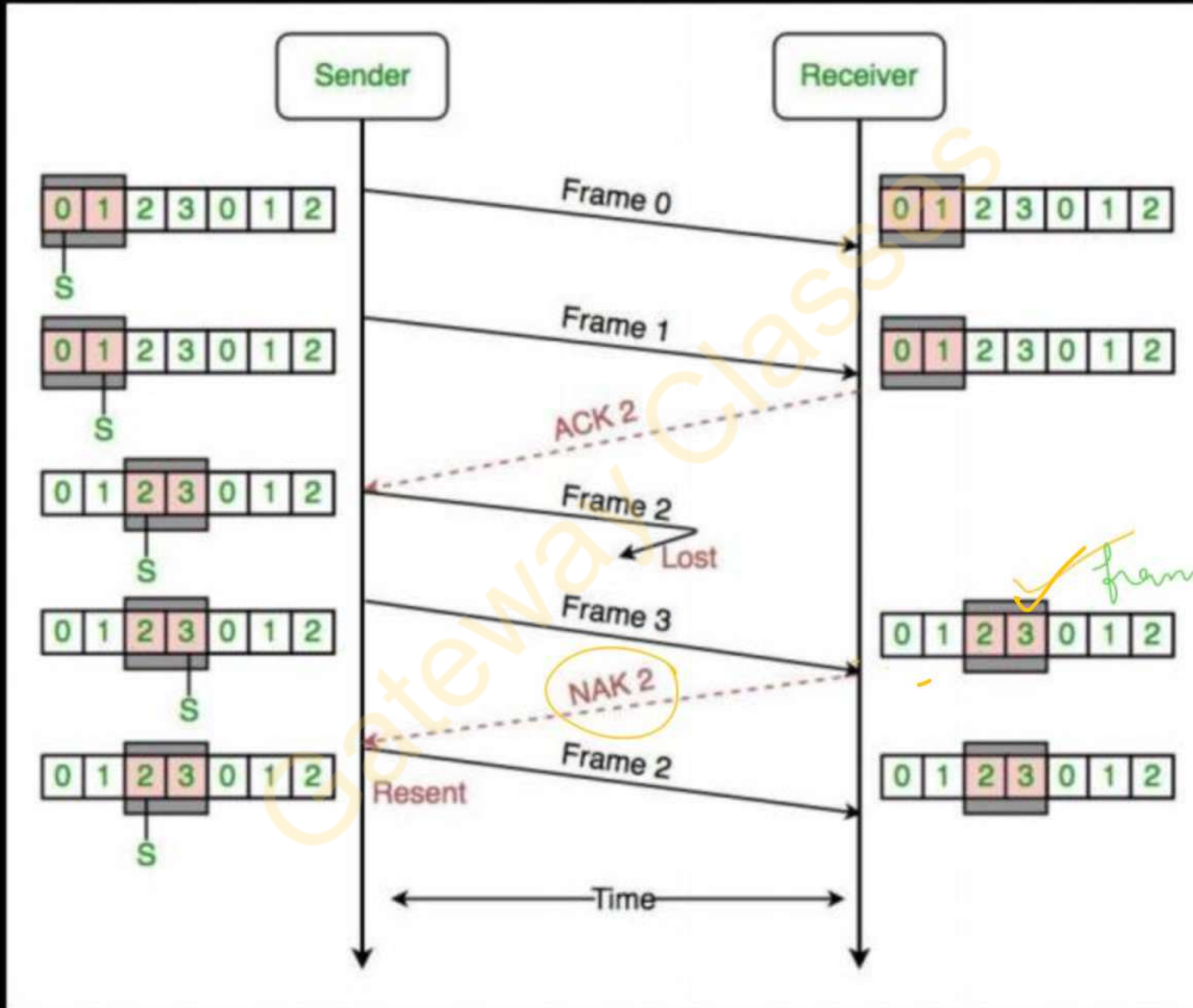
Why Selective Repeat Protocol?

- GBN works well if errors are less, otherwise it wastes a lot of bandwidth on retransmitted frames.
- Selective repeat protocol allow receiver to accept and buffer the frames following a damaged or lost one.
- Attempts to retransmit only those packets that are actually lost (due to errors) :
 - Receiver must be able to accept packets out of order.
 - Since receiver must release packets to higher layer in order, the receiver must be able to buffer some packets.

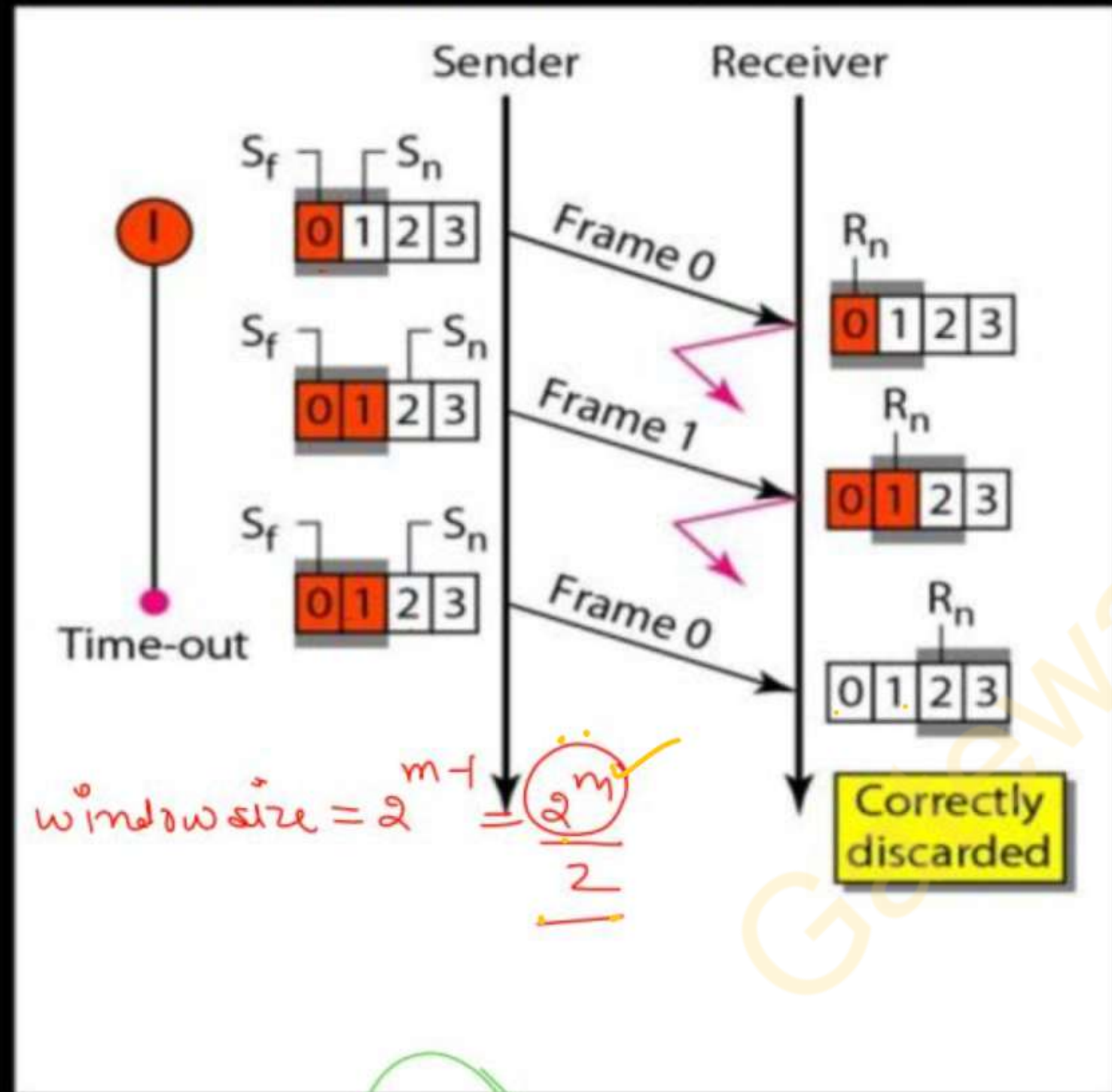
Selective Repeat ARQ

- In selective repeat, the sender sends several frames(window size).
- The receiver while keeping track of sequence numbers buffers the frames in memory and sends NACK for only frames which are missing or damaged.
- The sender will send/retransmit a packet for which NACK is received.
- Unlike GBN, only the lost or error frames are retransmitted, whereas correct frames are received and buffered.
- The retransmitted frame is received out of sequence.
 - **Sender's Windows (W_s) = Receiver's Windows (W_r)**

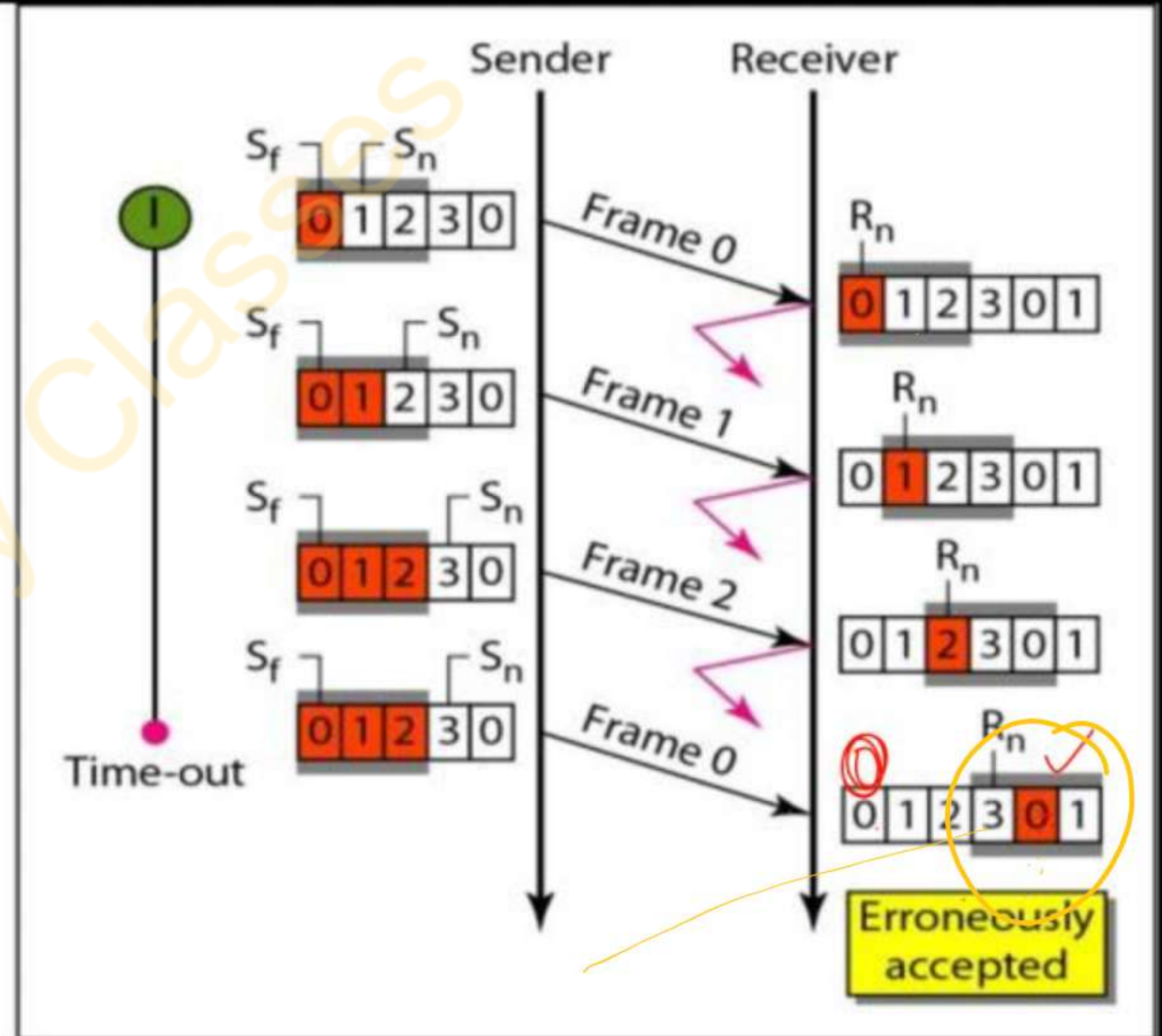
Selective Repeat ARQ Operation



- In SR ARQ, Window size should be less than or equal to half the sequence number to avoid packets being recognized incorrectly.



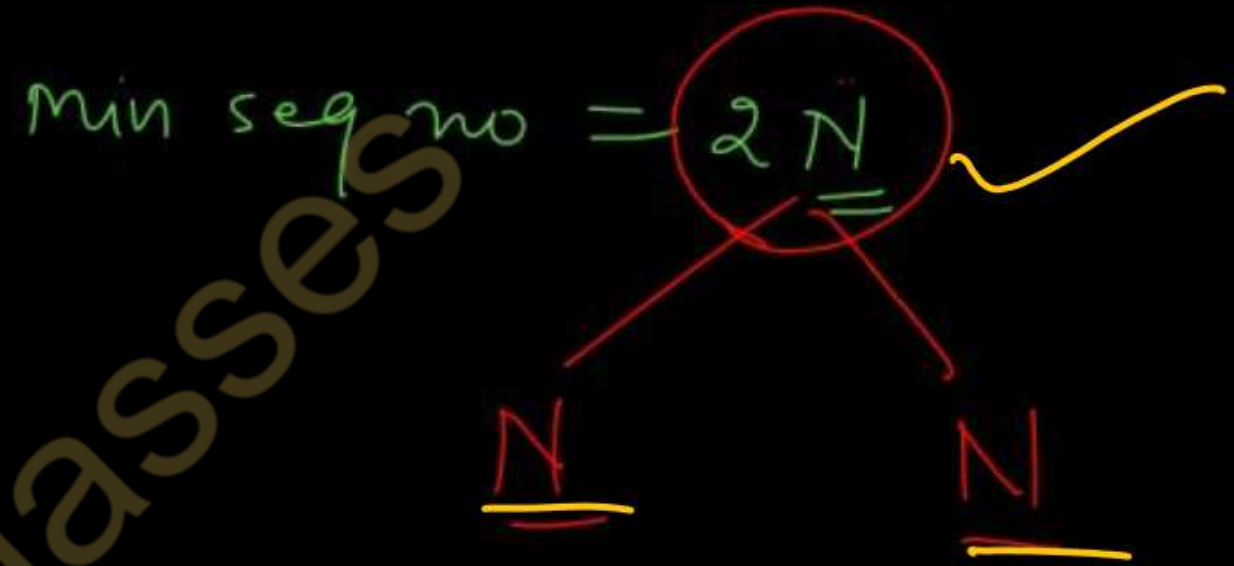
a. Window size = 2^{m-1}



b. Window size $> 2^{m-1}$

Summary of Selective Repeat ARQ

- Sender window size $W_s = N$
- Receiver window size $W_r = N$
- Sequence Number $\geq N + N$
- Uses only independent acknowledgement
- Can Accept out of order packets
- Packet Loss ? Resend only the lost packet after timeout
- Acknowledgement loss ? Resend if not receive before timeout
- Efficiency = $N/(1+2a)$ where $a = T_p / T_t$



Difference between the Go-Back-N ARQ and Selective Repeat ARQ

Go-Back-N Protocol	Selective Repeat Protocol
All the frames are re-transmitted from the lost packet to the last packet transmitted.	Only suspected frames are re-transmitted.
Sender window size: N	Sender window size :N
Receiver window size: 1	Receiver window size :N
Less complex.	More complex.
Acknowledgement is cumulative.	Acknowledgement is individual.
Out-of-Order packets are discarded	Out-of-Order packets are Accepted.
If Receives a corrupt packet, then also, the entire window is re-transmitted.	if Receives a corrupt packet, sends NACK and only selective packet is retransmitted.
Efficiency: $N/(1+2*a)$	Efficiency: $N/(1+2*a)$

AKTU PYQs

1. Illustrate the performance issues for GO-BACK-N data link protocol. (AKTU 2016-17)

Write short notes on following:

- i. Stop and Wait ARQ
- ii. Sliding Window Protocol
- iii. Go Back N ARQ

3. Explain transmission delay in flow control. (AKTU 2021-22)
4. Write a note on round trip time (RTT) in networking. (AKTU 2021-22)
5. List out and discuss the disadvantages in STOP N WAIT protocol (AKTU 2021-22)
6. Calculate the total number of transmissions that are required to send 10 data packets through GBN-3 and every 5th packet is lost. (AKTU 2021-22)
7. Find out window size and minimum sequence number in sliding window protocol, if Transmission delay (T_t) = 1 ms, Propagation delay (T_p) = 24.5 ms. (ms = milliseconds). (AKTU 2021-22)
8. Explain the following terms by taking real-world examples: (AKTU 2022-23)
 - i) Go Back-N
 - ii) Selective repeat.
9. Explain Noisy channel Protocols with explanation and diagram. (AKTU 2023-24)



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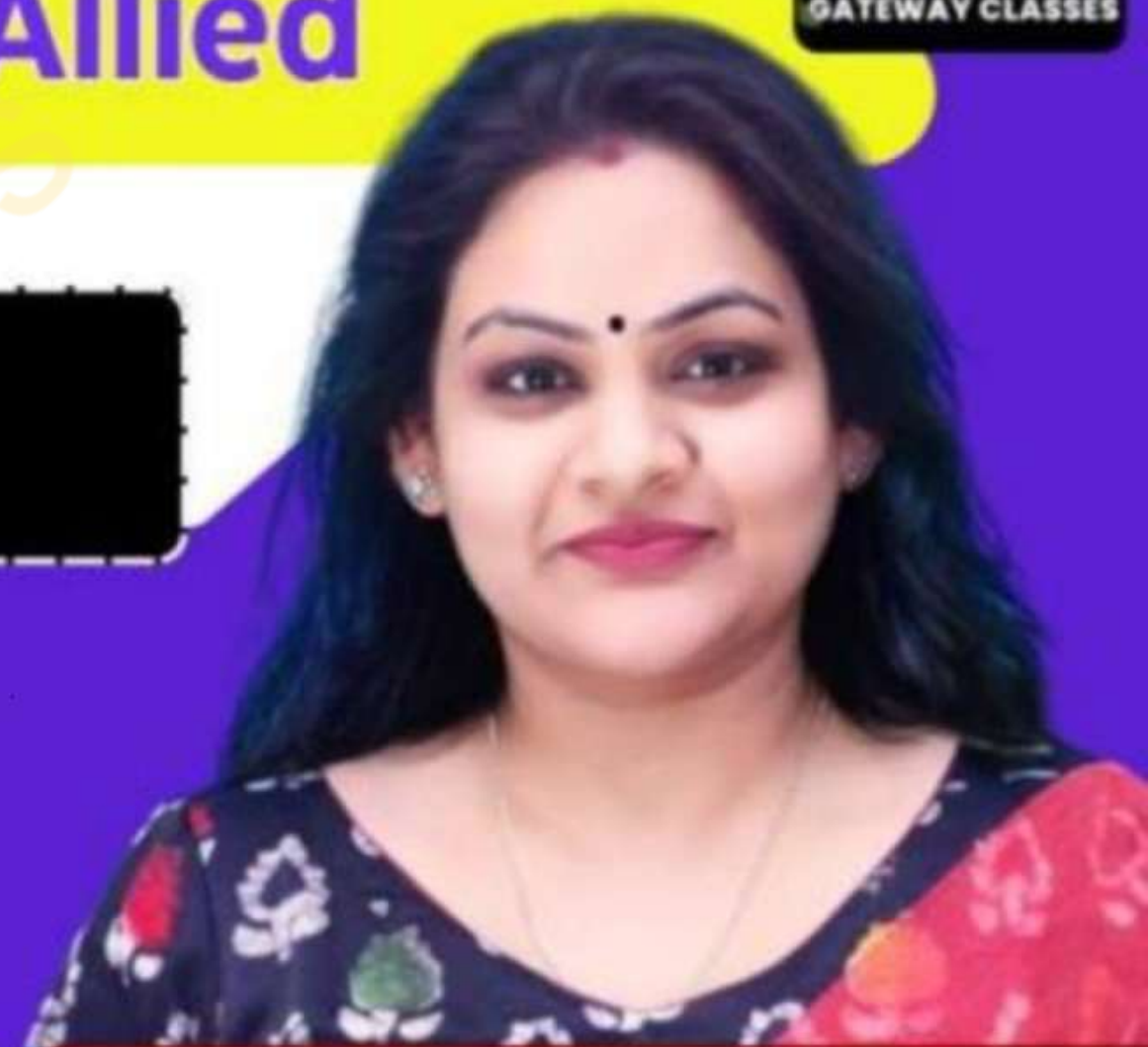


Computer Networks

Unit-2: Lecture 7

Today's Target

- Error Detection
 - Parity Bit ✓
 - Checksum ✓
 - Cyclic Redundancy Check (CRC) ✓
- AKTU PYQs



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- Net Qualified

Error Detection

- Data can be corrupted during transmission.
- Some applications can tolerate a small level of error. Such as errors in audio or video transmissions may be tolerable, but when we transfer text, we expect a very high level of accuracy.

Types of Errors

1. **Single Bit Error:** In a single-bit error, only 1 bit of a given data unit is changed, a 0 is changed to a 1 or a 1 to a 0.



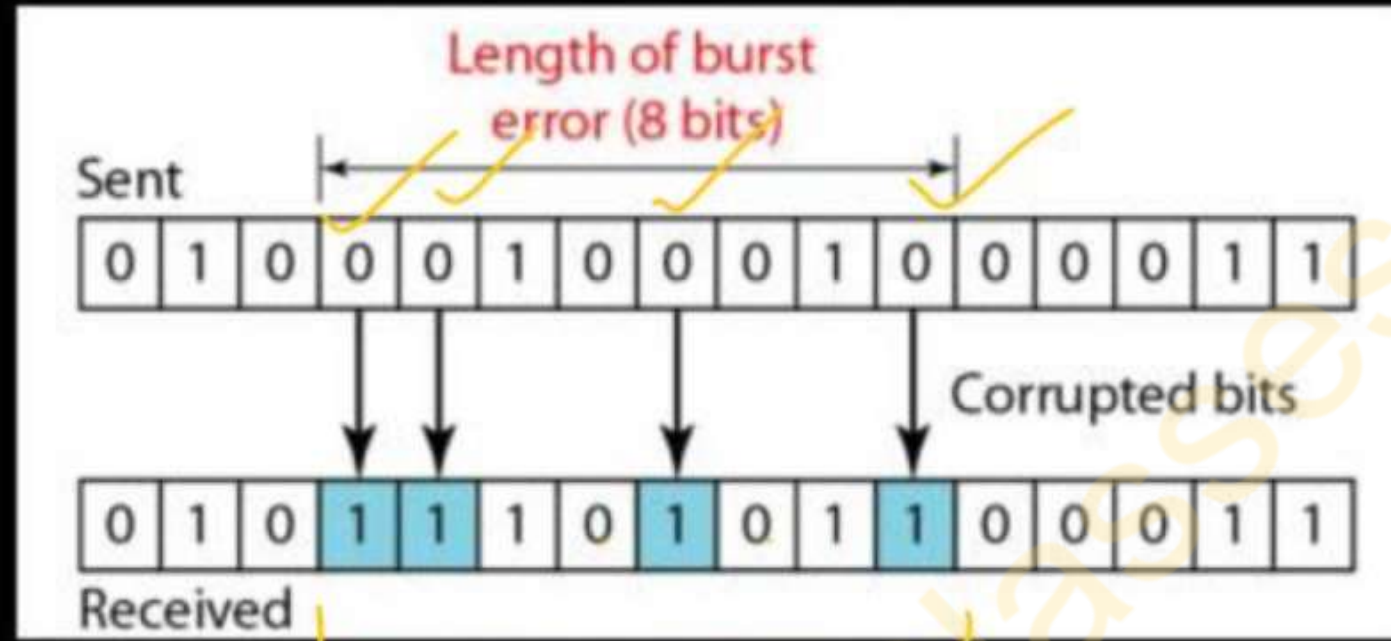
- Least likely type of error in serial data transmission.
- Suppose data is sent at 1 Mbps, means each bit lasts only $1/1,000,000$ s or 1 micro second. For a single-bit error to occur, the noise must have a duration of only 1 micro second, which is very rare.

$$\text{Data rate} = 1 \text{ Mbps}$$

$$1 \text{ sec} \rightarrow 10^6 \text{ bits}$$

$$\begin{aligned} \text{Time for which 1 bit lasts} &= \frac{1}{10^6} \text{ sec.} \\ &= \underline{1 \mu\text{sec}} \end{aligned}$$

2. Burst Error: In a burst error, 2 or more bits in the data unit have changed from 1 to 0 or from 0 to 1.



- Burst error is more likely to occur than a single-bit error.
- The duration of noise is normally longer than the duration of 1 bit, So noise affects a set of bits.
- The number of bits affected depends on the data rate and duration of noise.
- If data is sent at rate = 1Kbps then a noise of 1/100 sec can affect 10 bits. $(1/100 * 1000)$
- If same data is sent at rate = 1Mbps then a noise of 1/100 sec can affect 10,000 bits. $(1/100 * 10^6)$

Redundancy

To be able to detect or correct errors, we need to send some extra bits with our data.

These redundant bits are added by the sender and removed by the receiver.

Error Detection Vs Error Correction

- **Detection:** Checking whether error occurred
- **Correction:** Identifying the position of the errors : no. of errors, size

Forward Error Correction Vs Retransmission

- **FEC:** Receiver tries to guess the correct message using redundant bits
- **Retransmission:** detect the error and retransmit

Hamming Distance

- The Hamming distance between two words (of the same size) is the number of differences between the corresponding bits.
- Hamming distance between two words x and y as $d(x, y)$ can be found by applying the XOR operation on the two words and count the number of 1s in the result.

1. The Hamming distance $d(000, 011)$ is 2 because $000 \oplus 011$ is 011 (two 1s).

2. The Hamming distance $d(10101, 11110)$ is 3 because $10101 \oplus 11110$ is 01011 (three 1s).

①

$$\begin{array}{r} 000 \\ \oplus 011 \\ \hline \end{array}$$

011 2 1s are there

Hamming distance = 2

②

$$\begin{array}{r} 10101 \\ 11110 \\ \hline 01011 \end{array}$$

3

✓

There are 3 ways in which we can detect errors in the received message :

1. **Parity Bit**

2. **Checksum**

3. **Cyclic Redundancy Check (CRC)**

What is parity bit?

- A parity bit is an extra bit that is added to the message bits or data-word bits on the sender side.
- Data-word bits along with parity bits is called a codeword.
- There are two parity systems – even and odd parity checks.

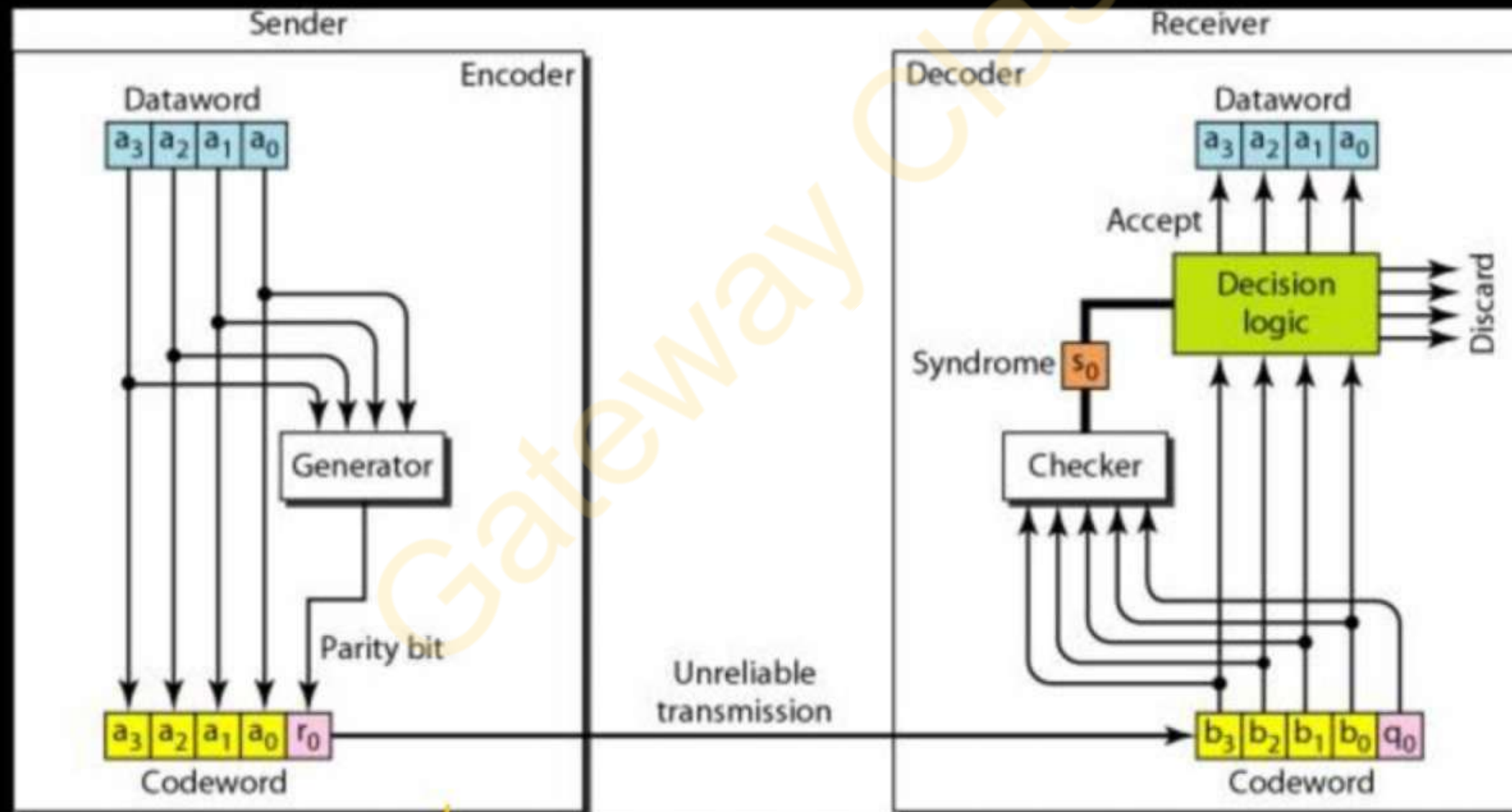
Even Parity: Total number of 1's in the given data bit should be even. So, if the total number of 1's in the data bit is odd then a single 1 will be appended to make total number of 1's even else 0 will be appended.

Odd Parity: If the total number of 1's in the given binary string (or data bits) are even then 1 is appended to make the total count of 1's as odd else 0 is appended.

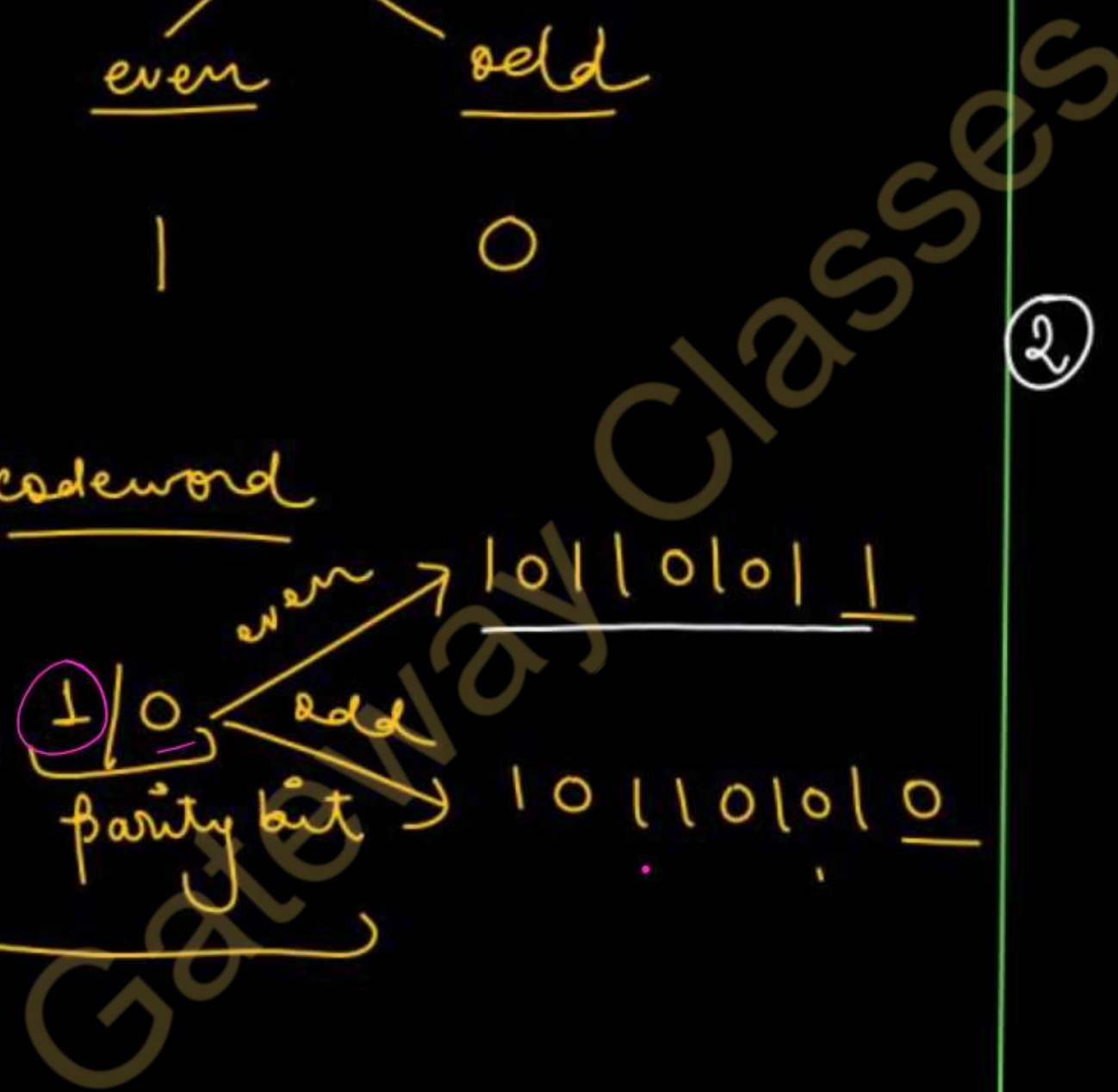
Simple Parity-Check Code

A k -bit data word is changed to an n -bit codeword where $n = k + 1$. The extra bit, called the parity bit.

A simple parity-check code is a single-bit error-detecting code in which $n = k + 1$ with $d_{\min} = 2$.



A Si



① 10110101

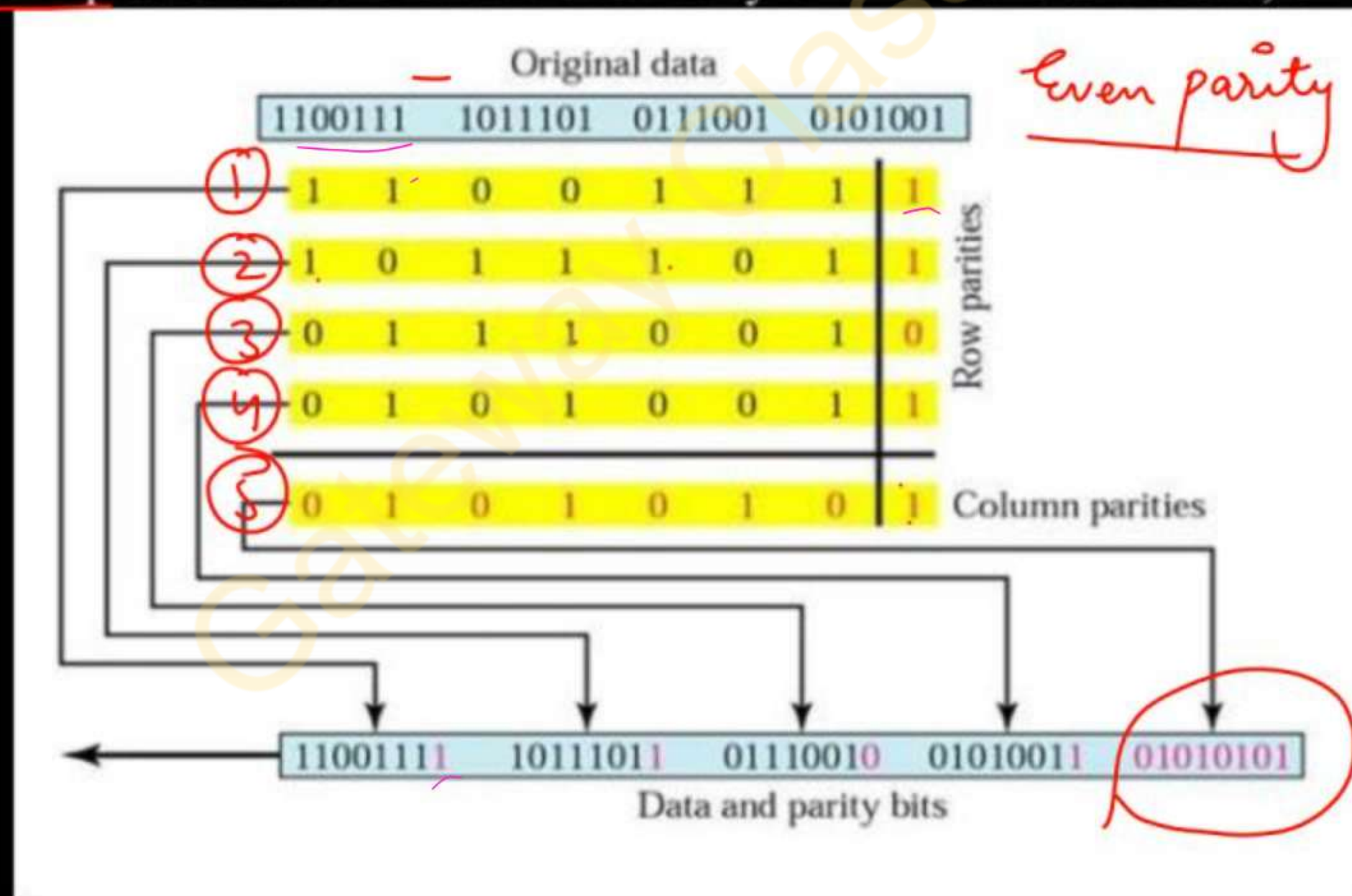
10110001

② 10110101

100101111 X niet detect

Two-dimensional parity check:

- Data word is organized in a table (rows and columns).
- Data are put in separate rows. For each row and each column, 1 parity bit is calculated.
- The whole table is sent to the receiver, which finds the syndrome for each row and each column. 2D parity check can detect up to three errors that occur anywhere in the table but, errors affecting 4 bits may not be detected.



➤ 2-D parity check can detect up to 3 errors but may not detect errors affecting 4 bits.

1	1	0	0	1	1	1	1
1	0	1	1	1	0	1	1
0	1	1	1	0	0	1	0
0	1	0	1	0	0	1	1
0	1	0	1	0	1	0	1

b. One error affects two parities

1	1	0	0	1	1	1	1
1	0	1	1	1	0	1	1
0	1	1	1	0	0	1	0
0	1	0	1	0	0	1	1
0	1	0	1	0	1	0	1

d. Three errors affect four parities

1	1	0	0	1	1	1	1
1	0	1	1	1	0	1	1
0	1	1	1	0	0	1	0
0	1	0	1	0	0	1	1
0	1	0	1	0	1	0	1

c. Two errors affect two parities

1	1	0	0	1	1	1	1
1	0	1	1	1	0	1	1
0	1	1	1	0	0	1	0
0	1	0	1	0	0	1	1
0	1	0	1	0	1	0	1

e. Four errors cannot be detected

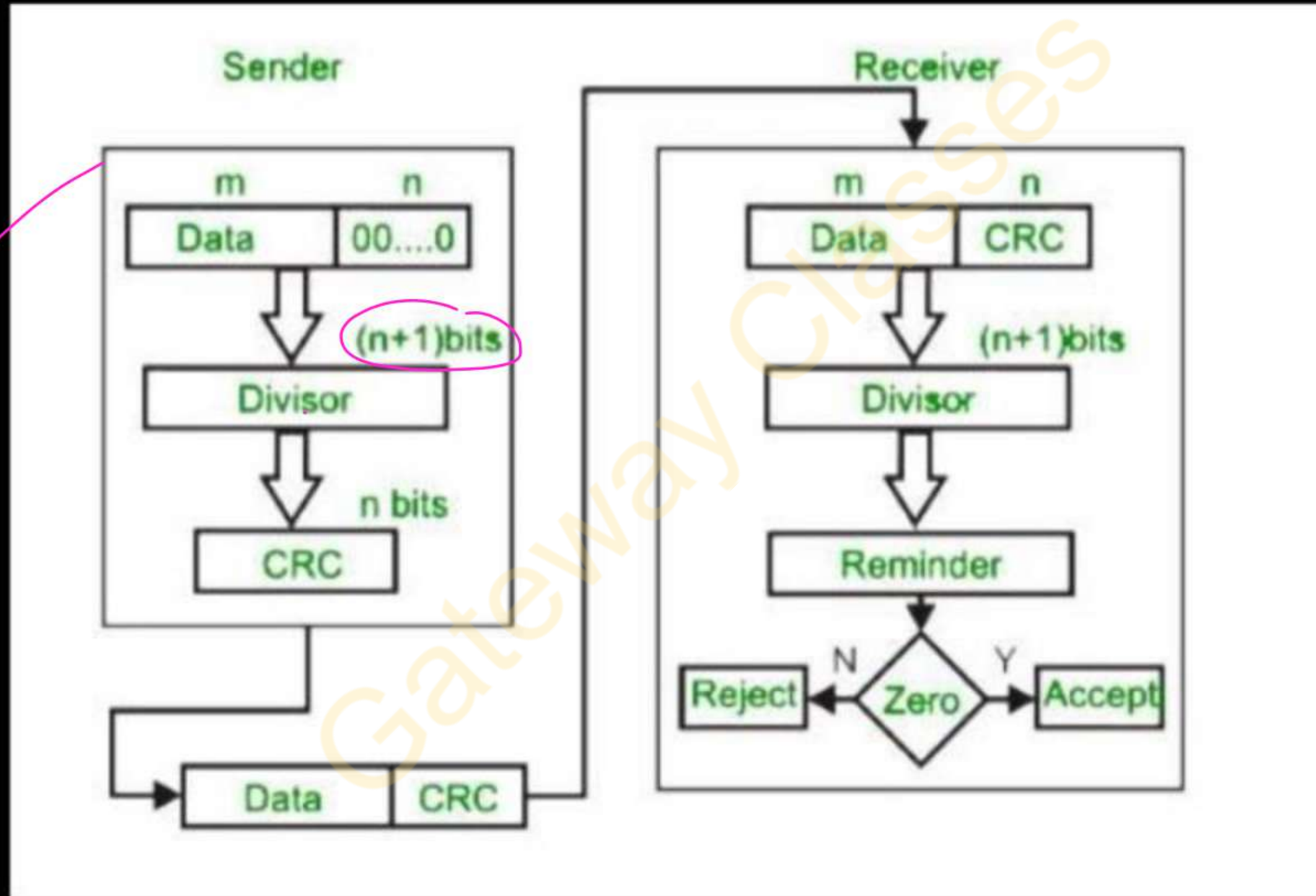
Cyclic Redundancy Check (CRC)

(M. Imp)

- A method of detecting errors in the communication channel.
- Uses **Generator Polynomial** available on both sender and receiver side.
- An example of generator polynomial like $x^3 + x + 1$ represents key 1011.
- Another example is $x^2 + 1$ that represents key 101.
- **n** : Number of bits in data to be sent from sender side.
- **k** : Number of bits in the key obtained from generator polynomial.
- **Sender Side (Generation of Encoded Data from Data and Generator Polynomial (or Key)):**
 - a. The binary data is first augmented by adding k-1 zeros in the end of the data.
 - b. Use *modulo-2 binary division* to divide binary data by the key and store remainder of division.
 - c. Append the remainder at the end of the data to form the encoded data and send the same.

➤ **Receiver Side (Check if there are errors introduced in transmission):**

Perform modulo-2 division again and if the remainder is 0, then there are no errors.



Requirements of CRC : (AKTU)

A CRC will be valid if and only if it satisfies the following requirements:

1. It should have exactly one less bit than divisor. ✓
2. Appending the CRC to the end of the data unit should result in the bit sequence which is exactly divisible by the divisor.

Imp

The polynomial generator should have following properties:

1. It should not be divisible by x .
 - This condition guarantees that all the burst errors of length equal to the length of polynomial are detected.
2. It should be divisible by $x+1$.
 - This condition guarantees that all the burst errors affecting an odd number of bits are detected.

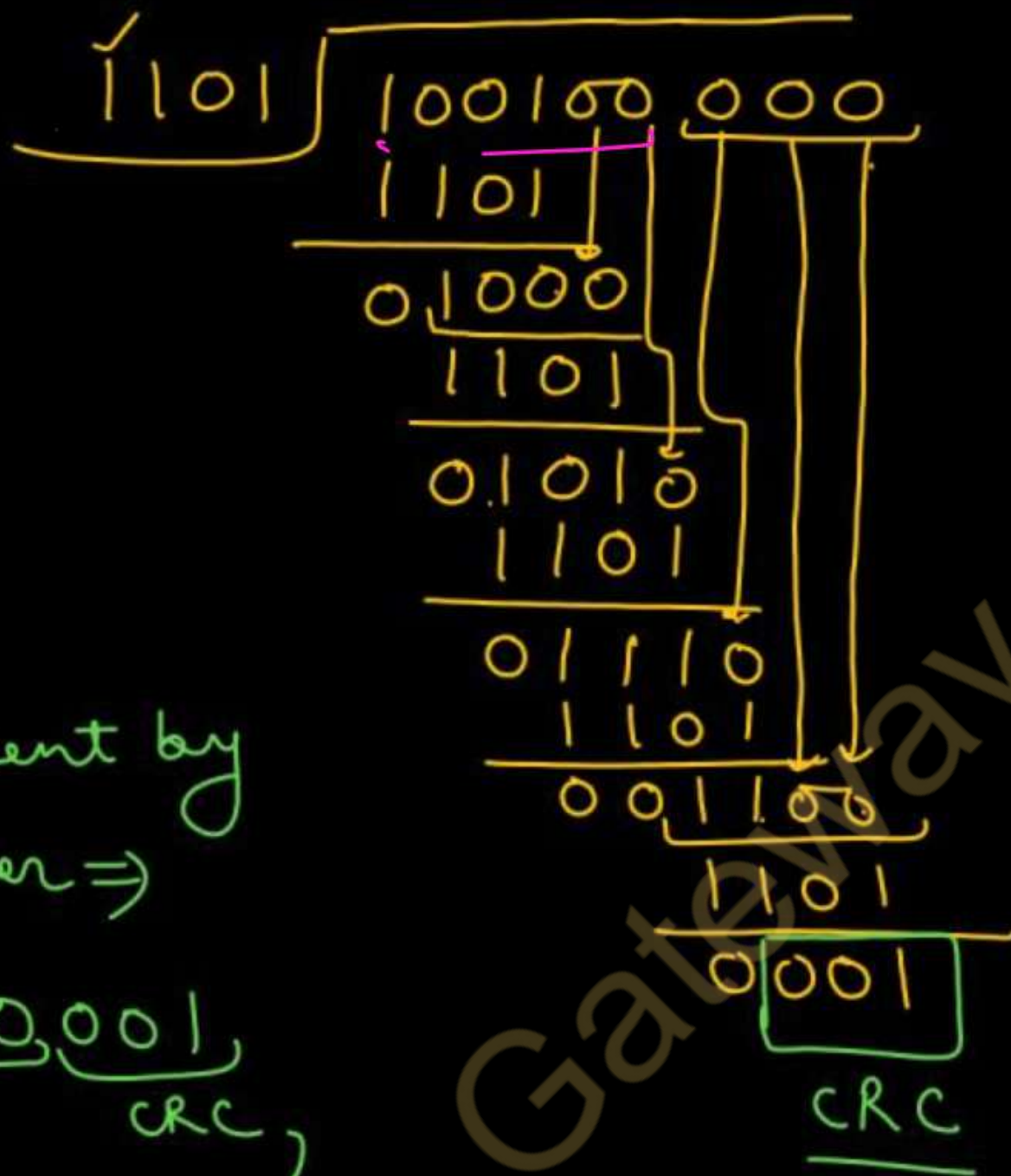
Advantages of CRC:

- CRC can detect all single-bit errors ✓
- Able to detect burst errors of length less than the degree of the polynomial. ✓
- CRC is also efficient and fast, making it ideal for real-time applications. ✓
- Easy to implement and widely supported across different systems and protocols, making it a versatile choice for various applications.

Limitations of CRC :

- While it can detect many types of errors, it cannot catch all possible errors. ✓
- This means that some rare or specific errors may still go undetected using CRC alone. ✓
- Effectiveness of CRC diminishes as the length of the message increases due to its fixed-length nature.

Example 1: Data word to be sent - 100100, Generator polynomial $x^3 + x^2 + 1$



Data being sent by the receiver $\Rightarrow$

100100001
data CRC
Codeword

$$x^3 + x^2 + 1$$

$$\begin{array}{cccc} 1 & 1 & 0 & 1 \\ \hline \end{array}$$

$$\underline{1.} \quad \underline{x^3} + \underline{1.} \quad \underline{x^2} + \underline{0.} \quad \underline{x} + \underline{1.} \quad \underline{1}$$

1101

$$x^4 + x + 1$$

10011

Sender

100100 001
data + CRC

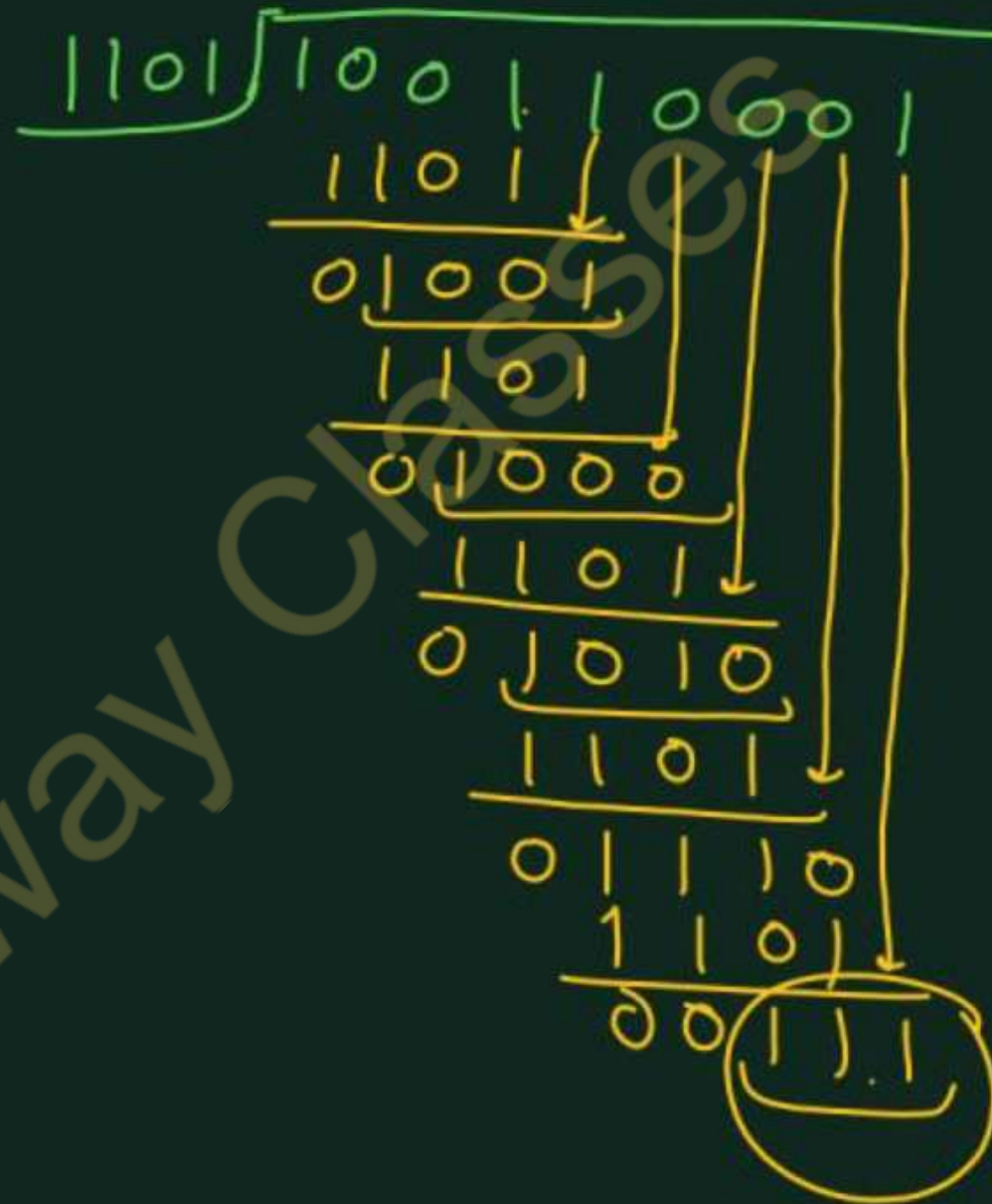
100100001

X

1001 10 001

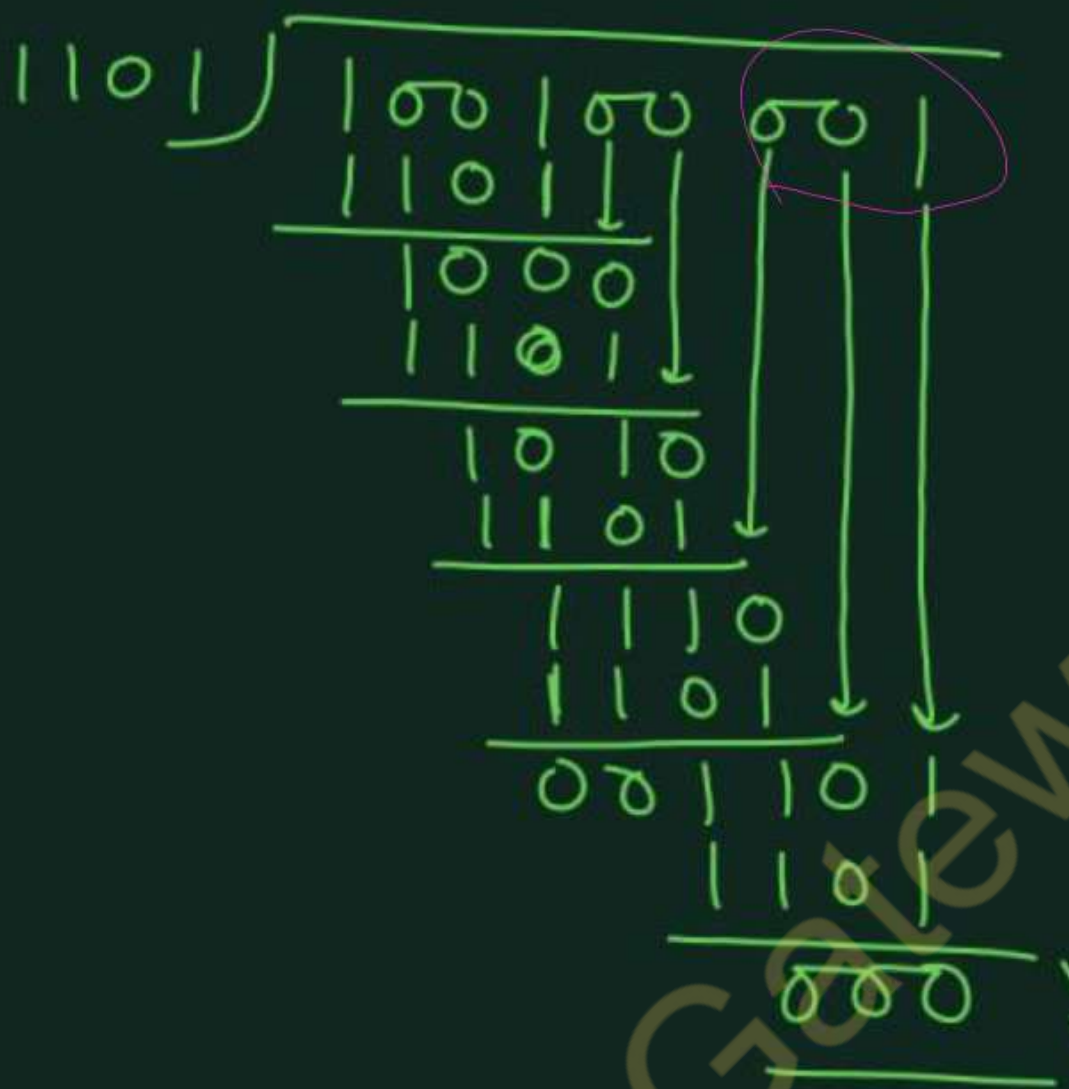
Received codeword

100110001



Non-zero value
tells the receiver
about errored
transmission

In case of correct transmission (without error)



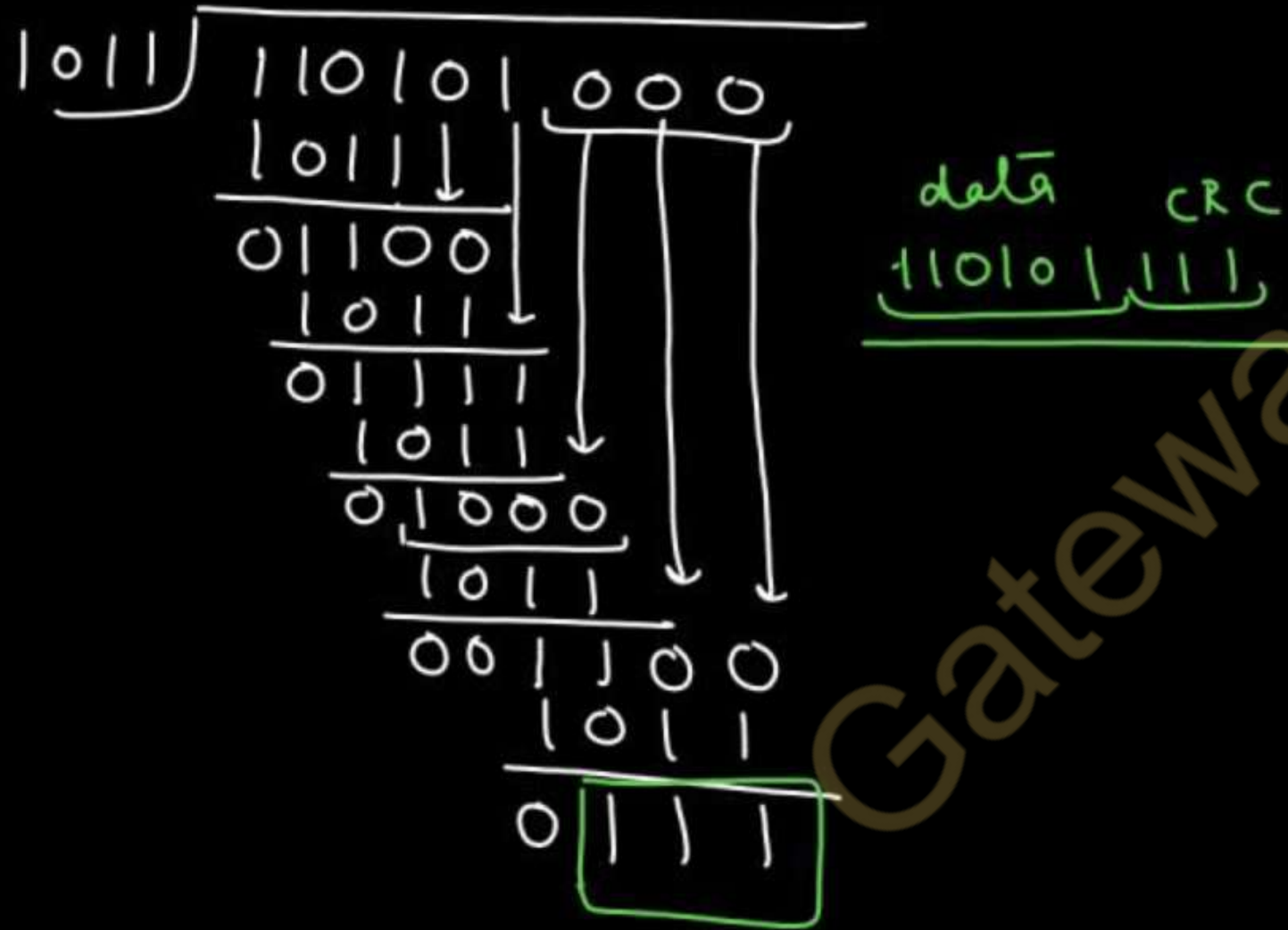
Successful transmission
Receiver will receive it

it is

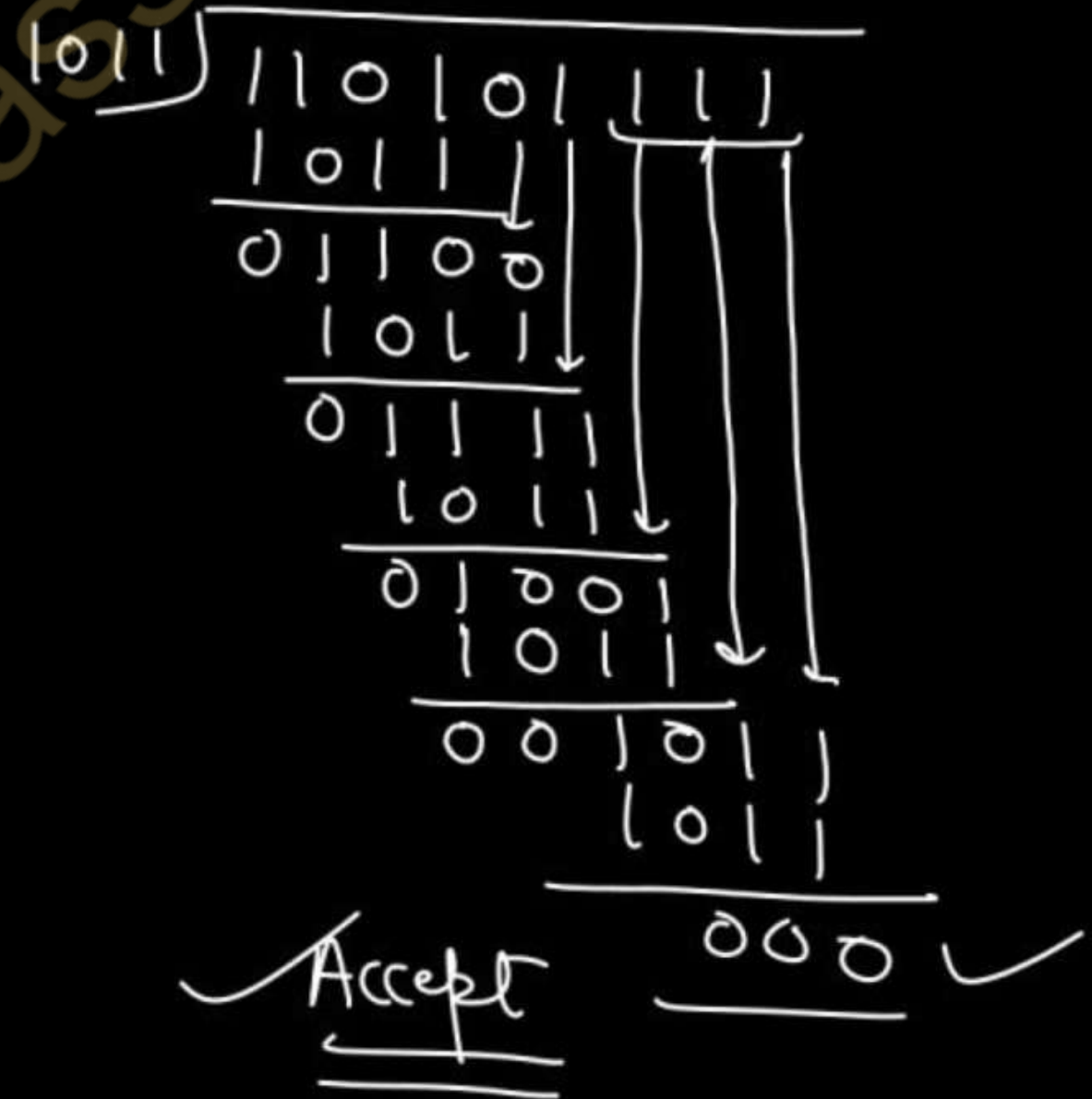
Q1: Senders' data $D = 110101$, CRC generator polynomial $= x^3 + x + 1$. Apply CRC algorithm and perform calculations both at sender and receiver end. (AKTU 2023-24)

Solⁿ. Generator Polynomial $= x^3 + x + 1$
 $= 1011$

Sender side



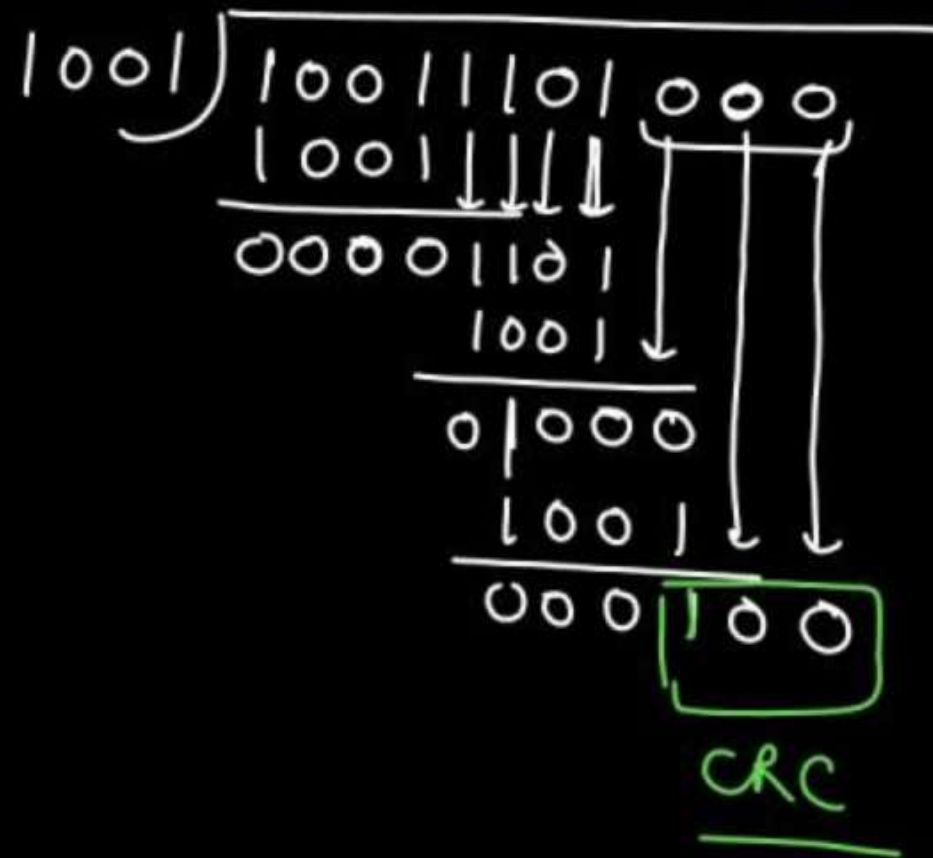
Receiver side



Q: A bit stream 10011101 is transmitted using x^3+1 generator polynomial. Generate the CRC code word for this message. (AKTU 2022-23)

Solⁿ. Generator polynomial = x^3+1
1 0 0 1

Sender Side



data

CRC

100111011000

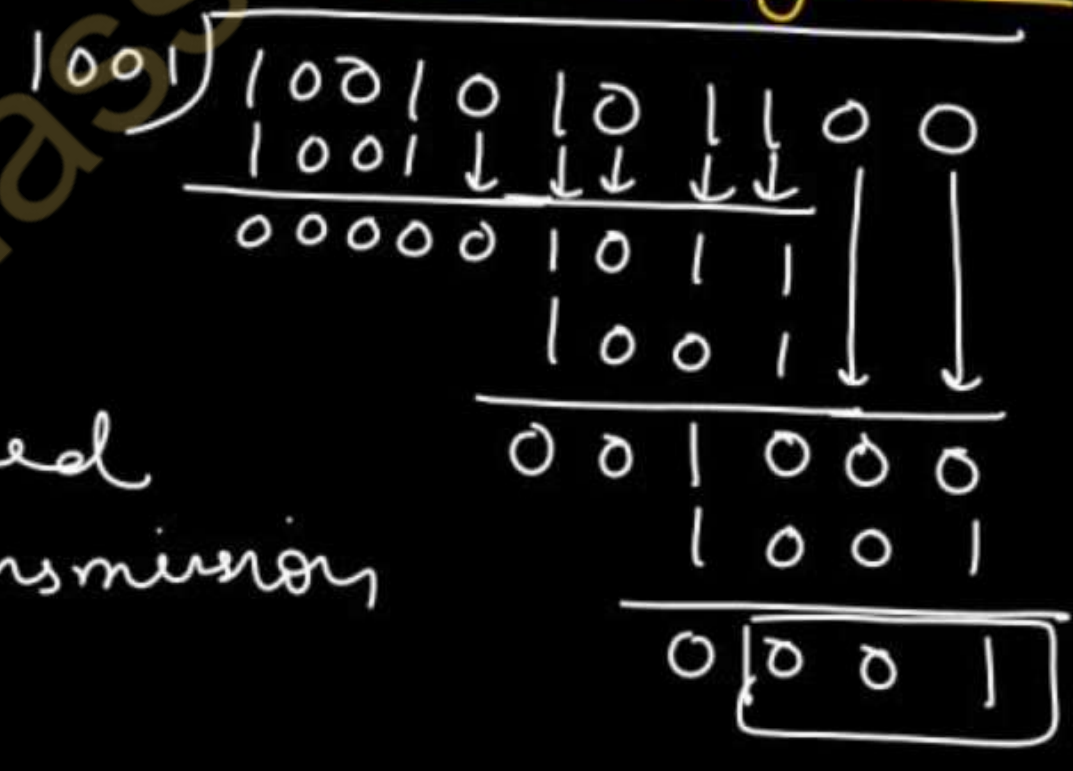
errored
Transmission

10010101100

x^3+1

$1 \cdot x^3 + 0 \cdot x^2 + 0 \cdot x + 1 \cdot 1$

Receiving Side

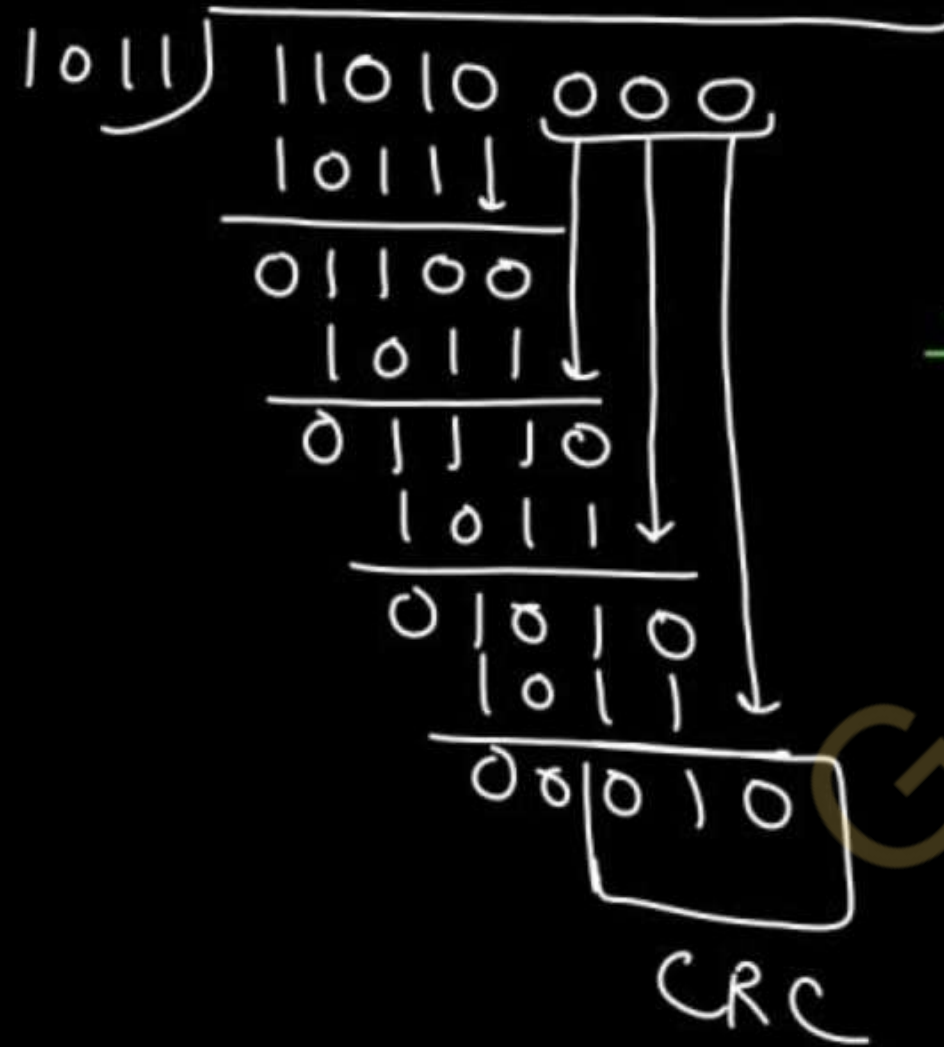


Reject

Q: Sender's data $D=11010$, CRC generator polynomial $= x^3+x+1$. Apply CRC algorithm and perform calculations both at sender and receiver end. (AKTU 2021-22)

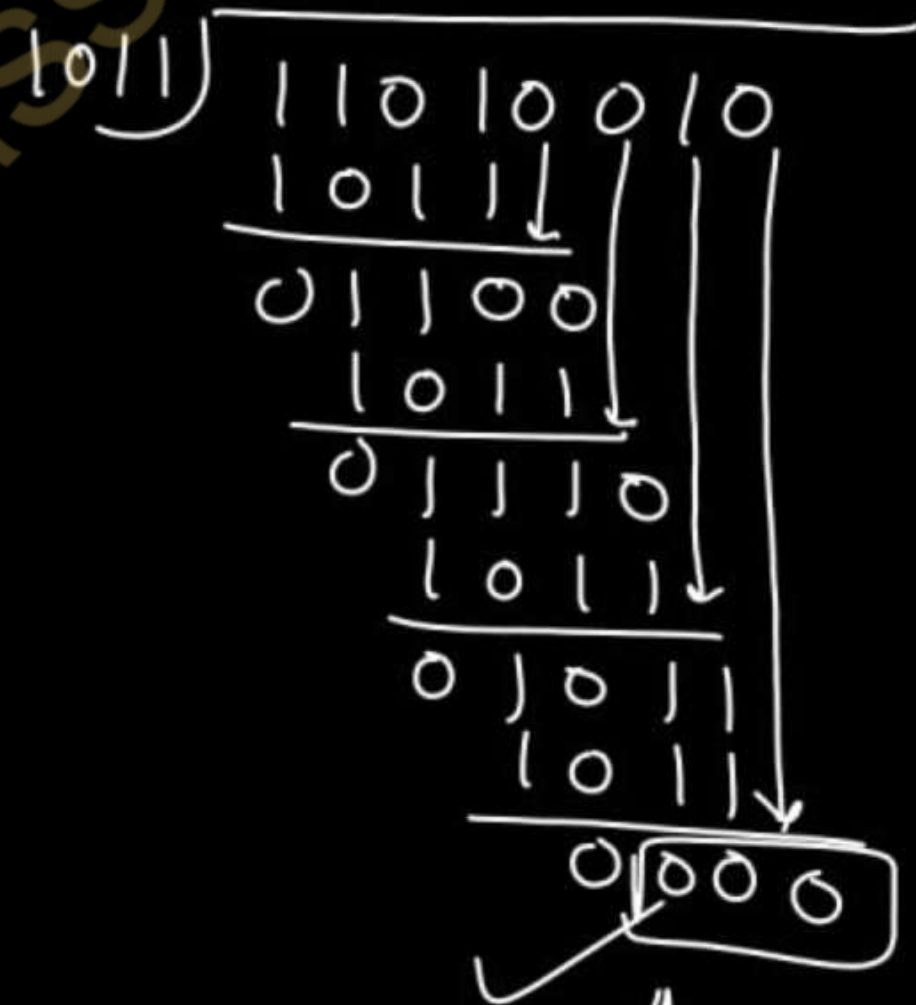
Solⁿ. Polynomial $= x^3+x+1$
 $= 1011$

Sender Side



data CRC
11010 010 →

Receiver Side



Checksum

- Checksum is an error detection method.
- Error detection using checksum method involves the following steps-

At sender side:

- If m bit checksum is used, the data unit to be transmitted is divided into segments of m bits.
- All the m bit segments are added.
- The result of the sum is then complemented using 1's complement arithmetic.
- The value so obtained is called as **checksum**.
- The data along with the checksum value is transmitted to the receiver.

At receiver side:

- If m bit checksum is being used, the received data unit is divided into segments of m bits.
- All the m bit segments are added along with the checksum value.
- The value so obtained is complemented and the result is checked.

If the result is zero

- Receiver assumes that no error occurred in the data during the transmission.
- Receiver accepts the data.

If the result is non-zero

- Receiver assumes that error occurred in the data during the transmission.
- Receiver discards the data and asks the sender for retransmission.

Section K Section 1

n bits $\dots$ n bits n bits

Section 1 n bits

Section 2 n bits

.....

.....

Section K n bits

Sum n bits

Complement

n bits

Checksum

Sender

Section k Section 1

n bits n bits $\dots$ n bits n bits

Checksum

Section 1 n bits

Section 2 n bits

.....

.....

Section K n bits

Checksum n bits

Sum

All 1s, accept
Otherwise, reject

Receiver

Consider the data unit to be transmitted is- 10011001 11100010 00100100 10000100

Consider 8-bit checksum is used.

At sender side:

$$\left. \begin{array}{l} 0+1=1 \\ 1+1=10 \\ 10+1=11 \\ 11+1=100 \end{array} \right\} \text{Binary Addition}$$

Data unit is divided into segments of 8 bits and all the segments are added and the result is obtained.

Sender

$$\begin{array}{r} 1111 \\ 10011001 \\ + 11100010 \\ 00100100 \\ 10000100 \\ \hline 1000100011 \\ + 10 \\ \hline 00100101 \end{array}$$

↓ 1's comp.

11011010 → checksum

data + checksum

Receiver

$$\begin{array}{r} 111111 \\ \text{data} \left\{ \begin{array}{l} 10011001 \\ 11100010 \\ 00100100 \\ 10000100 \end{array} \right. \\ + \\ \text{checksum} \left[\begin{array}{l} 11011010 \end{array} \right] \\ \hline 1011111101 \\ + 10 \\ \hline 11111111 \\ \downarrow \text{1's comp} \\ 00000000 \end{array}$$

Accept

Advantages of Checksum

- Detects all errors involving an odd number of bits.
- Detects most errors involving an even number of bits.
- Easy to implement making it ideal for real-time applications like network transfers.
- Requires minimal computing resources, so it doesn't slow down the system.

Disadvantages of Checksum

- If one or more bits of a segment are damaged and the corresponding bit or bits of opposite value in a second segment are also damaged, the sums of those columns will not change and the receiver will not detect a problem.
- It can only detect errors but cannot fix or recover corrupted data.
- Checksum algorithms are not designed for security and can be easily manipulated by attackers.

AKTU PYQs

1. State the requirements of CRC (AKTU 2016-17)
2. A bit stream 10011101 is transmitted using x^3+1 generator polynomial. Generate the CRC code word for this message. (AKTU 2022-23)
3. Senders' data $D=110101$, CRC generator polynomial $=x^3+x+1$. Apply CRC algorithm and Perform calculations both at sender and receiver end. (AKTU 2023-24)

Thank
you



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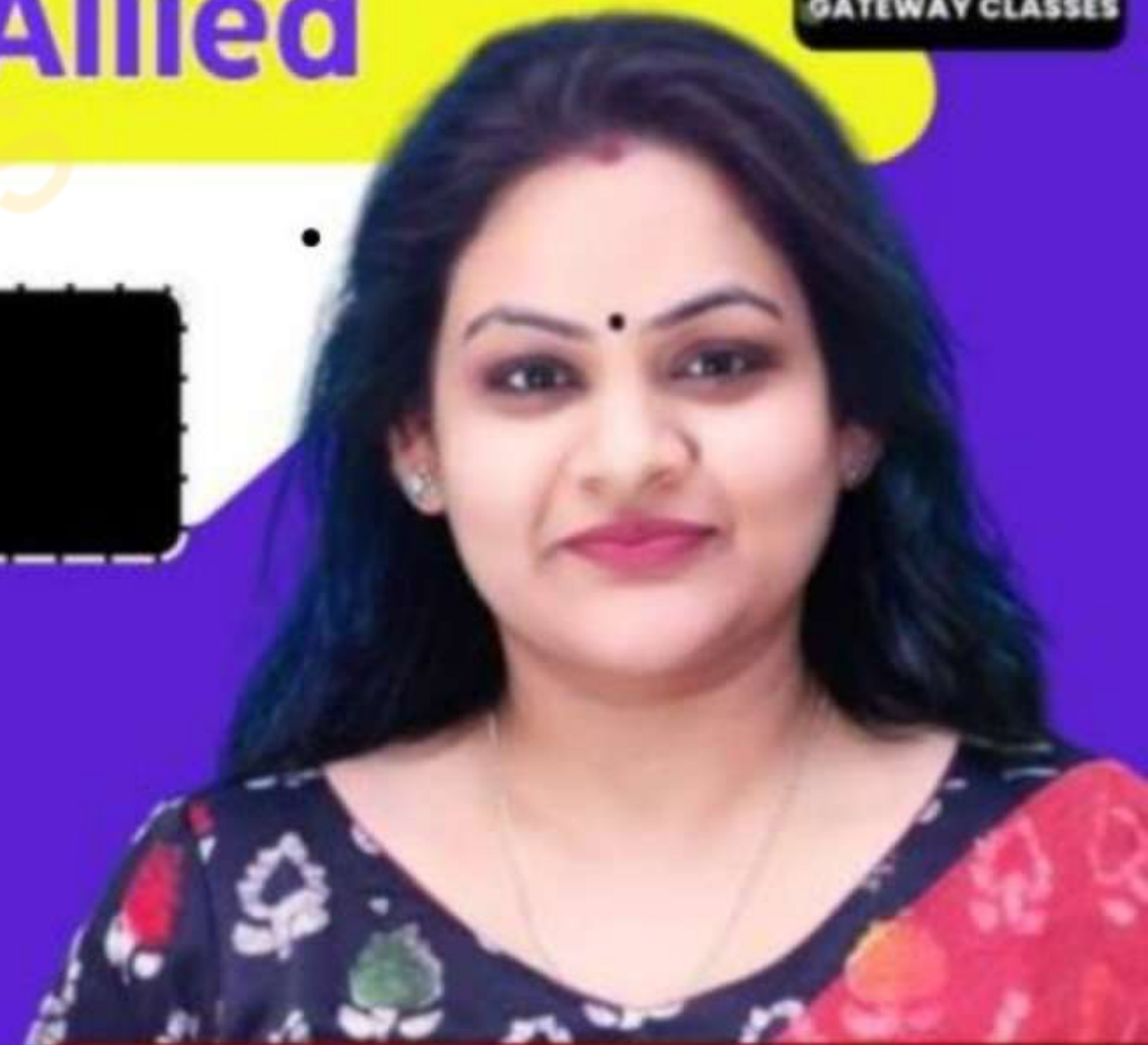


Computer Networks

Unit-2: Lecture 8

Today's Target

- Error Correction ✓
 - Hamming Code ✓
- AKTU PYQs



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- M.Tech Gold Medalist
- Net Qualified

Error Correction

It can be handled in two ways:

- 1) Receiver can have the sender retransmit the entire data unit.
- 2) Receiver can use an error-correcting code, which automatically corrects certain errors.

Single--bit error correction:

- To correct an error, the receiver reverses value of altered bit. For this, it must know which bit is in error.

Hamming Code

- It is an error-correcting code that detects and corrects the errors in data when moved from the sender to the receiver.
- It adds extra bits to the original data, allowing the system to detect and correct single-bit errors.
- It is constructed by adding a no. of parity bits to original data to be able to locate the bit position in which error occurs.
- Values (0 or 1) are assigned to the parity bits to make the hamming code have either even parity or odd parity & when an error occurs, the position no. will take on the value assigned to the location of the erroneous bit.

- Let data bits = m and Redundancy bits = r
- Total message sent = $m+r$
- The value of r must satisfy the following relation: $2^r \geq m+r+1$

Suppose the number of data bits is 7,

then the number of redundant bits can be calculated using: $2^4 \geq 7 + 4 + 1$.

Thus, the number of redundant bits is 4.

Q: Suppose the data to be transmitted is 1011001 from sender to receiver, Construct hamming code using even parity. Suppose in the above example the 6th bit is changed from 0 to 1 during data transmission. Show how hamming code detects and corrects the error.

Solⁿ

Data: 1011001

Step - 1

Determine the no. of parity bits.

$$2^r \geq m + r + 1$$

No. of Data bits $\downarrow$ No. of parity bits (redundant bits)

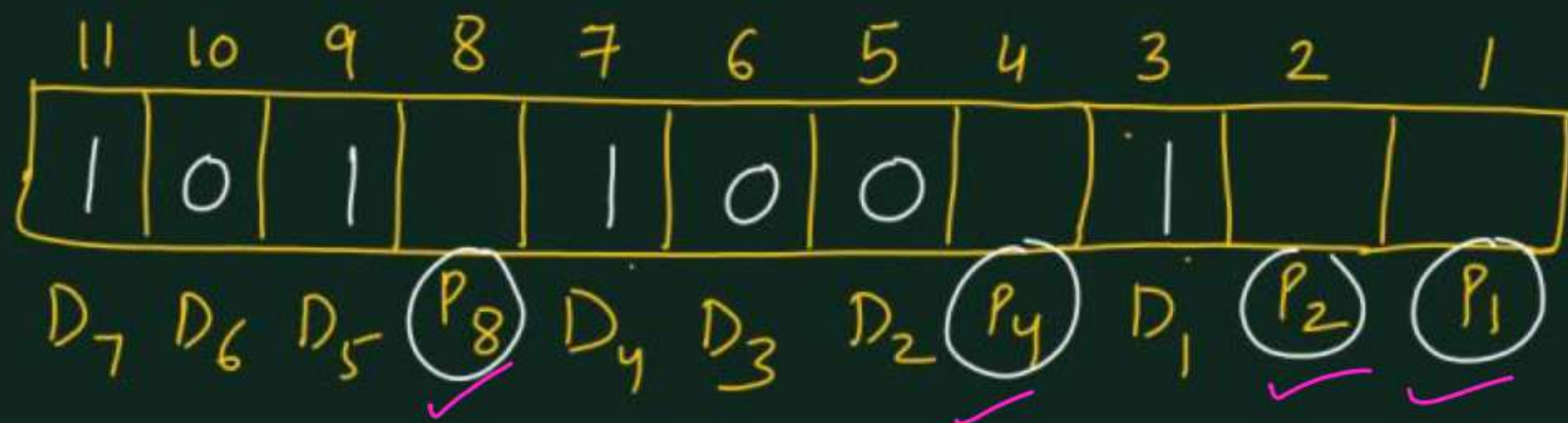
$m = 7$

$r = 2$	$\rightarrow$	$2^2 \geq 7 + 2 + 1$	$4 \geq 10$	X
$r = 3$	$\rightarrow$	$2^3 \geq 7 + 3 + 1$	$8 \geq 11$	X

$r = 4$

$2^4 \geq 7 + 4 + 1$	$16 \geq 12$	✓
----------------------	--------------	---

data + parity
7 + 4 = 11 bits will be sent by the sender.



Step-3

Determining bit positions for calculation of parity bits

Step-2

Determine the position of parity bits

$$P_1 \rightarrow 2^0 \rightarrow (1)$$

$$P_2 \rightarrow 2^1 \rightarrow (2)$$

$$P_4 \rightarrow 2^2 \rightarrow (4)$$

$$P_8 \rightarrow 2^3 \rightarrow (8)$$

bit positions

$$P_1 = 1, 3, 5, 7, 9, 11, 13, 15$$

$$P_2 = 2, 3, 6, 7, 10, 11, 14, 15$$

$$P_4 = 4, 5, 6, 7, 12, 13, 14, 15$$

$$P_8 = 8, 9, 10, 11, 12, 13, 14, 15$$

	P_8	P_4	P_2	P_1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

11	10	9	8	7	6	5	4	3	2	1
1	0	1	0	1	0	0	1	1	1	0
D_7	D_6	D_5	P_8	D_4	D_3	D_2	P_4	D_1	P_2	P_1

step-4 → Determine the values of parity bits. [Even Parity]

$$P_1 = 1, 3, 5, 7, 9, 11 \Rightarrow P_1 = 0$$

0 1 0 1 1 1

$$P_2 = 2, 3, 6, 7, 10, 11 \Rightarrow P_2 = 1$$

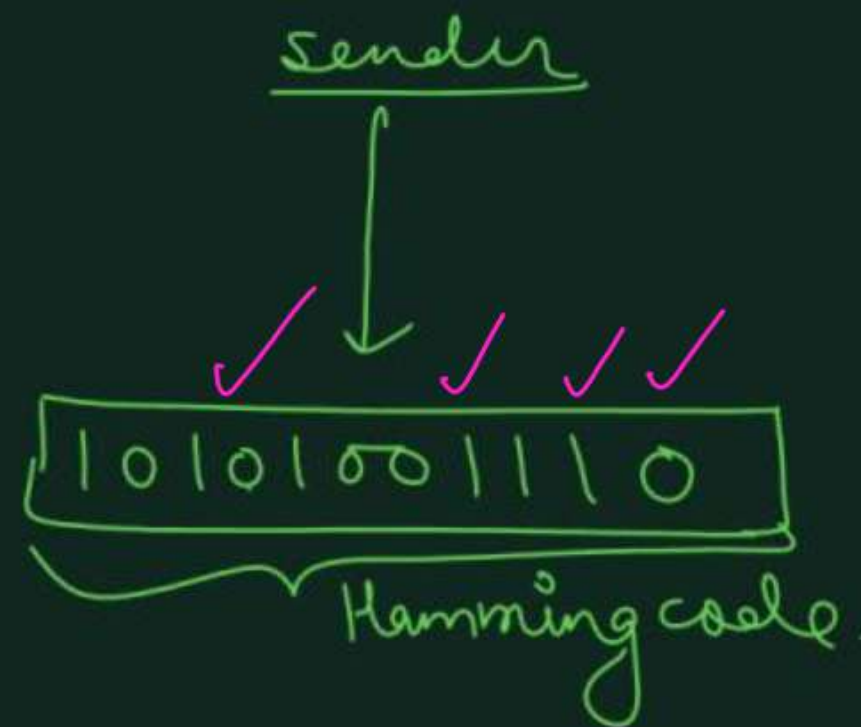
1 1 0 1 0 1

$$P_4 = 4, 5, 6, 7 \Rightarrow P_4 = 1$$

1 0 0 1

$$P_8 = 8, 9, 10, 11 \Rightarrow P_8 = 0$$

0 1 0 1



Receiver side — code Received

11	10	9	8	7	6	5	4	3	2	1
1	0	1	0	1	1	0	1	1	1	0
D_7	D_6	D_5	P_8	D_4	D_3	D_2	P_4	D_1	P_2	P_1

$$P_1 = 1, 3, 5, 7, 9, 11$$

$$0 \ 1 \ 0 \ 1 \ 1 \ 1$$

$$P_1 = 0$$

$$P_8 = 8, 9, 10, 11$$

$$0 \ 1 \ 0 \ 1$$

$$\Rightarrow P_8 = 0$$

$$P_2 = 2, 3, 6, 7, 10, 11$$

$$1 \ 1 \ 1 \ 1 \ 0 \ 1$$

$$P_2 = 1$$

$$P_4 = 4, 5, 6, 7$$

$$1 \ 0 \ 1 \ 1$$

$$\Rightarrow P_4 = 1$$

As all parities are not 0 so, the receiver will conclude that it has received the erroneous message.

which bit is erroneous?

$$P_4 \ P_3 \ P_2 \ P_1$$

$$0 \ 1 \ 1 \ 0$$

$$\downarrow$$

$$6$$

$P_8 \ P_4 \ P_2 \ P_1$
| 0 | 1 | 1 | 0 | $\Rightarrow$ 6th bit is error.

Receiver will change the bit from 1 to 0 and correct the message, then process it.

Forward Error Correction

Q: A 7-bit Hamming code is received as 1110101. Is there any error? If yes, locate the position of the error bit. Parity checks are created by odd parity. (AKTU 2022-23)

Solⁿ

$$2^r \geq m + r + 1$$

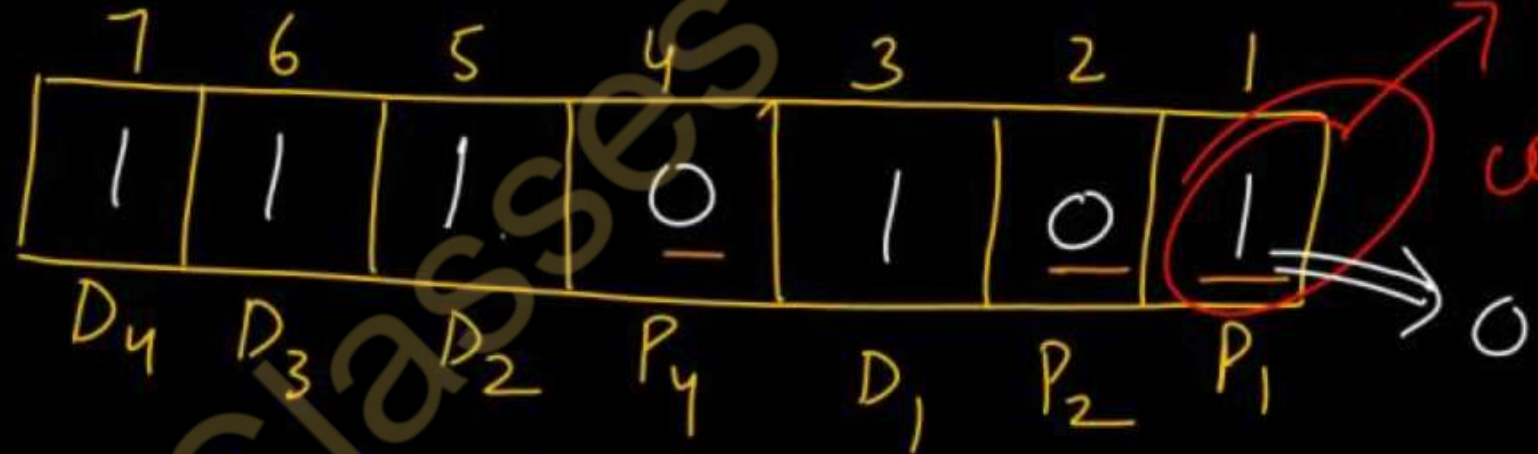
$$2^3 \geq \underline{4} + \underline{3} + 1$$

$$8 \geq 8$$

$$P_1 = \begin{matrix} 1, 3, 5, 7 \\ 1 \quad 1 \quad 1 \quad 1 \end{matrix} \Rightarrow \boxed{P_1 = 1}$$

$$P_2 = \begin{matrix} 2, 3, 6, 7 \\ 0 \quad 1 \quad 1 \quad 1 \end{matrix} \Rightarrow \boxed{P_2 = 0}$$

$$P_4 = \begin{matrix} 4, 5, 6, 7 \\ 0 \quad 1 \quad 1 \quad 1 \end{matrix} \Rightarrow \boxed{P_4 = 0}$$



As all parity bit are not 0 $\Rightarrow$ Errored Message.

$$\text{Erorred bit} = P_4 P_2 P_1 = \boxed{1 \quad 0 \quad 0 \quad 1} \Rightarrow 1$$

Bit at position - 1 is corrupted / errored $\Rightarrow$ correct & receive

Corrected code

1110100

AKTU PYQs

1. A 7-bit Hamming code is received as 1110101. Is there any error? If yes, locate the position of the error bit. Parity checks are created by odd parity (AKTU 2022-23)

Gateway Classes

Thank
you



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Computer Networks

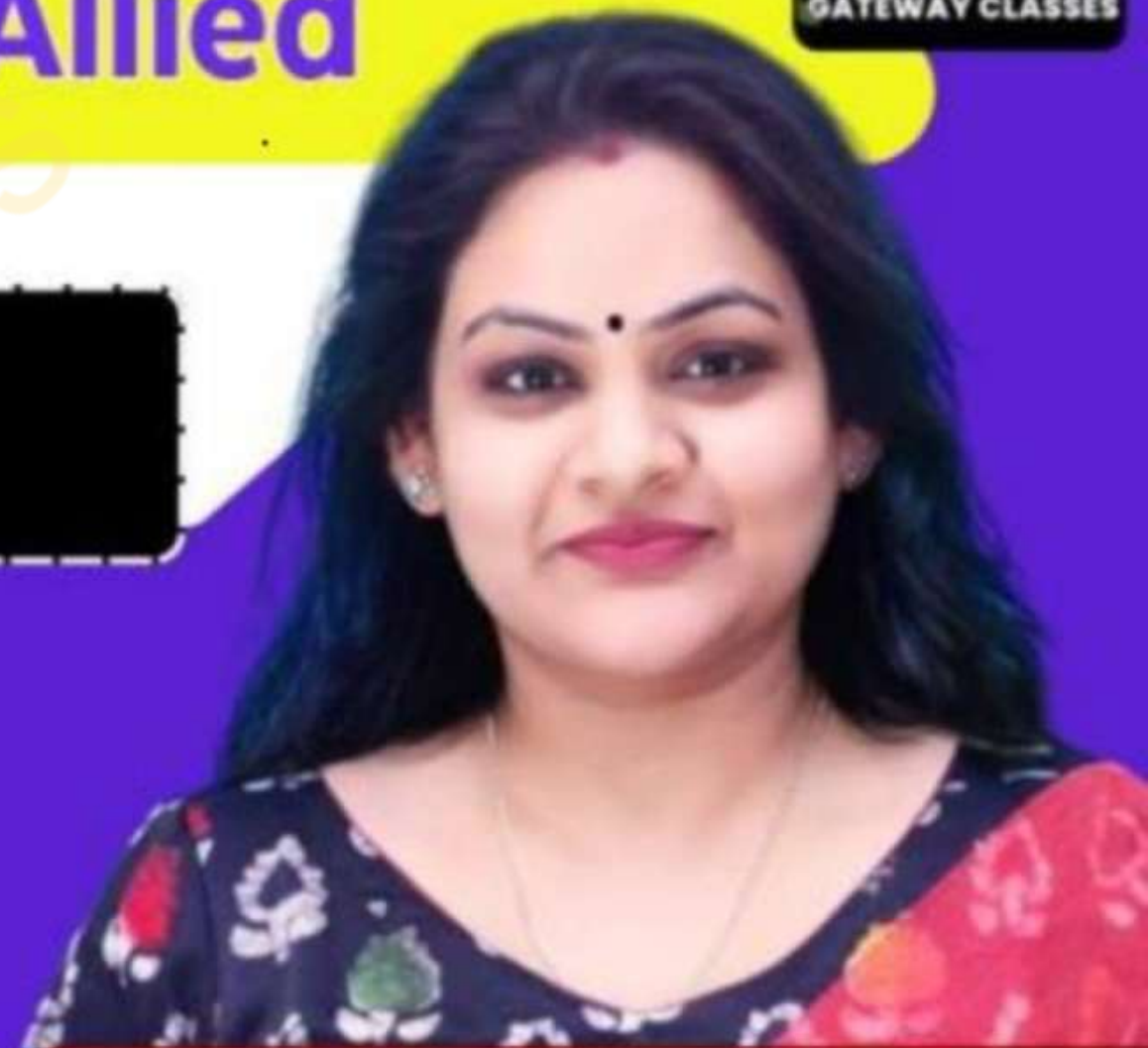
Unit-2: Lecture-9

Today's Target

- LAN standards ✓
- Ethernet(IEEE 802.3) ✓
- AKTU PYQs

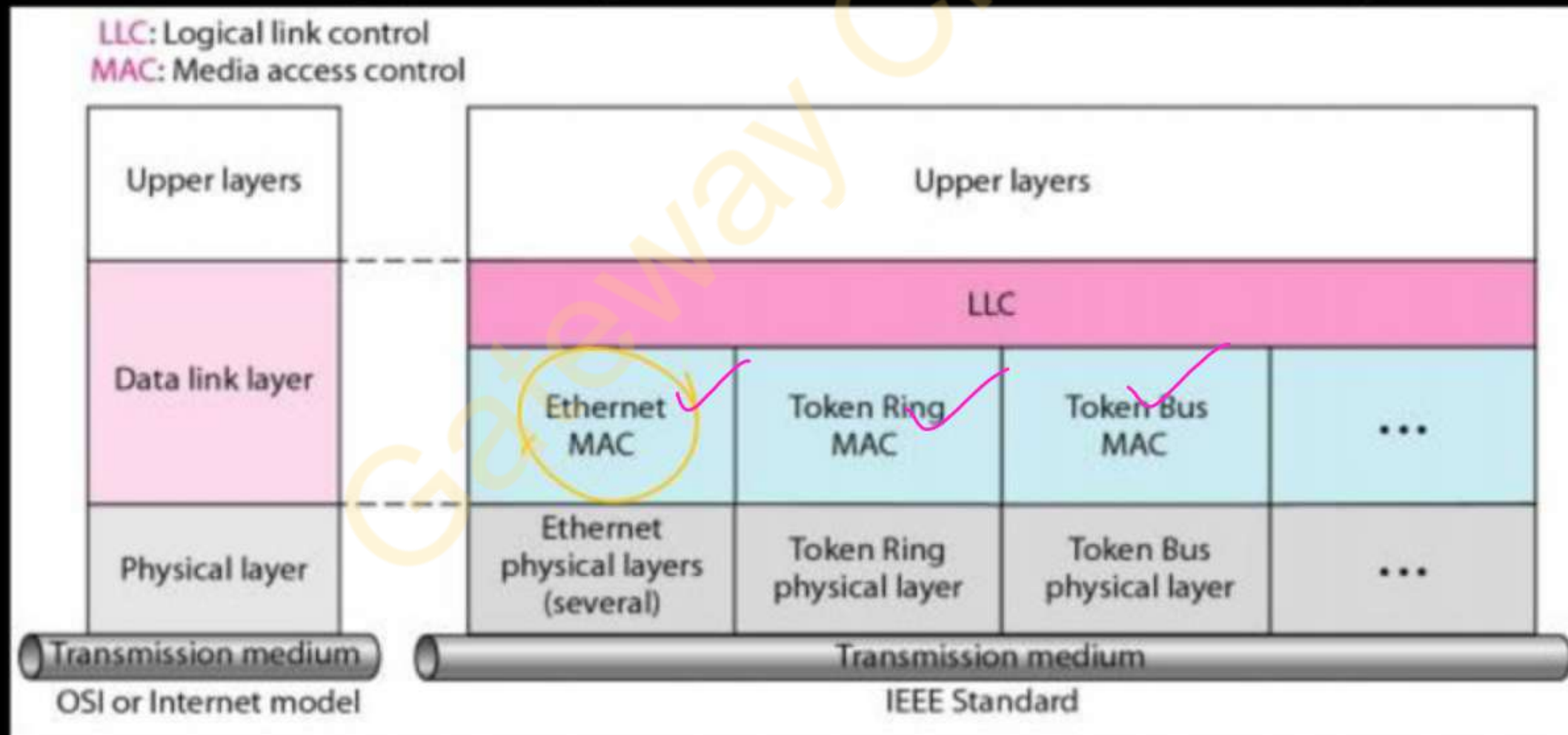
By Dr. Nidhi Parashar Ma'am

- Ph.D (CSE)
- M.Tech Gold Medalist
- Net Qualified



Wired LANs: Ethernet

- Computer Society of IEEE started a project 802, to set standards to enable intercommunication among equipment from a variety of manufacturers.
- Project 802 is a way of specifying functions of the physical layer and the data link layer of major LAN protocols.
- IEEE has also created several physical layer standards for different LAN protocols.



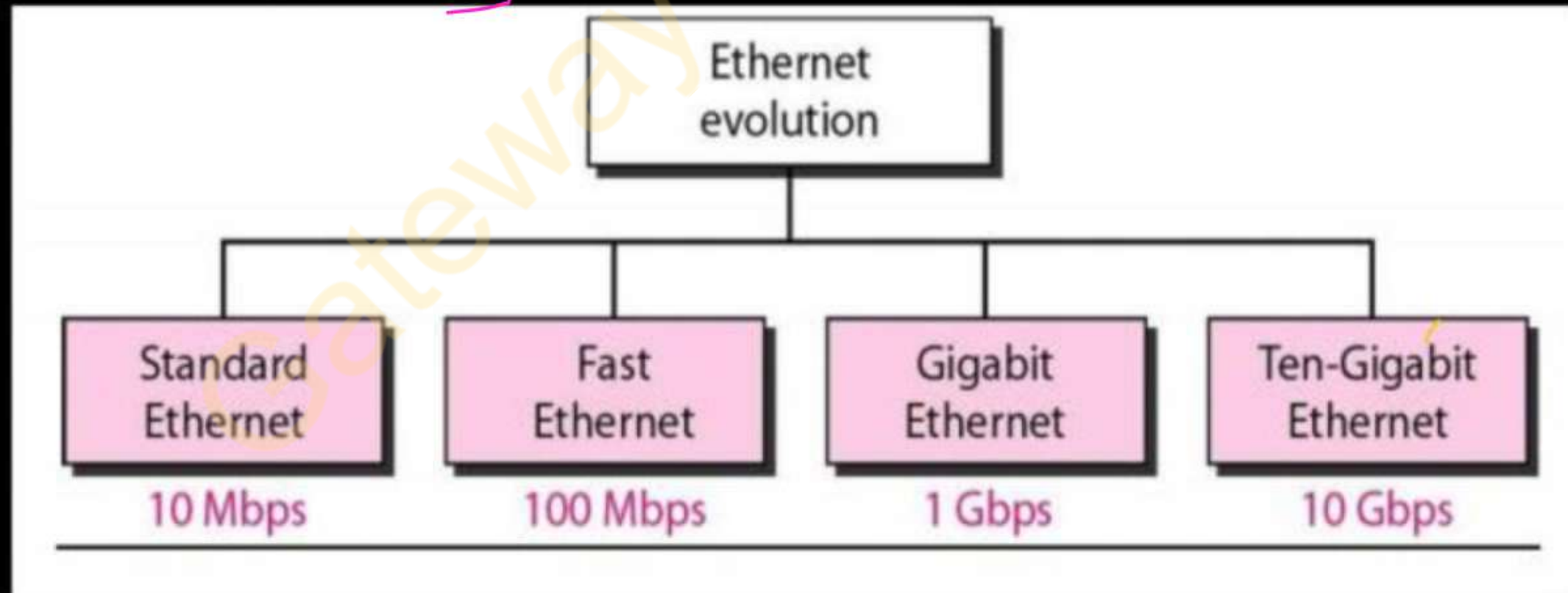
Ethernet

IEEE 802.3

The original Ethernet was created in 1976 at Xerox's Palo Alto Research Center (PARC).

It has gone through four generations:

1. Standard Ethernet (10 Mbps)
2. Fast Ethernet (100 Mbps)
3. Gigabit Ethernet (1 Gbps)
4. Ten-Gigabit Ethernet (10 Gbps)



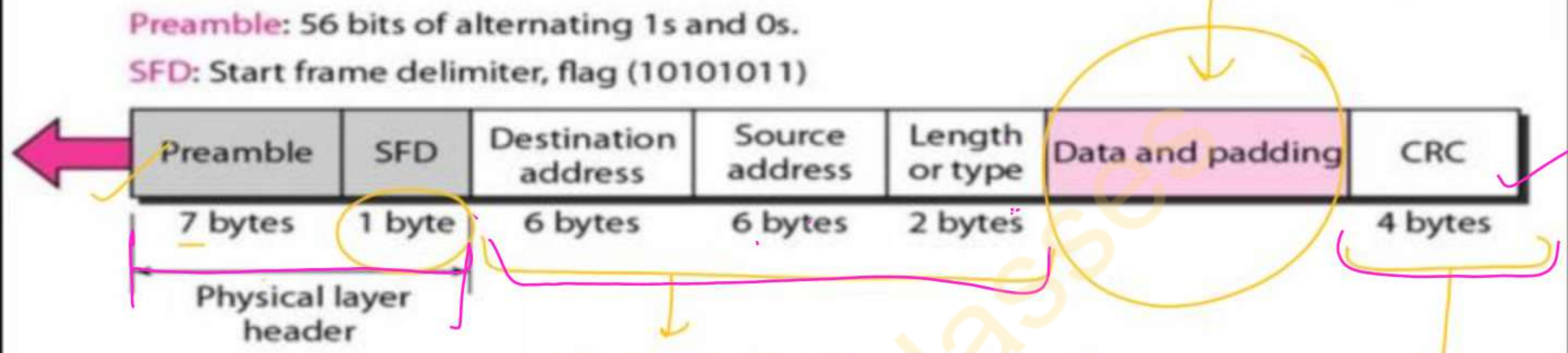
MAC Sublayer

- In Standard Ethernet, the MAC sublayer governs the operation of the access method.
- It also frames data received from the upper layer and passes them to the physical layer.

Frame Format (Imp)

- The Ethernet frame contains seven fields: preamble, SFD, DA, SA, length or type of protocol data unit (PDU), upper-layer data, and the CRC.
- Ethernet does not provide any mechanism for acknowledging received frames, making it what is known as an unreliable medium.
- Acknowledgments must be implemented at the higher layers.

802.3 MAC Frame Format



Preamble :

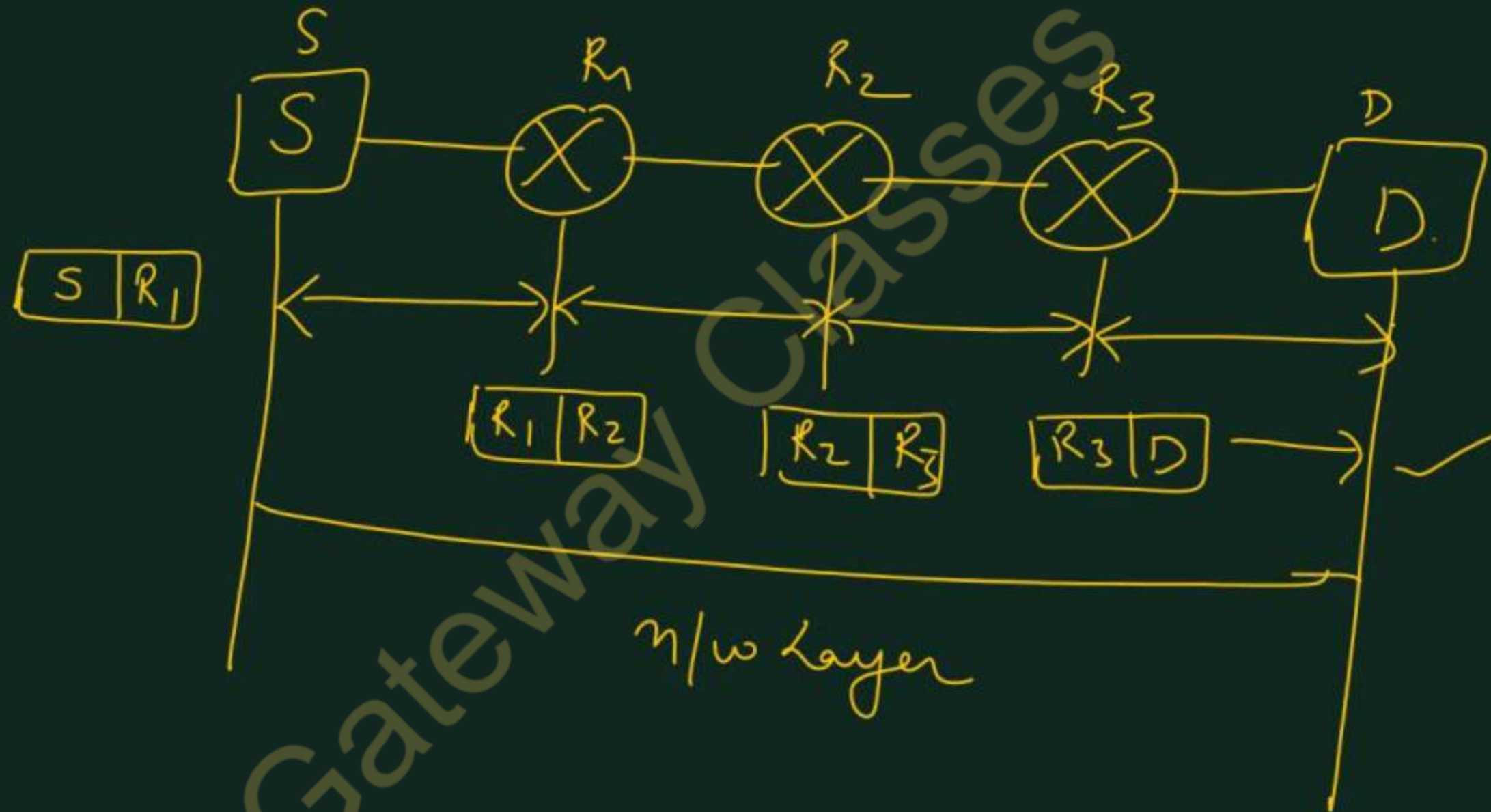
Data link layer header

Q.L.L. Trailer

- 7 bytes (56 bits) of alternating 0s and 1s that alerts the receiver to the coming frame and enables it to synchronize its input timing. (1 0 | 0 1 | 0 1 | 0 1 | 0 1 | 0 1 | 0 1)
- The preamble is actually added at the physical layer and is not (formally) part of the frame.

Start frame delimiter (SFD):

- The second field (1 byte: 10101011) signals the beginning of the frame. The SFD warns the station that this is the last chance for synchronization and alerts receiver that the next field is the destination address.



Destination address (DA):

6 bytes DA field contains the physical address of the destination station to receive the packet.

Source address (SA):

6 bytes contains the physical address of the sender of the packet.

Length or type: 2 byte $\Rightarrow$ 16 bits $(0 - 2^{16} - 1)$ $(0 - 65535)$

➤ Original Ethernet used this field as the type field to define the upper-layer protocol.

➤ The IEEE standard used it as the length field to define the number of bytes in the data field.

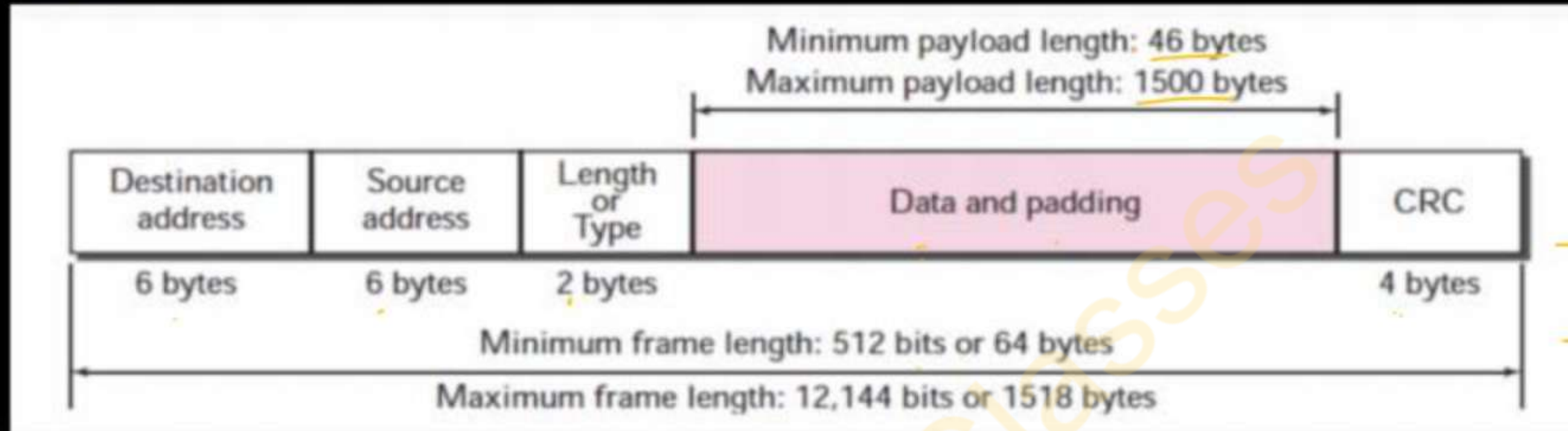
Data:

➤ This field carries data encapsulated from the upper-layer protocols. It is a minimum of 46 bytes and a maximum of 1500 bytes.

CRC: The last field contains error detection information, in this case a CRC-32.

Padding: Zeros are added to the data field to make the minimum data length = 46 bytes

Frame Length: Ethernet has imposed restrictions on both minimum and maximum lengths of a frame



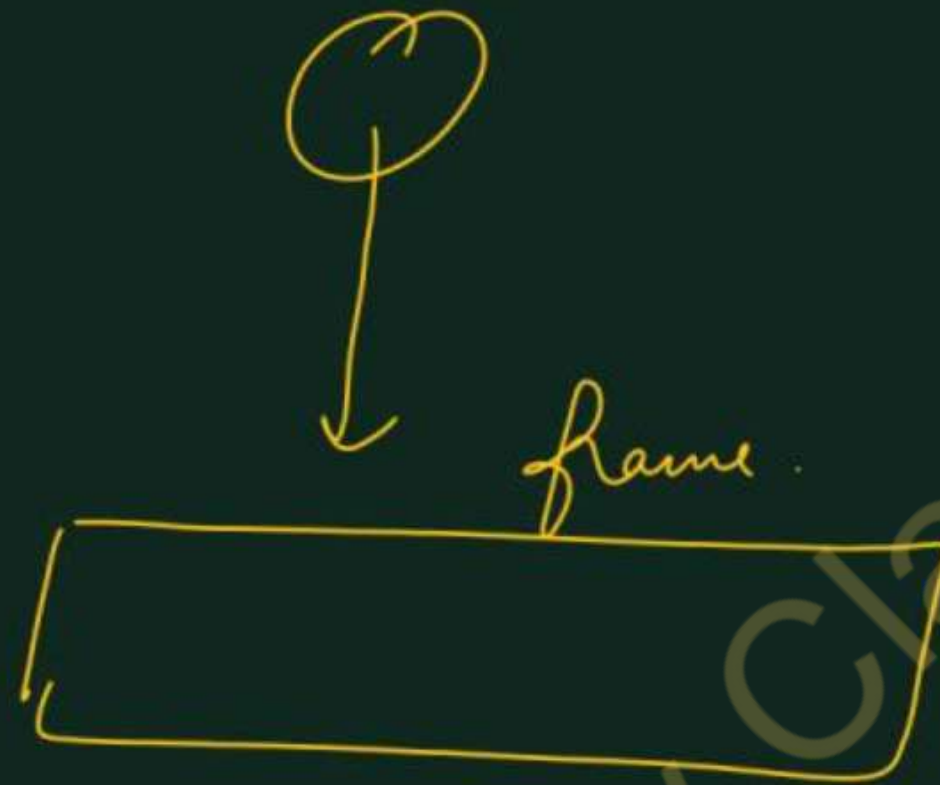
46
18

64 bytes Min

1500
+ 18

1518 bytes

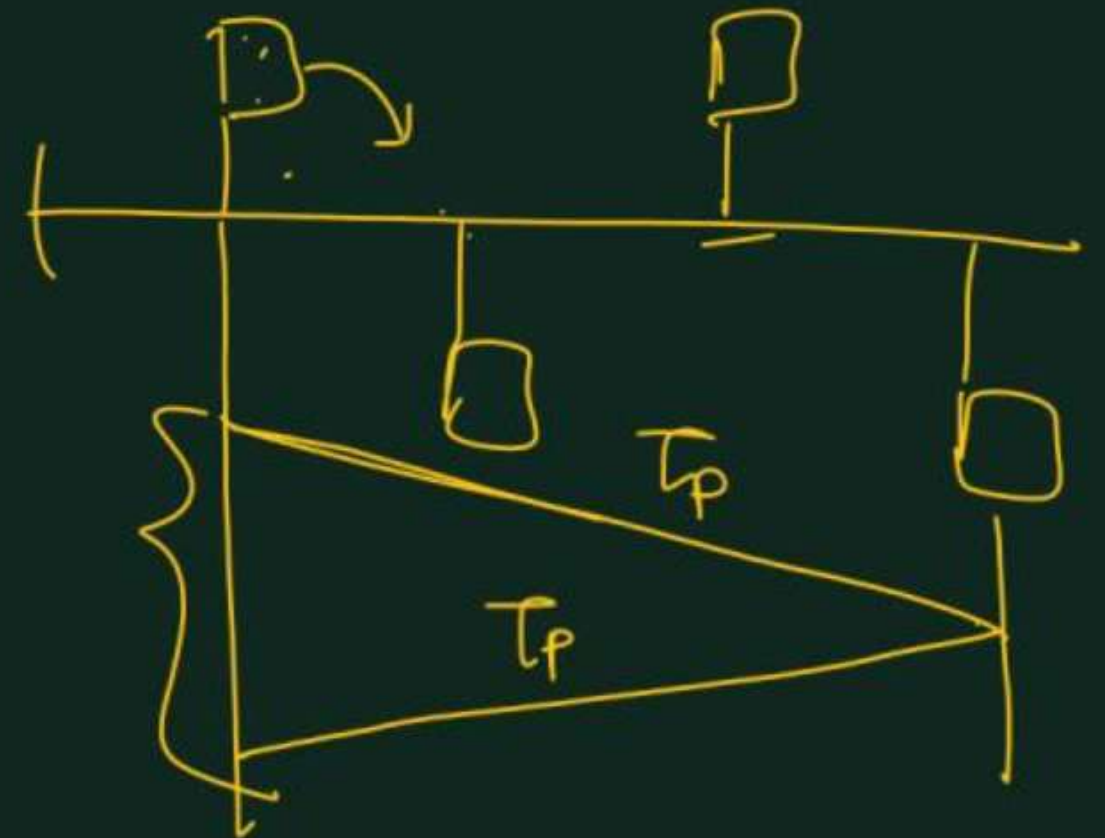
- Minimum length of Ethernet frame is 512 bits or 64 Bytes including header and trailer.
- Minimum length of data from the upper layer is $64 - 18 = 46$ bytes.
- If the upper-layer packet is less than 46 bytes, padding is added to make up the difference.
- Maximum length of a frame (without preamble and SFD field) is 1518 bytes (maximum payload is 1500 bytes).
- A max length restriction helped to reduce the size of buffer and prevents one station from monopolizing the shared medium, blocking other stations that have data to send.



1-persistent
CSMA/CD

frame size
bandwidth

$$T_{fr} \geq 2 \times T_p$$



Addressing :

The Ethernet address is 6 bytes (48 bits), normally written in hexadecimal notation, with a colon between the bytes.

Example:

✓✓✓✓✓✓
00:1A:2B:3C:4D:5E,

It is 12-digit hexadecimal number representing 48-bit hardware address assigned to a network interface

Unicast, Multicast, and Broadcast Addresses:

- A source address is always a unicast address-the frame comes from only one station.
- The destination address can be **unicast, multicast, or broadcast**.



- unicast ✓
- multicast
- Broadcast
↓
FF:FF:FF:FF:FF:FF

If least significant bit of first byte in a destination address is 0, the address is unicast; otherwise, it is multicast.

Unicast and multicast addresses:

- A unicast destination address defines only one recipient
- A multicast destination address defines a group of addresses
- The broadcast address is a special case of the multicast address, the recipients are all the stations on the LAN. A broadcast destination address is forty-eight 1s.

Q: Define the type of the following destination addresses:

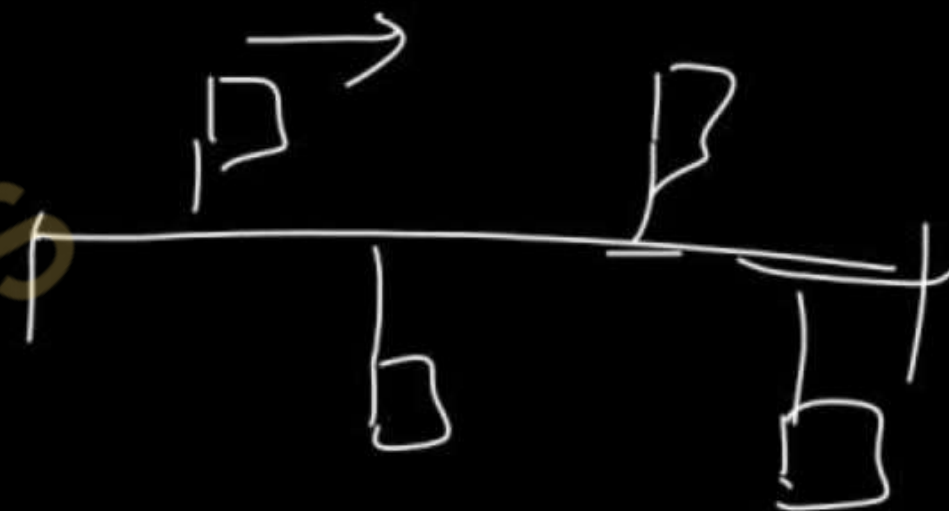
- a. 4A:30:10:21:10:1A $\Rightarrow$ Unicast address
 Even
- b. 47:20:1B:2E:08:EE $\Rightarrow$ Multicast
 odd
- c. FF:FF:FF:FF:FF:FF $\Rightarrow$ Broadcast

Solution: Look at the second hexadecimal digit from the left. If it is even, the address is unicast.

If it is odd, the address is multicast. If all digits are F's, the address is broadcast.

Access Method: CSMA/CD

- Standard Ethernet uses 1-persistent CSMA/CD ✓
- Encoding and Decoding: Manchester encoding. ✓



Slot Time In an Ethernet network:

Slot time = round-trip time + time required to send the jam sequence

- Slot time in Ethernet is time required for a station to send 512 bits. ✓
- This means that the actual slot time depends on the data rate;
- for traditional 10-Mbps Ethernet it is 51.2 micro sec. ✓

$$2 \times T_p + \text{---}$$

10 Mbps

Relation between Slot Time and Maximum Network Length (collision domain):

$$\frac{10 \times 10^6 \text{ bits/sec.}}{2 \times 10^8 \text{ m/s}}$$

It is dependent on the propagation speed of the signal in the particular medium.

In most transmission media, the signal propagates at $2 \times 10^8 \text{ m/s}$

For traditional Ethernet:

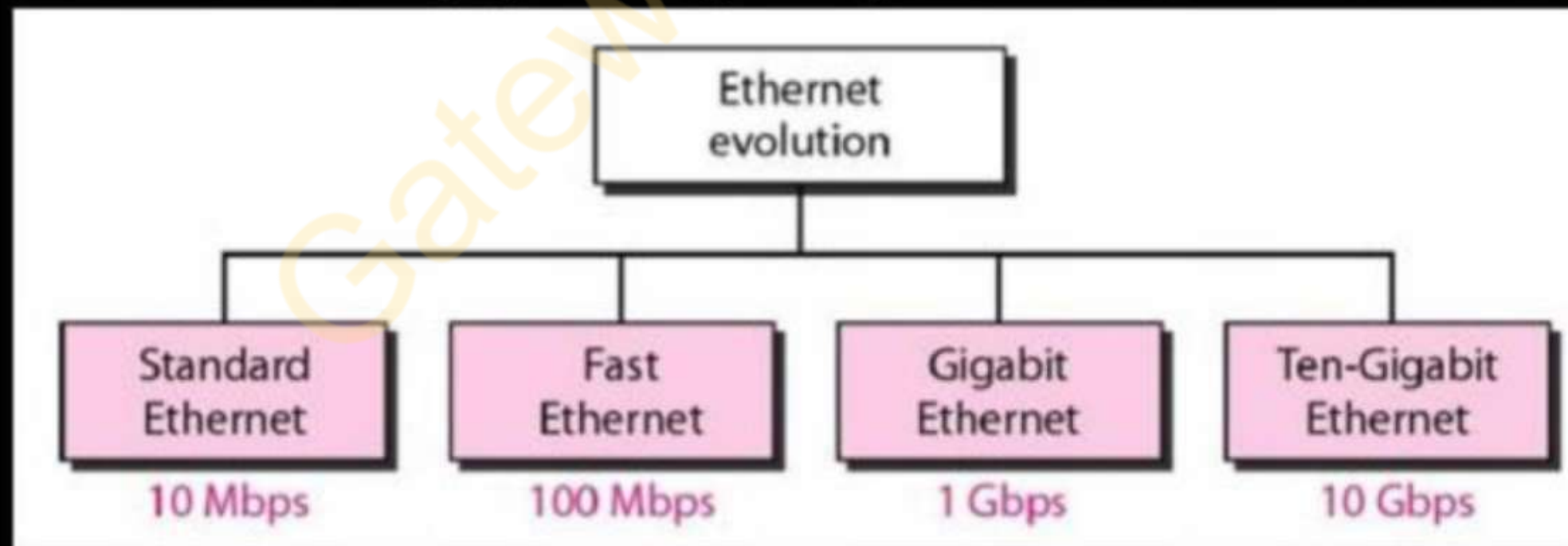
$$\text{Max Length} = \text{Propagation Speed} \times (\text{Slot Time}/2)$$

$$\text{Max Length} = (2 \times 10^8) \times (51.2 \times 10^{-6}) / 2 = 5120\text{m}$$

- We need to consider delay times in repeaters and interfaces, and time required to send the jam sequence. These reduce the maximum length of a traditional Ethernet network to 2500 m, just 48 percent of the theoretical calculation. Max Length=2500 m ✓

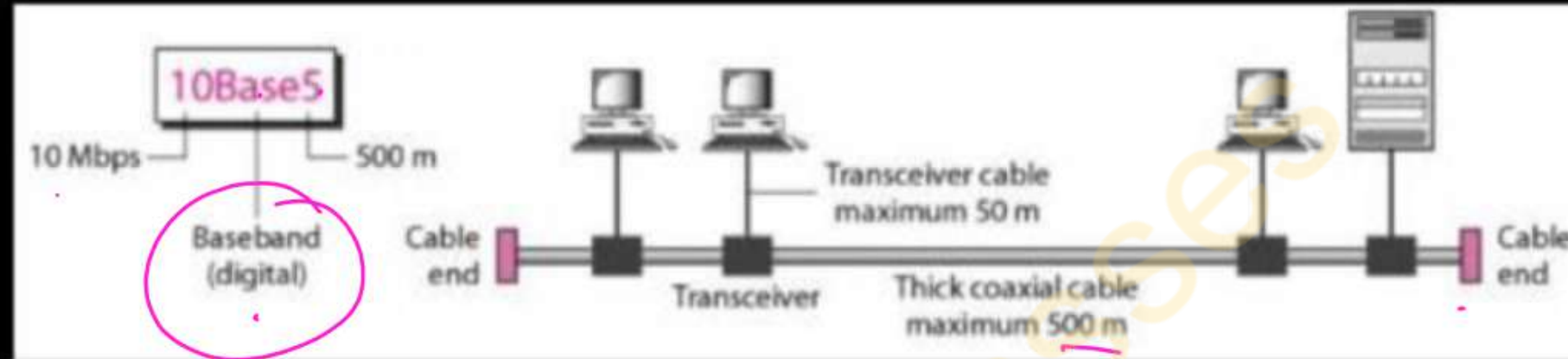
Physical Layer

The Standard Ethernet defines several physical layer implementations. Four most common are as follows:



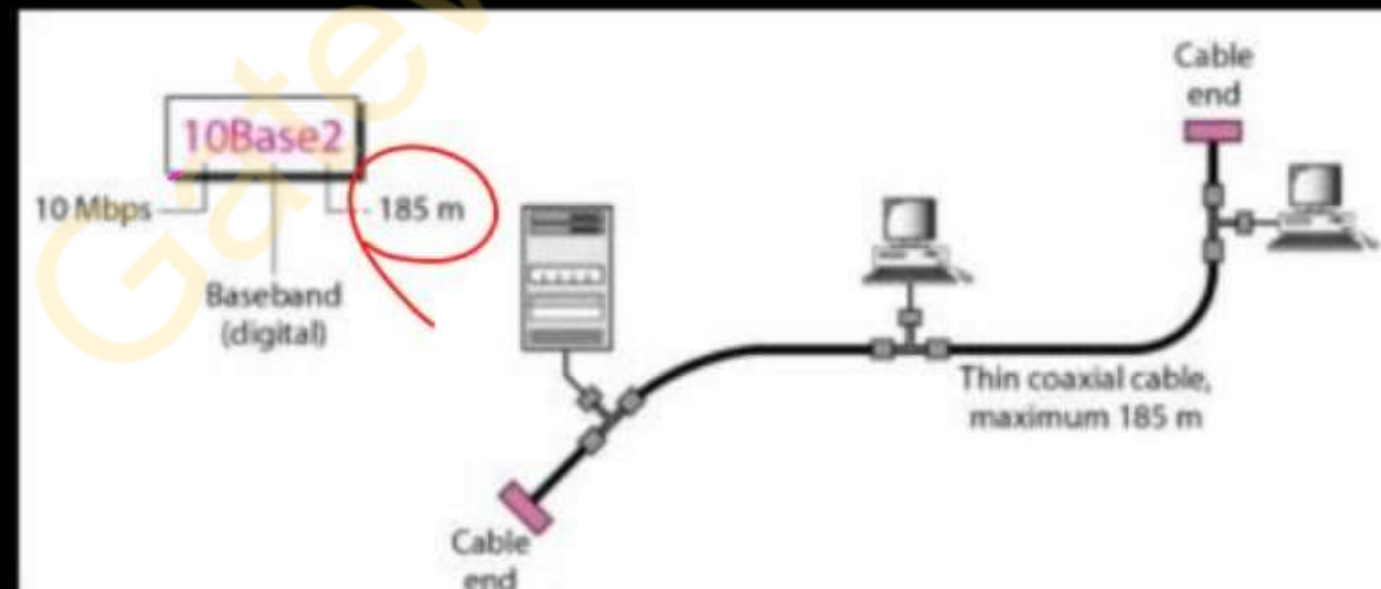
10Base5: Thick Ethernet (Thicknet)

- First Ethernet specification to use a bus topology with an external **transceiver** connected via a tap to a thick coaxial cable.



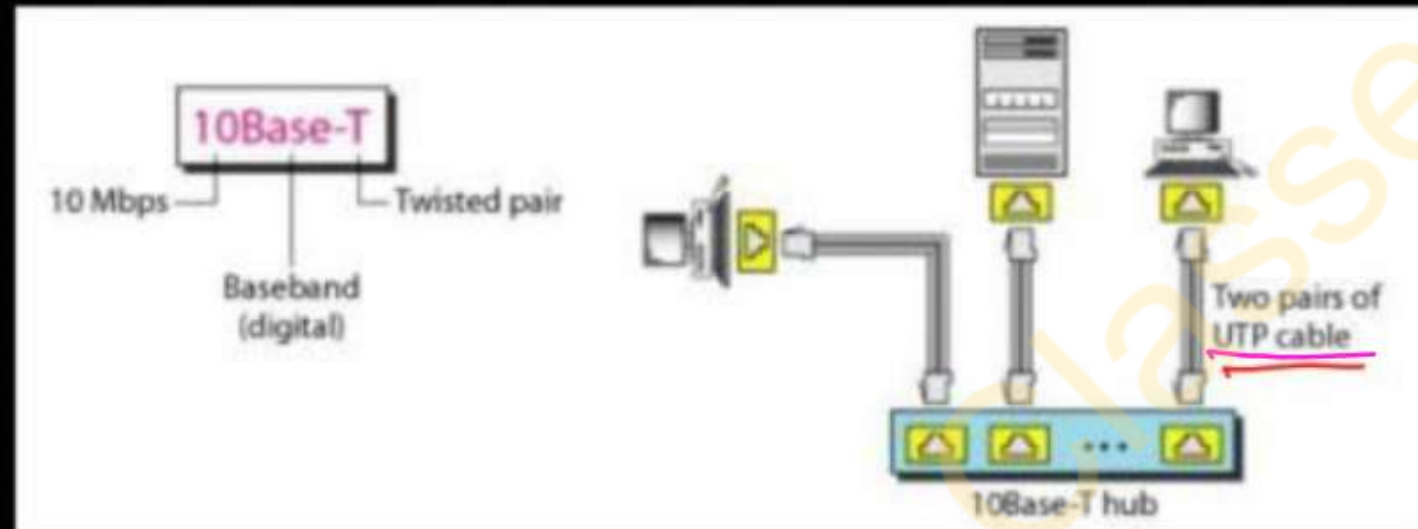
10Base2: Thin Ethernet

It also uses a bus topology, but the cable is much thinner and more flexible. Installation is simpler because the thin coaxial cable is very flexible. Length of each segment cannot exceed 185 m due to the high level of attenuation in thin coaxial cable.



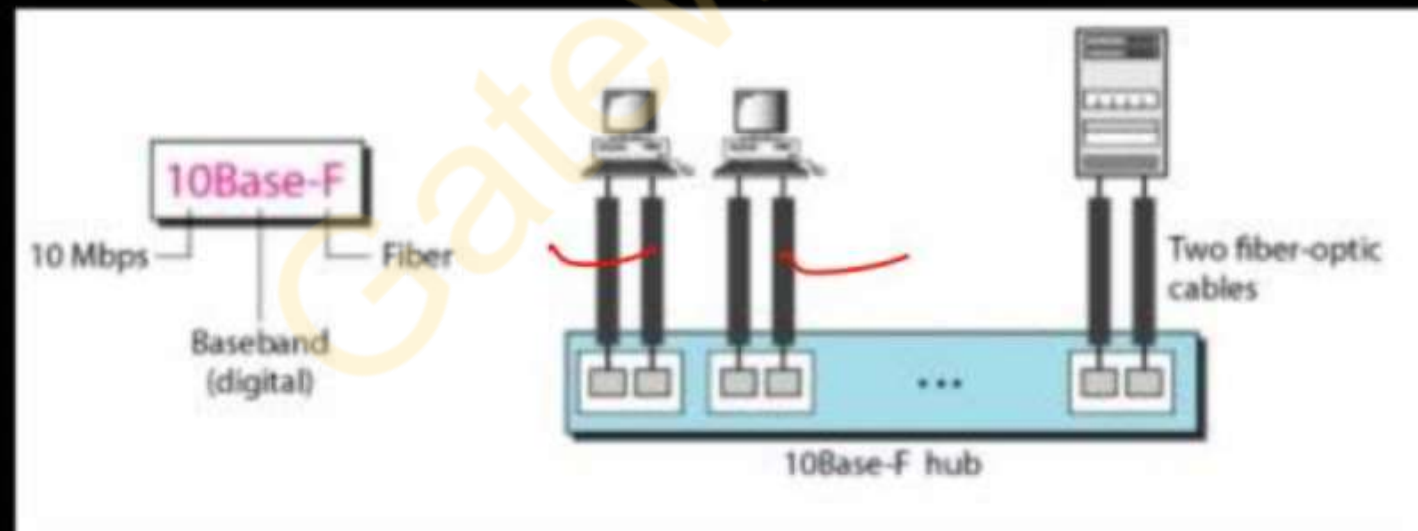
10Base-T: Twisted-Pair Ethernet:

It uses a physical star topology. The stations are connected to a hub via two pairs of twisted cable, the maximum length of the twisted cable is defined as 100 m, to minimize the effect of attenuation in the twisted cable.



10Base-F: Optical Fiber Ethernet:

Uses star topology to connect stations to a hub using two fiber-optic cables.



Summary of Standard Ethernet implementations

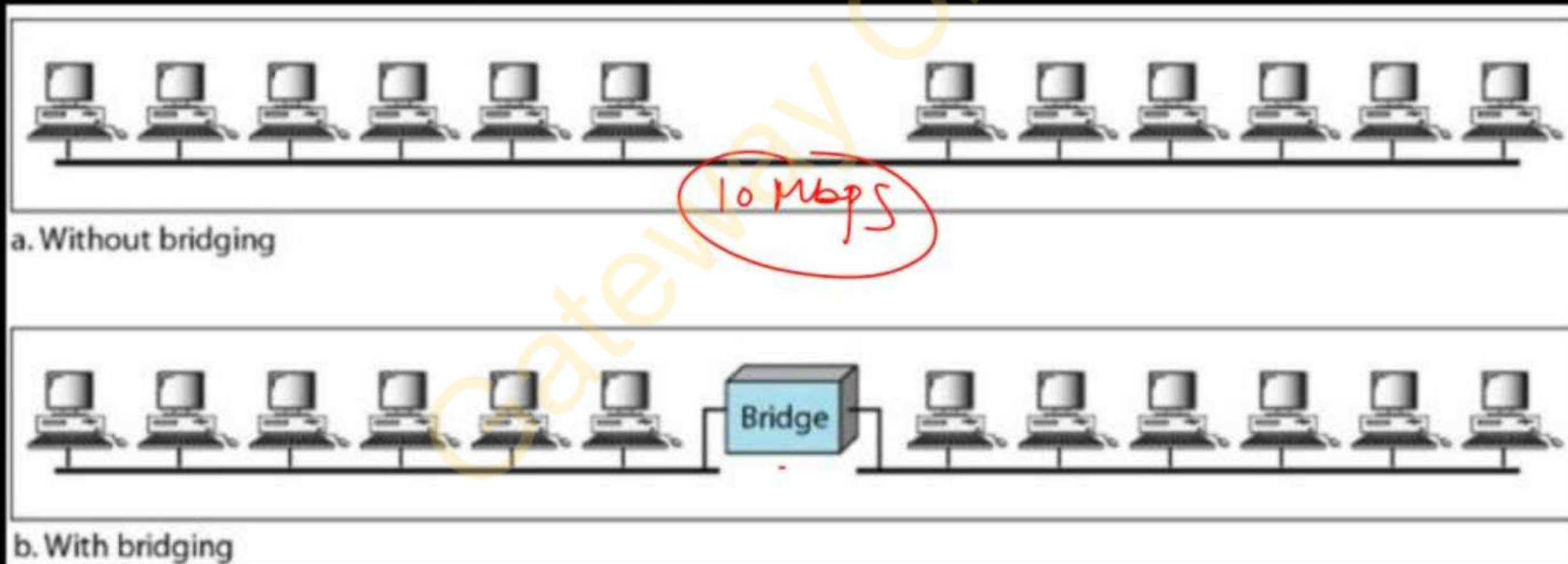
Characteristics	<i>10Base5</i>	<i>10Base2</i>	<i>10Base-T</i>	<i>10Base-F</i>
Media	Thick Coaxial Cable	Thin Coaxial Cable	2UTP	2Fiber
Maximum length	500m	185m	100m	2000m
Line encoding	Manchester	Manchester	Manchester	Manchester

Changes in the Standard: The 10-Mbps Standard Ethernet has gone through several changes to become compatible with other high-data-rate LANs.

1. **Bridged Ethernet:** A Bridge is a two port switch used to connect two segments of a LAN. Bridges raise the bandwidth and separate collision domains on an Ethernet LAN.

Raising the Bandwidth:

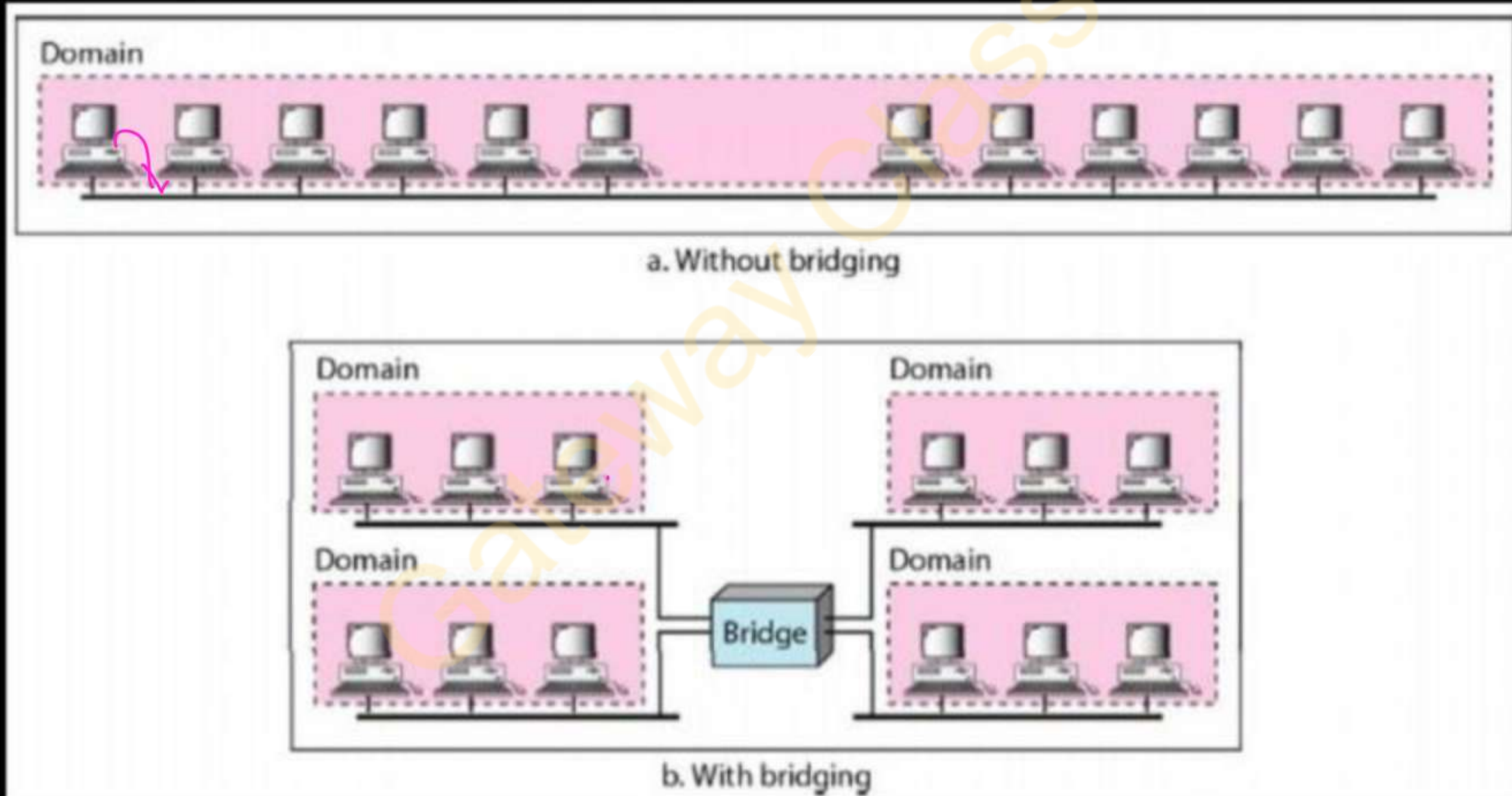
- In an unbridged Ethernet network, the total capacity (10 Mbps) is shared among all stations
- A bridge divides the network into two or more networks. Bandwidth-wise, each network is independent.
- Suppose a network with 12 stations is divided into two networks, each with 6 stations. Now each network has a capacity of 10 Mbps.
- If we further divide the network, we can gain more bandwidth for each segment.



Separating Collision Domains:

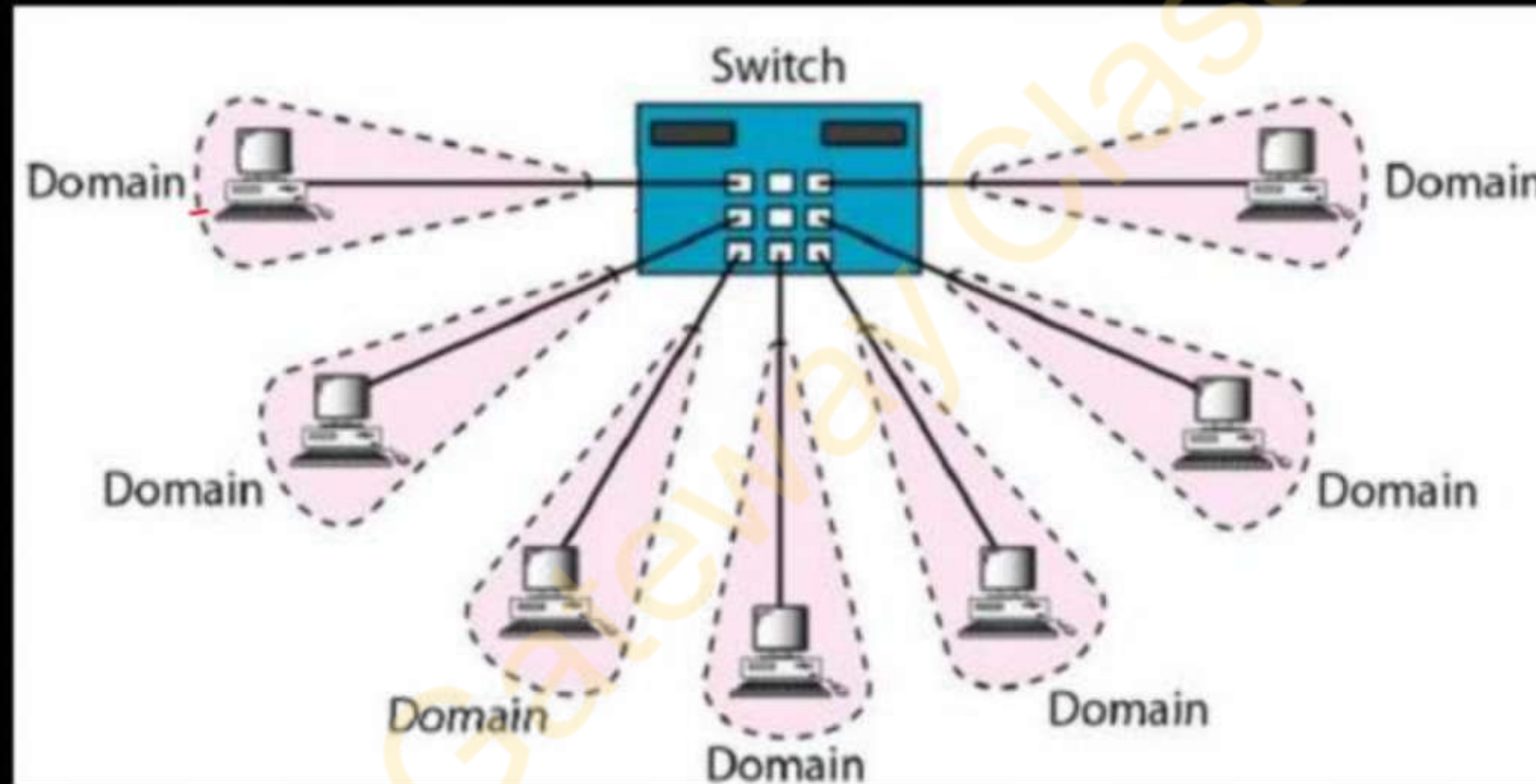
Collision domain becomes much smaller and the probability of collision is reduced tremendously.

Without bridging, 12 stations contend for access to the medium; with bridging only 3 stations contend for access to the medium.



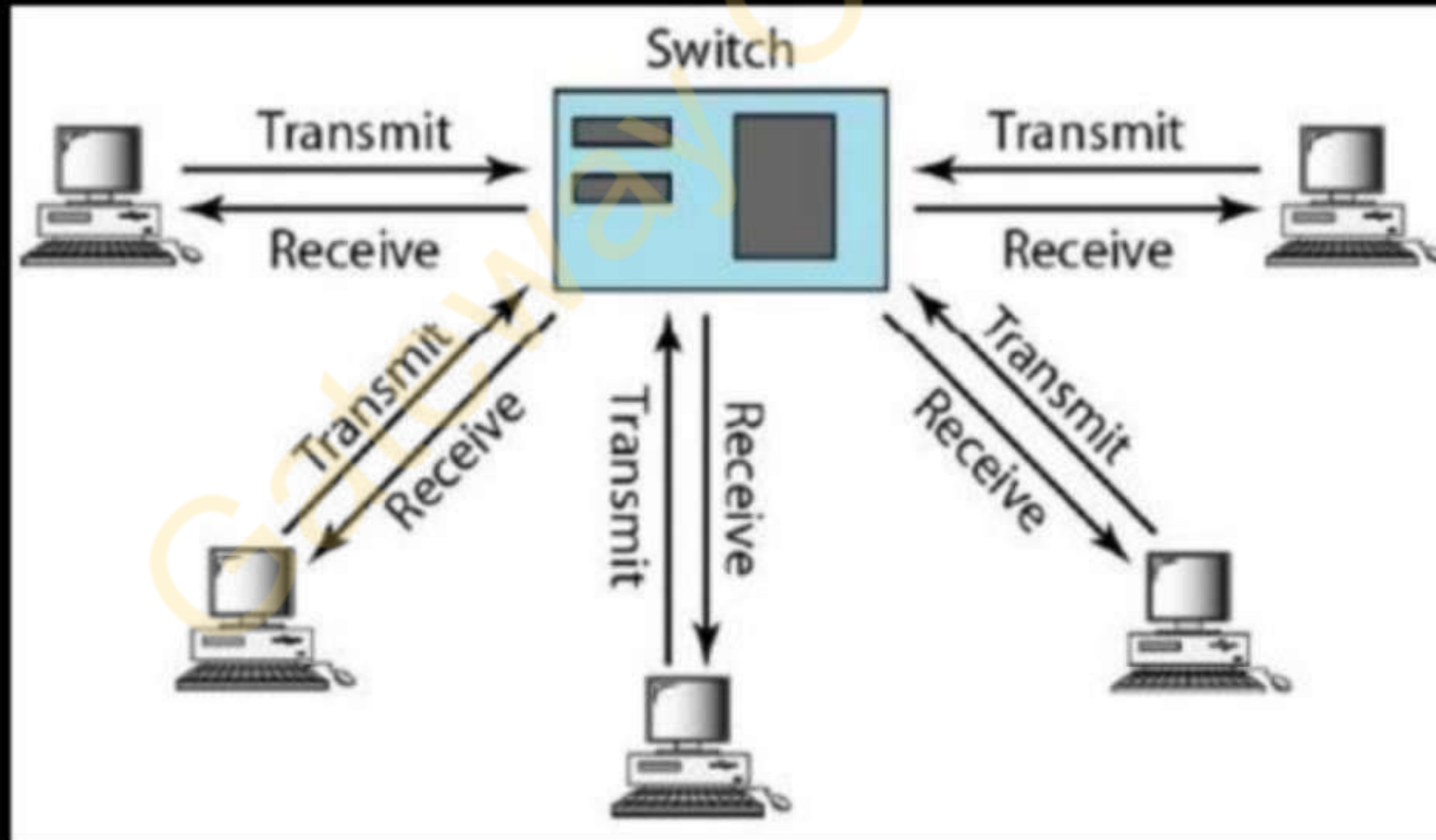
2. Switched Ethernet:

- The idea is to have N networks, where N is the number of stations on the LAN.
- Using a N -port switch, the bandwidth is shared only between the station and the switch.
- The collision domain is divided into N domains.



3. Full-Duplex Ethernet:

- One of the limitations of 10Base5 and 10Base2 is that communication is half-duplex (10Base-T is always full-duplex).
- The full-duplex mode increases the capacity of each domain from 10 to 20 Mbps.
- Instead of using one link between the station and the switch, the configuration uses two links: one to transmit and one to receive. No Need for CSMA/CD.

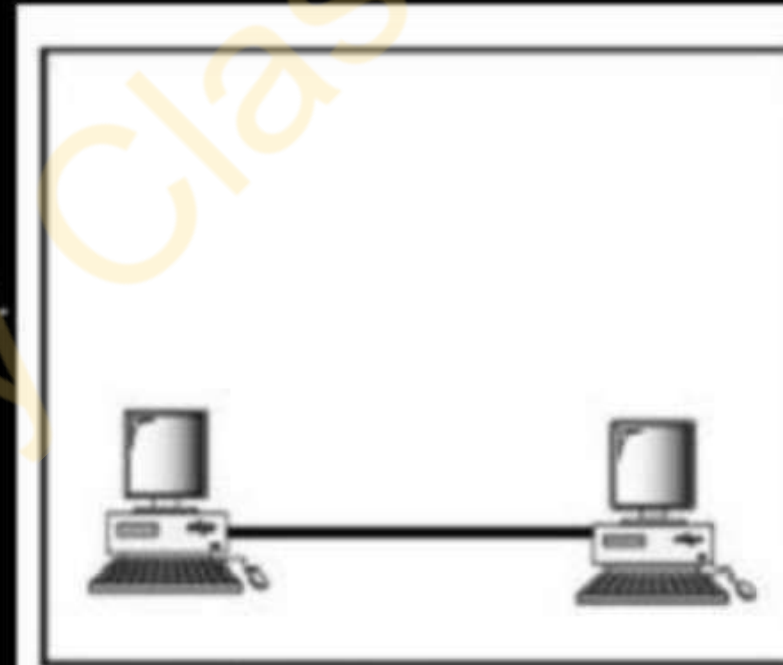


Fast Ethernet(IEEE 802.3u):

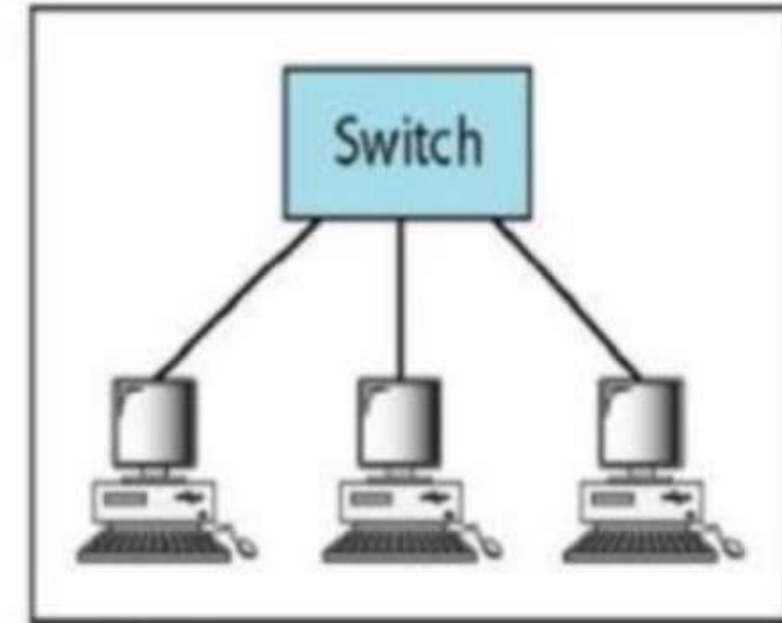
- Designed to compete with LAN protocols such as FDDI.
- Fast Ethernet is backward-compatible with Standard Ethernet, but it can transmit data 10 times faster at a rate of 100 Mbps. ✓

Goals of Fast Ethernet:

- Upgrade the data rate to 100 Mbps.
- Make it compatible with Standard Ethernet.
- Keep the same 48-bit address.
- Keep the same frame format.
- Keep the same minimum and maximum frame lengths.



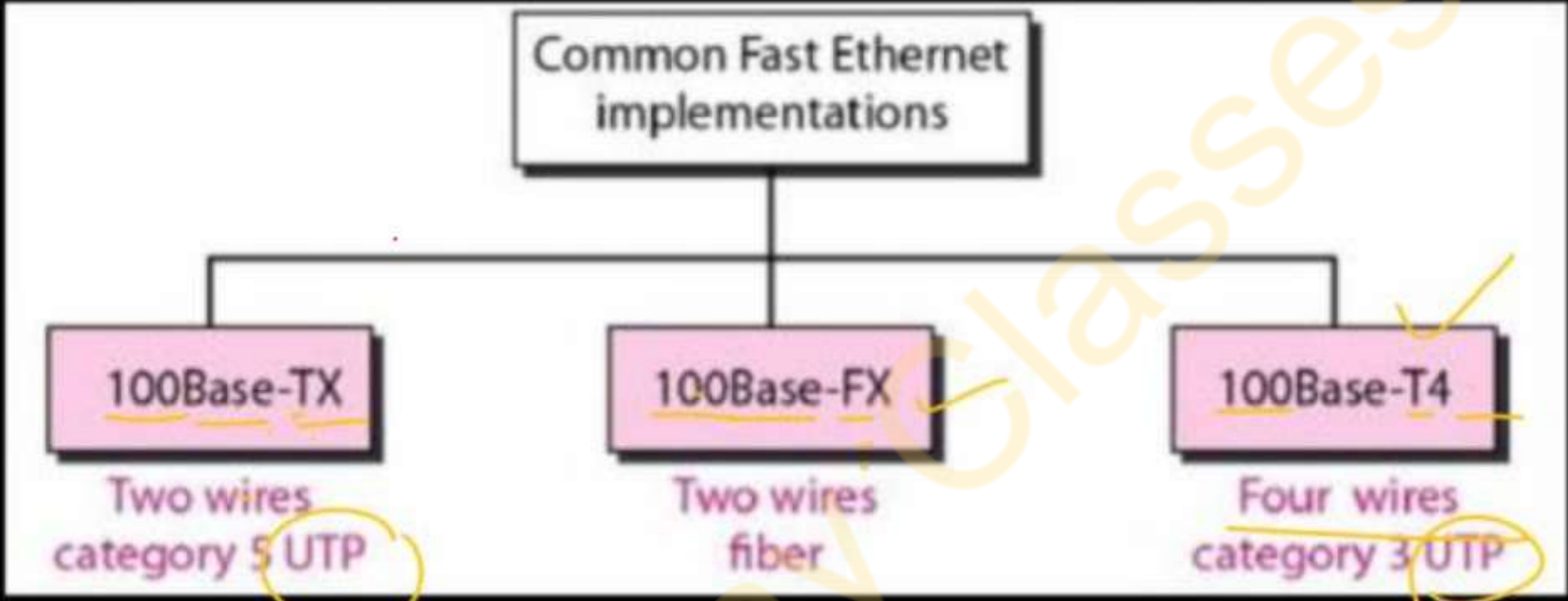
a. Point-to-point



b. Star

Fast Ethernet implementation:

- Two-wire implementation can be either 5 UTP (100Base-TX) or fiber optic cable (100Base-FX).
- The four-wire implementation is designed only for category 3 UTP (100Base-T4).



Characteristics	100Base-TX	100Base-FX	100Base-T4
Media	Cat 5 UTP or STP	Fiber	Cat 4 UTP
Number of wires	2	2	4
Maximum length	100m	100m	100m
Line encoding	MLT-3	NRZ-I	8B/6T

Gigabit Ethernet(IEEE 802.3z):

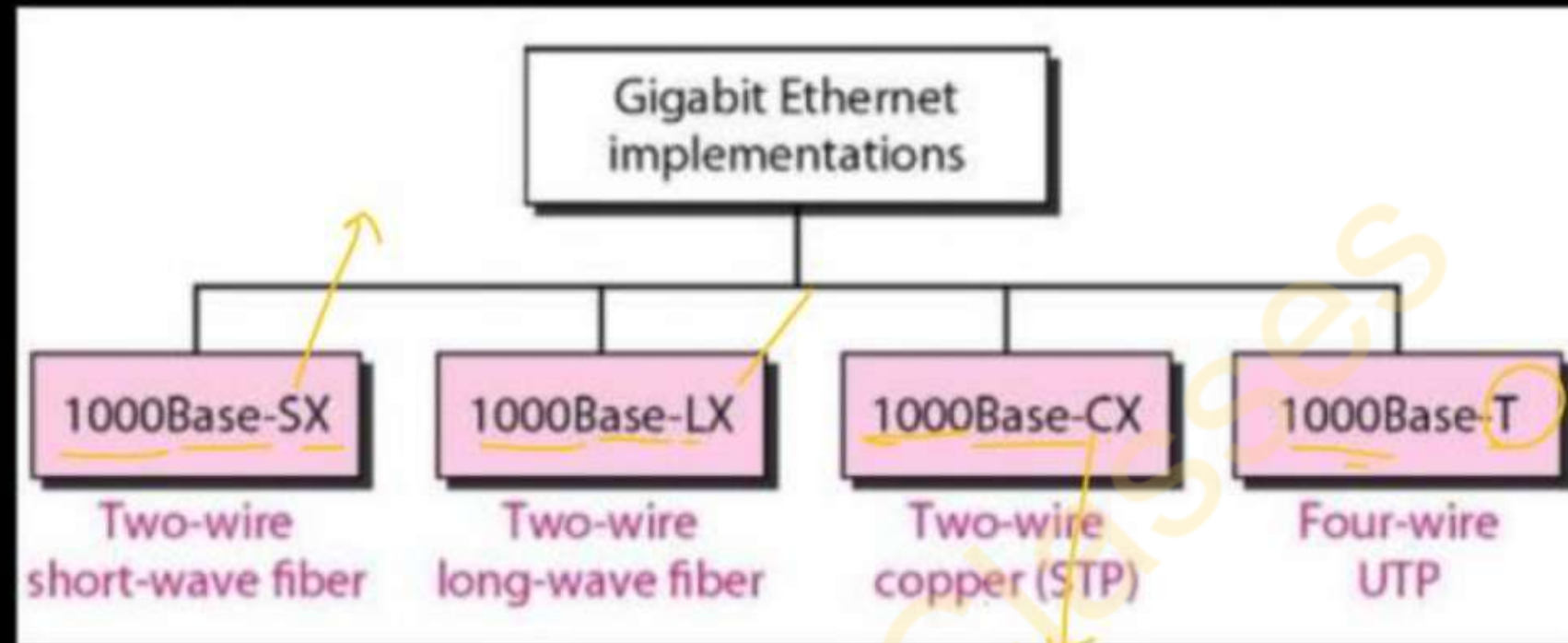
Higher data rate 1000 Mbps.

The goals of the Gigabit Ethernet:

- a. Upgrade the data rate to 1 Gbps. ✓
- b. Make it compatible with Standard or Fast Ethernet.
- c. Use the same 48-bit address.
- d. Use the same frame format.
- e. Keep the same minimum and maximum frame lengths.

Medium access: half-duplex and fullduplex(Mainly used).

Gigabit Ethernet Implementation



Characteristics	1000Base-SX	1000Base-LX	1000Base-CX	1000Base-T
Media	Fiber Short wave	Fiber Long wave	STP	CAT 5 UTP
Number of wires	2	2	2	4
Maximum length	550m	5000m	25m	100m
Line encoding	NRZ	NRZ	NRZ	4D-PAM5

Ten-Gigabit Ethernet(IEEE 802.3ae):

Goals of the Ten-Gigabit Ethernet :

- a. Upgrade the data rate to 10 Gbps.
 - b. Make it compatible with Standard, Fast, and Gigabit Ethernet.
 - c. Use the same 48-bit address.
 - d. Use the same frame format.
 - e. Keep the same minimum and maximum frame lengths.
 - f. Allow the interconnection of existing LANs into MAN or WAN.
 - g. Make Ethernet compatible with technologies such as Frame Relay and ATM.
- Ten-Gigabit Ethernet operates only in full duplex mode which means there is no need for contention; CSMA/CD is not used in Ten-Gigabit Ethernet.

Ten-Gigabit Ethernet implementations

Characteristics	10GBase-S	10GBase-L	10GBase-E
Media	Short-wave 850-nm multimode	Long-wave 1310-nm Single mode	Extended 1550-nm Single mode
Maximum Length	300m	10km	40km



AKTU

B.Tech 6th Semester
CS IT & Allied



Computer Networks

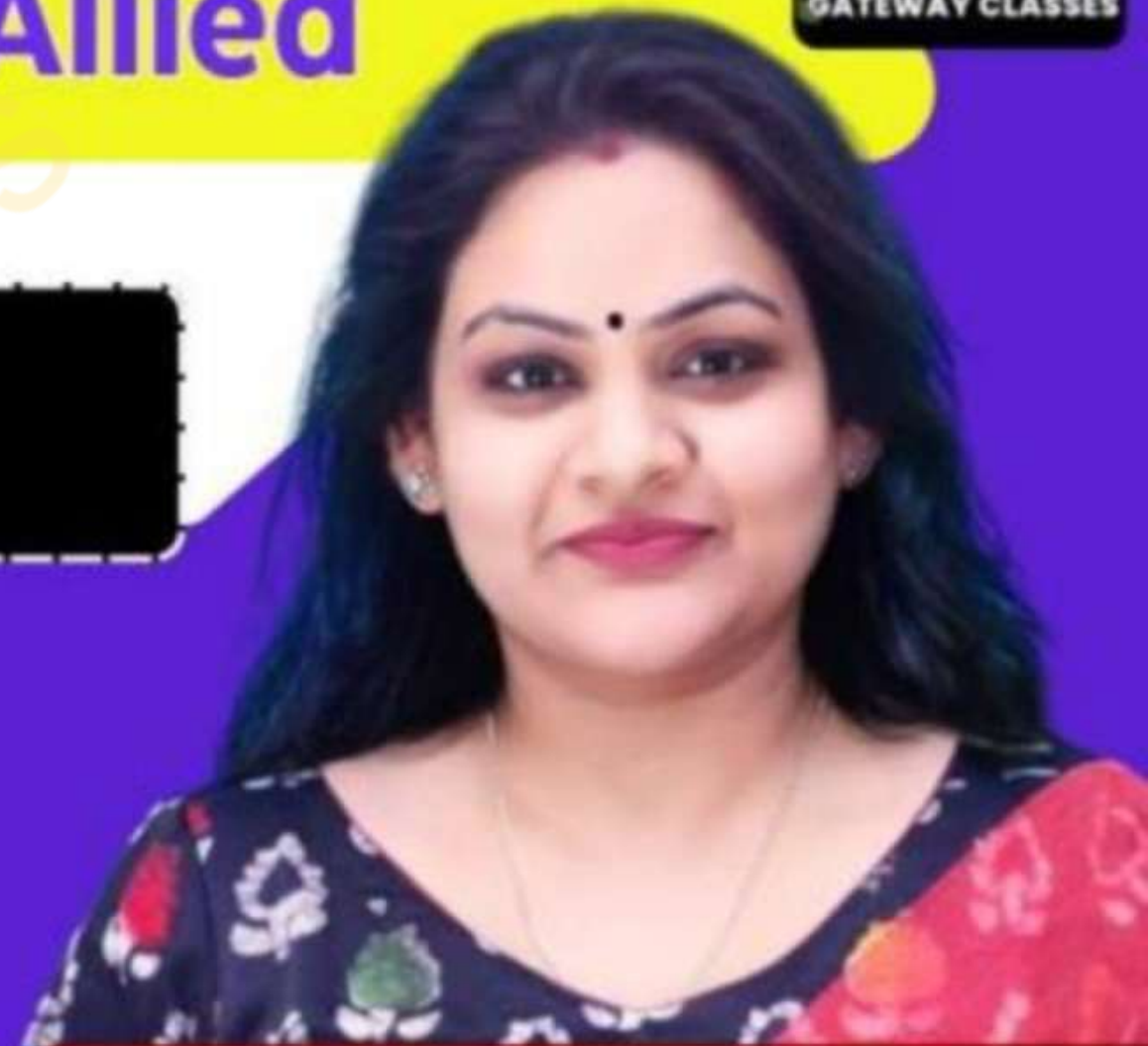
Unit-2: Lecture 10

Today's Target

- Token Bus(IEEE Standard 802.4) ✓
- Token Ring (IEEE Standard 802.5) ✓
- AKTU PYQs ✓

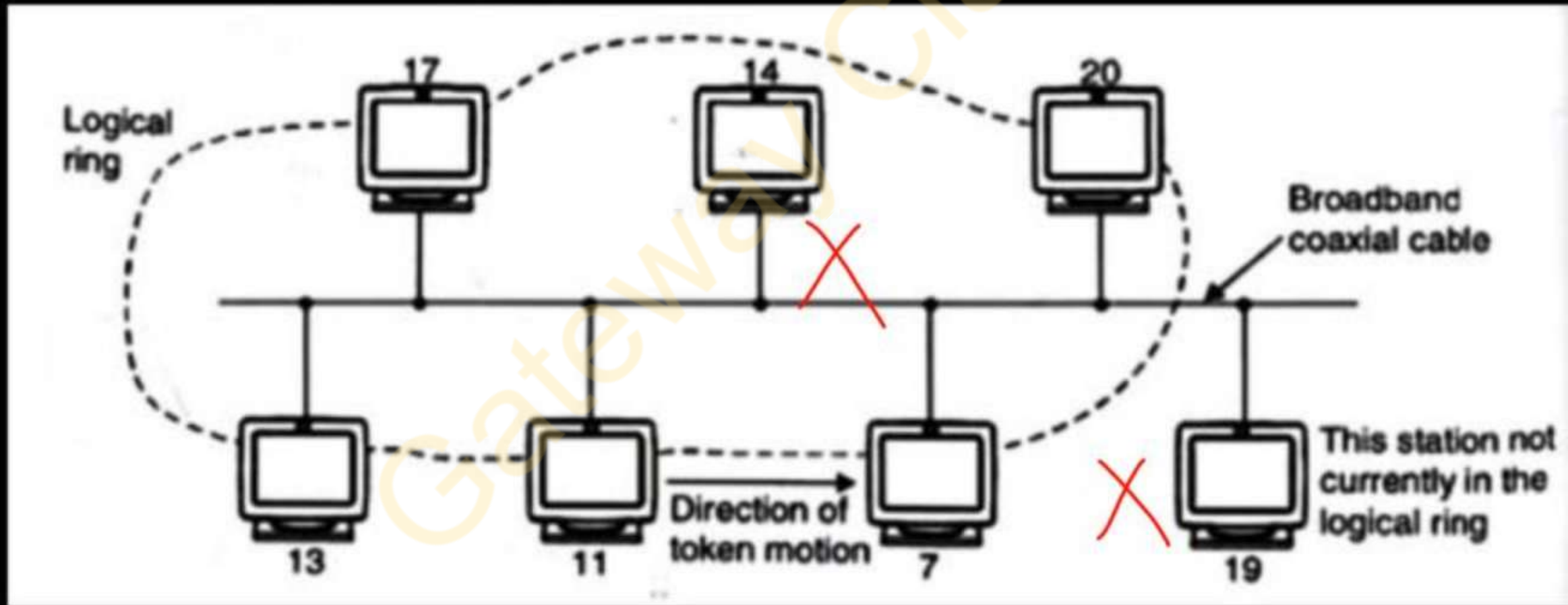
By Dr. Nidhi Parashar Ma'am

- Ph.D (CSE)
- M.Tech Gold Medalist
- Net Qualified



Token Bus (IEEE 802.4)

- Standard for deploying token rings in LANs over a virtual ring.
- The physical medium uses coaxial cables and has a bus or tree architecture.
- The nodes form a virtual ring, and the token is transmitted from one node to the next in a sequence along the virtual ring.



- Each node knows the address of the station before it and the station after it.
- When the logical ring is initialized, the highest numbered station may send the first frame.
- Whenever a station acquires the token, it can transmit frames for a specific amount of time.
- Then it passes the token (special control frame) to its immediate neighbor.
- Only the station holding a token is allowed to transmit data.
- Every station receives each frame; the station whose address is specified in the frame processes it and the other stations discard the frame.
- If a station has no data, it passes the token immediately upon receiving it.
- The token bus defines four priority classes, 0, 2, 4, and 6 for traffic, with 0 the lowest and 6 the highest.
- In running network, new stations may join the ring and older stations may leave the ring.



Token Bus Frame Format

1 byte	1 byte	1 byte	2-6 bytes	2-6 bytes	<u>0-8182 bytes</u>	4 bytes	1 byte
Preamble	Start Delimiter	Frame Control	Destination Address	Source Address	Data	Checksum	End Delimiter

used to maintain/control ring operations

- **Preamble:** Used for bit synchronization.
- **Start Delimiter:** Mark the beginning of the frame.
- **Frame Control:** Specifies the type of frame – data frame and control frames.
- **Destination Address/ Source Address :** Contains the destination address and the source address.
- **Data:** If 2-byte addresses are used then the field may be up to 8182 bytes and 8174 bytes in the case of 6-byte addresses.
- **Checksum:** Used to detect errors in the transmitted data.
- **End Delimiter:** This field marks the end of a frame.

Control Frames

Frame Control field	Name	Meaning
<u>00000000</u>	Claim_token	Claim token during ring initialization
00000001	Solicit_successor_1	Allow station to enter the ring
00000010	Solicit_successor_2	Allow stations to enter the ring
00000011	Who_follows	Recover from lost token. 1
00000100	Resolve_contention	Used when multiple stations want to enter.
00001000	Token	Pass the token
00001100	Set_successor	Allow station to leave the ring.

RING INITIALIZATION

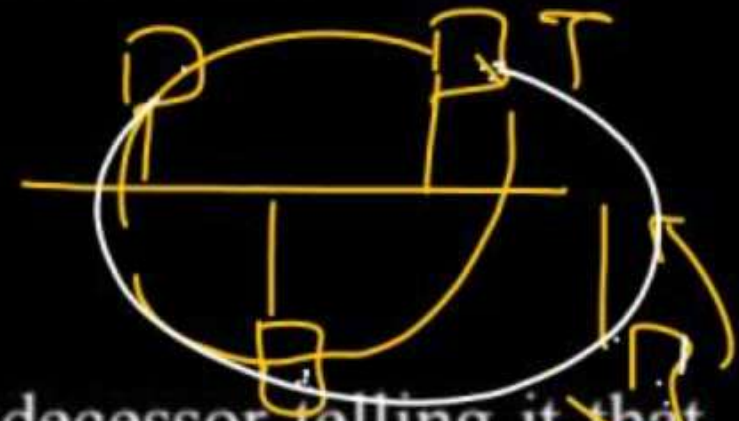
- When all the stations are powered off, there exists no logical ring.
- When the first station is powered-on, it notices no traffic on the network.
- This station then sends a claim_token control frame to hear from anyone alive in the network.
- If it does not listens to any competitions it creates a token and a logical ring that consists of itself only.
- As new stations are powered on, they join the logical ring by using the procedure used to add a station to the logical ring.

ADDITION OF STATION TO LOGICAL RING

- Station holding the token sends solicit_successor frame periodically, to solicit bid from computers that are wishing to enter into the logical ring.
- The station wishing to join the ring, passes its address to the token holder. The token holder makes it as a new Successor.
- If two or more stations want to enter into the logical ring, token holder sends resolve_contention control frame to allow exactly one of them to enter the ring.
- If no station responds for a particular time slot for entering in the ring, response window is closed.

2. REMOVAL OF STATION FROM A LOGICAL RING

- A station leaves the ring by sending a set_successor control frame to its predecessor telling it that from now onwards its successor is the station next to it and not the current station.



STATION FAILURE

- To avoid sending data to failed station, each station looks at station to which it has passed the token.
- If token receiver passes token to its successor or transmits its data, then it is considered all right, otherwise token is passed a second time to the same station.
- If there is no response even after this, the station transmits a who follows control frame with the address of its successor station.
- When a failed station's successor sees a who_follows frame naming its predecessor, it responds by sending a set_successor frame to the station whose successor failed naming itself as a new successor.
- In this way, the failed stations is removed from the ring.
- If the station next to the down station is also down, then the station holding the token sends solicit_successor_2 frame to see anyone alive in the network.

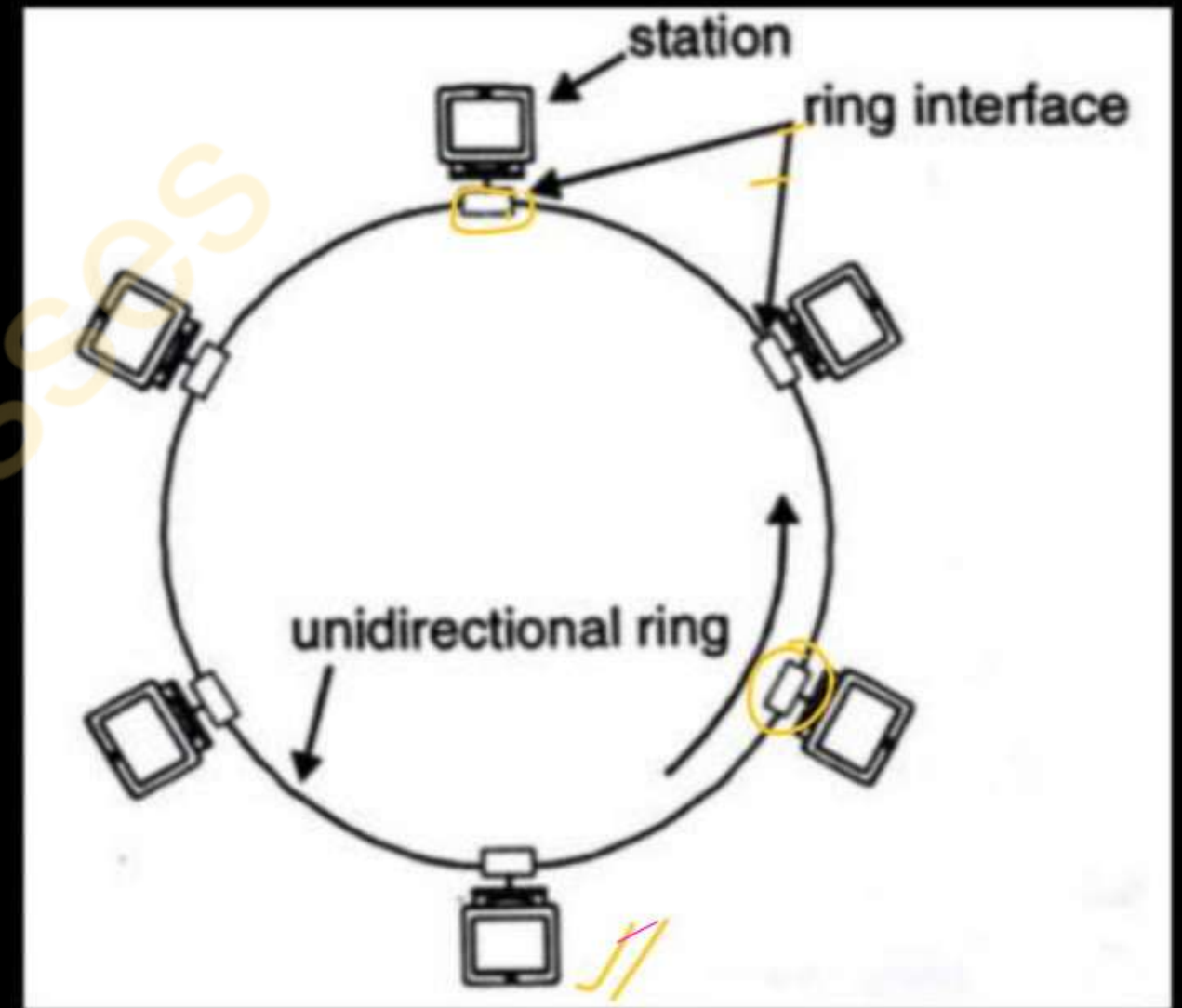
FAILURE OF THE STATION HOLDING THE TOKEN

- If token holder station goes down and takes the token with it, the ring initialization algorithm is run.
- Each station has a timer that is reset whenever frame appears on the network.
- When this timer expires and no token or frame is seen on the network, the Station issues claim_token frame.

Gateway Classes

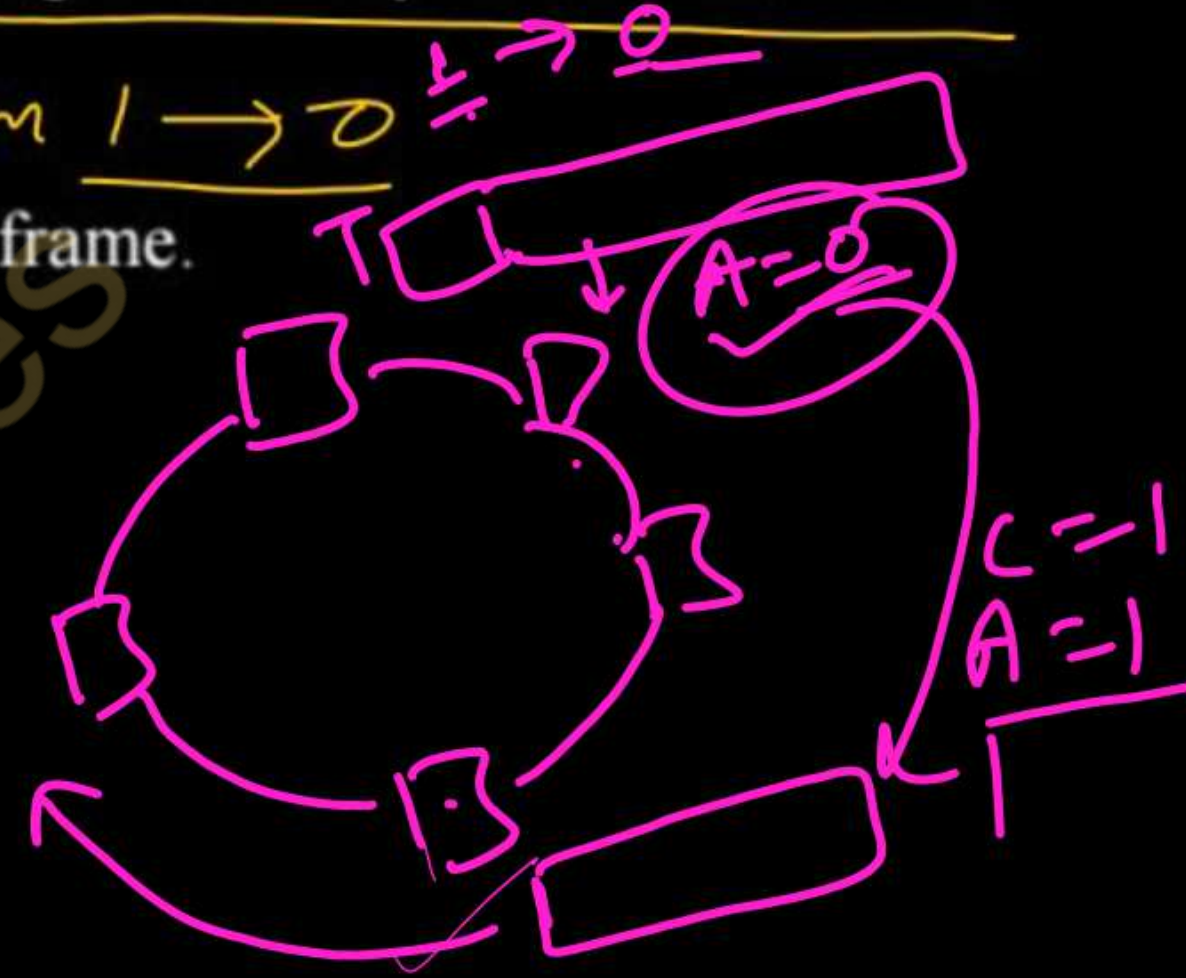
Token Ring(IEEE 802.5)

- It is a communication protocol used in LAN.
- The stations are connected to one another in a single ring.
- A ring consists of ring interfaces. Ring interface of one station is connected to ring interfaces of its left station as well as right station.
- Special 3-byte frame called “token” travels around a ring.
- Node that grabs token, will have right to transmit data.
- Frames are transmitted in the direction of the token. They circulate around the ring and reach the destination.
- Every node gets the token within some specified time so that starvation is avoided.



data frame

- Each bit arriving at an interface is copied into a 1-bit buffer, causing 1-bit delay at each interface.
- The token is seized by changing only one bit in the token. *from 1 → 0*
- Whenever the station transmits data, it adds this token to the data frame.
- No collisions.



Modes of operations of Ring Interface:

1. Listen: Incoming bits are simply transmitted to output.
2. Talk or Transmit: When station connected to ring interface has acquired a token.
3. By-pass Mode: when node is down, any data is just bypassed so no one-bit delay in this mode.

- The sender sends frames to receiver with ACK bit 0.
- The receiver on receiving frames, copies data into its buffer, verifies the checksum and set the ACK bit to 1.
- The verified frames come back to sender, where they are removed from the ring.
- After this transmission, sender creates a new token and places it on the ring.
- A station can hold token for a specific duration of time. During this time, it has to complete its transmission and regenerates the token in ring.
- Whenever a station finishes its transmissions, the other station grabs the token and start its own transmission.
- When there is no traffic on the network, the token keeps on circulating in the ring unless some station captures it by turning the value of a bit to 1 from 0.

Priority and Reservation

- In IEEE 802.5, each station has a priority code.
- As a token circulates on the ring, any station which wants to transmit the frame may reserve the token by entering its priority code in the Access Control (AC) field of the token frame or data frame.

Time Bounding

- A station having token can transmit frames for Token Holding Time (THT) around 10 ms.
- After the THT, the token frame must be handed over to some other station.

The Monitor Station

- The monitor station sets the timer on when each time the token passes.
- If the token does not regenerate in the allotted time, it is assumed to be lost, and then the monitor station generates a new token and sends it to the ring.
- If the monitor fails, a second station is designated as back-up.

Ring Latency – The time taken by a single bit to travel around the ring is known as ring latency.

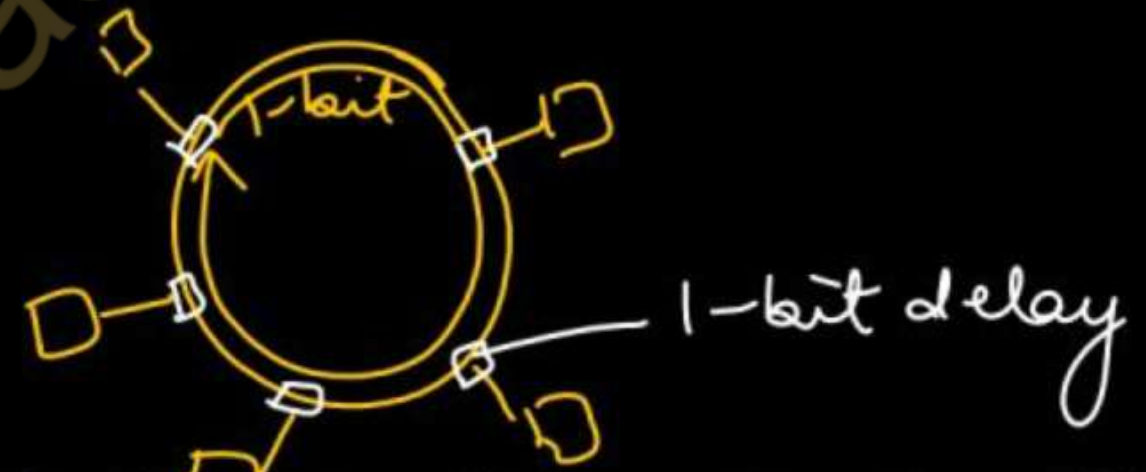
$$\text{Ring Latency} = \boxed{\frac{d}{v}} + \boxed{N \times b}$$

bit delay by each station

Time taken by a bit to travel along the ring

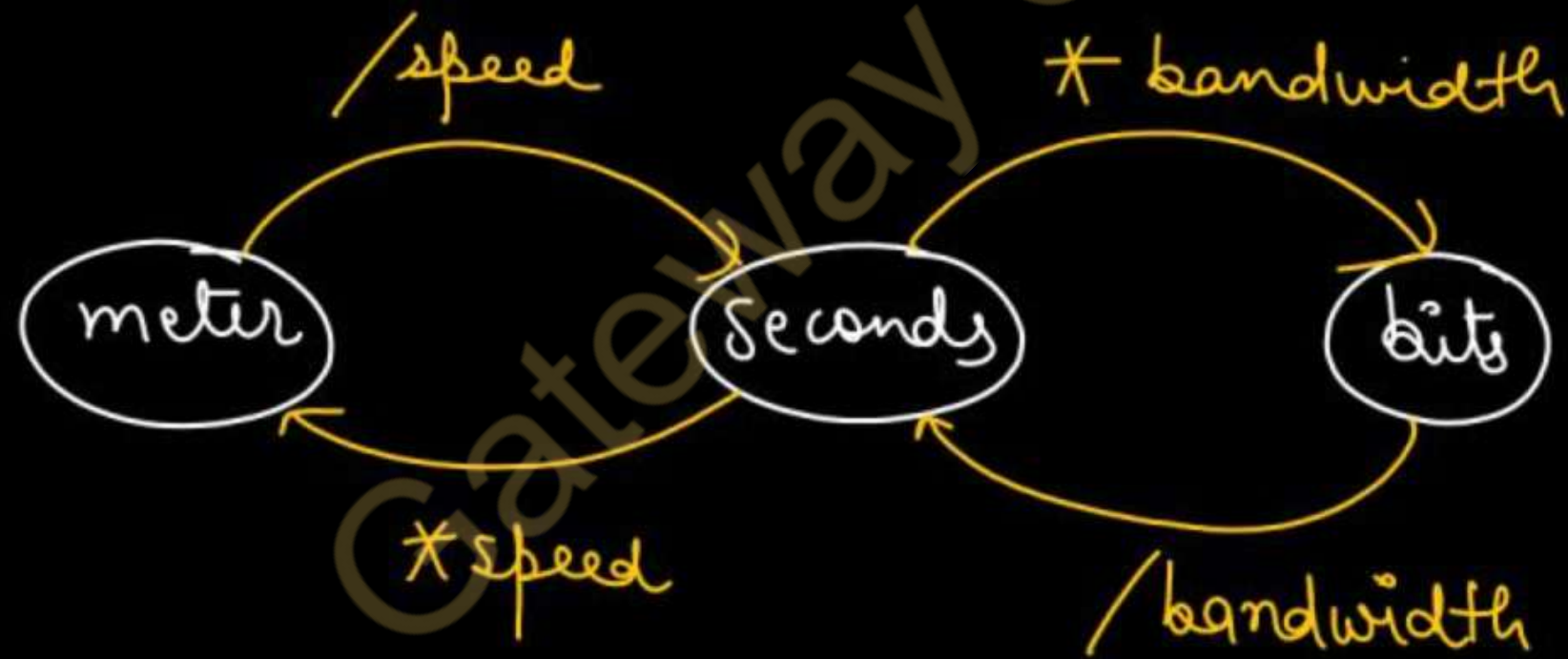
Time taken by stations to hold the bit

- Length of the ring = d
- Speed of the bit = v
- Number of stations = N
- Bit delay at each station = b



(Bit delay is the time for which a station holds the bit before transmitting to the other side)

- d / v is the propagation delay (T_p) expressed in seconds.
- Generally, bit delay is expressed in bits.
- So, both the terms (d / v and $N \times b$) have different units.
- While calculating the ring latency, both the terms are brought into the same unit.
- The conversion chart is used for conversion.



$$\begin{aligned} \text{Ring latency} &= \left(\frac{d}{v} + \frac{N \times b}{B} \right) \text{sec} \\ &= \boxed{\left(T_p + \frac{N \times b}{B} \right) \text{sec.}} \end{aligned}$$

OR

$$\begin{aligned} \text{Ring latency} &= \left(\frac{d \times B}{v} + N \times b \right) \text{bits} \\ &= \boxed{\left(T_p \times B + N \times b \right) \text{bits}} \end{aligned}$$

Cycle Time- Time taken by the token to complete one revolution of the ring is called as **cycle time**.

If-

- Length of the ring = d
- Speed of the bit = v
- Number of stations = N
- Token Holding Time = THT

$$\begin{aligned} \text{cycle time} &= \frac{d}{v} + N \times THT \\ &= T_p + N \times THT \end{aligned}$$

Efficiency-

In one cycle,

- Useful time = Sum of transmission delay of N stations since each station sends 1 data packet = $N \times T_t$
- Total Time = Cycle time = $T_p + N \times THT$

Thus,

$$\text{efficiency}(\eta) = \frac{N \times T_t}{T_p + N \times THT}$$

Imp

Token Passing Strategies:

1. Delayed Token Reinsertion (DTR)

2. Early Token Reinsertion (ETR)

1. Delayed Token Reinsertion:

- Station keeps holding the token until the last bit of the data packet transmitted by it takes the complete revolution of the ring and comes back to it.

$$\text{Token Holding Time (THT)} = \text{Transmission delay} + \text{Ring Latency}$$

$T_t \qquad + \qquad T_f$

We know,

- Ring Latency = $T_p + N \times \text{bit delay}$

- Assuming bit delay = 0 (in most cases), we get-

$$\text{Token Holding Time} = T_t + T_p$$

$$\text{efficiency } (\eta) = \frac{N \times T_t}{T_p + N \times (T_t + T_p)}$$

OR

$$a = \frac{T_p}{T_t}$$

$$\eta = \frac{1}{\frac{a}{N} + (1+a)}$$

2. Early Token Reinsertion-

- Station releases the token immediately after putting its data packet to be transmitted on the ring.

Token Holding Time (THT) = Transmission delay of data packet = T_t

$$\text{efficiency}(\eta) = \frac{N \times T_t}{T_p \times N \times T_t}$$

- **Topology** – Ring topology
- **Transmission** – Unidirectional
- **Encoding** – Differential Manchester encoding
- **Access control** – Token passing
- **Data rates** – 4 Mbps, 16 Mbps

Frame Control

Token Ring Frame Format

SD	AC	FC	Destination Address	Source Address	Data	Checksum	ED	FS
----	----	----	---------------------	----------------	------	----------	----	----

1 byte

1 byte

1 byte

2-6 bytes

2-6 bytes

4 bytes

1 bytes

1 byte

Access control field

unlimited

Data frame

frame status

<u>SD</u>	<u>AC</u>	<u>ED</u>
-----------	-----------	-----------

token

➤ **Start frame delimiter (SFD):** Alerts each station for arrival of token(or data frame) to synchronize clocks.

➤ **Access Control (AC) :**



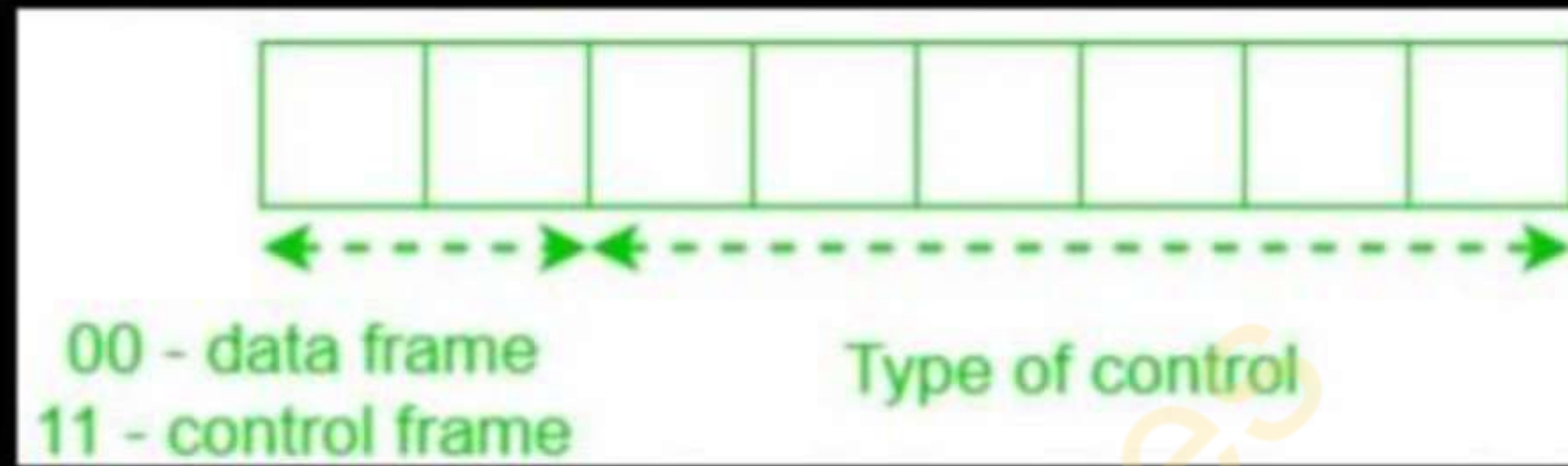
➤ **Priority bits** Determines the priority of a station for accessing token.

➤ **Reservation bits** Allows stations to reserve token for future transmission based on their priority.

➤ **Token Bit:** Indicates whether the frame is a token or a data frame.

➤ **Monitor Bit** is used by the active monitor station to detect continuously circulating frames on the ring.

➤ **Frame control (FC):**



- **DA:** Contains physical address of the frame's subsequent destination.
- **SA:** Contains the physical address of the sending station.
- **Data:** Data length can vary from 0 to maximum token holding time. Token ring imposes no lower bound on size of data i.e. an advantage over Ethernet.
- **Checksum:** Used for error detection.
- **End Delimiter (ED):** Used to mark the end of frame.
- **Frame Status:** Used to terminate a data frame and serves as the ACK. This field can be set by a receiver and indicates whether the MAC address was recognized and the frame was copied.

Frame status:

A	C	X	X	A	C	X	X
----------	----------	----------	----------	----------	----------	----------	----------

$A = 0$ and $C = 0$: Destination not present or not powered up.

$A = 1$ and $C = 0$: Destination present but frame not accepted.

$A = 1$ and $C = 1$: Destination present and frame copied

↓
Ack

↓
copy or not

$\square$ $A=0, C=0$

$\times \square$ $A=1, C=0$
 $A=1, C=1$

Q: A large FDDI ring has 100 stations and a token rotation time of 40 msec. The token holding time is 10 msec. What is the maximum achievable efficiency of the ring? (AKTU)

A:

$$N = \underline{100}$$

Time taken by the token to take a complete Revolution of the ring = $100 \times 10 + 40$
 $= 1040 \text{ ms.}$

$$\eta = \frac{100 \times 10}{1040} = \frac{1000}{1040} \approx \underline{96\%} \quad \underline{\text{Ans}}$$

AKTU PYQs

A large FDDI ring has 100 stations & a token rotation time of 40msec. The token holding time is 10msec. What is the maximum achievable efficiency of the ring?

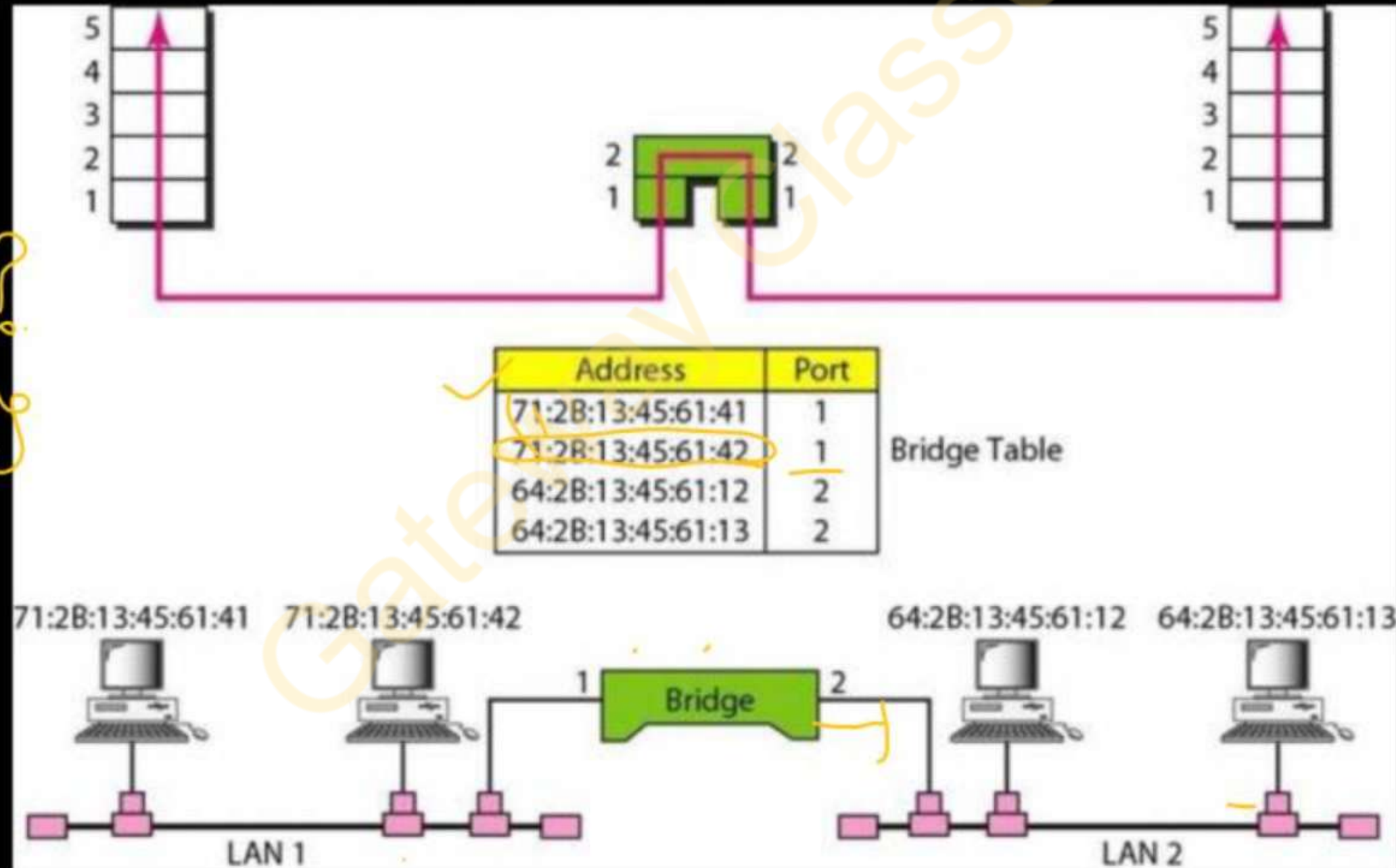
(AKTU 2023-24)

Gateway Classes

Bridges

- Operates in both the physical and the data link layer.
- As a physical layer device, it regenerates the signal it receives. ✓
- As a data link layer device, the bridge can check MAC addresses (source and destination) contained in the frame.

① Filtering?
② Forwarding



Filtering:

- A bridge has filtering capability. It can check the destination address of a frame and decide if the frame should be forwarded or dropped.
- If the frame is to be forwarded, the decision must specify the port.
- A bridge has a table that maps addresses to ports.
- Bridge does not change the physical addresses contained in the frame.

Learning:

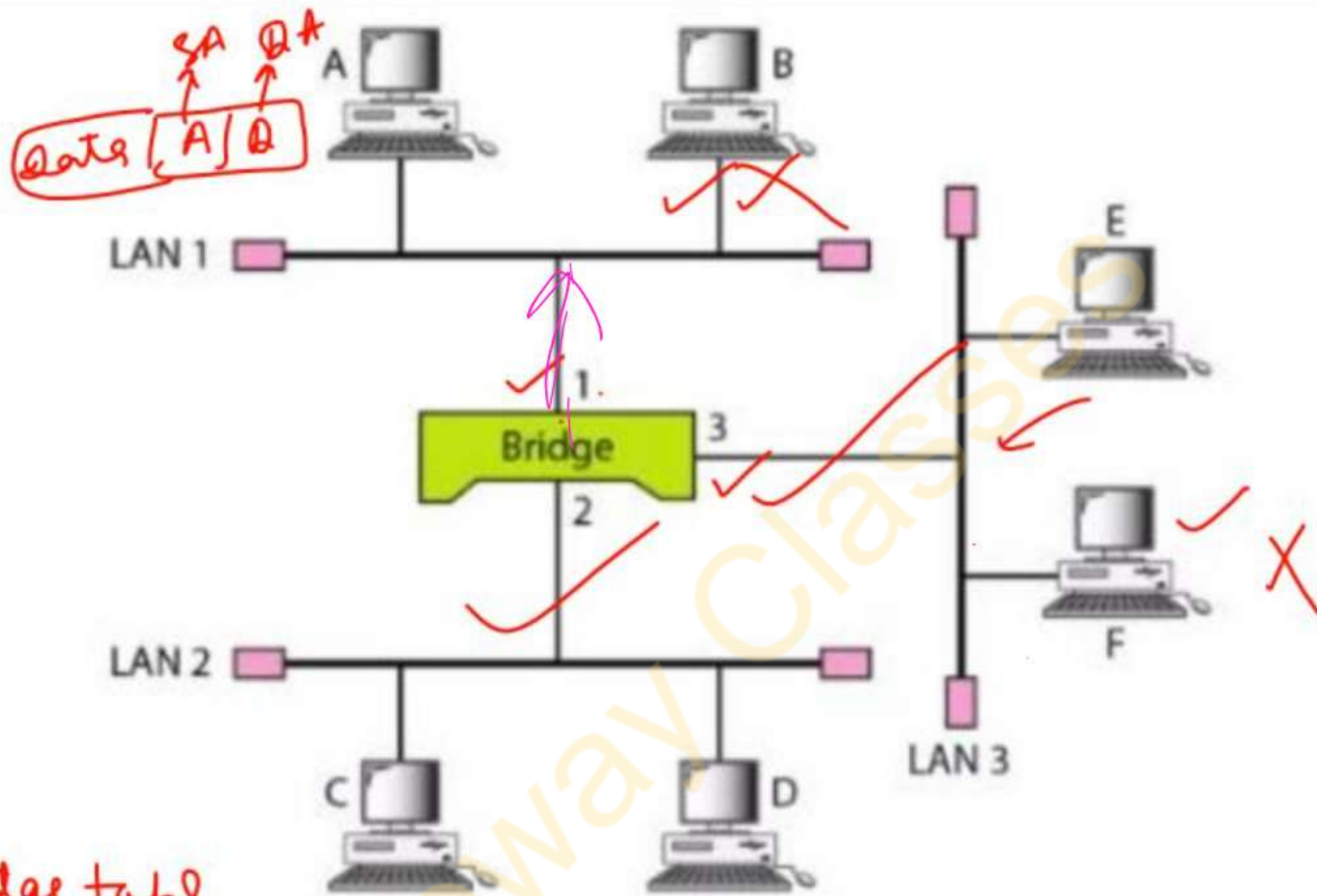
- The earliest bridges had forwarding tables that were static(manually entered by systems Administrator during bridge setup).
- If a station was added or deleted, or station's MAC address changed, the table had to be modified manually.
- A better solution is a dynamic table that maps addresses to ports automatically.

- To make a table dynamic, we need a bridge that gradually learns from the frame movements.

Transparent Bridges

- In a transparent bridge, stations are completely unaware of the bridge's existence.
- If a bridge is added or deleted from the system, reconfiguration of the stations is unnecessary.
- As per IEEE 802.1 d specification, a system equipped with transparent bridges must meet three criteria:
 1. Frames must be forwarded from one station to another.
 2. The forwarding table is automatically made by learning frame movements in the network.
 3. Loops in the system must be prevented.

How bridge learns?



bridge table

Address	Port

a. Original

Address	Port
A	1

b. After A sends a frame to D

Address	Port
A	1
E	3

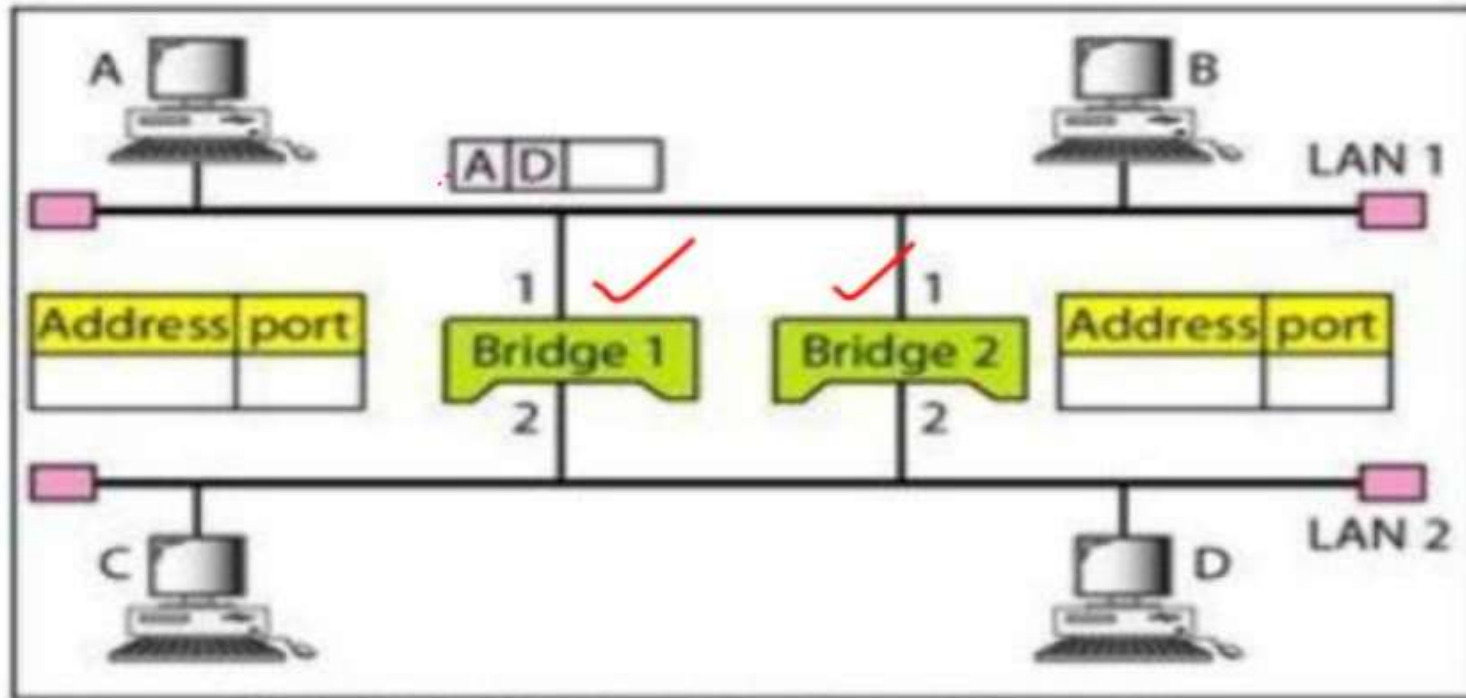
c. After E sends a frame to A

Address	Port
A	1
E	3
B	1

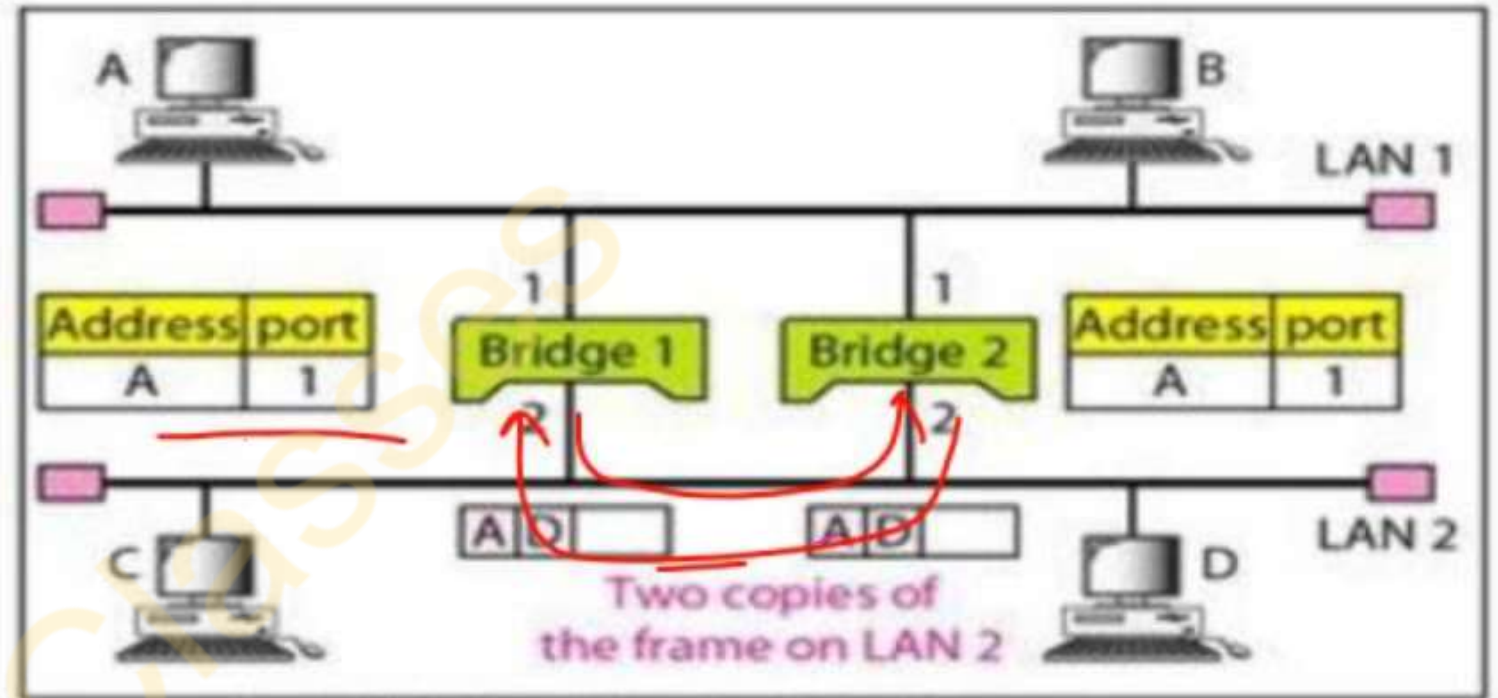
d. After B sends a frame to C

1. When station A sends a frame to station D, the bridge does not have an entry for either D or A. The frame goes out from all three ports. By looking at the source address, the bridge learns that station A must be located on the LAN connected to port 1. This means that frames destined for A, in the future, must be sent out through port 1. The bridge adds this entry to its table.
2. When station E sends a frame to station A, the bridge has an entry for A, so it forwards the frame only to port 1. There is no flooding. In addition, it uses the source address of the frame, E, to add a second entry to the table.
3. When station B sends a frame to C, the bridge has no entry for C, so once again it floods the network and adds one more entry to the table.
4. The process of learning continues as the bridge forwards frames.

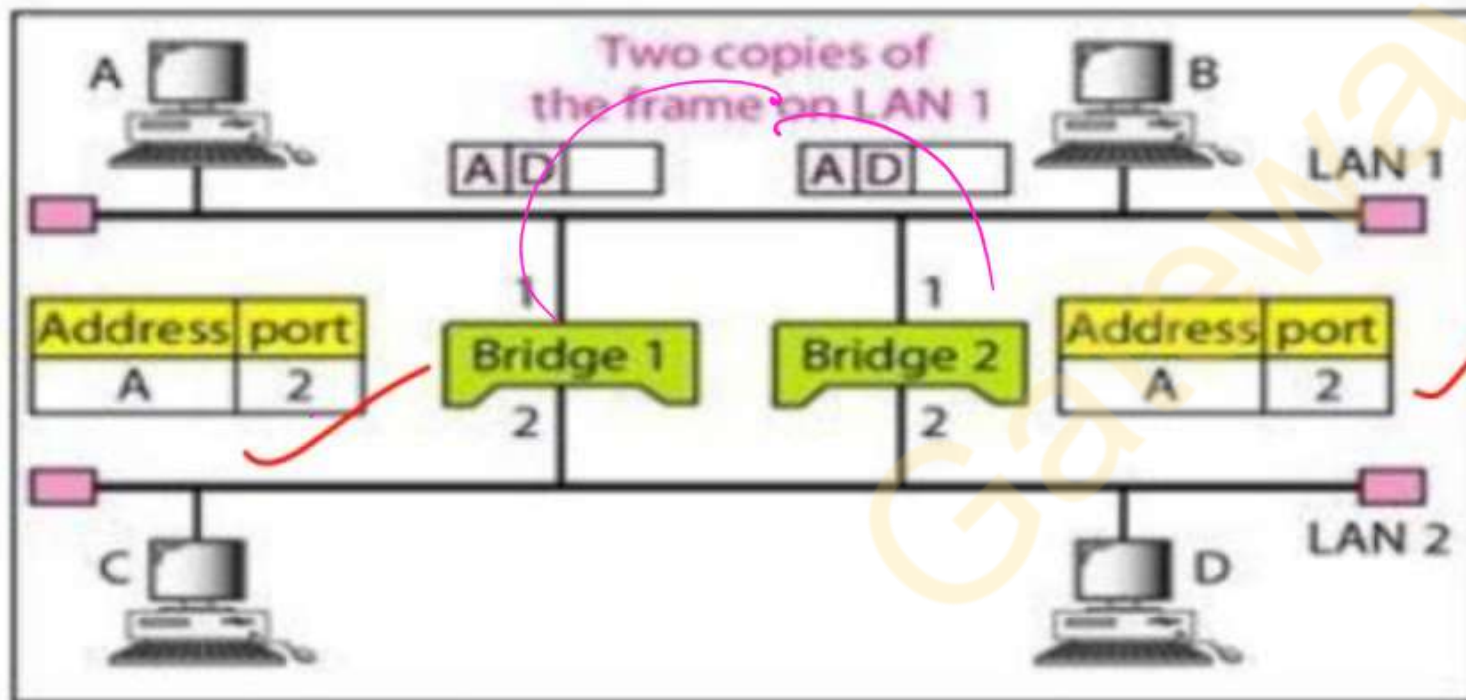
✓ Loop problem in a learning bridge



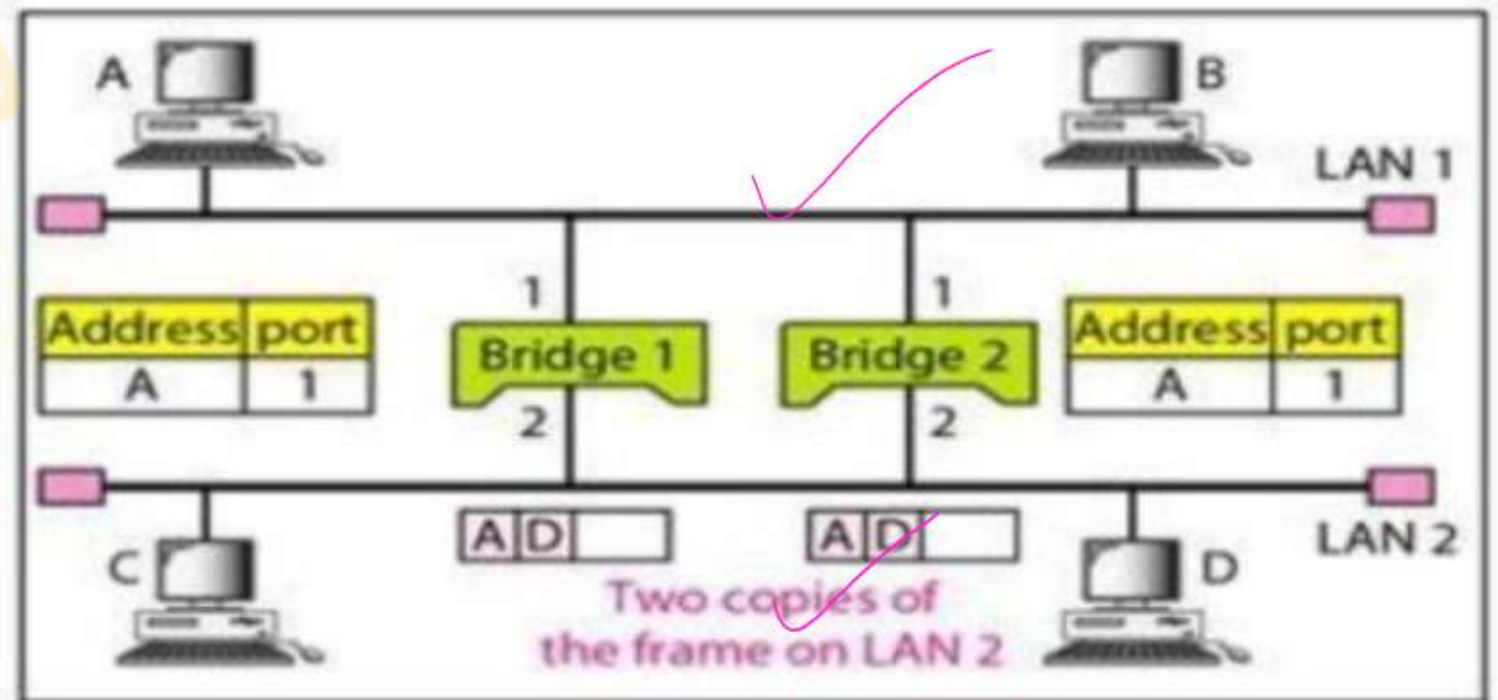
a. Station A sends a frame to station D



b. Both bridges forward the frame



c. Both bridges forward the frame



d. Both bridges forward the frame

Loop Problem:

- Transparent bridges work fine as long as there are no redundant bridges in the system.
- Redundancy can create loops in the system, which is very undesirable.

1. Station A sends a frame to station D. The tables of both bridges are empty. Both forward the frame and update their tables based on the source address A.
2. Now there are two copies of the frame on LAN 2. The copy sent out by bridge 1 is received by bridge 2, which does not have any information about the destination address D; it floods the bridge. The copy sent out by bridge 2 is received by bridge 1 and is sent out for lack of information about D.
3. Now there are two copies of the frame on LAN 1. Step 2 is repeated, and both copies flood the network.
4. The process continues on and on. Bridges are also repeaters and regenerate frames. So, in each iteration, there are newly generated fresh copies of the frames.

Steps for spanning tree Protocol

- 1) Set the lowest ID bridge as root bridge.
- 2) From every other bridge to root bridge $\Rightarrow$ find the least cost path and mark appropriate port as *.
- 3) From every LAN, select any one closest bridge and mark it as **.
- 4) Block all the ports which remained unmarked.

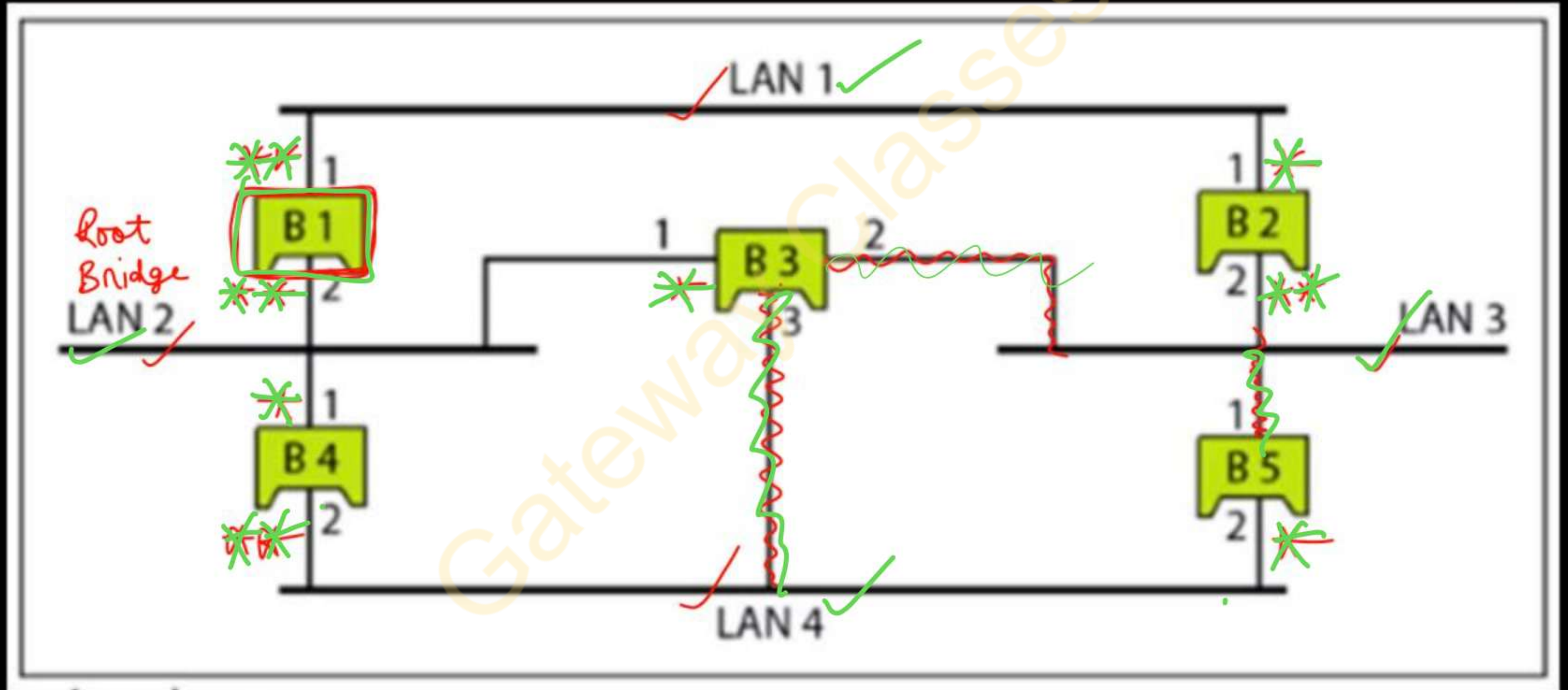
Spanning Tree

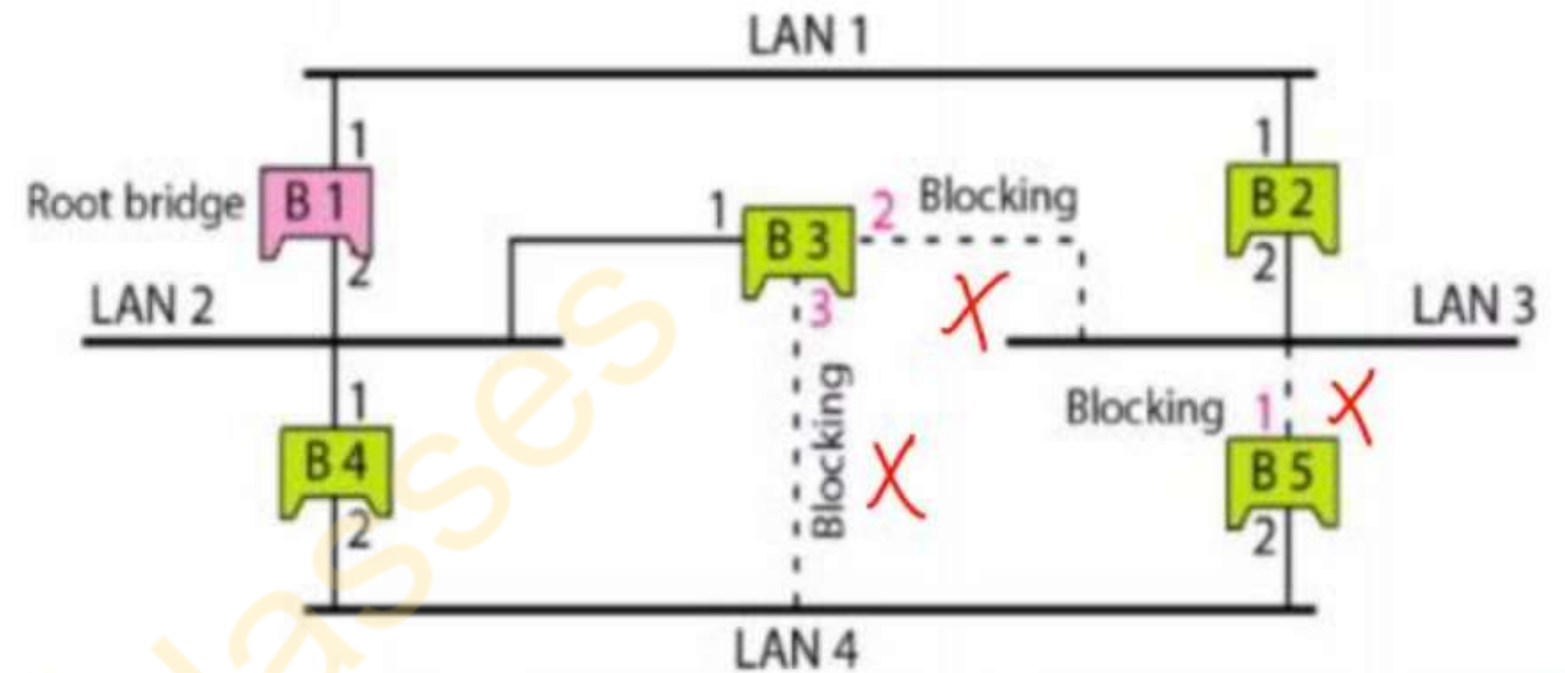
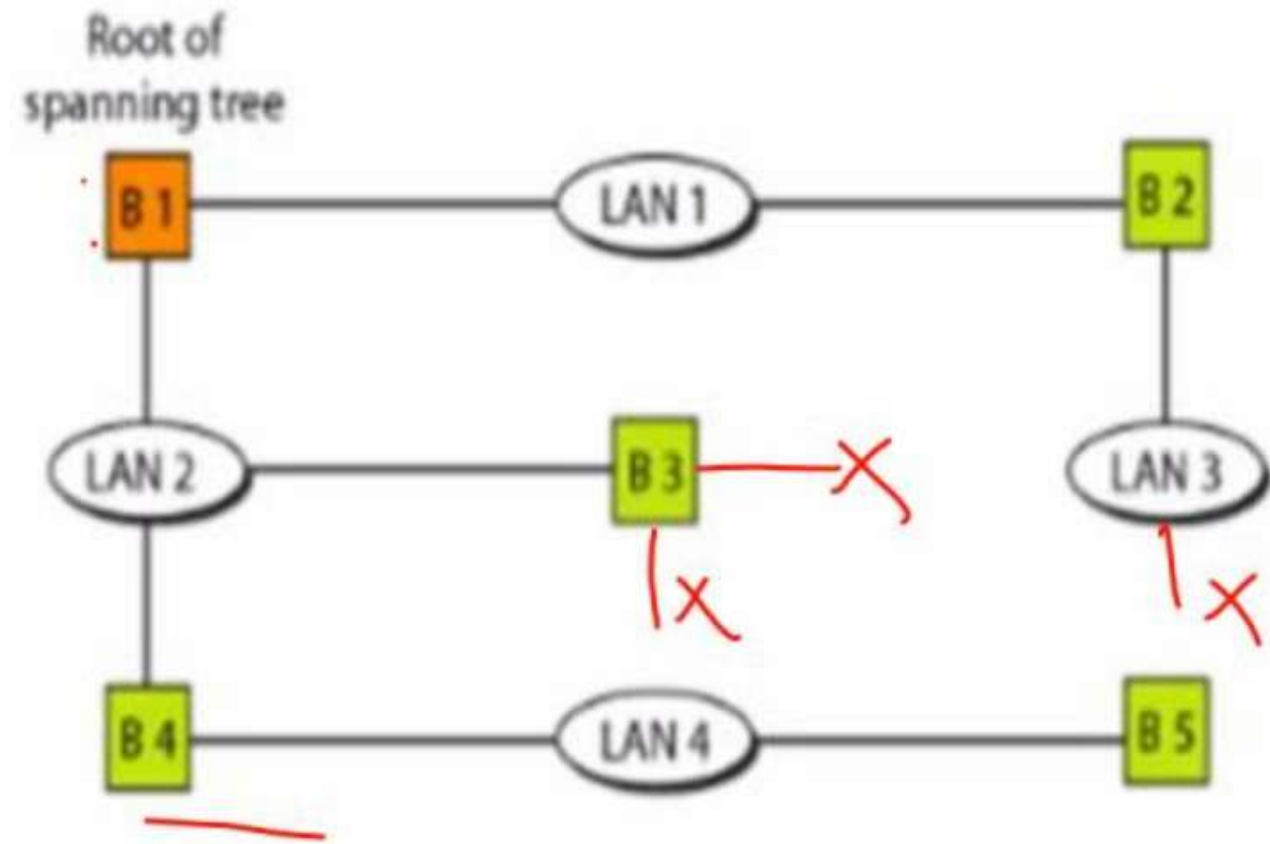
- **Spanning tree** is a graph in which there is no loop.
- In a bridged LAN, this means creating a topology in which each LAN can be reached from any other LAN through one path only (no loop).

The process to find the spanning tree involves three steps:

1. Every bridge has a built-in ID (unique serial number). Each bridge broadcasts this ID so that all bridges know which one has the smallest ID. The bridge with the smallest ID is selected as the *root* bridge (root of the tree)(B1 in example).
2. Algorithm tries to find the shortest path from the root bridge to by examining the total cost from the root bridge to the destination.
3. The combination of the shortest paths creates the shortest tree.

4. Based on spanning tree, we mark ports that are part of spanning tree, **forwarding ports**, which forward a frame that bridge receives. We also mark ports that are not part of the spanning tree, **blocking ports**, which block the frames received by the bridge.





Ports 2 and 3 of bridge B3 are blocking ports (no frame is sent out of these ports). Port 1 of bridge B5 is also a blocking port (no frame is sent out of this port).

- There is only one single path from any LAN to any other LAN.
- No loops are created.
- Each bridge is equipped with a software package to perform this process dynamically.
- The spanning tree is updated when there is a change in the system such as a failure of a bridge or an addition or deletion of bridges.

Source Routing Bridges:

- Another way to prevent loops in a system with redundant bridges.
- In a system that has source routing bridges, **filtering frames, forwarding, and blocking** are performed by the source station and, to some extent, the destination station.
- In source routing, a **sending station defines the bridges that the frame must visit.**
- The **addresses of these bridges are included in the frame.**
- The source gets these bridge addresses through the exchange of special frames with the destination prior to sending the data frame.
- Source routing bridges were designed by IEEE to be used with Token Ring LANs.
- These LANs are not very common today.

AKTU PYQs

1. Explain the working of Bridge. (AKTU 2023-24) ✓
2. Explain any five LAN standards. (AKTU 2021-22)

Gateway Classes

Thank
you